

R61500

528-channel, 262,144-color One-chip Driver
with RAM, Power Supply and Gate Circuits
for Amorphous TFT Panels

REJxxxxxxx-xxxx
Rev.1.0a'
18 November 2004

Description	5
Features	6
Liquid crystal power supply specifications.....	7
Block Diagram	8
Pin Function	9
PAD Arrangement	14
PAD Coordinate.....	15
BUMP Arrangement	17
Block Function	18
1. System Interface.....	18
2. External Display Interface	18
3. Bit Operations.....	19
4. Address Counter (AC)	19
5. Graphics RAM (GRAM)	19
7. Timing Generator	19
8. Oscillator (OSC).....	19
9. LCD Driver Circuit	20
10. LCD Drive Power Supply Circuit	20
GRAM Address MAP	21
Instructions	29
Outline.....	29
Index (IR)	31
Status Read (SR).....	31
Start Oscillation (R00h).....	31
Driver Output Control (R01h)	32
LCD Driving Waveform Control (R02h).....	35

Entry Mode (R03h) Compare register 1 (R04h) Compare register 2 (R05h).....	36
Display Control 1 (R07h).....	40
Display Control 2 (R08h).....	42
Display Control 3 (R09h).....	44
Frame Cycle Control (R0Bh).....	46
External Display Interface Control 1 (R0Ch).....	50
Driver Output Control 1 (R0Dh) Driver Output Control 2 (R0Eh).....	53
External Display Interface Control 2 (R0Fh).....	55
Power Control 1 (R10h) Power Control 2 (R11h).....	56
Power Control 3 (R12h) Power Control 4 (R13h).....	59
RAM Address Set (R21h).....	61
Write Data to GRAM (R22h).....	62
RAM Access via RGB-I/F and System I/F.....	67
Read Data Read from GRAM (R22h).....	68
RAM Write Data Mask (R23h) RAM Write Data Mask (R24h).....	71
γ Control (R30h to R39h).....	72
Gate Scan Position (R40h).....	73
Vertical Scroll Control (R41h).....	75
1st-Screen Drive Position (R42h), 2nd-Screen Drive Position (R43h).....	75
Horizontal RAM Address Position (R44h), Vertical RAM Address Position (R45h).....	76
 Instruction List.....	 77
 Reset Function.....	 78
 Interface Specifications.....	 80
 System Interface.....	 82
80-system 18-bit interface.....	83
80-system 16-bit interface.....	84
80-system 9-bit interface.....	85
Data transfer synchronizing in 9-bit bus interface mode.....	86
80-system 8-bit interface.....	87
Data transfer synchronization in 8-bit bus interface mode.....	89
Serial interface.....	90
 VSYNC Interface.....	 95
Notes to VSYNC interface.....	97
 External Display Interface.....	 99
RGB interface.....	99
ENABLE signal.....	100
RGB interface timing.....	101
Moving picture display.....	103
RAM access via a system interface in RGB-I/F mode.....	103
6-bit RGB interface.....	105
Data transfer synchronization in 6-bit RGB interface mode.....	106
16-bit RGB interface.....	107

R61500

18-bit RGB interface	108
Notes to external display interface	109
LCD panel Interfacing timing	111
RGB I/F mode	111
Internal clock operation mode	112
Scan Mode Setting	113
High-Speed RAM Write Function	114
Notes to high-speed RAM write function	116
High-speed RAM write operation in a window address area	117
Window Address Function	119
Graphics Operation Function	120
Write-data Mask Function	121
Graphics Operation Processing Examples	122
γ -Correction Function	126
Grayscale Amplifier unit configuration	127
γ -Correction Register	129
Ladder resistors and 8-to-1 selector	130
Variable resistors	130
8 to 1 selector	131
Relationship between RAM data and voltage output levels	136
8-color Display Mode	137
Power supply circuit Configuration	139
Specification of External Elements Connected to the R61500 Power Supply	141
Instruction Setting	142
Display ON/OFF	142
Standby and Sleep mode	143
Power Supply Setting	144
Pattern Diagram for Voltage Setting	145
Oscillator	146
n-line inversion AC drive	147
Interlaced scan	148
Alternating Timing	149

Frame-Frequency Adjustment Function	150
<i>Relationship between Liquid Crystal Drive Duty and Frame Frequency.....</i>	<i>150</i>
Partial Display Function	151
<i>Constraints in Setting the 1st/2nd Screen Drive Position Register.....</i>	<i>152</i>
Panel wiring example.....	155
Absolute Maximum Ratings	156
Electrical Characteristics	157
DC Characteristics	157
Step-up circuit Characteristics	159
AC Characteristics.....	159
80-system Bus Interface Timing Characteristics (18/16-bit interface mode).....	160
80-system Bus Interface Timing Characteristics (8/9-bit interface mode).....	161
Serial interface timing characteristics.....	161
Reset Timing Characteristics.....	162
RGB Interface timing characteristics	163
LCD driver output Characteristics.....	164
Notes to Electrical Characteristics.....	165
Example of test circuits.....	168
Timing characteristics diagram.....	168

Description

The R61500 is a 262,144-color, one-chip controller driver LSI for a TFT liquid crystal display of 176 RGB x 240 dots at maximum, comprising a 528-channel source driver, RAM for graphics data of 176 RGB x 240 dots at maximum, a gate driver and a power supply circuit.

The R61500 supports high-speed parallel interfaces via 8-, 9-, 16-, 18-bit ports and a function to write RAM data in high speed for transferring data efficiently and rewriting RAM graphics data in high speed. In addition, the R61500 supports 6-, 16-, 18-bit RGB interfaces (VSYNC, HSYNC, DOTCLK, ENABLE, and DB17-0) and a VSYNC interface (system interface + VSYNC) for displaying a moving picture, which, with use of window address function, enable the R61500 to display a moving picture easily at the position specified by setting and still pictures in other areas on the screen simultaneously. Since this combination allows transferring only moving picture data while retaining still picture data in the internal RAM, data transfer and the power consumption by the entire system can be minimized.

The R61500 can operate with low I/O interface power supply up to 1.65V, with an internal voltage follower circuit to generate voltage levels for driving the LCD. The R61500 also supports a function to display in 8 colors and the standby mode, allowing for precise power control by software. These features make the R61500 an ideal driver for the medium or small sized portable products supporting WWW browsers such as digital cellular phones or small PDAs, where long battery life is a measure concern.

Features

- A controller driver for a liquid crystal TFT display of 176RGB x 240dots graphics display in 262,144 colors
- Single chip solution for a liquid crystal TFT display
- System interface
 - High-speed interfaces via 8-, 9-, 16-, 18-bit parallel ports
 - Serial interface
- Interfaces for moving picture display
 - 6-, 16-, 18-bit RGB interfaces (VSYNC, HSYNC, DOTCLK, ENABLE, DB17-0)
 - VSYNC interface (System interface + VSYNC)
- High-speed RAM write function
- Window address function to specify a rectangular area on the internal RAM to write data
 - Facilitates moving picture display at the position specified by setting via a moving picture display interface
 - Limits the data rewriting area and reduce the total data transfer
 - Displays both moving and still pictures at a same time
- Bit operation functions for graphics data processing
 - Bit-unit write data mask function
 - Pixel-unit logical/conditional write function
- Abundant control functions for color display
 - γ -correction function for 262,144-color display
 - Line-unit vertical scrolling function
- Low -power consumption architecture
 - Low operating power supplies: $V_{cc} = 2.4V \sim 3.6 V$ (internal logic)
 $IOV_{cc} = 1.65V \sim 3.6 V$ (interface I/O)
 $V_{ci} = 2.5V \sim 3.3 V$ (analog)
 - Low voltage drive: $DDVDH = 4.5V \sim 5.5 V$
 - Power saving functions (standby mode etc.)
 - Partial drive function to display at two positions specified by setting
 - A voltage follower circuit to generate LCD driving voltage levels with a small direct current through bleeder resistors
- Internal step-up circuits to step up the LCD drive voltage level up to 6 times (x6)
- 95,040 byte internal RAM
- Internal 528-channel source driver and 240-channel gate driver
- n-line liquid crystal AC drive: invert polarity at an interval of n lines (n: 0 ~ 64)
- Internal oscillator and hardware reset
- Reversible source output shift direction
- For Cst only

Liquid crystal power supply specifications**Table 1**

No.	Item	R61500
1	TFT data lines	528 pins (176 x RGB)
2	TFT gate lines	240 pins
3	TFT display storage capacitor	Cst only (Common Vcom formula)
4	Liquid crystal drive output	S1~S528
		G1~240
		Vcom1, 11, 2, 21
		Change the VcomH-VcomL amplitude with electronic volume, or change the VcomH level with an external resistor (=VcomR)
5	Input voltage	IOVcc
		Vcc
		Vci
6	Internal step-up circuits	VLOUT1 (DDVDH)
		VLOUT2 (VGH)
		VLOUT3 (VGL)
		VLOUT4 (VCL)

Block Diagram

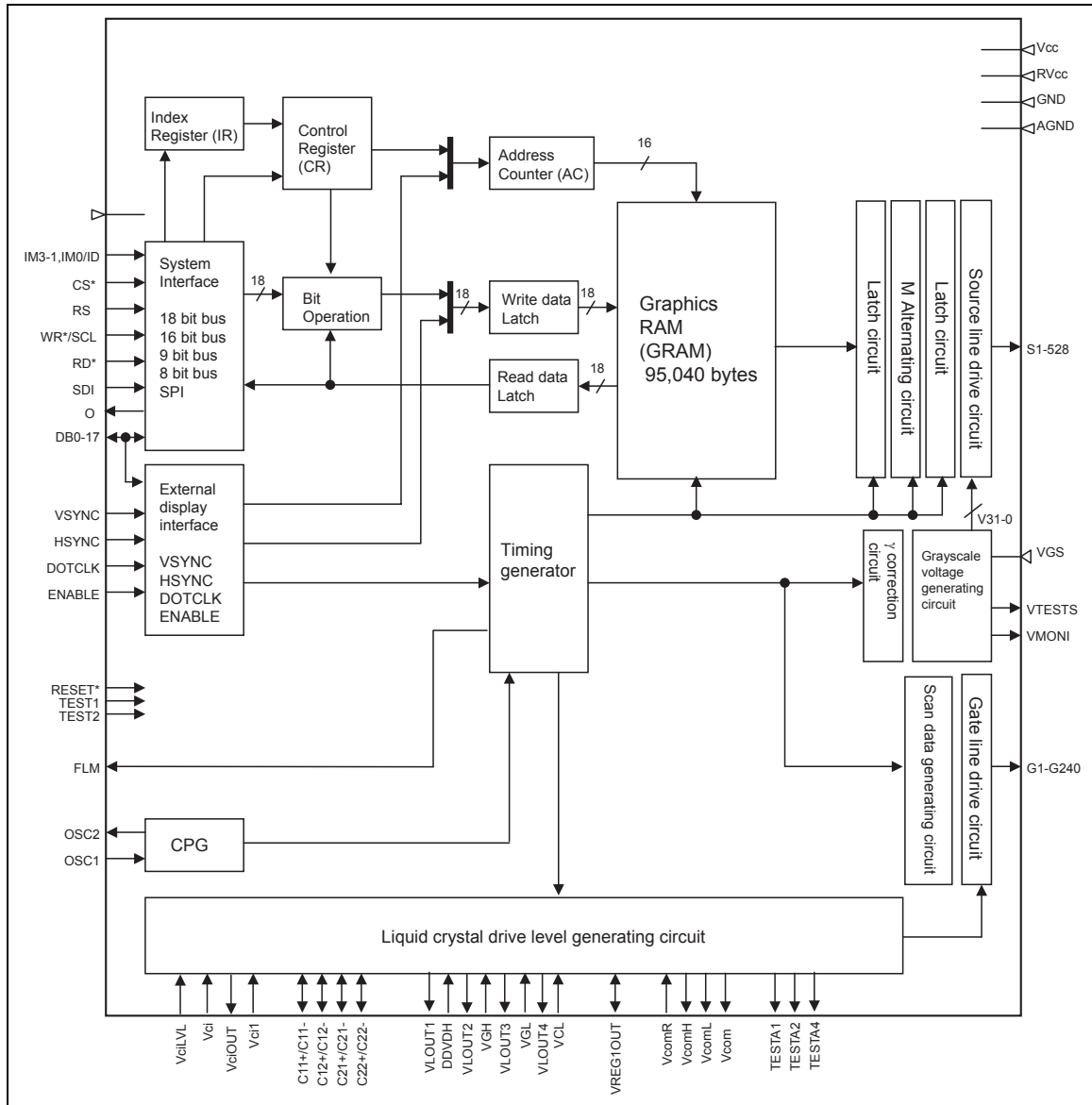


Figure 1

Pin Function

Table 2

Signal	No. of Pins	I/O	Connected to	Function																																																																								
IM3~1, IM0/ID	4	I	GND or IOVcc	Select a mode to interface to an MPU. In serial interface mode, the IM0 pin is used to set the ID of device code.																																																																								
				<table><tr><th>IM3</th><th>IM2</th><th>IM1</th><th>IM0/ID</th><th>Interface Mode</th><th>DB Pin</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>Setting disabled</td><td>—</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>Setting disabled</td><td>—</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>80-system 16-bit interface</td><td>DB17~10, DB8~1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>80-system 8-bit interface</td><td>DB17~10</td></tr><tr><td>0</td><td>1</td><td>0</td><td>ID</td><td>Serial interface</td><td>SDI, SDO</td></tr><tr><td>0</td><td>1</td><td>1</td><td>*</td><td>Setting disabled</td><td>—</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>Setting disabled</td><td>—</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>Setting disabled</td><td>—</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>80-system 18-bit interface</td><td>DB17~0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>80-system 9-bit interface</td><td>DB17~9</td></tr><tr><td>1</td><td>1</td><td>*</td><td>*</td><td>Setting disabled</td><td>—</td></tr></table>	IM3	IM2	IM1	IM0/ID	Interface Mode	DB Pin	0	0	0	0	Setting disabled	—	0	0	0	1	Setting disabled	—	0	0	1	0	80-system 16-bit interface	DB17~10, DB8~1	0	0	1	1	80-system 8-bit interface	DB17~10	0	1	0	ID	Serial interface	SDI, SDO	0	1	1	*	Setting disabled	—	1	0	0	0	Setting disabled	—	1	0	0	1	Setting disabled	—	1	0	1	0	80-system 18-bit interface	DB17~0	1	0	1	1	80-system 9-bit interface	DB17~9	1	1	*	*	Setting disabled	—
				IM3	IM2	IM1	IM0/ID	Interface Mode	DB Pin																																																																			
				0	0	0	0	Setting disabled	—																																																																			
				0	0	0	1	Setting disabled	—																																																																			
				0	0	1	0	80-system 16-bit interface	DB17~10, DB8~1																																																																			
				0	0	1	1	80-system 8-bit interface	DB17~10																																																																			
				0	1	0	ID	Serial interface	SDI, SDO																																																																			
				0	1	1	*	Setting disabled	—																																																																			
				1	0	0	0	Setting disabled	—																																																																			
				1	0	0	1	Setting disabled	—																																																																			
				1	0	1	0	80-system 18-bit interface	DB17~0																																																																			
				1	0	1	1	80-system 9-bit interface	DB17~9																																																																			
				1	1	*	*	Setting disabled	—																																																																			
CS*	1	I	MPU	Chip select signal. Amplitude: IOVcc-GND Low: the R61500 is selected and accessible High: the R61500 is not selected and not accessible Fix to the GND level when not in use.																																																																								
RS	1	I	MPU	Register select signal. Amplitude: IOVcc-GND Low: select Index or status register High: select control register Fix to either IOVcc or GND level when not in use.																																																																								
WR*/SCL	1	I	MPU	Write strobe signal in 80-system bus interface mode and enables the write operation when WR* is low. Synchronous clock signal in serial interface mode.																																																																								
RD*	1	I	MPU	Read strobe signal in 80-system bus interface mode and enables the read operation when RD* is low. In serial interface mode, fix the pin at either IOVcc or GND level.																																																																								
SDI	1	I	MPU	Serial data input (SDI) pin in serial interface mode. The data is input on the rising edge of the SCL signal. Fix the pin at either IOVcc or GND level when not in use.																																																																								
SDO	1	O	MPU	Serial data output (SDO) pin in serial interface mode. The data is output on the falling edge of the SCL signal. Fix the pin at either IOVcc or GND level when not in use.																																																																								

R61500

Signal	No. of Pins	I/O	Connected to	Function
DB0/DB17	1	I/O	MPU	18-bit parallel bi-directional data bus (Amplitude: IOVcc-GND). Unused pins must be fixed at either IOVcc or GND level. 6-bit I/F: DB17-DB12 are used. 8-bit I/F: DB17-DB10 are used. 9-bit I/F: DB17-DB9 are used. 16-bit I/F: DB17-DB10 and DB8-DB1 are used. 18-bit I/F: DB17-DB0 are used.
ENABLE	1	I	MPU	Data ENEABLE signal in RGB interface mode. Low: Select (access enabled) High: Not select (access inhibited) The EPL bit inverts the polarity of the ENABLE signal. Fix the pin at either IOVcc or GND level when not in use.
VSYN	1	I	MPU	Frame synchronous signal. When VSPL = 0: Active low. When VSPL = 1: Active high. Fix the pin at the IOVcc level when not in use.
HSYN	1	I	MPU	Line synchronous signal. When HSPL = 0: Active low. When HSPL = 1: Active high. Fix the pin at the IOVcc level when not in use
DOTCLK	1	I	MPU	Dot clock signal. When DPL = 0: Input data on the rising edge of DOTCLK When DPL = 1: Input data on the falling edge of DOTCLK Fix the pin at the IOVcc level when not in use
RESET*	1	I	MPU or external RC circuit	Reset signal. Initializes the R61500 when it is low. Be sure to execute a power-on reset after supplying power.
S1~S528	528	O	LCD	Output voltages applied to the LCD. To change the shift direction of output, use the SS bit. For example, if SS = 0, the data in the RAM address h00000 is output from S1. If SS = 1, the data in the RAM address h00000 is output from S528. S1, S4, S7, ... display red (R), S2, S5, S8, ... display green (G), and S3, S6, S9, ... display blue (B) when SS = 0.
G1~G240	240	O	LCD	Gate line output signals. VGH: gate line select level VGL: gate line non-select level
Vcom1, 11 Vcom2, 21	4	O	TFT panel common electrode	Supply voltages to the TFT panel's common electrode. Vcom1/11, Vcom2/21 alternates between the VcomH and VcomL levels. The alternating cycle is set by the M signal. Connect to the common electrode of TFT panel. All outputs come from the same node.
VcomH	1	O	Stabilizing capacitor	The higher level of Vcom1/11/2/21. Connect to a stabilizing capacitor.
VcomL	1	O	Stabilizing capacitor or open	The lower level of Vcom1/11/2/21. Adjust the VcomL level by internal register setting ([VDV4:0]). Connect to a stabilizing capacitor. To fix the VcomL level to GND, set VCOMG = 0. In this case, capacitor connection is not necessary.

R61500

Signal	No. of Pins	I/O	Connected to	Function
VcomR	1	I	Variable resistor or open	The VcomH reference level. When setting the VcomH level with an externally connected variable resistor, halt the internal VcomH adjusting circuit by setting the register and connect the resistor between VREG1OUT and GND. When setting the VcomH level by internal register setting, leave this pin open.
C11+, C11-	2	-	Step-up capacitor	Pins to connect capacitors for the internal step-up circuit 1. Leave the pins open when not using the step-up circuit 1.
C12+, C12-, C21+, C21-, C22+, C22-, C23+, C23-	8	-	Step-up capacitor	Pins to connect capacitors for the internal step-up circuit 2. Connect capacitors where they are required according to the step-up factor. Leave the pins open when not using the step-up circuit 2.
OSC1, OSC2	2	I O	Oscillation resistor	Pins to connect a resistor for RC oscillation.
FLM	1	O	MPU or open	Frame head pulse signal. Amplitude; Vcc-GND. The FLM signal is used when synchronizing RAM data write operation with the start of a frame. Leave the pin open when not in use.
Vci	1	I	Power supply	A supply voltage to the analog circuit. Connect to an external power supply of 2.5V ~ 3.3V.
VciLVL	1	I	Power supply	A reference level to generate the VciOUT/REGP level according to the step-up factor set with the VC2-0 bits. Make sure to connect VciLVL with Vci on the FPC to prevent noise.
VciOUT	1	O	Stabilizing capacitor, Vci1	Internal reference voltage level of amplitude VciLVL-GND. Connect a stabilizing capacitor between GND.
Vci1	1	I/O	VciOUT	Reference voltage input to the step-up circuit 1. When not using the internal reference voltage, connect to an external power supply of 2.75V or less.
VLOUT1	1	O	Stabilizing capacitor, DDVDH	An output voltage from the step-up circuit 1, of twice the Vci1 level. Connect a stabilizing capacitor between GND. Connect a schottky diode between Vci. See Power supply circuit. VLOUT1 = 4.5V ~ 5.5V
DDVDH	1	I	VLOUT1	A power supply to the source driver LCD output unit and the reference level for the step-up circuit 2.
VLOUT2	1	O	Stabilizing capacitor, VGH	An output voltage from the step-up circuit 2, of 4 ~ 6 times the Vci1 level. The step-up factor is set with the BT[2:0] bits. Connect a stabilizing capacitor between GND. Connect a schottky diode between Vci. See Power supply circuit. VLOUT2 = max 16.5V
VGH	1	I	VLOUT2	Supply voltage to drive the gate lines of the TFT panel. Connect to VLOUT2.
VLOUT3	1	O	Stabilizing capacitor, VGL	An output voltage from the step-up circuit 2, of -3 ~ -5 times the Vci1 level. The step-up factor is set with the BT[2:0] bits. Connect a stabilizing capacitor between GND. Connect a schottky diode between GND. See Power supply circuit. VLOUT3 = min -16.5V
VGL	1	I	VLOUT3	Supply voltage to drive the gate lines of the TFT panel. Connect to VLOUT3.

R61500

Signal	No. of Pins	I/O	Connected to	Function
VLOUT4	1	O	Stabilizing capacitor, VCL	An output voltage from the step-up circuit 2, of -1 time the Vci1 level. Connect to a stabilizing capacitor. VLOUT4 = 0V ~ -3.3V
VCL	1	I	VLOUT4	Supply voltage to generate the VcomL level. Connect to VLOUT4.
VREG1OUT	1	I/O	Stabilizing capacitor or power supply	Reference voltage DDVDH-GND, generated from the reference level Vci-GND according to the factor set with the VRH[3:0] bits. VREG1OUT is used for (1) source driver grayscale reference voltage VDH, (2) VcomH level reference voltage, and (3) Vcom amplitude reference voltage. Connect to a stabilizing capacitor. VREG1OUT = 3.0V ~ (DDVDH - 0.5) V
Vcc	1	-	Power supply	Power supply to the internal logic circuits: Vcc = 2.4V~3.6V
RVcc	1	-	Power supply	The Vcc level input to the internal RAM. RVcc and Vcc must be at the same electrical potential.
IOVcc	1	-	Power supply	Power supply to the interface pins (IOVcc = 1.65V ~ 3.6V). IOVcc and the internal logic voltage Vcc must be supplied at the same timing. In case of COG, connect to Vcc on the FPC if IOVcc=Vcc, to prevent noise.
GND	1	-	Power supply	GND for the logic circuits: GND = 0V.
AGND	1	-	Power supply	GND for the analog circuits: AGND = 0V. In case of COG, connect to GND on the FPC to prevent noise.
RGND	1	-	Power supply	GND for the internal RAM: RGND = 0V. In case of COG, connect to GND on the FPC to prevent noise.
TEST1	1	I	GND	Test pin. Connect to GND.
TEST2	1	I	GND	Test pin. Connect to GND.
VGS	1	I	GND or external resistor	Reference level for the grayscale voltage generating circuit. The VGS level can be changed with an external resistor.
VTESTS	1	I/O	Open	Test pin. Leave it open.
TESTA1	1	I/O	Open	Test pin for VcomH output. Leave it open.
TESTA2	1	I/O	Open	Test pin for VcomL output. Leave it open.
TESTA4	1	I/O	Open	Test pin for VcomL output. Leave it open.
VMONI	1	O	Open	Test pin. Leave it open.
IOVccDUM1, IOVccDUM2	2	O	Input pins	Internal IOVcc level. When adjacent pins are fixed to IOVcc, short-circuit with them.
IOGND DUM 1~4	4	O	Input pins (power supply)	Internal GND level. When adjacent pins are fixed to GND, short-circuit with them. When IOVcc > Vcc, it is recommended to connect them to GND on the FPC.
TESTO1, 2	2	O	Open	Test pins. Leave them open.
VDOUT DUMMY	1	O	Open	Test pin. Leave it open.
MOUT DUMMY	1	O	Open	Test pin. Leave it open.

R61500

Signal	No. of Pins	I/O	Connected to	Function
DUMMY3-19	17	-	Open	Dummy pads. Leave them open.
DUMMY20, DUMMY21	2	-	Open	Dummy pads. Leave them open.
DUMMY 22- 33	12	-	-	Dummy pads. Connectable to COG panel wiring.
DUMMYR 1-10	10	-	-	Dummy pads. Use them when measuring COG contact resistance. DUMMYR1 and DUMMYR10, DUMMYR2 and DUMMYR9, DUMMYR3 and DUMMYR4, DUMMYR5 and DUMMYR8, DUMMYR6 and DUMMYR7 are short-circuited within the chip.
RVcc DUMMY1, 2	2	O	(Power supply)	Internal RVcc level. It is recommended to connect with Vcc on the FPC.
RGND DUMMY1, 2	2	O	(Power supply)	Internal RGND level. It is recommended to connect with GND on the FPC.
VGLDUM1-4	4	-	-	Internal VGL level. Use them as dummy gate output.

Patents of dummy pin which is used to fix pin to VCC or GND are pending or granted.

PATENT ISSUED: United States Patent No. 6,323,930

PATENT PENDING: Japanese Application No. 10-514484, Korean Application No. 19997002322

Taiwanese Application No.086103756, (PCT/JP96/02728(W098/12597))

PAD Arrangement

- Chip size: 23.51mm×1.99mm
- Chip thickness: 550μm(typ.)
- PAD coordinates: PAD center
- Coordinates origin: Chip center

● Au bump size:
(1) 59μm×91μm
No.1~No.224

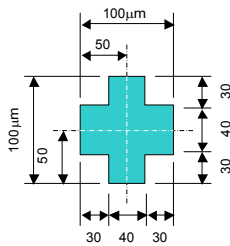
(2) 27μm×87μm
No.226~No.1023

(3) 56μm×42μm
No.225, No.1024

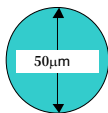
- Au bump pitch: See PAD coordinates
- Au bump height: 15μm(typ.)
- No. in the figure corresponds to No. in the PAD coordinates

- Alignment Mark

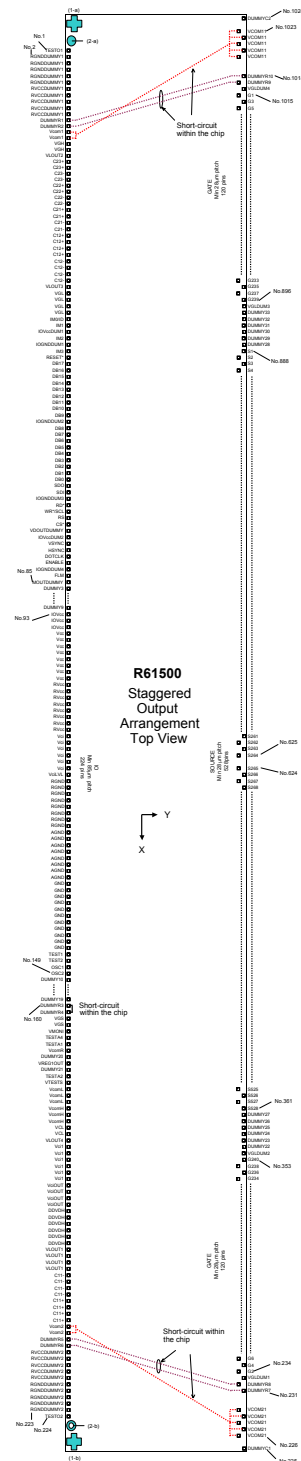
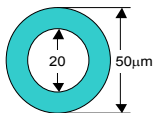
(1-a), (1-b)



(2-a)



(2-b)



Rev0.13 2004.03.16

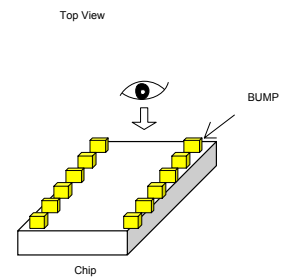


Figure 2

R61500

PAD Coordinate

Table 3

R61500PAD Coordinates (No.1) (Unit: μm)												2004.04.09 rev.2			
pad No	pad name	X	Y	pad No	pad name	X	Y	pad No	pad name	X	Y	pad No	pad name	X	Y
1	TEST01	-11346.7	-859.8	131	AGND	1426.2	-859.8	261	G56	10528.0	873.3	391	S498	6552.0	873.3
2	RGND01	-11261.6	-859.8	132	AGND	1511.4	-859.8	262	G58	10500.0	751.3	392	S497	6524.0	751.3
3	RGND02	-11176.5	-859.8	133	AGND	1596.5	-859.8	263	G60	10472.0	873.3	393	S496	6496.0	873.3
4	RGND03	-11091.4	-859.8	134	AGND	1681.6	-859.8	264	G62	10444.0	751.3	394	S495	6468.0	751.3
5	RGND04	-11006.2	-859.8	135	GND	1815.7	-859.8	265	G64	10416.0	873.3	395	S494	6440.0	873.3
6	RGND05	-10921.1	-859.8	136	GND	1900.8	-859.8	266	G66	10388.0	751.3	396	S493	6412.0	751.3
7	RVCC01	-10836.0	-859.8	137	GND	1985.9	-859.8	267	G68	10360.0	873.3	397	S492	6384.0	873.3
8	RVCC02	-10750.9	-859.8	138	GND	2120.0	-859.8	268	G70	10332.0	751.3	398	S491	6356.0	751.3
9	RVCC03	-10665.8	-859.8	139	GND	2205.1	-859.8	269	G72	10304.0	873.3	399	S490	6328.0	873.3
10	RVCC04	-10580.6	-859.8	140	GND	2290.2	-859.8	270	G74	10276.0	751.3	400	S489	6300.0	751.3
11	RVCC05	-10495.5	-859.8	141	GND	2375.4	-859.8	271	G76	10248.0	873.3	401	S488	6272.0	873.3
12	DUMMYR1	-10407.8	-859.8	142	GND	2460.5	-859.8	272	G78	10220.0	751.3	402	S487	6244.0	751.3
13	DUMMYR2	-10322.5	-859.8	143	GND	2545.6	-859.8	273	G80	10192.0	873.3	403	S486	6216.0	873.3
14	Vcom1	-10186.4	-859.8	144	GND	2630.7	-859.8	274	G82	10164.0	751.3	404	S485	6188.0	873.3
15	Vcom1	-10103.3	-859.8	145	GND	2715.8	-859.8	275	G84	10136.0	873.3	405	S484	6160.0	873.3
16	VGH	-9969.2	-859.8	146	TEST1	2842.5	-859.8	276	G86	10108.0	751.3	406	S483	6132.0	751.3
17	VGH	-9884.1	-859.8	147	TEST2	2927.6	-859.8	277	G88	10080.0	873.3	407	S482	6104.0	873.3
18	VLOUT2	-9774.5	-859.8	148	OSC1	3067.8	-859.8	278	G90	10052.0	751.3	408	S481	6076.0	751.3
19	C23+	-9619.4	-859.8	149	OSC2	3152.9	-859.8	279	G92	10024.0	873.3	409	S480	6048.0	873.3
20	C23+	-9534.2	-859.8	150	DUMMY10	3233.8	-859.8	280	G94	9996.0	751.3	410	S479	6020.0	751.3
21	C23+	-9449.1	-859.8	151	DUMMY11	3408.9	-859.8	281	G96	9968.0	873.3	411	S478	5992.0	873.3
22	C23+	-9364.0	-859.8	152	DUMMY12	3494.0	-859.8	282	G98	9940.0	751.3	412	S477	5964.0	751.3
23	C22+	-9278.9	-859.8	153	DUMMY13	3579.1	-859.8	283	G100	9912.0	873.3	413	S476	5936.0	873.3
24	C22+	-9193.8	-859.8	154	DUMMY14	3664.2	-859.8	284	G102	9884.0	751.3	414	S475	5908.0	873.3
25	C22+	-9108.6	-859.8	155	DUMMY15	3749.3	-859.8	285	G104	9856.0	873.3	415	S474	5880.0	873.3
26	C22+	-9023.5	-859.8	156	DUMMY16	3834.5	-859.8	286	G106	9828.0	751.3	416	S473	5852.0	751.3
27	C21+	-8938.4	-859.8	157	DUMMY17	3919.6	-859.8	287	G108	9800.0	873.3	417	S472	5824.0	873.3
28	C21+	-8853.3	-859.8	158	DUMMY18	4004.7	-859.8	288	G110	9772.0	751.3	418	S471	5796.0	751.3
29	C21+	-8768.2	-859.8	159	DUMMY19	4089.8	-859.8	289	G112	9744.0	873.3	419	S470	5768.0	873.3
30	C21+	-8683.0	-859.8	160	DUMMYR3	4223.9	-859.8	290	G114	9716.0	751.3	420	S469	5740.0	751.3
31	C12+	-8597.9	-859.8	161	DUMMYR4	4309.0	-859.8	291	G116	9688.0	873.3	421	S468	5712.0	873.3
32	C12+	-8462.0	-859.8	162	VGS	4452.9	-859.8	292	G118	9660.0	751.3	422	S467	5684.0	751.3
33	C12+	-8375.0	-859.8	163	VGS	4579.3	-859.8	293	G120	9632.0	873.3	423	S466	5656.0	873.3
34	C12+	-8289.9	-859.8	164	VOMNI	4708.0	-859.8	294	G122	9604.0	751.3	424	S465	5628.0	873.3
35	C12+	-8152.2	-859.8	165	TESTA4	4856.8	-859.8	295	G124	9576.0	873.3	425	S464	5600.0	873.3
36	C12+	-8067.0	-859.8	166	TESTA1	5139.7	-859.8	296	G126	9548.0	751.3	426	S463	5572.0	873.3
37	C12+	-7981.9	-859.8	167	VcomR	5266.1	-859.8	297	G128	9520.0	873.3	427	S462	5544.0	873.3
38	C12+	-7896.8	-859.8	168	DUMMY20	5362.5	-859.8	298	G130	9492.0	751.3	428	S461	5516.0	751.3
39	VLOUT3	-7762.7	-859.8	169	VREGOUT	5518.9	-859.8	299	G132	9464.0	873.3	429	S460	5488.0	873.3
40	VGL	-7638.6	-859.8	170	DUMMY21	5645.8	-859.8	300	G134	9436.0	751.3	430	S459	5460.0	751.3
41	VGL	-7494.6	-859.8	171	TESTA2	5771.7	-859.8	301	G136	9408.0	873.3	431	S458	5432.0	873.3
42	VGL	-7350.5	-859.8	172	VTESTS	5898.1	-859.8	302	G138	9380.0	751.3	432	S457	5404.0	751.3
43	VCL	-7226.4	-859.8	173	VcomL	6067.9	-859.8	303	G140	9352.0	873.3	433	S456	5376.0	873.3
44	IM01D	-7080.3	-859.8	174	Vcom	6228.5	-859.8	304	G142	9324.0	751.3	434	S455	5348.0	873.3
45	IM1	-6995.2	-859.8	175	VcomL	6410.1	-859.8	305	G144	9296.0	873.3	435	S454	5320.0	873.3
46	IOVCCDUM1	-6850.6	-859.8	176	VcomH	6554.1	-859.8	306	G146	9268.0	751.3	436	S453	5292.0	873.3
47	IM2	-6730.9	-859.8	177	VcomH	6735.7	-859.8	307	G148	9240.0	873.3	437	S452	5264.0	873.3
48	IOGNDUM1	-6586.3	-859.8	178	VcomH	6917.3	-859.8	308	G150	9212.0	751.3	438	S451	5236.0	873.3
49	IM3	-6466.6	-859.8	179	VCL	7056.3	-859.8	309	G152	9184.0	873.3	439	S450	5208.0	873.3
50	RESET*	-6381.5	-859.8	180	VCL	7141.4	-859.8	310	G154	9156.0	751.3	440	S449	5180.0	873.3
51	DB17/PD17	-6296.3	-859.8	181	VLOUT4	7226.6	-859.8	311	G156	9128.0	873.3	441	S448	5152.0	873.3
52	DB16/PD16	-6211.2	-859.8	182	VGL1	7364.3	-859.8	312	G158	9100.0	751.3	442	S447	5124.0	751.3
53	DB15/PD15	-6126.1	-859.8	183	VCL1	7449.4	-859.8	313	G160	9072.0	873.3	443	S446	5096.0	873.3
54	DB14/PD14	-6041.0	-859.8	184	VCL1	7534.5	-859.8	314	G162	9044.0	751.3	444	S445	5068.0	873.3
55	DB13/PD13	-5955.8	-859.8	185	VCL1	7619.7	-859.8	315	G164	9016.0	873.3	445	S444	5040.0	873.3
56	DB12/PD12	-5870.7	-859.8	186	VCL1	7704.8	-859.8	316	G166	8988.0	751.3	446	S443	5012.0	751.3
57	DB11/PD11	-5785.6	-859.8	187	VCL1	7789.9	-859.8	317	G168	8960.0	873.3	447	S442	4984.0	873.3
58	DB10/PD10	-5700.5	-859.8	188	VLOUT	7875.1	-859.8	318	G170	8932.0	751.3	448	S441	4956.0	873.3
59	DB9/PD9	-5615.4	-859.8	189	VLOUT	8016.5	-859.8	319	G172	8904.0	873.3	449	S440	4928.0	873.3
60	IOGNDUM2	-5530.3	-859.8	190	VLOUT	8157.9	-859.8	320	G174	8876.0	751.3	450	S439	4900.0	873.3
61	DB8/PD8	-5351.0	-859.8	191	VLOUT	8243.0	-859.8	321	G176	8848.0	873.3	451	S438	4872.0	873.3
62	DB7/PD7	-5265.9	-859.8	192	DOVH	8384.5	-859.8	322	G178	8820.0	751.3	452	S437	4844.0	873.3
63	DB6/PD6	-5180.8	-859.8	193	DOVH	8469.6	-859.8	323	G180	8792.0	873.3	453	S436	4816.0	873.3
64	DB5/PD5	-5095.7	-859.8	194	DOVH	8554.7	-859.8	324	G182	8764.0	751.3	454	S435	4788.0	873.3
65	DB4/PD4	-5010.6	-859.8	195	DOVH	8639.8	-859.8	325	G184	8736.0	873.3	455	S434	4760.0	873.3
66	DB3/PD3	-4925.4	-859.8	196	DOVH	8725.0	-859.8	326	G186	8708.0	751.3	456	S433	4732.0	873.3
67	DB2/PD2	-4840.3	-859.8	197	DOVH	8810.1	-859.8	327	G188	8680.0	873.3	457	S432	4704.0	873.3
68	DB1/PD1	-4755.2	-859.8	198	VLOUT1	8895.5	-859.8	328	G190	8652.0	751.3	458	S431	4676.0	873.3
69	DB0/PD0	-4670.1	-859.8	199	VLOUT1	9036.6	-859.8	329	G192	8624.0	873.3	459	S430	4648.0	873.3
70	SDO	-4585.0	-859.8	200	VLOUT1	9121.8	-859.8	330	G194	8596.0	751.3	460	S429	4620.0	873.3
71	SD1	-4499.8	-859.8	201	VLOUT1	9206.9	-859.8	331	G196	8568.0	873.3	461	S428	4592.0	873.3
72	IOGNDUM3	-4414.7	-859.8	202	C11										

R61500

Table 4

R61500PAD Coordinates (No.2) (Unit: μm)			
pad No	pad name	X	Y
521	S368	2912.0	873.3
522	S367	2884.0	751.3
523	S366	2856.0	873.3
524	S365	2828.0	751.3
525	S364	2800.0	873.3
526	S363	2772.0	751.3
527	S362	2744.0	873.3
528	S361	2716.0	751.3
529	S360	2688.0	873.3
530	S359	2660.0	751.3
531	S358	2632.0	873.3
532	S357	2604.0	751.3
533	S356	2576.0	873.3
534	S355	2548.0	751.3
535	S354	2520.0	873.3
536	S353	2492.0	751.3
537	S352	2464.0	873.3
538	S351	2436.0	751.3
539	S350	2408.0	873.3
540	S349	2380.0	751.3
541	S348	2352.0	873.3
542	S347	2324.0	751.3
543	S346	2296.0	873.3
544	S345	2268.0	751.3
545	S344	2240.0	873.3
546	S343	2212.0	751.3
547	S342	2184.0	873.3
548	S341	2156.0	751.3
549	S340	2128.0	873.3
550	S339	2100.0	751.3
551	S338	2072.0	873.3
552	S337	2044.0	751.3
553	S336	2016.0	873.3
554	S335	1988.0	751.3
555	S334	1960.0	873.3
556	S333	1932.0	751.3
557	S332	1904.0	873.3
558	S331	1876.0	751.3
559	S330	1848.0	873.3
560	S329	1820.0	751.3
561	S328	1792.0	873.3
562	S327	1764.0	751.3
563	S326	1736.0	873.3
564	S325	1708.0	751.3
565	S324	1680.0	873.3
566	S323	1652.0	751.3
567	S322	1624.0	873.3
568	S321	1596.0	751.3
569	S320	1568.0	873.3
570	S319	1540.0	751.3
571	S318	1512.0	873.3
572	S317	1484.0	751.3
573	S316	1456.0	873.3
574	S315	1428.0	751.3
575	S314	1400.0	873.3
576	S313	1372.0	751.3
577	S312	1344.0	873.3
578	S311	1316.0	751.3
579	S310	1288.0	873.3
580	S309	1260.0	751.3
581	S308	1232.0	873.3
582	S307	1204.0	751.3
583	S306	1176.0	873.3
584	S305	1148.0	751.3
585	S304	1120.0	873.3
586	S303	1092.0	751.3
587	S302	1064.0	873.3
588	S301	1036.0	751.3
589	S300	1008.0	873.3
590	S299	980.0	751.3
591	S298	952.0	873.3
592	S297	924.0	751.3
593	S296	896.0	873.3
594	S295	868.0	751.3
595	S294	840.0	873.3
596	S293	812.0	751.3
597	S292	784.0	873.3
598	S291	756.0	751.3
599	S290	728.0	873.3
600	S289	700.0	751.3
601	S288	672.0	873.3
602	S287	644.0	751.3
603	S286	616.0	873.3
604	S285	588.0	751.3
605	S284	560.0	873.3
606	S283	532.0	751.3
607	S282	504.0	873.3
608	S281	476.0	751.3
609	S280	448.0	873.3
610	S279	420.0	751.3
611	S278	392.0	873.3
612	S277	364.0	751.3
613	S276	336.0	873.3
614	S275	308.0	751.3
615	S274	280.0	873.3
616	S273	252.0	751.3
617	S272	224.0	873.3
618	S271	196.0	751.3
619	S270	168.0	873.3
620	S269	140.0	751.3
621	S268	112.0	873.3
622	S267	84.0	751.3
623	S266	56.0	873.3
624	S265	28.0	751.3
625	S264	-0.0	751.3
626	S263	-28.0	873.3
627	S262	-56.0	751.3
628	S261	-84.0	873.3
629	S260	-112.0	751.3
630	S259	-140.0	873.3
631	S258	-168.0	751.3
632	S257	-196.0	873.3
633	S256	-224.0	751.3
634	S255	-252.0	873.3
635	S254	-280.0	751.3
636	S253	-308.0	873.3
637	S252	-336.0	751.3
638	S251	-364.0	873.3
639	S250	-392.0	751.3
640	S249	-420.0	873.3
641	S248	-448.0	751.3
642	S247	-476.0	873.3
643	S246	-504.0	751.3
644	S245	-532.0	873.3
645	S244	-560.0	751.3
646	S243	-588.0	873.3
647	S242	-616.0	751.3
648	S241	-644.0	873.3
649	S240	-672.0	751.3
650	S239	-700.0	873.3

pad No	pad name	X	Y
651	S238	-756.0	751.3
652	S237	-784.0	873.3
653	S236	-812.0	751.3
654	S235	-840.0	873.3
655	S234	-868.0	751.3
656	S233	-896.0	873.3
657	S232	-924.0	751.3
658	S231	-952.0	873.3
659	S230	-980.0	751.3
660	S229	-1008.0	873.3
661	S228	-1036.0	751.3
662	S227	-1064.0	873.3
663	S226	-1092.0	751.3
664	S225	-1120.0	873.3
665	S224	-1148.0	751.3
666	S223	-1176.0	873.3
667	S222	-1204.0	751.3
668	S221	-1232.0	873.3
669	S220	-1260.0	751.3
670	S219	-1288.0	873.3
671	S218	-1316.0	751.3
672	S217	-1344.0	873.3
673	S216	-1372.0	751.3
674	S215	-1400.0	873.3
675	S214	-1428.0	751.3
676	S213	-1456.0	873.3
677	S212	-1484.0	751.3
678	S211	-1512.0	873.3
679	S210	-1540.0	751.3
680	S209	-1568.0	873.3
681	S208	-1596.0	751.3
682	S207	-1624.0	873.3
683	S206	-1652.0	751.3
684	S205	-1680.0	873.3
685	S204	-1708.0	751.3
686	S203	-1736.0	873.3
687	S202	-1764.0	751.3
688	S201	-1792.0	873.3
689	S200	-1820.0	751.3
690	S199	-1848.0	873.3
691	S198	-1876.0	751.3
692	S197	-1904.0	873.3
693	S196	-1932.0	751.3
694	S195	-1960.0	873.3
695	S194	-1988.0	751.3
696	S193	-2016.0	873.3
697	S192	-2044.0	751.3
698	S191	-2072.0	873.3
699	S190	-2100.0	751.3
700	S189	-2128.0	873.3
701	S188	-2156.0	751.3
702	S187	-2184.0	873.3
703	S186	-2212.0	751.3
704	S185	-2240.0	873.3
705	S184	-2268.0	751.3
706	S183	-2296.0	873.3
707	S182	-2324.0	751.3
708	S181	-2352.0	873.3
709	S180	-2380.0	751.3
710	S179	-2408.0	873.3
711	S178	-2436.0	751.3
712	S177	-2464.0	873.3
713	S176	-2492.0	751.3
714	S175	-2520.0	873.3
715	S174	-2548.0	751.3
716	S173	-2576.0	873.3
717	S172	-2604.0	751.3
718	S171	-2632.0	873.3
719	S170	-2660.0	751.3
720	S169	-2688.0	873.3
721	S168	-2716.0	751.3
722	S167	-2744.0	873.3
723	S166	-2772.0	751.3
724	S165	-2800.0	873.3
725	S164	-2828.0	751.3
726	S163	-2856.0	873.3
727	S162	-2884.0	751.3
728	S161	-2912.0	873.3
729	S160	-2940.0	751.3
730	S159	-2968.0	873.3
731	S158	-2996.0	751.3
732	S157	-3024.0	873.3
733	S156	-3052.0	751.3
734	S155	-3080.0	873.3
735	S154	-3108.0	751.3
736	S153	-3136.0	873.3
737	S152	-3164.0	751.3
738	S151	-3192.0	873.3
739	S150	-3220.0	751.3
740	S149	-3248.0	873.3
741	S148	-3276.0	751.3
742	S147	-3304.0	873.3
743	S146	-3332.0	751.3
744	S145	-3360.0	873.3
745	S144	-3388.0	751.3
746	S143	-3416.0	873.3
747	S142	-3444.0	751.3
748	S141	-3472.0	873.3
749	S140	-3500.0	751.3
750	S139	-3528.0	873.3
751	S138	-3556.0	751.3
752	S137	-3584.0	873.3
753	S136	-3612.0	751.3
754	S135	-3640.0	873.3
755	S134	-3668.0	751.3
756	S133	-3696.0	873.3
757	S132	-3724.0	751.3
758	S131	-3752.0	873.3
759	S130	-3780.0	751.3
760	S129	-3808.0	873.3
761	S128	-3836.0	751.3
762	S127	-3864.0	873.3
763	S126	-3892.0	751.3
764	S125	-3920.0	873.3
765	S124	-3948.0	751.3
766	S123	-3976.0	873.3
767	S122	-4004.0	751.3
768	S121	-4032.0	873.3
769	S120	-4060.0	751.3
770	S119	-4088.0	873.3
771	S118	-4116.0	751.3
772	S117	-4144.0	873.3
773	S116	-4172.0	751.3
774	S115	-4200.0	873.3
775	S114	-4228.0	751.3
776	S113	-4256.0	873.3
777	S112	-4284.0	751.3
778	S111	-4312.0	873.3
779	S110	-4340.0	751.3
780	S109	-4368.0	873.3

pad No	pad name	X	Y
781	S108	-4396.0	751.3
782	S107	-4424.0	873.3
783	S106	-4452.0	751.3
784	S105	-4480.0	873.3
785	S104	-4508.0	751.3
786	S103	-4536.0	873.3
787	S102	-4564.0	751.3
788	S101	-4592.0	873.3
789	S100	-4620.0	751.3
790	S99	-4648.0	873.3
791	S98	-4676.0	751.3
792	S97	-4704.0	873.3
793	S96	-4732.0	751.3
794	S95	-4760.0	873.3
795	S94	-4788.0	751.3
796	S93	-4816.0	873.3
797	S92	-4844.0	751.3
798	S91	-4872.0	873.3
799	S90	-4900.0	751.3
800	S89	-4928.0	873.3
801	S88	-4956.0	751.3
802	S87	-4984.0	873.3
803	S86	-5012.0	751.3
804	S85	-5040.0	873.3
805	S84	-5068.0	751.3
806	S83	-5096.0	873.3
807	S82	-5124.0	751.3
808	S81	-5152.0	873.3
809	S80	-5180.0	751.3

BUMP Arrangement

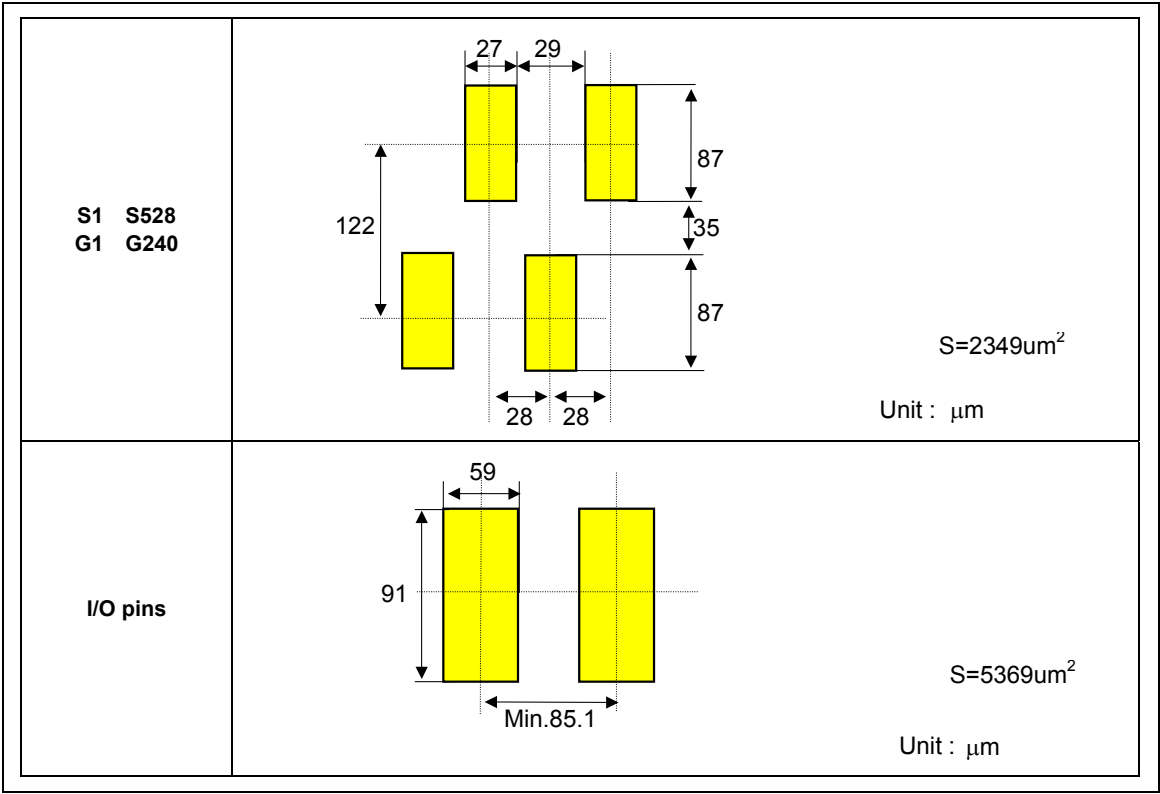


Figure 3

Block Function

1. System Interface

The R61500 supports 2-system high-speed interfaces: 80-system high-speed interfaces via 8-, 9-, 16-, 18-bit parallel ports and a serial interface. The interface mode is selected by setting the IM3-0 pins.

The R61500 has a 16-bit index register (IR); an 18-bit write-data register (WDR); and an 18-bit read-data register (RDR). The IR is the register to store index information from control registers and the internal GRAM. The WDR is the register to temporarily store data to be written to control registers and the internal GRAM. The RDR is the register to temporarily store data read from the GRAM. The data from the MPU to be written to the internal GRAM is first written to the WDR and then automatically written to the internal GRAM in internal operation. The data is read via the RDR from the internal GRAM. Therefore, invalid data is sent to the data bus when the R61500 performs the first read operation from the internal GRAM. Valid data is read out after the R61500 performs the second and subsequent read operation.

The instruction execution time except starting oscillation takes 0 clock cycle and instructions can be written consecutively.

Table 5 Register Selection (80-system 8/9/16/18-bit Parallel Interface)

80-system I/F			Function
WR*	RD*	Rs	
0	1	0	Write index to IR
1	0	0	Read internal status
0	1	1	Write to control registers or internal GRAM via WDR
1	0	1	Read from the internal GRAM via RDR

Table 6 Register Selection (serial interface)

Start byte (serial interface)

R/W	RS	Function
0	0	Write index to IR
1	0	Read internal status
0	1	Write to control registers or internal GRAM via WDR
1	1	Read from the internal GRAM via RDR

2. External Display Interface

The R61500 supports the RGB interface and the VSYNC interface as the external interface for moving picture display. When the RGB interface is selected, the display operation is synchronized with externally supplied signals, VSYNC, HSYNC, and DOTCLK. In RGB interface mode, the data (DB17-0) is written in synchronization with these signals according to the polarity of the enable signal (ENABLE) to prevent flicker while updating display data.

R61500

In VSYNC interface mode, the display operation is synchronized with the internal clock except frame synchronization using VSYNC. The display data is written to the internal GRAM via system interface. In this case, there are constraints in speed and method in writing data to the internal RAM. For details, see the External Display Interface section.

The R61500 allows for switching between the external display interface and the system interface using instruction so that an optimum interface is selected for the picture on the screen (still and/or moving picture(s)). The display data transferred via RGB interface is written to the internal RAM. With this structure, it becomes possible to write data only when updating frames and thereby reduce the total data transfer and power consumption for moving picture display.

3. Bit Operations

The R61500 supports a write data mask function for selectively writing data to the internal RAM in units of bits and a logical/compare operation to write data to the GRAM only when a condition is met as a result of comparing the data and the compare register bits. For details, see Graphics Operation Function.

4. Address Counter (AC)

The address counter (AC) gives an address to the internal GRAM. When the index of the register for setting a RAM address in the AC is written to the IR, the address information is sent from the IR to the AC. As writing data to the internal GRAM, the address in the AC is automatically updated plus or minus 1. The window address function enables writing data only in the rectangular area specified by setting on the GRAM.

5. Graphics RAM (GRAM)

GRAM is graphics RAM storing bit-pattern data of 95,040 (176 x 240 x 18/8) bytes, using 18 bits per pixel.

6. Grayscale Voltage Generating Circuit

The grayscale voltage generating circuit generates a liquid crystal drive voltage according to the grayscale data in the γ -correction register to enable 262,144-color display. For details, see the γ -Correction Register section.

7. Timing Generator

The timing generator generates a timing signal for the operations of internal circuits such as the internal GRAM. The R61500 generates a timing signal for the display operation such as the RAM read operation and a timing signal for the internal operation such as the access from the MPU separately to avoid mutual interference. Also FLM is generated internally and output from the timing generator.

8. Oscillator (OSC)

The R61500 generates the RC oscillation clock just by connecting an external oscillation resistor between the OSC1 and OSC2 pins. The oscillation frequency can be changed by changing the resistance of the external resistor. Adjust the oscillation frequency according to the operating voltage or the frame

R61500

frequency. During standby mode, RC oscillation is halted to reduce power consumption. For details, see Oscillator.

9. LCD Driver Circuit

The LCD driver circuit of the R61500 consists of a 528-output source driver (S1 ~ S528) and a 240-output gate driver (G1~G240). The display pattern data is latched when 528-bits are input. The latched data control the source driver by generating driving waveforms. The gate driver for scanning gate lines outputs either VGH or VGL level. The shift direction of 528-bit source outputs from the source driver is set with the SS bit and the shift direction of gate outputs from the gate driver is set with the GS bit. The scan mode by the gate driver is set with the SM bit. These bit settings allows selecting an optimum scan method for the LCD module.

10. LCD Drive Power Supply Circuit

The LCD drive power supply circuit generates the voltage levels VREG1OUT, VGH, VGL and Vcom for driving the LCD.

GRAM Address MAP

Table 7 GRAM address and display panel position (SS = 0, BGR = 0)

S/G pin		S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12		S517	S518	S519	S520	S521	S522	S523	S524	S525	S526	S527	S528
GS=0	GS=1	DB17-0			DB17-0			DB17-0			DB17-0				DB17-0			DB17-0			DB17-0			DB17-0		
G1	G240	"0000"H			"0001"H			"0002"H			"0003"H				"00AC"H			"00AD"H			"00AE"H			"00AF"H		
G2	G239	"0100"H			"0101"H			"0102"H			"0103"H				"01AC"H			"01AD"H			"01AE"H			"01AF"H		
G3	G238	"0200"H			"0201"H			"0202"H			"0203"H				"02AC"H			"02AD"H			"02AE"H			"02AF"H		
G4	G237	"0300"H			"0301"H			"0302"H			"0303"H				"03AC"H			"03AD"H			"03AE"H			"03AF"H		
G5	G316	"0400"H			"0401"H			"0402"H			"0403"H				"04AC"H			"04AD"H			"04AE"H			"04AF"H		
G6	G235	"0500"H			"0501"H			"0502"H			"0503"H				"05AC"H			"05AD"H			"05AE"H			"05AF"H		
G7	G234	"0600"H			"0601"H			"0602"H			"0603"H				"06AC"H			"06AD"H			"06AE"H			"06AF"H		
G8	G233	"0700"H			"0701"H			"0702"H			"0703"H				"07AC"H			"07AD"H			"07AE"H			"07AF"H		
G9	G232	"0800"H			"0801"H			"0802"H			"0803"H				"08AC"H			"08AD"H			"08AE"H			"08AF"H		
G10	G231	"0900"H			"0901"H			"0902"H			"0903"H				"09AC"H			"09AD"H			"09AE"H			"09AF"H		
G11	G230	"0A00"H			"0A01"H			"0A02"H			"0A03"H				"0AAC"H			"0AAD"H			"0AAE"H			"0AAF"H		
G12	G229	"0B00"H			"0B01"H			"0B02"H			"0B03"H				"0BAC"H			"0BAD"H			"0BAE"H			"0BAF"H		
G13	G228	"0C00"H			"0C01"H			"0C02"H			"0C03"H				"0CAC"H			"0CAD"H			"0CAE"H			"0CAF"H		
G14	G227	"0D00"H			"0D01"H			"0D02"H			"0D03"H				"0DAC"H			"0DAD"H			"0DAE"H			"0DAF"H		
G15	G226	"0E00"H			"0E01"H			"0E02"H			"0E03"H				"0EAC"H			"0EAD"H			"0EAE"H			"0EAF"H		
G16	G225	"0F00"H			"0F01"H			"0F02"H			"0F03"H				"0FAC"H			"0FAD"H			"0FAE"H			"0FAF"H		
G17	G224	"1000"H			"1001"H			"1002"H			"1003"H				"10AC"H			"10AD"H			"10AE"H			"10AF"H		
G18	G223	"1100"H			"1101"H			"1102"H			"1103"H				"11AC"H			"11AD"H			"11AE"H			"11AF"H		
G19	G222	"1200"H			"1201"H			"1202"H			"1203"H				"12AC"H			"12AD"H			"12AE"H			"12AF"H		
G20	G221	"1300"H			"1301"H			"1302"H			"1303"H				"13AC"H			"13AD"H			"13AE"H			"13AF"H		
:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
G233	G8	"E800"H			"E 801"H			"E 802"H			"E803"H				"E8AC"H			"E8AD"H			"E8AE"H			"E 8AF"H		
G234	G7	"E900"H			"E 901"H			"E 902"H			"E903"H				"E9AC"H			"E9AD"H			"E9AE"H			"E 9AF"H		
G235	G6	"EA00"H			"E A01"H			"E A02"H			"EA03"H				"EAAC"H			"EAAD"H			"EAAE"H			"E AAF"H		
G236	G5	"EB00"H			"E B01"H			"E B02"H			"EB03"H				"EBAC"H			"EBAD"H			"EBAE"H			"E BAF"H		
G237	G4	"EC00"H			"E C01"H			"E C02"H			"EC03"H				"ECAC"H			"ECAD"H			"ECAE"H			"E CAF"H		
G238	G3	"ED00"H			"E D01"H			"E D02"H			"ED03"H				"EDAC"H			"EDAD"H			"EDAE"H			"E DAF"H		
G239	G2	"EE00"H			"E E01"H			"E E02"H			"EE03"H				"EEAC"H			"EEAD"H			"EEAE"H			" E EAF"H		
G240	G1	"EF00"H			"E F01"H			"E F02"H			"EF03"H				"EFAC"H			"EFAD"H			"EFAE"H			" EFAF"H		

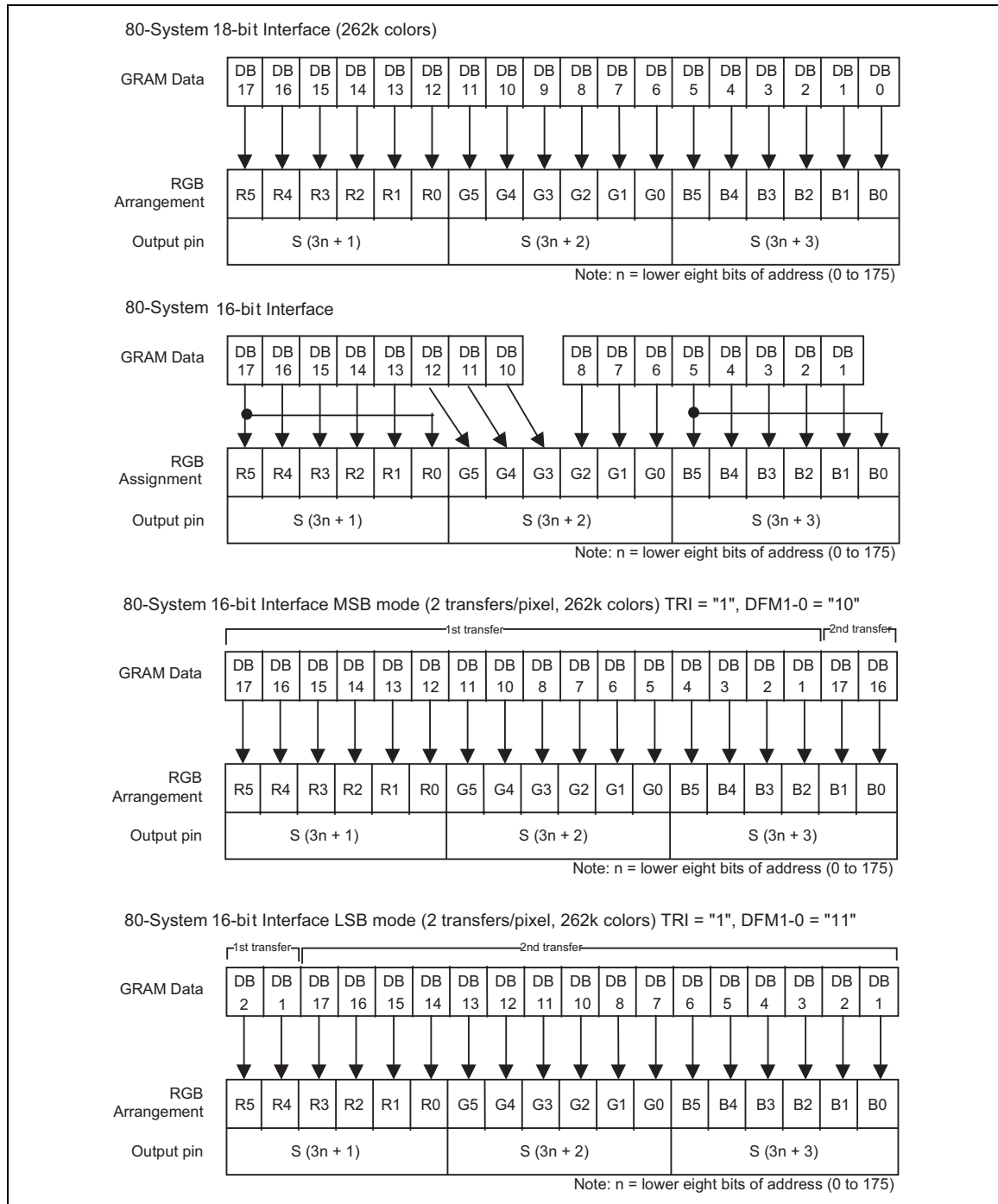
Table 8 GRAM data and display data: system interface (SS = 0, BGR = 0)

Table 9 GRAM data and display data: system interface (SS = 0, BGR = 0)

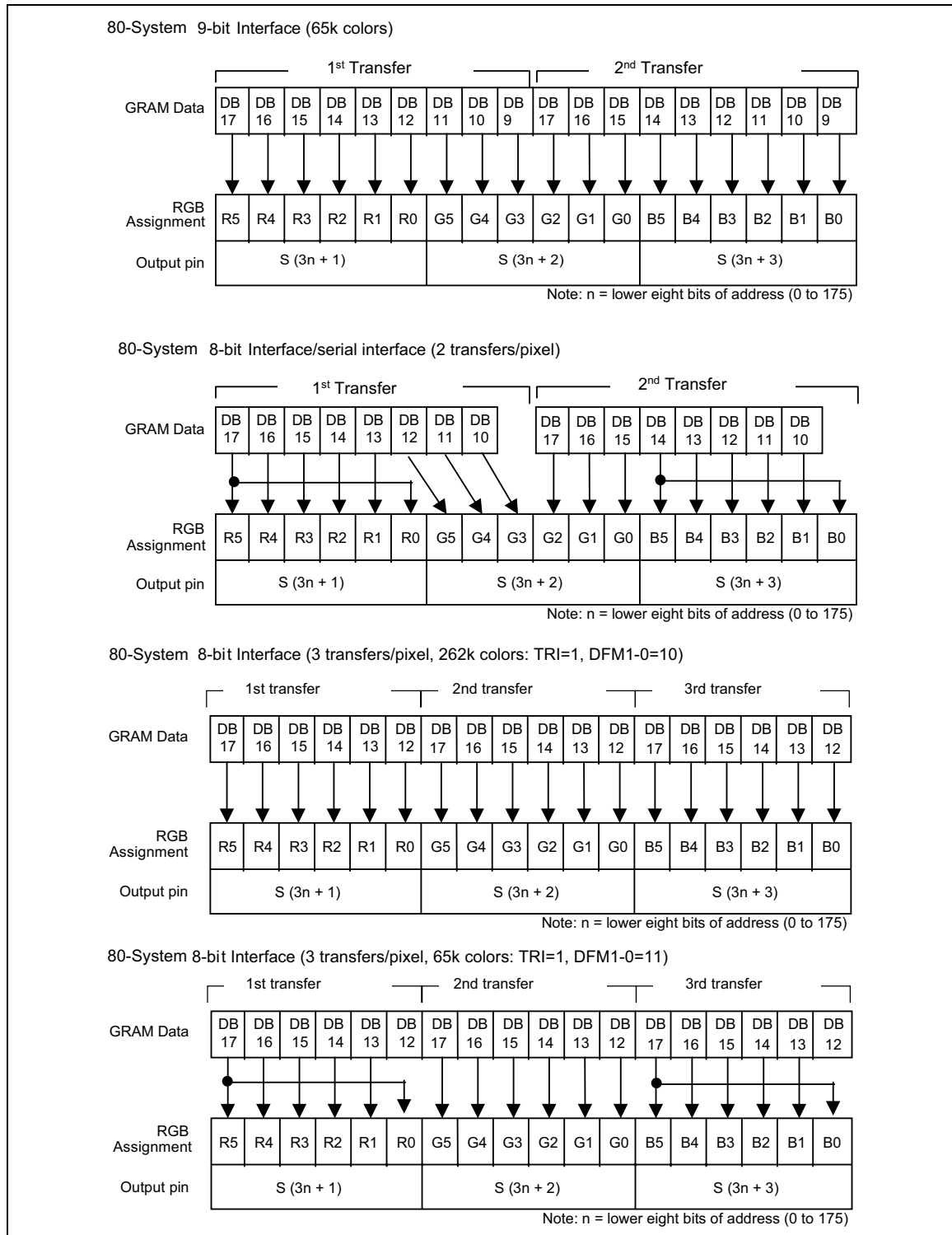


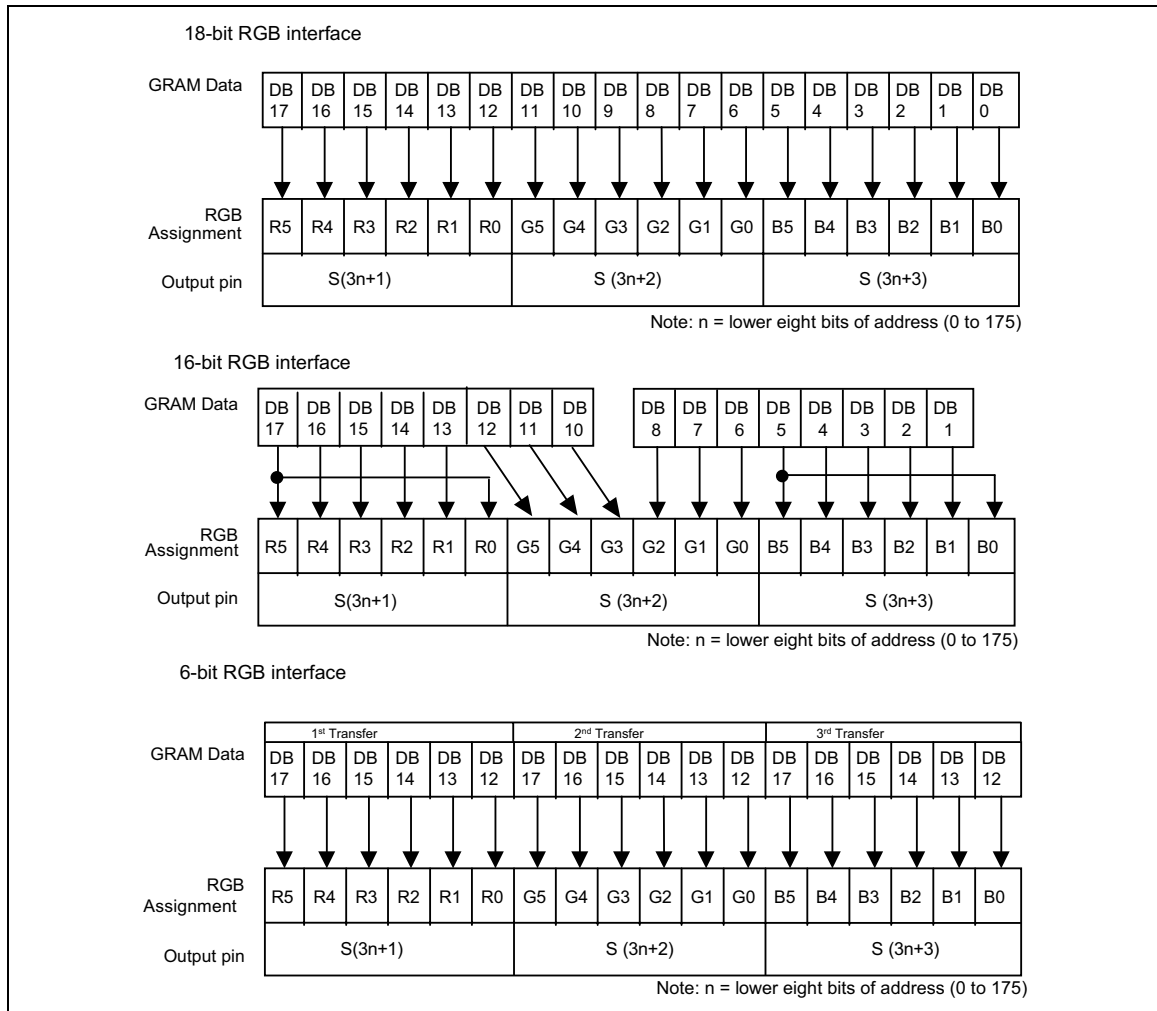
Table 10 GRAM data and display data: RGB interface (SS = 0, BGR = 0)

Table 11 GRAM address and display panel position (SS = 1, BGR = 1)

S/G p i n		S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12		S517	S518	S519	S520	S521	S522	S523	S524	S525	S526	S527	S528
GS=0	GS=1	DB17-0			DB17-0			DB17-0			DB17-0				DB17-0			DB17-0			DB17-0			DB17-0		
G1	G240	"00AF"H			"00AE"H			"00AD"H			"00AC"H				"0003"H			"0002"H			"0001"H			"0000"H		
G2	G239	"01AF"H			"01AE"H			"01AD"H			"01AC"H				"0103"H			"0102"H			"0101"H			"0100"H		
G3	G238	"02AF"H			"02AE"H			"02AD"H			"02AC"H				"0203"H			"0202"H			"0201"H			"0200"H		
G4	G237	"03AF"H			"03AE"H			"03AD"H			"03AC"H				"0303"H			"0302"H			"0301"H			"0300"H		
G5	G236	"04AF"H			"04AE"H			"04AD"H			"04AC"H				"0403"H			"0402"H			"0401"H			"0400"H		
G6	G235	"05AF"H			"05AE"H			"05AD"H			"05AC"H				"0503"H			"0502"H			"0501"H			"0500"H		
G7	G234	"06AF"H			"06AE"H			"06AD"H			"06AC"H				"0603"H			"0602"H			"0601"H			"0600"H		
G8	G233	"07AF"H			"07AE"H			"07AD"H			"07AC"H				"0703"H			"0702"H			"0701"H			"0700"H		
G9	G232	"08AF"H			"08AE"H			"08AD"H			"08AC"H				"0803"H			"0802"H			"0801"H			"0800"H		
G10	G231	"09AF"H			"09AE"H			"09AD"H			"09AC"H				"0903"H			"0902"H			"0901"H			"0900"H		
G11	G230	"0AAF"H			"0AAE"H			"0AAD"H			"0AAC"H				"0A03"H			"0A02"H			"0A01"H			"0A00"H		
G12	G229	"0BAF"H			"0BAE"H			"0BAD"H			"0BAC"H				"0B03"H			"0B02"H			"0B01"H			"0B00"H		
G13	G228	"0CAF"H			"0CAE"H			"0CAD"H			"0CAC"H				"0C03"H			"0C02"H			"0C01"H			"0C00"H		
G14	G227	"0DAF"H			"0DAE"H			"0DAD"H			"0DAC"H				"0D03"H			"0D02"H			"0D01"H			"0D00"H		
G15	G226	"0EAF"H			"0EAE"H			"0EAD"H			"0EAC"H				"0E03"H			"0E02"H			"0E01"H			"0E00"H		
G16	G225	"0FAF"H			"0FAE"H			"0FAD"H			"0FAC"H				"0F03"H			"0F02"H			"0F01"H			"0F00"H		
G17	G224	"10AF"H			"10AE"H			"10AD"H			"10AC"H				"1003"H			"1002"H			"1001"H			"1000"H		
G18	G223	"11AF"H			"11AE"H			"11AD"H			"11AC"H				"1103"H			"1102"H			"1101"H			"1100"H		
G19	G222	"12AF"H			"12AE"H			"12AD"H			"12AC"H				"1203"H			"1202"H			"1201"H			"1200"H		
G20	G221	"13AF"H			"13AE"H			"13AD"H			"13AC"H				"1303"H			"1302"H			"1301"H			"1300"H		
:	:	:			:			:			:			:	:			:			:			:		
:	:	:			:			:			:			:	:			:			:			:		
G233	G8	"E8AF"H			"E8AE"H			"E8AD"H			"E8AC"H				"E803"H			"E802"H			"E801"H			"E800"H		
G234	G7	"E9AF"H			"E9AE"H			"E9AD"H			"E9AC"H				"E903"H			"E902"H			"E901"H			"E900"H		
G235	G6	"EAAF"H			"EAAE"H			"EAAD"H			"EAAC"H				"EA03"H			"EA02"H			"EA01"H			"EA00"H		
G236	G5	"EBAF"H			"EBAE"H			"EBAD"H			"EBAC"H				"EB03"H			"EB02"H			"EB01"H			"EB00"H		
G237	G4	"ECAF"H			"ECAE"H			"ECAD"H			"ECAC"H				"EC03"H			"EC02"H			"EC01"H			"EC00"H		
G238	G3	"EDAF"H			"EDAE"H			"EDAD"H			"EDAC"H				"ED03"H			"ED02"H			"ED01"H			"ED00"H		
G239	G2	"EEAF"H			"EEAE"H			"EEAD"H			"EEAC"H				"EE03"H			"EE02"H			"EE01"H			"EE00"H		
G240	G1	"EFAF"H			"EFAE"H			"EFAD"H			"EFAC"H				"EF03"H			"EF02"H			"EF01"H			"EF00"H		

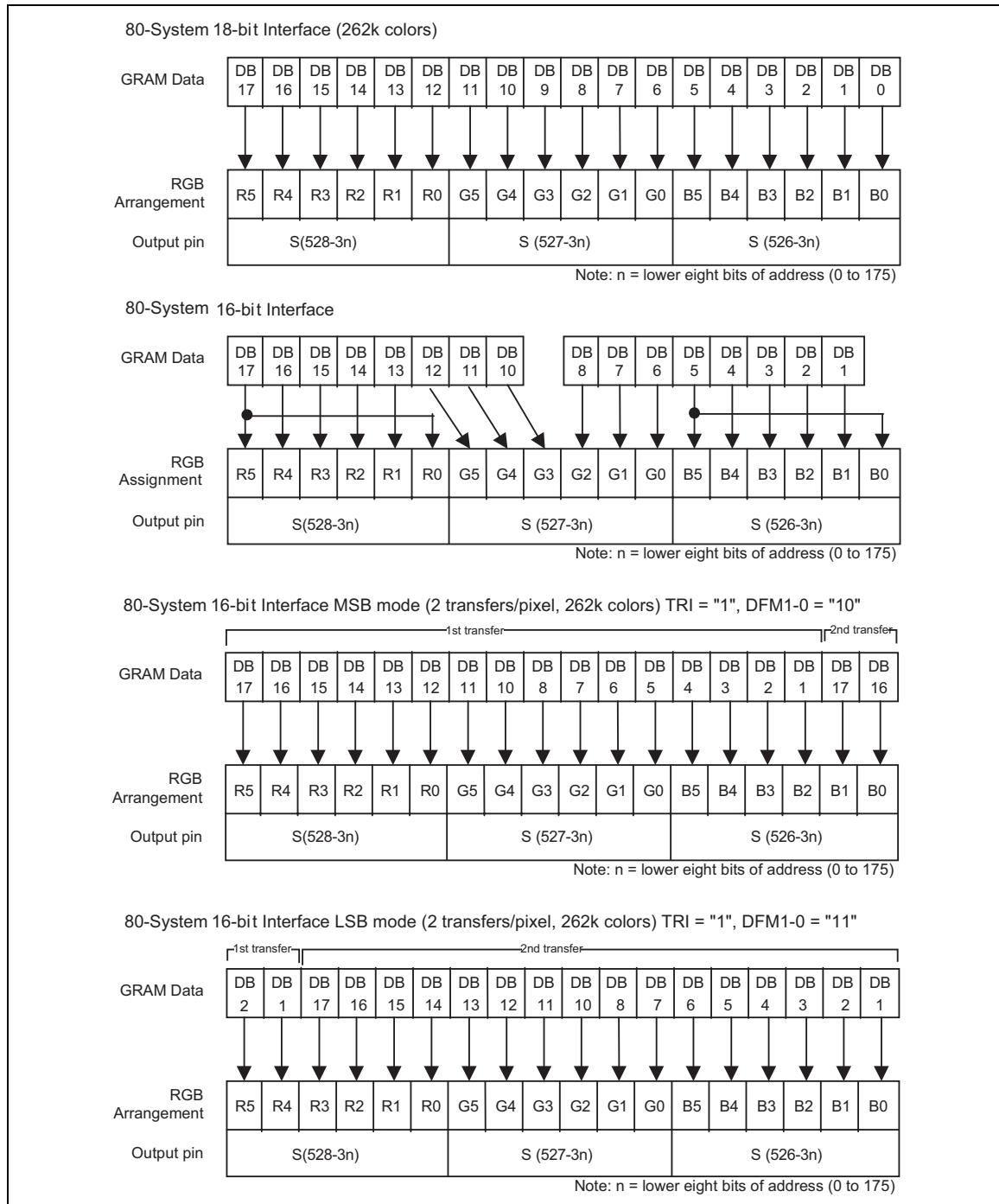
Table 12 GRAM data and display data: system interface (SS = 1, BGR = 1)

Table 13 GRAM data and display data: system interface (SS = 1, BGR = 1)

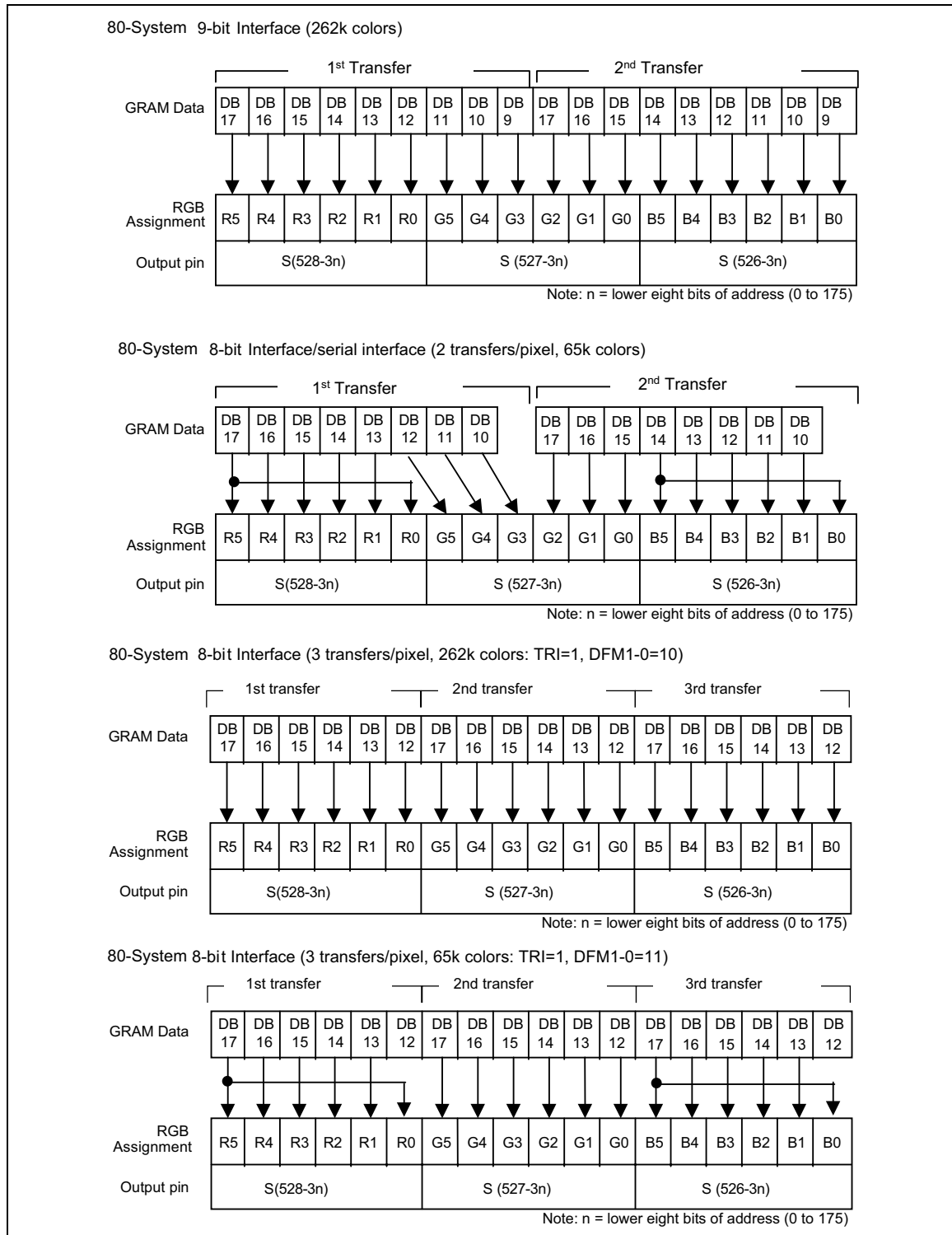
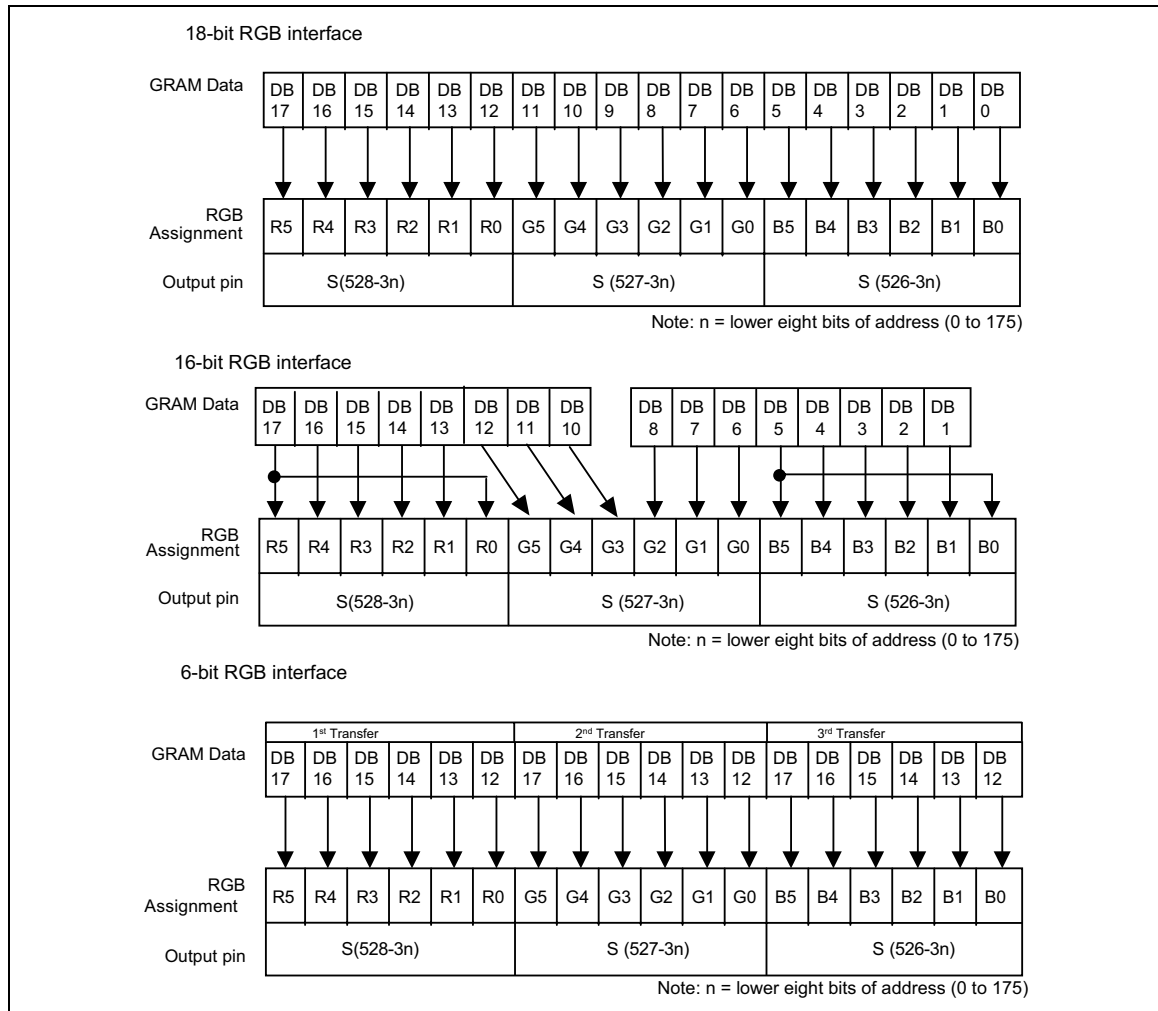


Table 14 GRAM data and display data: RGB interface (SS = 1, BGR = 1)

Instructions

Outline

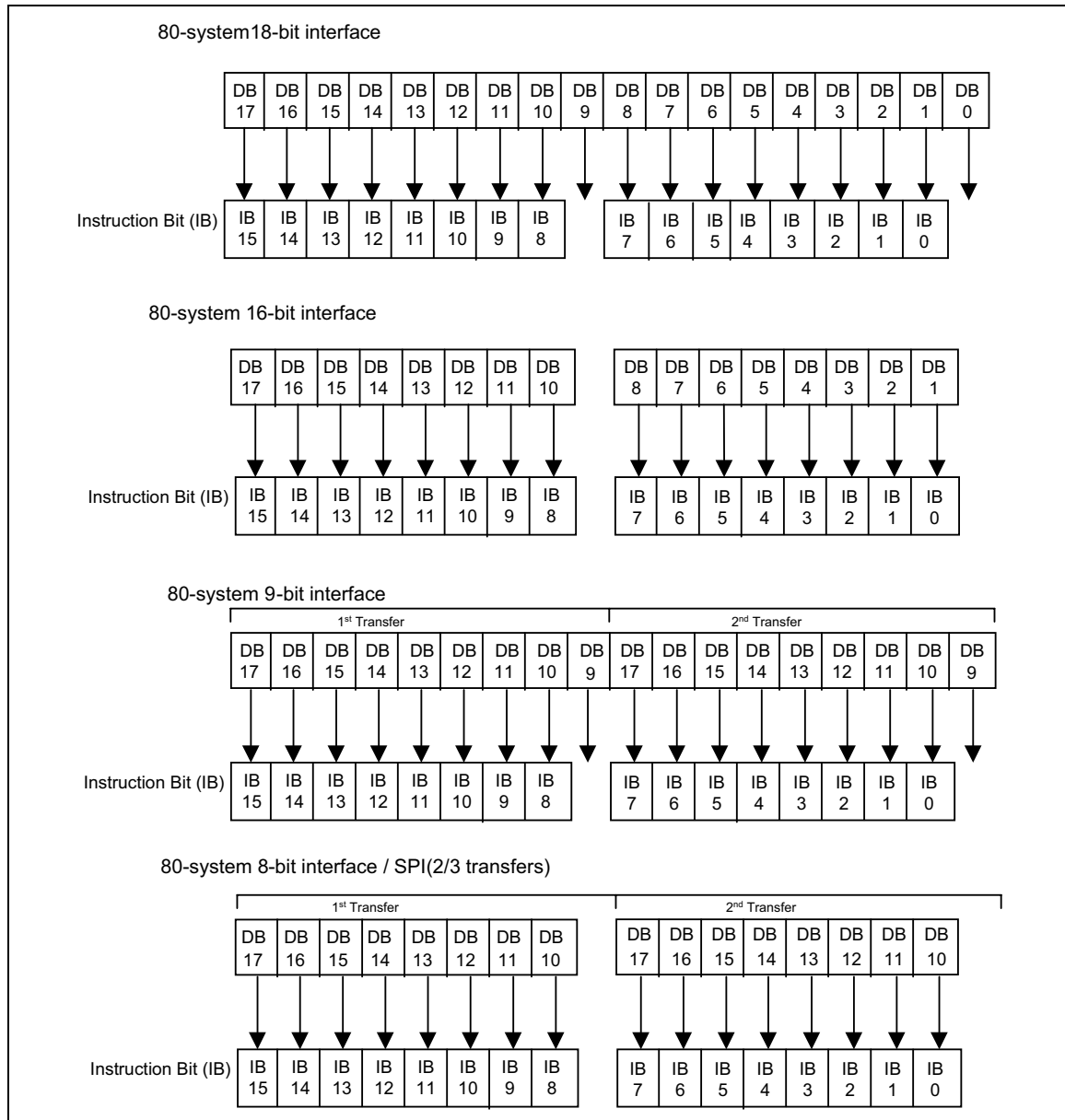
The R61500 adopts 18-bit bus architecture to interface to a high-performance microcomputer. The R61500 starts internal processing after storing control information of externally sent 18-/16-/9-/8-bit data in the instruction register (IR) and the data register (DR). Since the R61500's internal operation is controlled by the signals sent from the microcomputer, the register selection signal (RS), the read/write signal (R/W), and the internal 16-bit data bus signals (IB15 to IB0) are called instructions. The R61500 uses the 18-bit format internally for operations involving internal GRAM access. The instructions of the R61500 are categorized into the following groups.

1. Specify index
2. Read status
3. Display control
4. Power management Control
5. Graphics data processing
6. Set internal GRAM address
7. Transfer data to and from the internal GRAM
8. Internal grayscale γ -correction

Normally, the instruction to write data to the internal GRAM is used the most often. Since the R61500 can update the internal GRAM address automatically as it writes data to the internal GRAM, the total data transfer can be minimized by using the window address function, and thereby mitigates the load on the microcomputer. Since instructions are executed within the cycle when they are written (0 cycle), it is possible to write instructions consecutively.

As the following figure shows, the way of assigning data to the 16 instruction bits (IB15-0) varies for each interface. Instructions must be transferred in accordance with the following data transfer format.

Table 15 Instruction bits



R61500

Explanation of each instruction

The following are detailed explanations of instructions with illustrations of instruction bits (IB15-0) assigned to each interface.

Index (IR)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	0	*	*	*	*	*	*	*	*	*	ID6	ID5	ID4	ID3	ID2	ID1	ID0

The index register represents the index (R00h ~ RFFh) of the control register or the RAM control register to be accessed using a binary number from 0000_0000 to 1111_1111. Accessing the register as well as the instruction bits in the register is prohibited unless the index is specified by this register.

Status Read (SR)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
R	0	L7	L6	L5	L4	L3	L2	L1	L0	0	0	0	0	0	0	0	0

The SR bits represent the internal status of the R61500.

L7-0: Specifies the line currently driving liquid crystal.

Start Oscillation (R00h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	1
R	1	0	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0

The start oscillation instruction restarts the oscillator at a halt in standby mode. After executing this instruction, wait at least 10 ms for stabilizing oscillator before issuing another instruction.

The device code “1500”H is read out when reading out this register forcibly.

Driver Output Control (R01h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	VSPL	HSPL	DPL	EPL	SM	GS	SS	0	0	0	NL4	NL3	NL2	NL1	NL0

SS: Selects the shift direction of output from the source driver.

When SS = 0, output from S1 to S528.

When SS = 1, output from S528 to S1.

The combination of SS and BGR bits controls the order of assigning RGB dots to each source driver pin.

When SS = 0 and BGR = 0, one dot is assigned to one pin interchangeably from S1 to S528.

When SS = 1 and BGR = 1, one dot is assigned to one pin interchangeably from S528 to S1.

When changing SS or BGR bits, RAM data must be rewritten.

GS: Sets the shift direction of output from the gate driver in accordance with the LC module.

SM: Sets the gate driver pin assignment in combination with the GS bit in accordance with the LC module.
See Scan Mode Setting for details.

EPL: Sets the polarity of the signal from the ENABLE pin in RGB interface mode. .

EPL = 0	: ENABLE = Low / Write data to DB17-0
	: ENABLE = High / Disenable data write operation
EPL = 1	: ENABLE = High / Write data to DB17-0
	: ENABLE = Low / Disenable data write operation

The following table shows the relationship between the EPL, ENABLE bits, and RAM access.

Table 16

EPL	ENABLE	RAM write	RAM address
0	0	Enabled	Updated
0	1	Disenabled	Retained
1	0	Disenabled	Retained
1	1	Enabled	Updated

VSPL: Inverts the polarity of signals from the VSYNC pin.

VSPL = 0 : Low active.

VSPL = 1 : High active.

R61500

HSPL: Inverts the polarity of signals from the HSYNC pin.

HSPL = 0 : Low active.

HSPL = 1 : High active.

DPL: Inverts the polarity of signals from the DOTCLK pin.

DPL = 0 : The data is read on the rising edge of the DOTCLK.

DPL = 1 : The data is read on the falling edge of the DOTCLK.

R61500

NL4-0: Sets the number of gate lines for driving the liquid crystal display panel at an interval of 8 lines as the following table. The GRAM address mapping is not affected by the number of gate lines set with the NL4-0 bits. Select the number of gate lines that is equal to or more than the number of gate lines necessary for driving the panel.

Table 17

NL4	NL3	NL2	NL1	NL0	Display Size	Lines	Driven gate lines
0	0	0	0	0	Setting disabled	Setting disabled	Setting disabled
0	0	0	0	1	528 x 16 dots	16	G1 ~ G16
0	0	0	1	0	528 x 24 dots	24	G1 ~ G24
0	0	0	1	1	528 x 32 dots	32	G1 ~ G32
0	0	1	0	0	528 x 40 dots	40	G1 ~ G40
0	0	1	0	1	528 x 48 dots	48	G1 ~ G48
0	0	1	1	0	528 x 56 dots	56	G1 ~ G56
0	0	1	1	1	528 x 64 dots	64	G1 ~ G64
0	1	0	0	0	528 x 72 dots	72	G1 ~ G72
0	1	0	0	1	528 x 80 dots	80	G1 ~ G80
0	1	0	1	0	528 x 88 dots	88	G1 ~ G88
0	1	0	1	1	528 x 96 dots	96	G1 ~ G96
0	1	1	0	0	528 x 104 dots	104	G1 ~ G104
0	1	1	0	1	528 x 112 dots	112	G1 ~ G112
0	1	1	1	0	528 x 120 dots	120	G1 ~ G120
0	1	1	1	1	528 x 128 dots	128	G1 ~ G128
1	0	0	0	0	528 x 136 dots	136	G1 ~ G136
1	0	0	0	1	528 x 144 dots	144	G1 ~ G144
1	0	0	1	0	528 x 152 dots	152	G1 ~ G152
1	0	0	1	1	528 x 160 dots	160	G1 ~ G160
1	0	1	0	0	528 x 168 dots	168	G1 ~ G168
1	0	1	0	1	528 x 176 dots	176	G1 ~ G176
1	0	1	1	0	528 x 184 dots	184	G1 ~ G184
1	0	1	1	1	528 x 192 dots	192	G1 ~ G192
1	1	0	0	0	528 x 200 dots	200	G1 ~ G200
1	1	0	0	1	528 x 208 dots	208	G1 ~ G208
1	1	0	1	0	528 x 216 dots	216	G1 ~ G216
1	1	0	1	1	528 x 224 dots	224	G1 ~ G224
1	1	1	0	0	528 x 232 dots	232	G1 ~ G232
1	1	1	0	1	528 x 240 dots	240	G1 ~ G240

Note: A back porch period (set with BP bits) and a front porch period (set with FP bits) are inserted as blank periods before and after driving all gate lines, respectively.

LCD Driving Waveform Control (R02h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	LEQ2	LEQ1	LEQ0	0	FLD1	FLD0	B/C	EOR	0	0	NW5	NW4	NW3	NW2	NW1	NW0

NW5-0: Specifies n, the number of gate lines from 1 to 64, to set the interval of polarity inversion when the R61500 is set to generate a C-pattern waveform (B/C = 1). The LCD is driven with polarity inversion at an interval of n+1 gate line periods.

EOR: When EOR = 1, the polarity is inverted according to the result of EOR (exclusive OR) operation performed on the signal to select either odd or even frames and the signal to invert polarity in units of n lines when the R61500 is set to generate a C-pattern waveform (B/C = 1). This instruction is used when the number of gate lines for driving the LCD panel is at odds with the n-line interval set for inverting polarity. For details, see n-line inversion AC drive.

B/C: When the R61500 is set to generate a field-inversion waveform (B/C = 0), polarity is inverted at an interval of fields. The R61500 inverts the polarity at an interval of n lines, when a C-pattern waveform is generated (B/C = 1) according to NW and EOR bits. For details, see n line-inversion AC drive.

FLD1-0: Sets the number of fields for n-field interlaced scan. See Interlaced scan for details. The FLD bits are disabled in external display interface mode. When using the external display interface, set FLD1-0= 01.

Table 18

FLD1	FLD0	Number of fields
0	0	Setting disabled
0	1	1 field (= 1 frame)
1	0	Setting disabled
1	1	3 fields

LEQ2-0: Sets the equalizing period as follows to improve the quality of display when driving the LCD panel with a field-inversion AC waveform.

Table 19

LEQ2	LEQ1	LEQ0	Internal clock (1clk=internal clock)	RGB interface (clk=DOTCLK)	LEQ2	LEQ1	LEQ0	Internal clock (1clk=internal clock)	RGB interface (clk=DOTCLK)
0	0	0	No equalize	No equalize	1	0	0	8 clocks	64 clocks
0	0	1	2 clocks	16 clocks	1	0	1	10 clocks	80 clocks
0	1	0	4 clocks	32 clocks	1	1	0	12 clocks	96 clocks
0	1	1	6 clocks	48 clocks	1	1	1	14 clocks	112 clocks

R61500

Entry Mode (R03h)

Compare register 1 (R04h)

Compare register 2 (R05h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	TRI	DFM1	DFM0	BGR	0	0	HWM	LHWM	0	0	I/D1	I/D0	AM	LG2	LG1	LG0
W	1	0	0	CP11	CP10	CP9	CP8	CP7	CP6	0	0	CP5	CP4	CP3	CP2	CP1	CP0
W	1	0	0	0	0	0	0	0	0	0	0	CP17	CP16	CP15	CP14	CP13	CP12

The R61500 modifies data sent from the microcomputer before writing them to the internal GRAM in order to write the GRAM data in high speed and reduce the load on the microcomputer. See Graphics Operation Function for details.

TRI: When TRI = 1, the data is transferred to the internal RAM in 3 transfers via 8-bit interface to realize display in 262k colors. It is also possible to realize 262k-color display with 16-bit or serial interface by setting TRI and DFM bits. Make sure to set TRI = 0 when using other data transfer modes.

DFM1-0: Sets the data format when TRI = 1 in 8-bit/16-bit/serial interface mode. See the following figures for details.

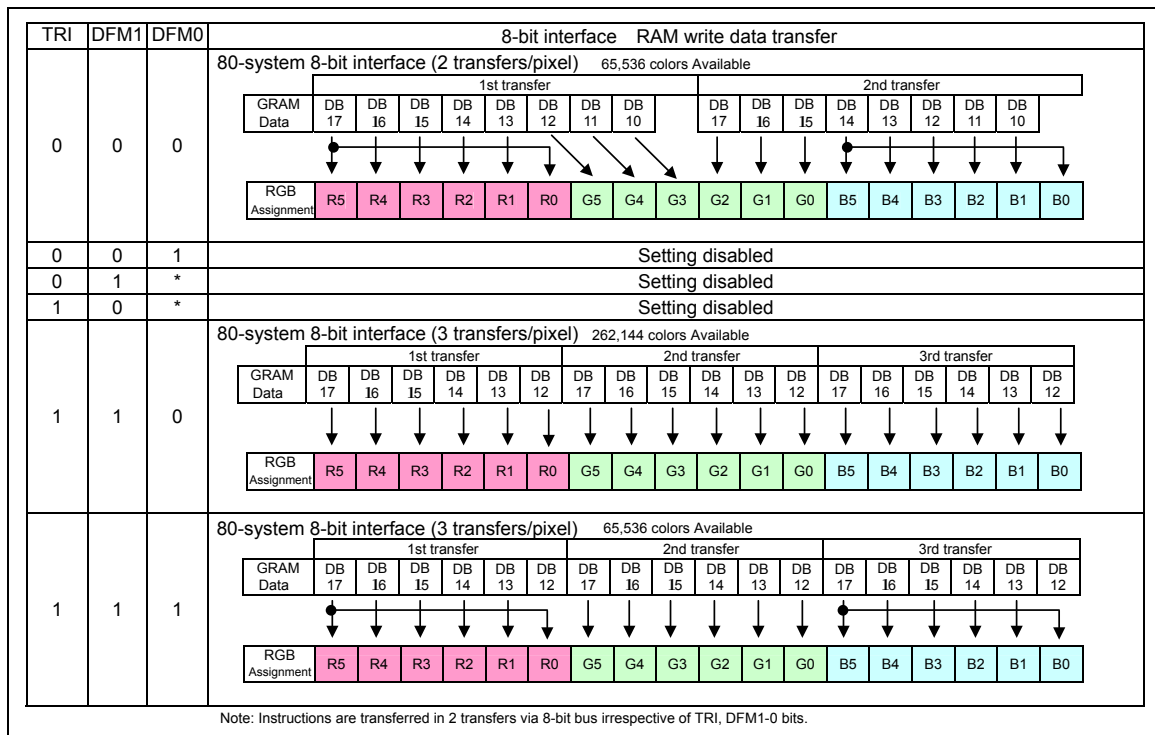


Figure 4

R61500

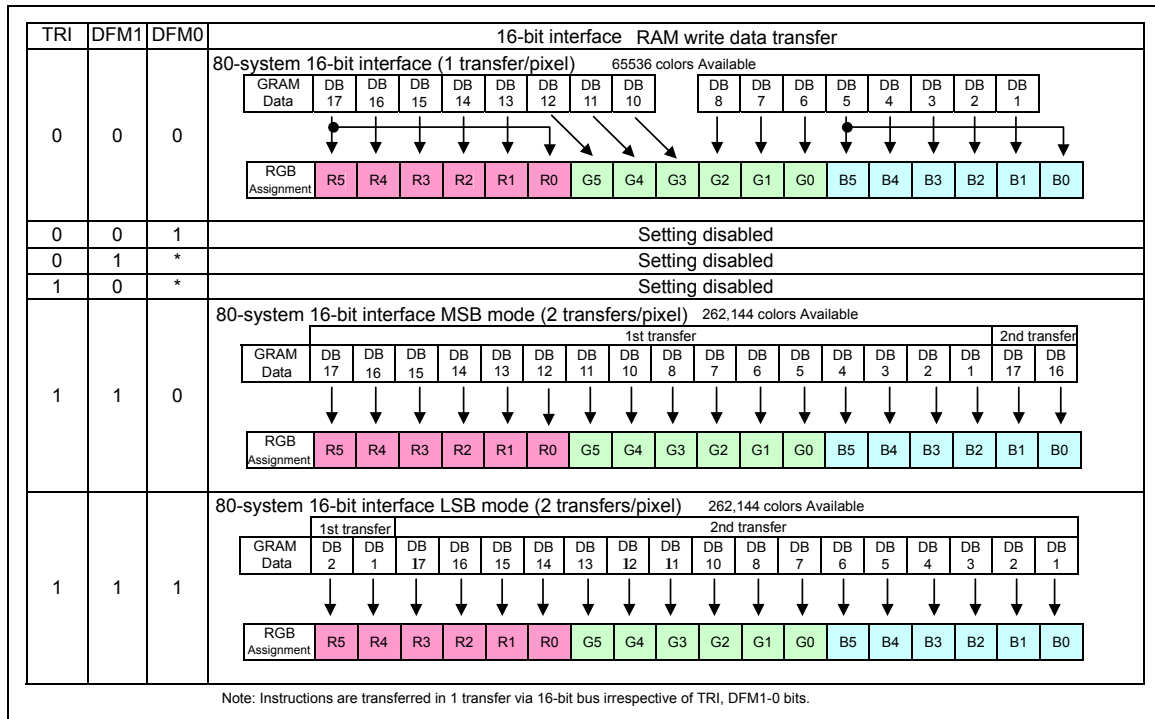


Figure 5

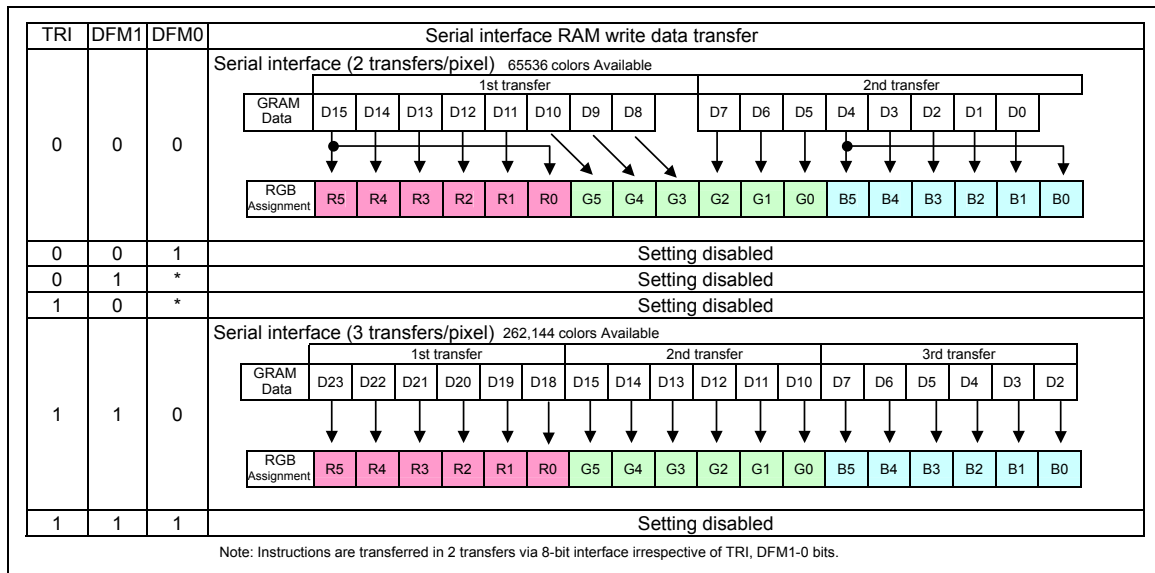


Figure 6

R61500

HWM: When HWM= 1, the high-speed write mode is enabled and the R61500 writes 4 words at a time to the internal GRAM, after writing data to the internal register 4 times. If the write operation to the register is terminated before reaching 4 times, the write operation to the GRAM will fail. Make sure to conclude the write operation at every 4 times. For this reason, the lower 2 bits of the AD15-0 bits must be set to 0 when executing RAM address set (R21h). For details, see High-Speed RAM Write Function section.

LHWM: Enables the low-power-consumption high-speed write mode. In this mode, the data to be written in each address line is written at a time to the internal GRAM. When writing data to the internal register, make sure that the data is written to the end of the address line. If the write operation is terminated in the middle of the line, the data for that address line is not written to the GRAM. When making a window address area, the end of address line is defined with the HEA7-0 bits.

Note: The high speed write functions are not used when writing data in vertical direction (AM = 1).
Make sure to set AM = 0 when using a high-speed write function.

Table 20

HWM	LHWM	Write mode
0	0	Normal write mode
0	1	1-line HWM
1	0	4-word HWM
1	1	Setting disabled

I/D1-0: The address counter is automatically incremented by 1 as writing data to the internal GRAM when I/D1-0 = 1. The address counter is automatically decremented by 1 as writing data to the internal GRAM when I/D1-0 = 0. The increment/decrement can be set to the upper (AD15-8) and the lower (AD7-0) byte of the address separately. The vertical or horizontal transition direction of the address when writing data to the internal GRAM is set with the AM bit.

AM: Sets the direction of updating address in the address counter (AC) automatically when writing data to the internal RAM. When AM = 0, the address is updated in horizontal writing direction. When AM = 1, the address is updated in vertical writing direction. The R61500 writes data within the window address area when making a window address setting in the writing direction set with I/D1-0 and AM bits.

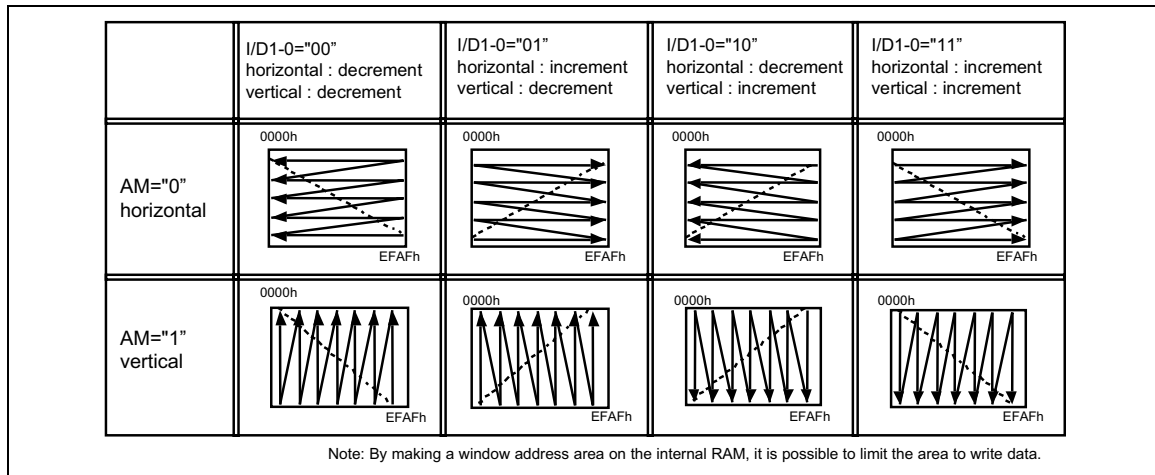


Figure 7 Address transition direction

LG2-0: Compares the data from the microcomputer with the compare register bits (CP17-0) before writing the data to the internal RAM. For details, see the Graphics Operation Function.

CP17-0: Compare register bits, with which the data read out from the internal GRAM or the data to be written to the internal GRAM from the microcomputer are compared.

BGR: Reverses the order of RGB to BGR when writing 18-bit pixel data to the internal GRAM. The RGB order is automatically reversed with regard to WM17-0 and CP17-0 bits upon setting BGR = 1.

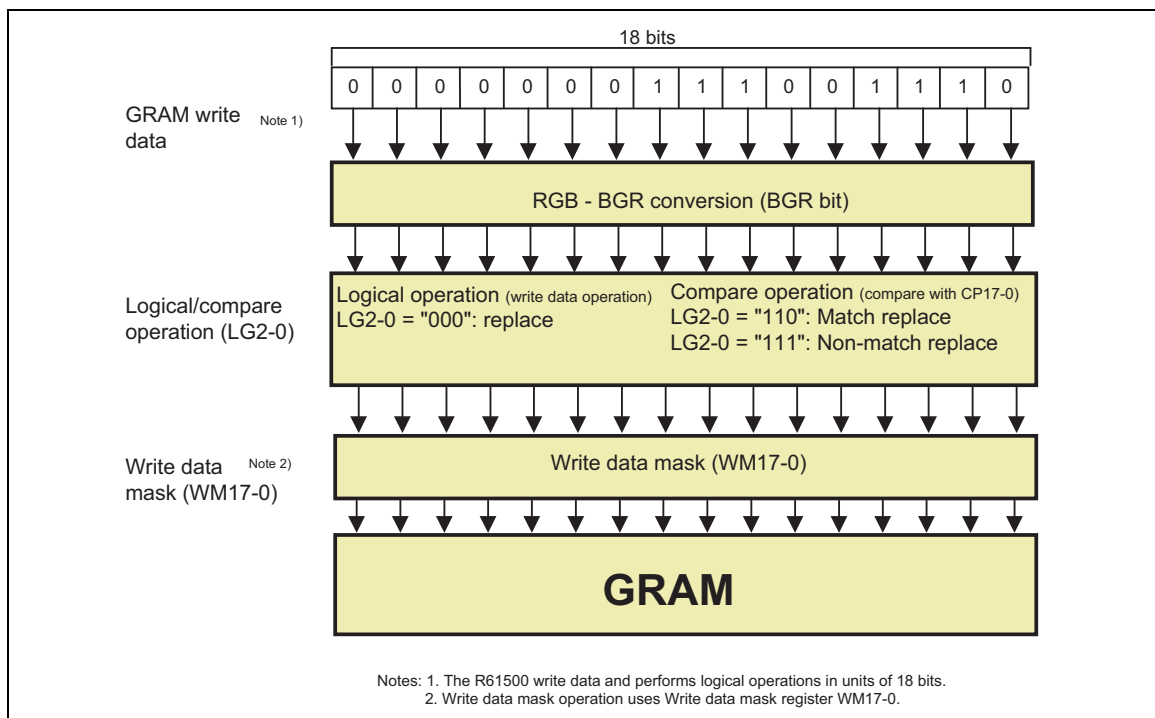


Figure 8 Bit operation

Display Control 1 (R07h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	PT2	PT1	PT0	VLE2	VLE1	SPT	0	0	GON	DTE	CL	REV	D1	D0

PT2-0: Sets the kind of source output in non-display areas in partial display mode. For details, see the Partial Display Function section.

Table 21

PT2	PT1	PT0	Source Output in Non-display Area		Operating grayscale amplifier in non-display area
			Positive Polarity	Negative Polarity	
0	0	0	V31	V0	V0 ~ V31
0	0	1	Setting disabled	Setting disabled	-
0	1	0	GND	GND	V0 ~ V31
0	1	1	High impedance	High impedance	V0 ~ V31
1	0	0	V31	V0	V0, V31
1	0	1	Setting disabled	Setting disabled	-
1	1	0	GND	GND	V0, V31
1	1	1	High impedance	High impedance	V0, V31

VLE2-1: When VLE1 = 1, the first display is scrolled in vertical direction. When VLE2 = 1, the second display is scrolled in vertical direction. The first and second displays cannot be scrolled simultaneously. This function is not available with external display interface. In this case, set VLE2-1 = 00.

Table 22

VLE2	VLE1	2nd display (image)	1st display (image)
0	0	Fixed	Fixed
0	1	Fixed	Scrolled
1	0	Scrolled	Fixed
1	1	Setting disabled	Setting disabled

CL: When CL = 1, 8-color display mode is selected. For details, see the 8-color Display Mode section. The 8-color display mode is disabled in external interface mode.

Table 23

CL	Colors
0	262,144
1	8

R61500

SPT: When SPT = 1, the R61500 drives 2 split screens. For details, see the Partial Display Function section. This function is disabled in external display interface mode. In this case, set SPT = 0.

REV: By setting REV = 1, the grayscale levels for V0 ~V31 can be inverted. It enables both normally black and normally white panels to display a same image from the same data. The source output level during front and back porch periods and a blank period in partial display mode is set with the PT2-0 bits.

Table 24

REV	GRAM Data	Source Output in Display Area*	
		Positive Polarity	Negative Polarity
0	18'h00000	V31	V0
	18'h3FFFF	V0	V31
1	18'h00000	V0	V31
	18'h3FFFF	V31	V0

GON, DTE: Sets the output level from gate lines G1 ~ G240 as follows.

Table 25

GON	DTE	Gate Output G1~240
0	0	VGH
0	1	VGH
1	0	VGL
1	1	VGH/VGL

R61500

D1-0: The graphics display appears on the screen when D1 = 1, and it is turned off upon setting D1 = 0. When setting D1 = 0, the graphics display data is retained in the internal GRAM and the display appears on the screen upon setting D1 to 1. When the D1 bit is 0, i.e. while no display is shown on the screen, all source output levels becomes the GND level to reduce charging/discharging current on liquid crystal cells, which is generated when applying voltages with polarity inversion to the LCD.

Upon setting D1-0 = 00, the display is turned off and the internal display operation is halted completely.

In combination with the GON, DTE bit, the D1-0 bits controls ON/OFF of graphics display. For details, see the flowcharts in the Instruction Setting section.

Table 26

D1	D0	Source, Vcom Output	R61500 Internal Operations	Clock signals for internal display operation (CL1, FLM, M)
0	0	GND	Halt	OFF
0	1	GND	Operate	ON
1	0	Non-lit display	Operate	ON
1	1	Display	Operate	ON

Note 1) The data write operation from the microcomputer is not affected by the setting in the D1-0 bits.

Note 2) When D1-0 = 00, the R61500's internal state becomes the same as the standby mode. This does not mean the D1-0 bits are written over to 00 upon setting the standby mode.

Display Control 2 (R08h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	FP3	FP2	FP1	FP0	0	0	0	0	BP3	BP2	BP1	BP0

FP3-0/BP3-0: Sets the blank periods made at the beginning and the end of a display (front porch and back porch, respectively). The FP3-0 and BP3-0 bits specify the number of lines for the front and back porch periods, respectively. In setting, be sure:

$$BP + FP \leq 16 \text{ lines}$$

$$FP \geq 2 \text{ lines}$$

$$BP \geq 2 \text{ lines}$$

In external display interface mode, a back porch (BP) period starts on the falling edge of the VSYNC signal, followed by a display operation period. After driving the number of lines set with NL bits, a front porch period starts. After the front porch period, a blank period continues until the next VSYNC input.

Table 27

FP3	FP2	FP1	FP0	Number of lines for the Front Porch
BP3	BP2	BP1	BP0	Number of lines for the Back Porch
0	0	0	0	Setting disabled
0	0	0	1	Setting disabled
0	0	1	0	2 lines
0	0	1	1	3 lines
0	1	0	0	4 lines
0	1	0	1	5 lines
0	1	1	0	6 lines
0	1	1	1	7 lines
1	0	0	0	8 lines
1	0	0	1	9 lines
1	0	1	0	10 lines
1	0	1	1	11 lines
1	1	0	0	12 lines
1	1	0	1	13 lines
1	1	1	0	14 lines
1	1	1	1	Setting disabled

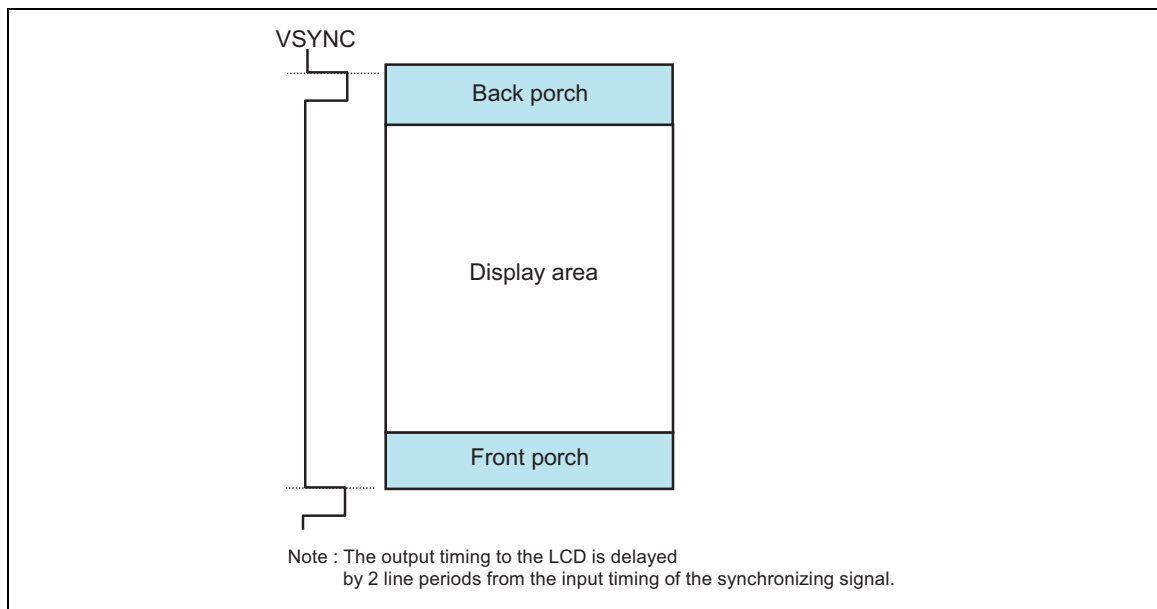


Figure 9 Back/Front porch periods

R61500

BP3-0, FP3-0 Setting

Set the BP3-0, FP3-0 bits as follows in each operation mode.

Table 28

Operation of internal clock	FLD1-0 = 01	BP ≥ 2 lines	FP ≥ 2 lines	FP +BP ≤ 16 lines
	FLD1-0 = 11	BP = 3 lines	FP = 5 lines	
RGB interface		BP ≥ 2 lines	FP ≥ 2 lines	FP +BP ≤ 16 lines
VSYNC interface		BP ≥ 2 lines	FP ≥ 2 lines	FP +BP = 16 lines

Display Control 3 (R09h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	0	0	PTG1	PTG0	ISC3	ISC2	ISC1	ISC0

PTG1-0: Sets the scan mode by the gate driver in non-display area.

Table 29

PTG1	PTG0	Gate output in non-display area	Source output in non-display area	Vcom output
0	0	Normal scan	Set with the PT2-0 bits	VcomH/VcomL amplitude
0	1	VGL (Fixed)	Set with the PT2-0 bits	VcomH/VcomL amplitude
1	0	Interval scan	Set with the PT2-0 bits	VcomH/VcomL amplitude
1	1	Setting Disabled	-	-

ISC3-0: Sets the scan cycle by the gate driver when the PTG1-0 bits are set to the interval scan mode in non-display area. The scan cycle can be set in units of n frames, where n is an odd number from 0 to 31. In this case, polarity is inverted when scanning gate lines.

R61500

Table 30

ISC3	ISC2	ISC1	ISC0	Scan cycle	(fFLM) = 60Hz
0	0	0	0	0 frame	—
0	0	0	1	3 frames	50 mS
0	0	1	0	5frames	84 mS
0	0	1	1	7 frames	117 mS
0	1	0	0	9 frames	150 mS
0	1	0	1	11 frames	184 mS
0	1	1	0	13 frames	217 mS
0	1	1	1	15 frames	251 mS
1	0	0	0	17 frames	284 mS
1	0	0	1	19 frames	317 mS
1	0	1	0	21 frames	351 mS
1	0	1	1	23 frames	384 mS
1	1	0	0	25 frames	418 mS
1	1	0	1	27 frames	451 mS
1	1	1	0	29 frames	484 mS
1	1	1	1	31 frames	518 mS

Frame Cycle Control (R0Bh)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	NO0	NO1	SDT1	SDT0	EQ1	EQ0	DIV1	DIV0	VEM	0	0	0	RTN3	RTN2	RTN1	RTN0

RTN3-0: Sets the 1H (1 line) period.

Table 31

RTN3	RTN2	RTN1	RTN0	Clock Cycles per line
0	0	0	0	16 clocks
0	0	0	1	17 clocks
0	0	1	0	18 clocks
0	0	1	1	19 clocks
0	1	0	0	20 clocks
0	1	0	1	21 clocks
0	1	1	0	22 clocks
0	1	1	1	23 clocks
1	0	0	0	24 clocks
1	0	0	1	25 clocks
1	0	1	0	26 clocks
1	0	1	1	27 clocks
1	1	0	0	28 clocks
1	1	0	1	29 clocks
1	1	1	0	30 clocks
1	1	1	1	31 clocks

DIV1-0: The internal operation is synchronized with the divided clock. The division ratio of the clock is set with the DIV1-0 bits. By setting the RTN and DIV bits, frame frequency can be adjusted. If the number of lines to drive the LCD is changed, the frame frequency must also be adjusted. See Frame-Frequency Adjustment Function. In RGB interface mode, the DIV1-0 bits are disabled.

Table 32

DIV1	DIV0	Division Ratio	Internal Operation Clock Frequency
0	0	1	$f_{osc} / 1$
0	1	2	$f_{osc} / 2$
1	0	4	$f_{osc} / 4$
1	1	8	$f_{osc} / 8$

f_{osc} = RC oscillation frequency

R61500

Formula to calculate frame frequency

$$\text{Frame frequency} = \frac{f_{osc}}{\text{Clock cycles per line} \times \text{division ratio} \times (\text{Line} + \text{BP} + \text{FP})} \quad [\text{Hz}]$$

f_{osc} : frequency of RC oscillation

Line: number of lines for driving liquid crystal (NL bits)

Division ratio: DIV bits

Clock cycles per line: RTN bits

FP: the number of lines for the front porch period

BP: the number of lines for the back porch period

VEM: Sets either one of the following low power consumption modes when driving the LCD with Vcom. The period for the low power consumption mode is set with the EQ1-0 bits

Table 33

VEM	Mode
0	Equalization
1	Low power consumption drive

Note 1: The power saving effect depends on the size of the panel and the kind of display on the screen. It is recommended to check the quality of display on the panel in use.

Note 2: When using the equalization mode, make sure $V_{comL} \geq GND$.

Note 3: When using the low power consumption mode, make sure $V_{ci} < V_{comH}$, $GND > V_{comL}$.

EQ1-0: Extends the period for equalization and low power consumption drive by the number of clocks. The signals for equalization and low power consumption drive are output at the polarity inversion timing.

Table 34

		Equalization period	
EQ1	EQ0	Internal Operation (reference clock: internal oscillator)	RGB I/F Operation (reference clock: DOTCLK)
0	0	Not equalized	Not equalized
0	1	1 clock	8 clocks
1	0	2 clocks	16 clocks
1	1	3 clocks	24 clocks

R61500

SDT1-0: Sets the source output delay from the falling edge of gate output signal by the number of clocks as follows.

Table 35

Source output delay			
SDT1	SDT0	Internal Operation (reference clock: internal oscillator)	RGB I/F Operation (reference clock: DOTCLK)
0	0	1 clock	8 clocks
0	1	2 clocks	16 clocks
1	0	3 clocks	24 clocks
1	1	4 clocks	32 clocks

NO1-0: Sets the gate output non-overlap period.

Table 36

Gate output non-overlap period			
NO1	NO0	Internal Operation (reference clock: internal oscillator)	RGB I/F Operation (reference clock: DOTCLK)
0	0	0 clock	0 clock
0	1	4 clocks	32 clocks
1	0	6 clocks	48 clocks
1	1	8 clocks	64 clocks

Note 1) The signals in the parentheses in the following figure are internal signals.

Note 2) The non-overlap period is measured from the falling edge of the CL1.

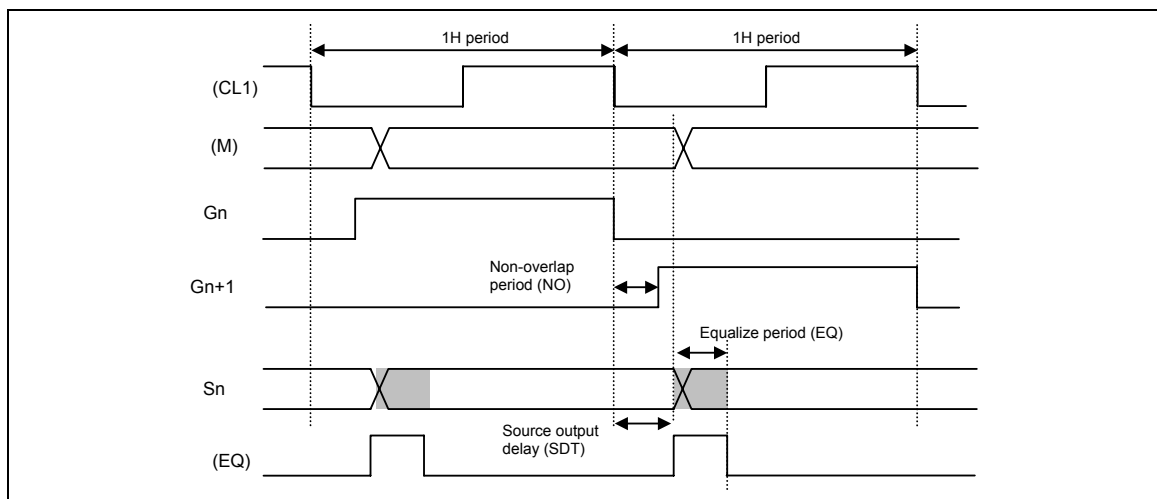


Figure 10

R61500

Note that the clock mentioned in the description of EQ1-0, SDT1-0 and NO1-0 refers to different clocks according to the interface in use as follows.

Table 37

Interface mode in use	Reference clock
Internal operation mode	Internal clock
RGB interface more	DOTCLK
VSYNC interface mode	Internal clock

External Display Interface Control 1 (R0Ch)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	ENC1	ENC0	0	0	0	RM	0	0	DM1	DM0	0	0	RIM1	RIM0

RIM1-0: Selects one of the following RGB interface modes when the RGB interface mode is selected with the RM and DM bits. Make this setting before starting display operation via external display interface. Do not make changes to the setting during display operation.

DM1-0: Sets the display operation mode. By setting DM[1:0] as follows, it is possible to switch between the internal clock operation mode and the external display interface mode. Do not switch RGB interface and VSYNC interface.

Table 38 RIM1-0 bits

RIM1	RIM0	RGB Interface Mode
0	0	18-bit RGB interface (one transfer/pixel)
0	1	16-bit RGB interface (one transfer/pixel)
1	0	6-bit RGB interface (three transfers/pixel)
1	1	Setting disabled

Table 39 DM1-0 bits

DM1	DM0	Display Interface
0	0	Internal clock operation
0	1	RGB interface
1	0	VSYNC interface
1	1	Setting disabled

Note 1) Instructions are set only via system interface.

Note 2) make sure that data transfer operation and DOTCLK input are performed in units of dots in 6-bit RGB interface mode.

RM: Selects the interface to access the R61500's internal GRAM. The RAM access is possible only via the interface selected with the RM bit. Set RM to 1 when writing display data via the RGB interface. The R61500 allows for setting the RM bit not constrained by the mode used for the display operation. This means it is possible to rewrite display data via system interface by setting RM = 0 even while the display operation is performed via RGB interface.

ENC1-0: Sets the RAM data write cycle in RGB interface mode.

Table 40 RM bit

RM	Interface for RAM Access
0	System interface/VSYNC interface
1	RGB interface

Table 41 ENC1-0 bits

ENC1	ENC0	RAM data write cycle
0	0	Write data to RAM every frame cycle
0	1	Setting disabled
1	0	Write data to RAM every 2 frame cycles
1	1	Write data to RAM every 4 frame cycles

R61500

As the following table, the R61500 allows selecting an optimum interface for the display by setting external display interface control register.

When displaying a moving picture via RGB interface or VSYNC interface, use a high-speed write function (either HWM = 1 or LHWM = 1) so that the internal GRAM is accessed in high speed with low power consumption.

Table 42

Display State	Operation Mode	RAM Access (RM)	Display Operation Mode (DM1-0)
Still pictures	Internal clock operation	System interface (RM = 0)	Internal clock operation (DM1-0 = 00)
Moving pictures	RGB interface (1)	RGB interface (RM = 1)	RGB interface (DM1-0 = 01)
Rewrite still picture area while displaying moving pictures.	RGB interface (2)	System interface (RM = 0)	RGB interface (DM1-0 = 01)
Moving pictures	VSYNC interface	System interface (RM = 0)	VSYNC interface (DM1-0 = 10)

Note 1) Instructions are set only via system interface.

Note 2) RGB interface and VSYNC interface cannot be used simultaneously.

Note 3) Do not make changes to the RGB interface setting (RIM-0) while it is in operation.

Note 4) See the External Display Interface section for the sequences when switching from one mode to another.

Note 5) Use a high-speed write function (HWM/LHWM = 1) when writing data via RGB or VSYNC interface.

Internal clock operation mode

The display operation is synchronized with the signal generated from the internal operating clock in this mode. No input via external display interface is valid. The internal RAM is accessible only via system interface.

RGB interface mode (1)

In RGB interface mode, the display operation is synchronized with the frame synchronous signal (VSYNC), the line synchronous signal (HSYNC), and the dot clock (DOTCLK). These signals must be supplied throughout the display period via RGB interface.

The display data is transferred in units of pixels via DB17-0. The display data is stored in the internal RAM. In combination with high-speed RAM write function and window address function, RGB interface can display both data in the moving picture area and the data in the internal RAM at a time. This structure enables R61500 minimize the total data transfer by transferring the data for rewriting moving picture frames.

The front porch (FP) and back porch (BP) periods, and the display duration period (NL) are automatically calculated inside the R61500 by counting the number of line synchronous signal clocks (HSYNC) from the falling edge of the frame synchronous signal (VSYNC). Take this into consideration when transferring data via DB17-0.

R61500

RGB interface mode (2)

The R61500 enables rewriting RAM data via system interface while RGB interface is selected for the display operation. In this case, make sure to write RAM data while data transfer via RGB interface is not performed (ENABLE = High). To return to the display data transfer mode via RGB interface, change the ENABLE bit first and then set a new address (AD15-0) in the AC and the index register to R22h.

VSYNC interface mode

In VSYNC interface mode, the R61500's internal display operation is synchronized with the frame synchronous signal (VSYNC). In this mode, a moving picture can be displayed via system interface by writing data to the internal RAM at a calculated minimum or higher speed from the falling edge of frame synchronous signal (VSYNC). In this case, there are constraints in the RAM writing speed and method. For details, see External Display Interface.

No external signal input except VSYNC input is accepted in VSYNC interface mode.

The timings and durations of front porch (FP), back porch (BP) periods and display period (NL) are automatically calculated from the falling edge of the frame synchronization signal (VSYNC) according to the register setting.

Driver Output Control 1 (R0Dh)**Driver Output Control 2 (R0Eh)**

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	VSEL 1	VSEL 0	0	CPB W4	CPB W3	CPB W2	CPB W1	CPB W0	0	0	0	CPA W4	CPA W3	CPA W2	CPA W1	CPA W0
W	1	0	0	0	CNB W4	CNB W3	CNB W2	CNB W1	CNB W0	0	0	0	CNA W4	CNA W3	CNA W2	CNA W1	CNA W0

CPAW4-0: Sets the source output period of positive polarity.

CPBW4-0: Sets the source output period of positive polarity.

CNAW4-0: Sets the source output period of negative polarity.

CNBW4-0: Sets the source output period of negative polarity.

Make sure: $CPAW + CPBW \leq (1H - 1\text{-clock period})$
 $CNAW + CNBW \leq (1H - 1\text{-clock period})$

Table 43

CPAW4	CPAW3	CPAW2	CPAW1	CPAW0	Source output period
CPBW4	CPBW3	CPBW2	CPBW1	CPBW0	
CNAW4	CNAW3	CNAW2	CNAW1	CNAW0	
CNBW4	CNBW3	CNBW2	CNBW1	CNBW0	
0	0	0	0	0	Setting disabled
0	0	0	0	1	1 clock
0	0	0	1	0	2 clocks
0	0	0	1	1	3 clocks
0	0	1	0	0	4 clocks
:	:	:	:	:	:
0	1	1	0	0	28 clocks
1	1	1	0	1	29 clocks
1	1	1	1	0	30 clocks
1	1	1	1	1	31 clocks

Note: Adjustment will be required to the number of clocks for RGB interface according to the number of DOTCLK during HSYNC high period.

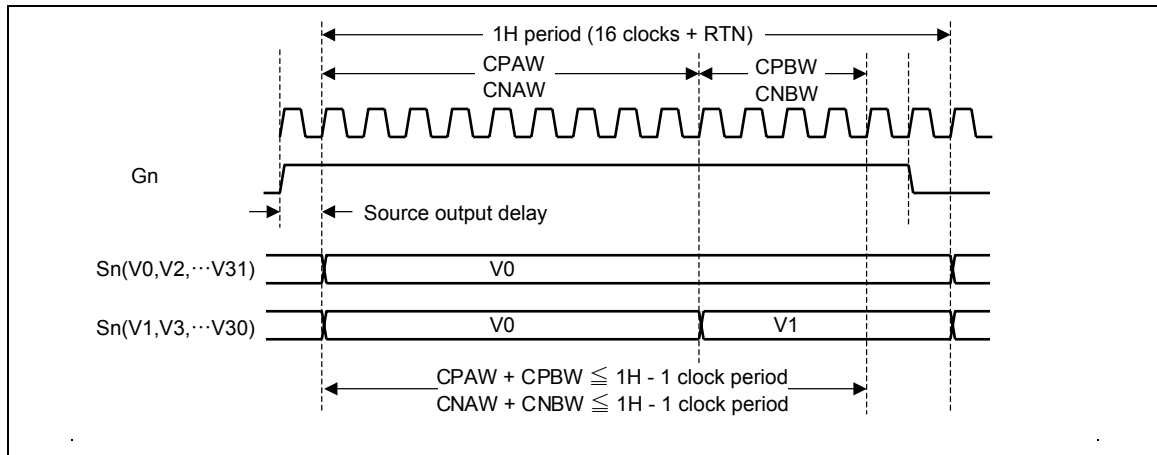


Figure 11

VSEL1-0: Sets the grayscale output order as follows.

Table 44

VSEL1	VSEL0	Grayscale levels output in the CPAW, CNAW periods	Grayscale levels output in the CPBW, CNBW periods
0	0	V0, V2, V4, V6, V8, V10, V12, V14, V17, V19, V21, V23, V25, V27, V29, V31	V1, V3, V5, V7, V9, V11, V13, V15, V16, V18, V20, V22, V24, V26, V28, V30
0	1	V0, V2, V4, V6, V8, V10, V12, V14, V16, V18, V20, V22, V24, V26, V28, V30	V1, V3, V5, V7, V9, V11, V13, V15, V17, V19, V21, V23, V25, V27, V29, V31
1	0	V1, V3, V5, V7, V9, V11, V13, V15, V17, V19, V21, V23, V25, V27, V29, V31	V0, V2, V4, V6, V8, V10, V12, V14, V16, V18, V20, V22, V24, V26, V28, V30
1	1	V1, V3, V5, V7, V9, V11, V13, V15, V16, V18, V20, V22, V24, V26, V28, V30	V0, V2, V4, V6, V8, V10, V12, V14, V17, V19, V21, V23, V25, V27, V29, V31

External Display Interface Control 2 (R0Fh)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	0	0	0	CSTP 4	CSTP 3	CSTP 2	CSTP 1	CSTP 0

CSTP4-0: Set the number of clock in 1H period for RGB interface operation according to the following table.

Table 45

DOTCLK in HSYNC H period	CSTP4-0	Internal clocks in 1H period
132 ~ 139 clocks	10000	16 clocks
140 ~ 147 clocks	10001	17 clocks
148 ~ 155 clocks	10010	18 clocks
156 ~ 163 clocks	10011	19 clocks
164 ~ 171 clocks	10100	20 clocks
172 ~ 179 clocks	10101	21 clocks
180 ~ 187 clocks	10110	22 clocks
188 ~ 195 clocks	10111	23 clocks
196 ~ 203 clocks	11000	24 clocks
204 ~ 211 clocks	11001	25 clocks
212 ~ 219 clocks	11010	26 clocks
220 ~ 227 clocks	11011	27 clocks
228 ~ 235 clocks	11100	28 clocks
236 ~ 243 clocks	11101	29 clocks
244 ~ 251 clocks	11110	30 clocks
252 ~ 255 clocks	11111	31 clocks

Power Control 1 (R10h)**Power Control 2 (R11h)**

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	SAP2	SAP1	SAP0	0	BT2	BT1	BT0	0	AP2	AP1	AP0	0	DK	SLP	STB
W	1	0	0	0	0	0	DC12	DC11	DC10	0	DC02	DC01	DC00	0	VC2	VC1	VC0

SAP2-0: Adjusts the constant current in the constant current source in the operational amplifier circuit for the source driver. Setting a larger constant current stabilizes the operation of amplifiers, but increases current consumption. Adjust the constant current taking the trade-off between the display quality and the current consumption into account. During no display period, set SAP2-0 = 000 to halt the operational amplifier circuit to reduce current consumption.

Table 46

SAP2	SAP1	SAP0	Constant current (for source driver)	SAP2	SAP1	SAP0	Constant current (for source driver)
0	0	0	Halt	1	0	0	1
0	0	1	Setting disabled	1	0	1	1.25
0	1	0	0.62	1	1	0	1.43
0	1	1	0.71	1	1	1	Setting disabled

Note: The constant current in the table is the ratio to the constant current when SAP[2:0] = 100.

BT2-0: Changes the factor applied to the step-up circuit. Adjust the step-up factor according to the voltage in use. To reduce current consumption, set a smaller step-up factor.

DC02-00: Selects the operating frequency of the step-up circuit 1. The higher step-up operating frequency enhances the drivability of the step-up circuit and the quality of display. Adjust the frequency taking the trade-off between the display quality and the current consumption into account.

DC12-10: Selects the operating frequency of the step-up circuit 2. The higher step-up operating frequency enhances the drivability of the step-up circuit and the quality of display. Adjust the frequency taking the trade-off between the display quality and the current consumption into account.

Note: in setting step-up cycles of step-up circuits 1/2, make sure the step-up cycle of the step-up circuit 1 is more than that of the step-up circuit 2.

AP2-0: Adjusts the constant current in the constant current source in the operational amplifier circuit for the LCD power supply. Setting a larger constant current stabilizes the operation of amplifiers, but increases current consumption. Adjust the constant current taking the trade-off between the display quality and the current consumption into account. During no display period, set AP2-0 = 000 to halt the operational amplifier circuit and the step-up circuits to reduce current consumption.

R61500

DK: Controls the operation of step-up circuit 1. In supplying power to the R61500, stop generating VLOUT1 for a moment, and wait until the VLOUT2 level is stabilized. Then start generating the VLOUT1 level. For details, see the Power Supply Setting section.

SLP: When SLP = 1, the R61500 enters the sleep mode. In sleep mode, the internal display operation except RC oscillation is halted to reduce current consumption. In sleep mode, only the power control register instructions, BT2-0, DC02-00, DC12-10, AP2-0, SLP, STB, VC2-0, VRH4-0, VCOMG, VDV4-0, and VCM4-0, are accepted. Changes to the GRAM data or other instruction sets are not accepted. In sleep mode, the GRAM data and the instruction sets written before the sleep mode are retained.

STB: When STB = 1, the R61500 enters the standby mode. In standby mode, the display operation is completely halted, and the R61500's internal operation including internal RC oscillation and reception of external clocks is halted. See the Instruction Setting section for the sequence. Only the instructions to exit standby mode (STB = 0) or to start oscillators are accepted during standby mode. There are cases that GRAM data and instruction sets written before the standby mode are destroyed. Set them again after exiting standby mode.

VC2-0: Sets the factor applied to VciLVL to generate the VREG1OUT level and the reference voltage for Vci1.

Table 47

AP2	AP1	AP0	Constant current (for LCD power supply)	AP2	AP1	AP0	Constant current (for LCD power supply)
0	0	0	Halt	1	0	0	1
0	0	1	Setting disabled	1	0	1	1.25
0	1	0	0.5	1	1	0	1.5
0	1	1	0.75	1	1	1	Setting disabled

Note: The constant current in the table is the ratio to the constant current when AP[2:0] = 100.

Table 48

DC02	DC01	DC00	Step-up circuit 1 step-up cycle (fDCDC1)	DC12	DC11	DC10	Step-up circuit 2 step-up cycle (fDCDC2)
0	0	0	Oscillation clock / 8	0	0	0	Oscillation clock / 16
0	0	1	Oscillation clock / 16	0	0	1	Oscillation clock / 32
0	1	0	Oscillation clock / 32	0	1	0	Oscillation clock / 64
0	1	1	Oscillation clock / 64	0	1	1	Oscillation clock / 128
1	0	0	Oscillation clock / 128	1	0	0	Oscillation clock / 256
1	0	1	Setting disabled	1	0	1	Setting disabled
1	1	0	Setting disabled	1	1	0	Setting disabled
1	1	1	Setting disabled	1	1	1	Setting disabled

Note: Make sure fDCDC1 ≥ fDCDC2 when setting DC02-00, DC12-10.

R61500

Table 49

BT2	BT1	BT0	VLOUT1 output (DDVDH)	VLOUT4 output (VCL)	VLOUT2 output (VGH) see note 4)	VLOUT3 output (VGL)	Capacitor connection pins
0	0	0	$V_{ci1} \times 2$ [x2]	$V_{ci1} \times -1$ [x-1]	$DDVDH \times 3$ [x 6]	$-(V_{ci1} + DDVDH \times 2)$ [x -5]	DDVDH, VGH, VGL, VCL, C11±, C12±, C21±, C22±
0	0	1	$V_{ci1} \times 2$ [x2]	$V_{ci1} \times -1$ [x-1]	$DDVDH \times 3$ [x 6]	$-(DDVDH \times 2)$ [x -4]	DDVDH, VGH, VGL, VCL, C11±, C12±, C21±, C22±
0	1	0	$V_{ci1} \times 2$ [x2]	$V_{ci1} \times -1$ [x-1]	$DDVDH \times 3$ [x 6]	$-(V_{ci1} + DDVDH)$ [x -3]	DDVDH, VGH, VGL, VCL, C11±, C12±, C21±, C22±
0	1	1	$V_{ci1} \times 2$ [x2]	$V_{ci1} \times -1$ [x-1]	$V_{ci1} + DDVDH \times 2$ [x 5]	$-(V_{ci1} + DDVDH \times 2)$ [x -5]	DDVDH, VGH, VGL, VCL, C11±, C12±, C21±, C22±
1	0	0	$V_{ci1} \times 2$ [x2]	$V_{ci1} \times -1$ [x-1]	$V_{ci1} + DDVDH \times 2$ [x 5]	$-(DDVDH \times 2)$ [x -4]	DDVDH, VGH, VGL, VCL, C11±, C12±, C21±, C22±
1	0	1	$V_{ci1} \times 2$ [x2]	$V_{ci1} \times -1$ [x-1]	$V_{ci1} + DDVDH \times 2$ [x 5]	$-(V_{ci1} + DDVDH)$ [x -3]	DDVDH, VGH, VGL, VCL, C11±, C12±, C21±, C22±
1	1	0	$V_{ci1} \times 2$ [x2]	$V_{ci1} \times -1$ [x-1]	$DDVDH \times 2$ [x 4]	$-(DDVDH \times 2)$ [x -4]	DDVDH, VGH, VGL, VCL, C11±, C12±, C21±, C22±
1	1	1	$V_{ci1} \times 2$ [x2]	$V_{ci1} \times -1$ [x-1]	$DDVDH \times 2$ [x 4]	$-(V_{ci1} + DDVDH)$ [x -3]	DDVDH, VGH, VGL, VCL, C11±, C12±, C21±

Note 1) The step-up factor from the V_{ci1} level is shown in the bracket [] in the above table.

Note 2) Connect capacitors for step-up operation where they are required when using DDVDH, VCL, VGH and VGL.

Note 3) Make sure the following voltages are within the limits:

DDVDH = max. 5.5V, VCL = min. -3.3V, VGH = max. 16.5V, VGL = min. -16.5V.

Table 50

VC2	VC1	VC0	V_{ciOUT} output voltage (REGP)
0	0	0	V_{ciLVL} *see Note
0	0	1	$0.92 \times V_{ciLVL}$
0	1	0	$0.87 \times V_{ciLVL}$
0	1	1	$0.83 \times V_{ciLVL}$
1	0	0	$0.76 \times V_{ciLVL}$
1	0	1	$0.73 \times V_{ciLVL}$
1	1	0	Setting disabled
1	1	1	Setting disabled

Table 51

DK	Operation of step-up circuit 1
0	Operate
1	Halt

R61500

Power Control 3 (R12h)

Power Control 4 (R13h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	0	0	0	PON	VRH3	VRH2	VRH1	VRH0
W	1	0	0	VCOMG	VDV4	VDV3	VDV2	VDV1	VDV0	0	0	0	VCM4	VCM3	VCM2	VCM1	VCM0

PON: Controls ON/OFF of VLOUT3 output. To stop VLOUT3 output, set PON = 0. To start VLOUT3 output, set PON = 1.

VRH3-0: Sets the amplifying factor (1.33 ~ 1.90) applied to REGP to generate the VREG1OUT level. VREG1OUT is a reference level for VCOM and the grayscale voltage level.

VCOMG: When VCOMG = 1, the R61500 can output a negative voltage level for VcomL (1.0V ~ -Vci+0.5V Max.). When VCOMG = 0, the R61500 halts the amplifier for negative voltage to save power. When VCOMG = 0, the VDV4-0 bits are disabled. In this case, adjust the amplitude of Vcom with VcomH setting (VCM[4:0]). VCOMG = 1 is valid when PON = 1.

VDV4-0: Sets the amplitude of Vcom when driving the LCD with Vcom. The VDV4-0 bits can set the Vcom amplitude 0.6 ~ 1.23 times the VREG1OUT level. When VCOMG = 0, the VDV4-0 bits are disabled.

VCM4-0: Sets the VcomH level (the higher level of Vcom). The VCM4-0 bits can set the VcomH level 0.4 ~ 0.98 times the VREG1OUT level. To stop adjusting VcomH with the internal electronic volume and adjust it with an external resistor from VcomR, set VCM4-0 = 1111.

Table 52 VRH3-0

VRH3	VRH2	VRH1	VRH0	VREG1OUT voltage
0	0	0	0	REGP x 1.33
0	0	0	1	REGP x 1.45
0	0	1	0	REGP x 1.55
0	0	1	1	REGP x 1.65
0	1	0	0	REGP x 1.75
0	1	0	1	REGP x 1.80
0	1	1	0	REGP x 1.85
0	1	1	1	Halt

VRH3	VRH2	VRH1	VRH0	VREG1OUT voltage
1	0	0	0	REGP x 1.90
1	0	0	1	Setting disabled
1	0	1	0	Setting disabled
1	0	1	1	Setting disabled
1	1	0	0	Setting disabled
1	1	0	1	Setting disabled
1	1	1	0	Setting disabled
1	1	1	1	Setting disabled

Table 53 VCM4-0

VCM4	VCM3	VCM2	VCM1	VCM0	VcomH
0	0	0	0	0	VREG1OUT x 0.40
0	0	0	0	1	VREG1OUT x 0.42
0	0	0	1	0	VREG1OUT x 0.44
0	0	0	1	1	VREG1OUT x 0.46
0	0	1	0	0	VREG1OUT x 0.48
0	0	1	0	1	VREG1OUT x 0.50
0	0	1	1	0	VREG1OUT x 0.52
0	0	1	1	1	VREG1OUT x 0.54
0	1	0	0	0	VREG1OUT x 0.56
0	1	0	0	1	VREG1OUT x 0.58
0	1	0	1	0	VREG1OUT x 0.60
0	1	1	1	1	VREG1OUT x 0.62
0	1	1	0	0	VREG1OUT x 0.64
0	1	1	0	1	VREG1OUT x 0.66
0	1	1	1	0	VREG1OUT x 0.68
0	1	1	1	1	Halt internal volume. Adjust with a variable external resistor from VcomR.
1	0	0	0	0	VREG1OUT x 0.70
1	0	0	0	1	VREG1OUT x 0.72
1	0	0	1	0	VREG1OUT x 0.74
1	0	0	1	1	VREG1OUT x 0.76
1	0	1	0	0	VREG1OUT x 0.78
1	0	1	0	1	VREG1OUT x 0.80
1	0	1	1	0	VREG1OUT x 0.82
1	0	1	1	1	VREG1OUT x 0.84
1	1	0	0	0	VREG1OUT x 0.86
1	1	0	0	1	VREG1OUT x 0.88
1	1	0	1	0	VREG1OUT x 0.90
1	1	0	1	1	VREG1OUT x 0.92
1	1	1	0	0	VREG1OUT x 0.94
1	1	1	0	1	VREG1OUT x 0.96
1	1	1	1	0	VREG1OUT x 0.98
1	1	1	1	1	Halt internal volume adjustment. Adjust with a variable external resistor from VcomR.

Table 54 VDV4-0

VDV4	VDV3	VDV2	VDV1	VDV0	Vcom amplitude
0	0	0	0	0	VREG1OUT x 0.60
0	0	0	0	1	VREG1OUT x 0.63
0	0	0	1	0	VREG1OUT x 0.66
0	0	0	1	1	VREG1OUT x 0.69
0	0	1	0	0	VREG1OUT x 0.72
0	0	1	0	1	VREG1OUT x 0.75
0	0	1	1	0	VREG1OUT x 0.78
0	0	1	1	1	VREG1OUT x 0.81
0	1	0	0	0	VREG1OUT x 0.84
0	1	0	0	1	VREG1OUT x 0.87
0	1	0	1	0	VREG1OUT x 0.90
0	1	0	1	1	VREG1OUT x 0.93
0	1	1	0	0	VREG1OUT x 0.96
0	1	1	0	1	VREG1OUT x 0.99
0	1	1	1	0	VREG1OUT x 1.02
0	1	1	1	1	Setting disabled
1	0	0	0	0	VREG1OUT x 1.05
1	0	0	0	1	VREG1OUT x 1.08
1	0	0	1	0	VREG1OUT x 1.11
1	0	0	1	1	VREG1OUT x 1.14
1	0	1	0	0	VREG1OUT x 1.17
1	0	1	0	1	VREG1OUT x 1.20
1	0	1	1	0	VREG1OUT x 1.23
1	0	1	1	1	Setting disabled
1	1	0	0	0	Setting disabled
1	1	0	0	1	Setting disabled
1	1	0	1	0	Setting disabled
1	1	0	1	1	Setting disabled
1	0	1	0	0	Setting disabled
1	0	1	0	1	Setting disabled
1	0	1	1	0	Setting disabled
1	1	*	*	*	Setting disabled

Note 1) Adjust VREG1OUT and VCM0-4 so that VcomH becomes within the range 3.0V ~ (DDVDH – 0.5)V

Note 2) Adjust VREG1OUT and VDV0-4 so that the amplitude of Vcom becomes 6.0V or less.

R61500

RAM Address Set (R21h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	AD 15	AD 14	AD 13	AD 12	AD 11	AD 10	AD 9	AD 8	AD 7	AD 6	AD 5	AD 4	AD 3	AD 2	AD 1	AD 0

AD15–0: Represents the GRAM address initially set in the AC (Address Counter). The address in the AC is automatically updated in accordance with the AM, I/D bits as the data is written to the internal GRAM. The automatic address update enables consecutive data write without resetting an address in the AC. The address is not automatically updated when reading data from the internal GRAM.

RAM address set is disabled when the R61500 is in standby mode. Also make sure to set an address within the window address area.

Note 1) When RGB interface is selected (RM = 1), the address AD15-0 is set in the address counter for each frame on the falling edge of VSYNC.

Note 2) When either internal clock operation or VSYNC interface operation is selected (RM = 0), the address AD15-0 is set when executing the instruction.

Table 55 GRAM Address Range

AD15–AD0	GRAM Setting
"0000"H – "00AF"H	Bitmap data for G1
"0100"H – "01AF"H	Bitmap data for G2
"0200"H – "02AF"H	Bitmap data for G3
"0300"H – "03AF"H	Bitmap data for G4
:	:
"EC00"H – "ECAF"H	Bitmap data for G237
"ED00"H – "EDAF"H	Bitmap data for G238
"EE00"H – "EEAF"H	Bitmap data for G239
"EF00"H – "EFAF"H	Bitmap data for G240

R61500

Write Data to GRAM (R22h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	RAM write data (WD17-0) The assignment to DB17-0 pins differs for each interface (see below).															

WD17-0: If the data consists of less than 18 bits in unit, the R61500 expands the data into 18 bits internally before written to the internal GRAM. How the data is expanded into 18 bits differs for the interface and the data transfer mode.

The grayscale level is selected according to GRAM data. The GRAM address is automatically updated according to the AM and I/D bits as the data is written to the internal GRAM. In standby mode, no access to the internal GRAM is allowed. When the 8 or 16 bit interface mode is selected, the data is expanded into 18 bits internally by writing the MSBs of R and B dots to the LSBs of R and B dots respectively.

When writing data to the GRAM via system interface while using RGB interface, make sure that the writing operations via two interfaces (RGB and system interfaces) are not conflicting with each other. When 18-bit RGB interface is selected, the 18-bit data is written via DB17-0 and 262,144 colors are available. When 16-bit RGB interface is selected, the MSBs of R and B dots are written to the LSBs of R and B dots respectively, and 65,536 colors are available.

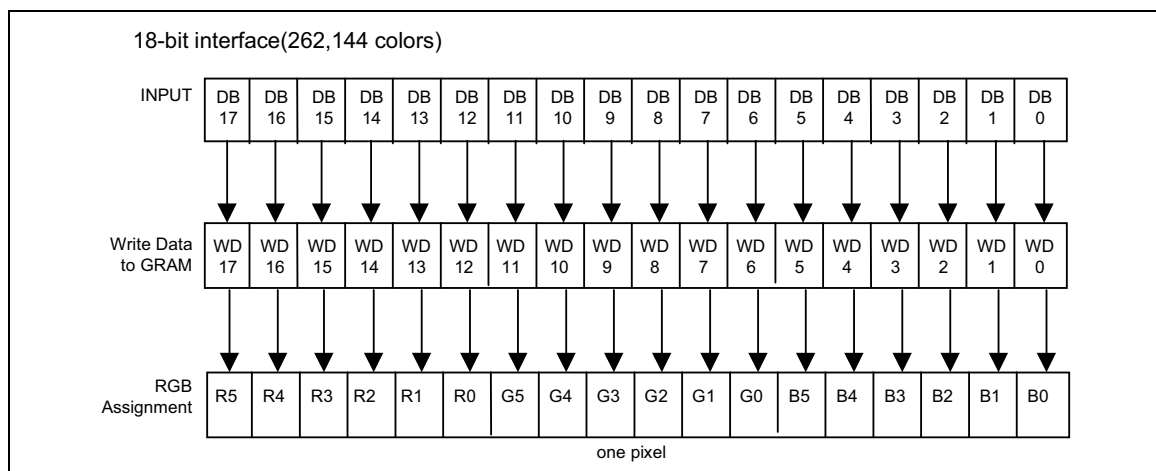


Figure 12 Write data to GRAM in 18-bit interface mode

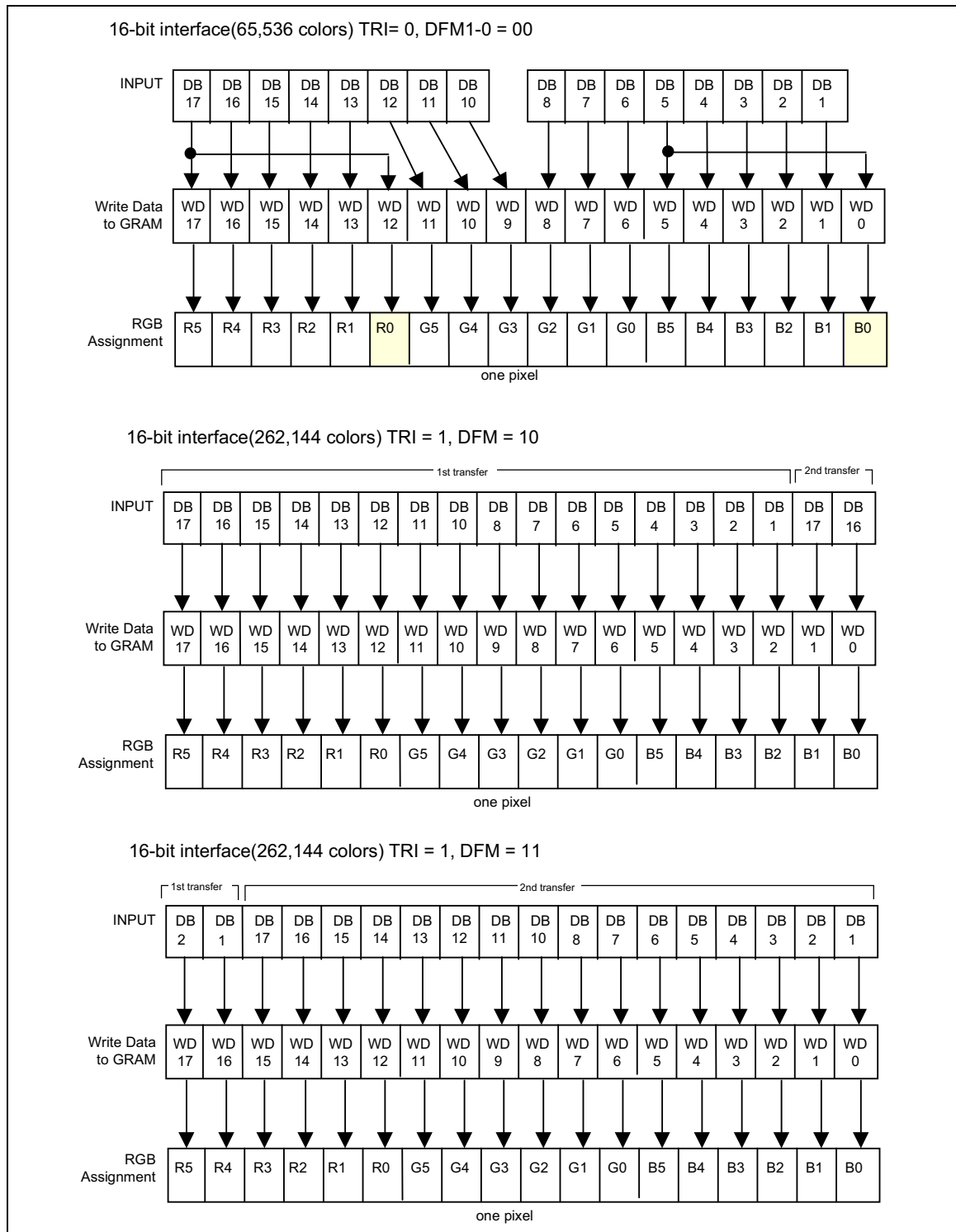


Figure 13 Write data to GRAM in 16-bit interface mode

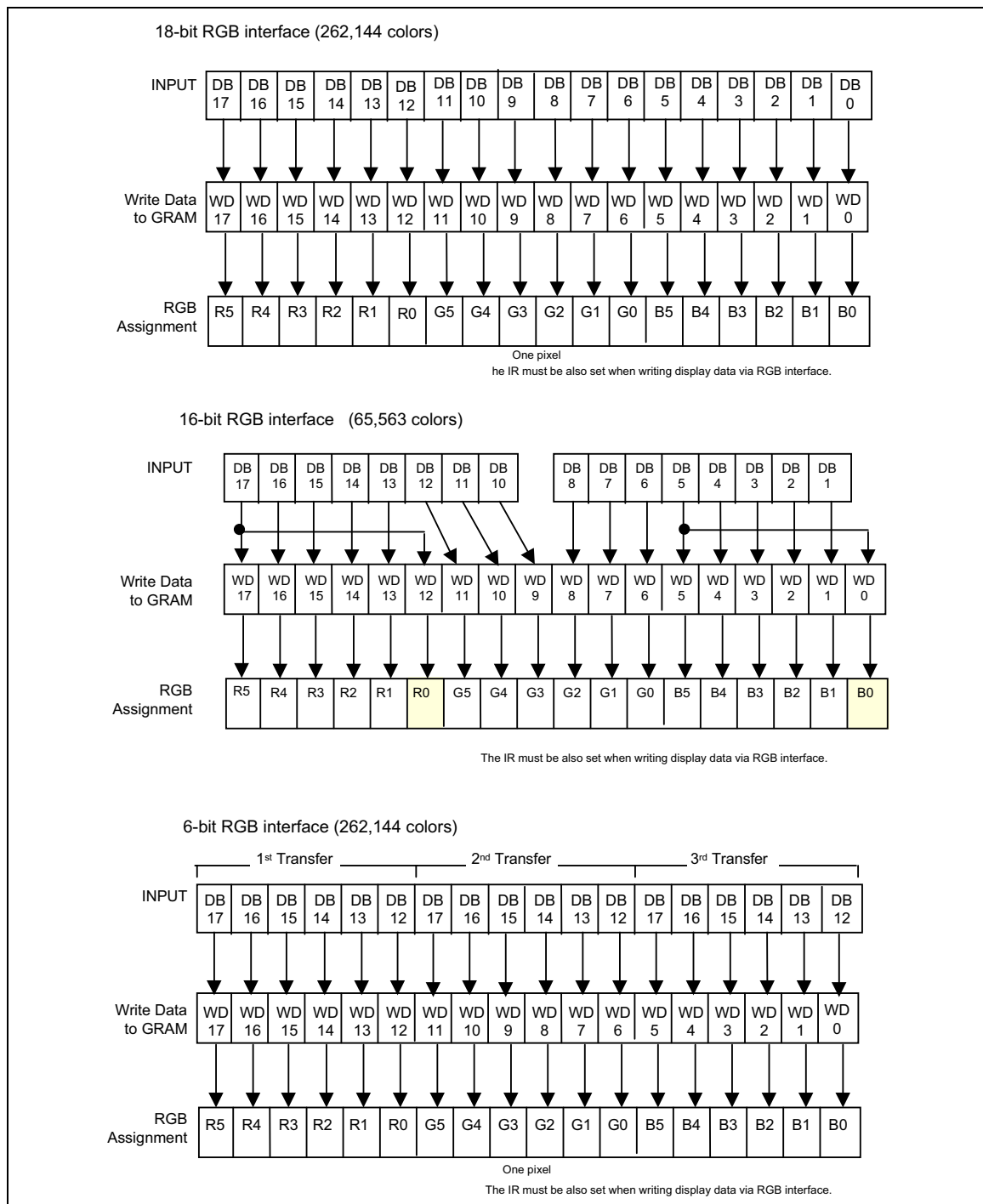


Figure 15 Write data to GRAM in 18/16/6-bit RGB interface mode

Table 56 GRAM data and LCD output level

GRAM data settings	Grayscale	
	Negative	Positive
000000	V0	V31
000001	$(V0+V1)/2$	$(V30+V31)/2$
000010	V1	V30
000011	$(V1+V2)/2$	$(V29+V30)/2$
000100	V2	V29
000101	$(V2+V3)/2$	$(V28+V29)/2$
000110	V3	V28
000111	$(V3+V4)/2$	$(V27+V28)/2$
001000	V4	V27
001001	$(V4+V5)/2$	$(V26+V27)/2$
001010	V5	V26
001011	$(V5+V6)/2$	$(V25+V26)/2$
001100	V6	V25
001101	$(V6+V7)/2$	$(V24+V25)/2$
001110	V7	V24
001111	$(V7+V8)/2$	$(V23+V24)/2$
010000	V8	V23
010001	$(V8+V9)/2$	$(V22+V23)/2$
010010	V9	V22
010011	$(V9+V10)/2$	$(V21+V22)/2$
010100	V10	V21
010101	$(V10+V11)/2$	$(V20+V21)/2$
010110	V11	V20
010111	$(V11+V12)/2$	$(V19+V20)/2$
011000	V12	V19
011001	$(V12+V13)/2$	$(V18+V19)/2$
011010	V13	V18
011011	$(V13+V14)/2$	$(V17+V18)/2$
011100	V14	V17
011101	$(V14+V15)/2$	$(V16+V17)/2$
011110	V15	V16
011111	$(V15+V16)/2$	$(V15+V16)/2$

GRAM data settings	Grayscale	
	Negative	Positive
100000	V16	V15
100001	$(V16+V17)/2$	$(V14+V15)/2$
100010	V17	V14
100011	$(V17+V18)/2$	$(V13+V14)/2$
100100	V18	V13
100101	$(V18+V19)/2$	$(V12+V13)/2$
100110	V19	V12
100111	$(V19+V20)/2$	$(V11+V12)/2$
101000	V20	V11
101001	$(V20+V21)/2$	$(V10+V11)/2$
101010	V21	V10
101011	$(V21+V22)/2$	$(V9+V10)/2$
101100	V22	V9
101101	$(V22+V23)/2$	$(V8+V9)/2$
101110	V23	V8
101111	$(V23+V24)/2$	$(V7+V8)/2$
110000	V24	V7
110001	$(V24+V25)/2$	$(V6+V7)/2$
110010	V25	V6
110011	$(V25+V26)/2$	$(V5+V6)/2$
110100	V26	V5
110101	$(V26+V27)/2$	$(V4+V5)/2$
110110	V27	V4
110111	$(V27+V28)/2$	$(V3+V4)/2$
111000	V28	V3
111001	$(V28+V29)/2$	$(V2+V3)/2$
111010	V29	V2
111011	$(V29+V30)/2$	$(V1+V2)/2$
111100	V30	V1
111101	$(V30+V31)/2$	$(V0+V1)/2$
111110	$(V30+V31 \times 2)/3$	$(V0 \times 2 + V1)/3$
111111	V31	V0

RAM Access via RGB-I/F and System I/F

In RGB interface mode, the R61500 stores the display data in the internal RAM in order to transfer moving picture data when updating moving picture frames. In combination with high speed write function (HWM = 1 or LHWM = 1) and window address function, RGB interface enables the R61500 to access the internal RAM in high speed and thereby reduce power consumption for moving picture display. While not updating moving picture frames, it is possible to rewrite still picture data outside the moving picture area via system interface.

In RGB interface mode, the R61500 writes data to the internal RAM in synchronization with DOTCLK during ENABLE = Low. To access the internal RAM via system interface while using RGB interface for the display operation, set ENABLE High to stop writing via RGB interface. To start accessing the internal RAM via RGB interface after accessing the RAM via system interface, wait at least for a write/read bus cycle time. The data write operation will fail if writing operations via two interfaces (RGB and system interfaces) are conflicting with each other.

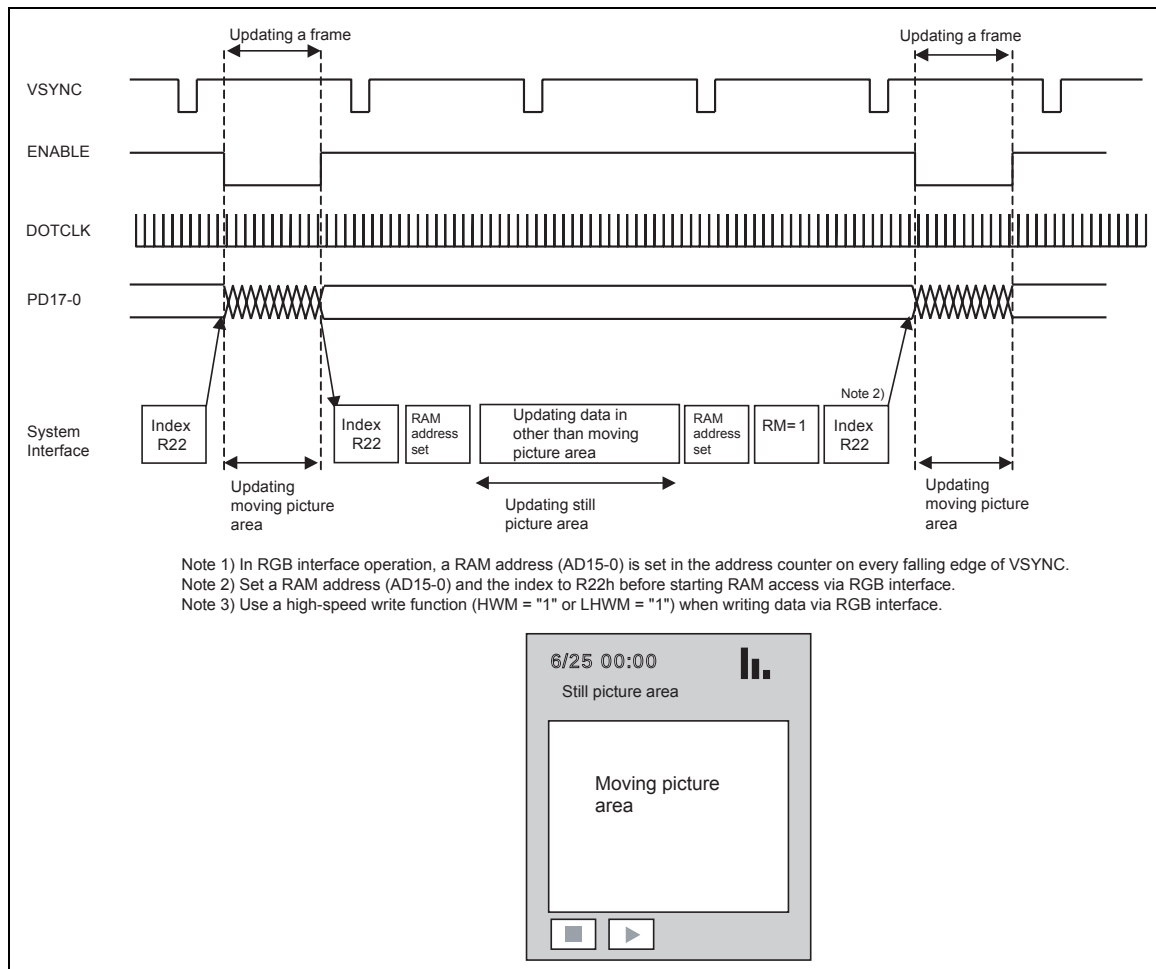


Figure 16

R61500

Read Data Read from GRAM (R22h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
R	1	RAM Read data (RD17-0) The assignment to DB17-0 pins differs for each interface (see below).															

RD17-0: 18-bit data read from the GRAM. The bit assignment between the data read from the GRAM and the DB[17:0] pins differs for each interface.

When the data is read from the GRAM to the microcomputer, the first word read immediately after executing RAM address set is taken in the internal read data latch and invalid data is sent to the data bus DB17-0. Valid data is sent to the data bus as the R61500 reads the second and subsequent words from the internal GRAM.

The first word sent to the internal read data latch is used when a bit operation (logical/compare operation) is performed inside the R61500 and the bit operation can be performed with one read operation. Note that the R61500 performs a bit operation on the 18-bit data.

When 8- or 16-bit interface is selected, the LSBs of R and B dots are not read.

Note: This register is disabled in RGB interface operation.

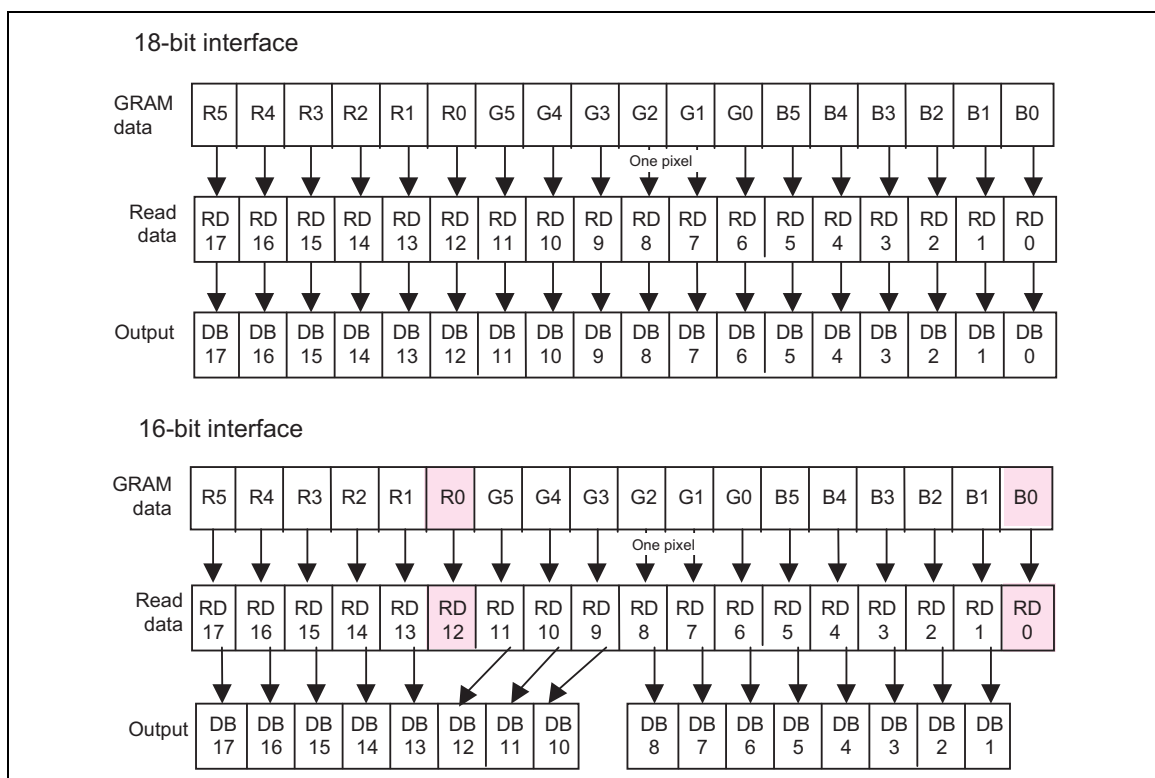


Figure 17 Read data from GRAM in 18/16-bit interface mode

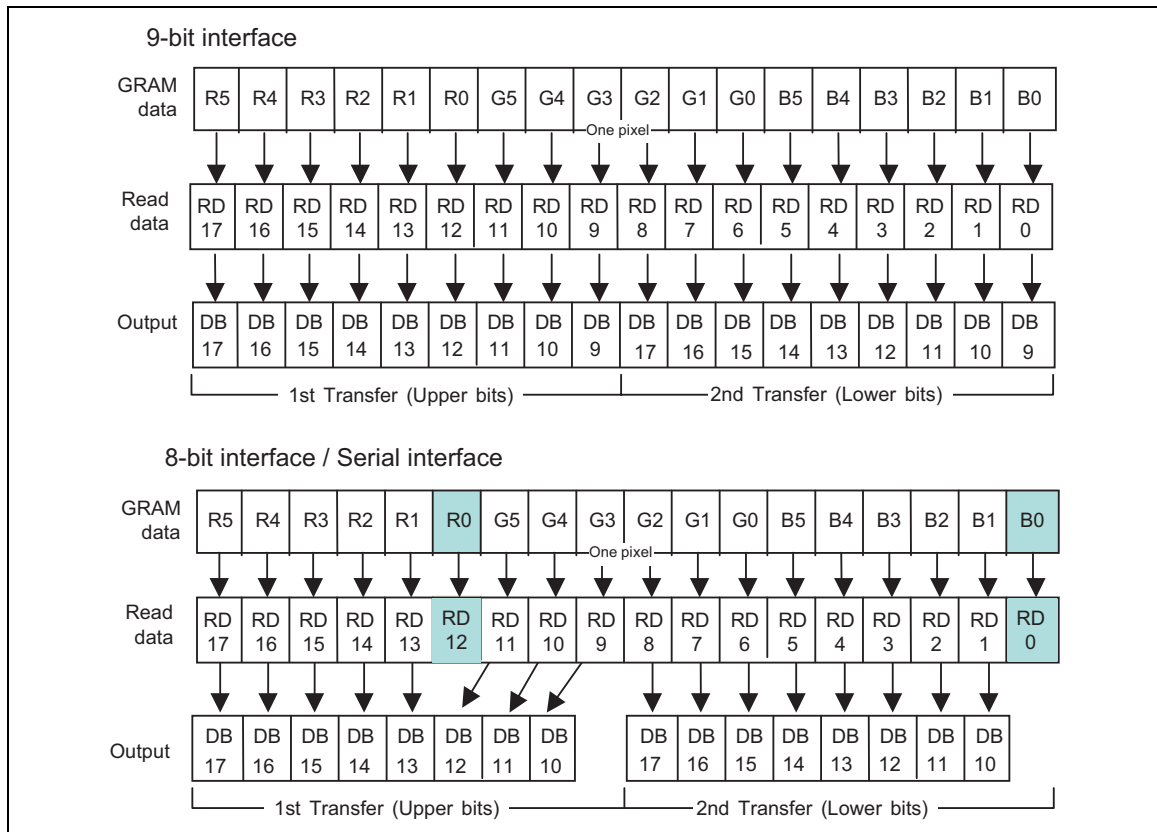


Figure 18 Read data from GRAM in 9/8-bit interface, serial interface mode

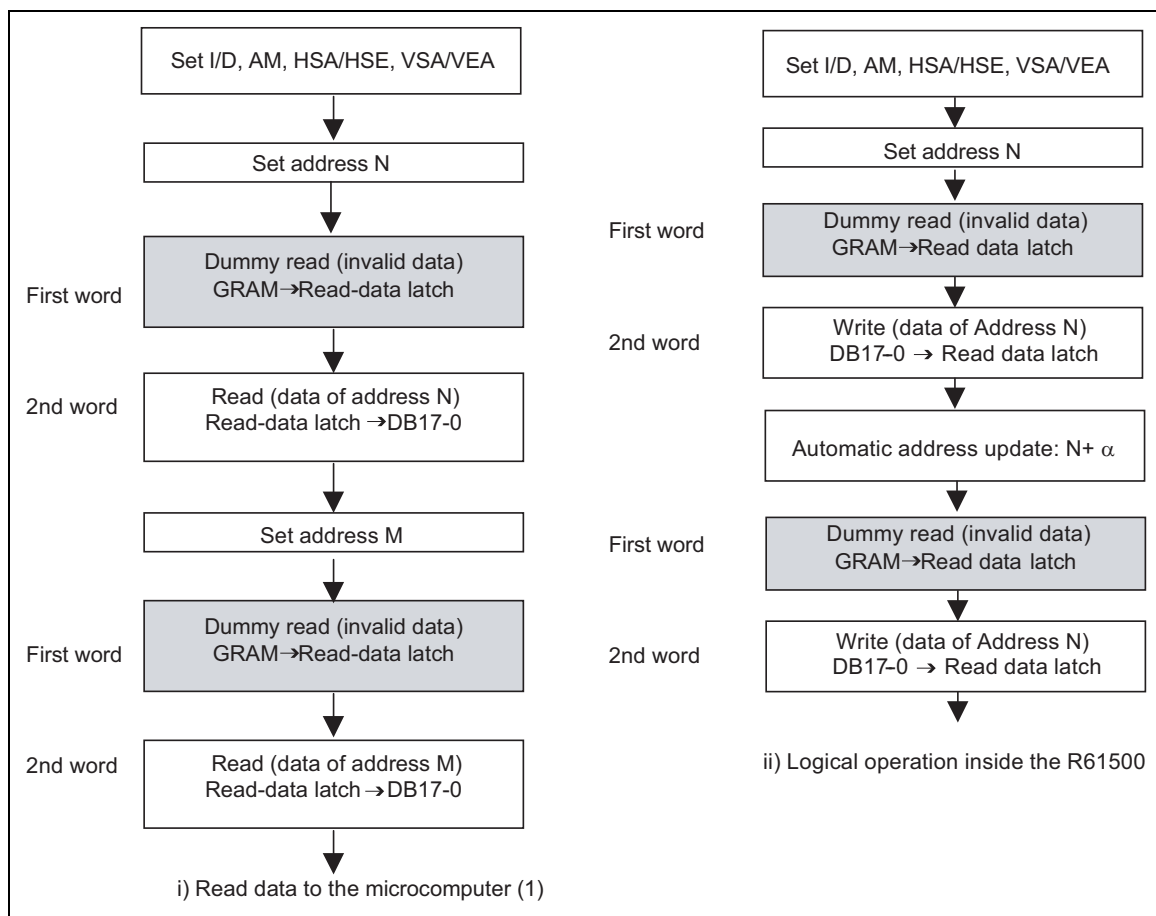


Figure 19 GRAM read sequence

R61500

RAM Write Data Mask (R23h)

RAM Write Data Mask (R24h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	WM11	WM10	WM9	WM8	WM7	WM6	0	0	WM5	WM4	WM3	WM2	WM1	WM0
W	1	0	0	0	0	0	0	0	0	0	0	WM17	WM16	WM15	WM14	WM13	WM12

WM17–0: Masks the data in units of bits before writing the 18-bit data to the GRAM. For example, if WM17 = 1, the MSB of the write data is masked so that the MSB is not written over to the GRAM data. In the same manner, when the WM16~WM0 bits are set to 1, the corresponding bits of the write data are masked and not written over to the GRAM data. For details, see the Graphics Operation Function section.

Note that the write mask function is always performed on the 18-bit GRAM write data. This function is disabled in RGB interface operation.

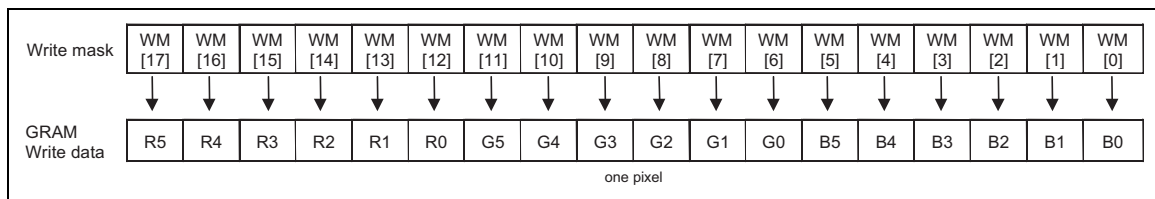


Figure 20 RAM write data mask

γ Control (R30h to R39h)

	R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
R30	W	1	0	0	0	0	0	PKP 12	PKP 11	PKP 10	0	0	0	0	0	PKP 02	PKP 01	PKP 00
R31	W	1	0	0	0	0	0	PKP 32	PKP 31	PKP 30	0	0	0	0	0	PKP 22	PKP 21	PKP 20
R31	W	1	0	0	0	0	0	PKP 52	PKP 51	PKP 50	0	0	0	0	0	PKP 42	PKP 41	PKP 40
R33	W	1	0	0	0	0	0	PRP 12	PRP 11	PRP 10	0	0	0	0	0	PRP 02	PRP 01	PRP 00
R34	W	1	0	0	0	0	0	PKN 12	PKN 11	PKN 10	0	0	0	0	0	PKN 02	PKN 01	PKN 00
R35	W	1	0	0	0	0	0	PKN 32	PKN 31	PKN 30	0	0	0	0	0	PKN 22	PKN 21	PKN 20
R36	W	1	0	0	0	0	0	PKN 52	PKN 51	PKN 50	0	0	0	0	0	PKN 42	PKN 41	PKN 40
R37	W	1	0	0	0	0	0	PRN 12	PRN 11	PRN 10	0	0	0	0	0	PRN 02	PRN 01	PRN 00
R38	W	1	0	0	0	VRP 14	VRP 13	VRP 12	VRP 11	VRP 10	0	0	0	0	VRP 03	VRP 02	VRP 01	VRP 00
R39	W	1	0	0	0	VRN 14	VRN 13	VRN 12	VRN 11	VRN 10	0	0	0	0	VRN 03	VRN 02	VRN 01	VRN 00

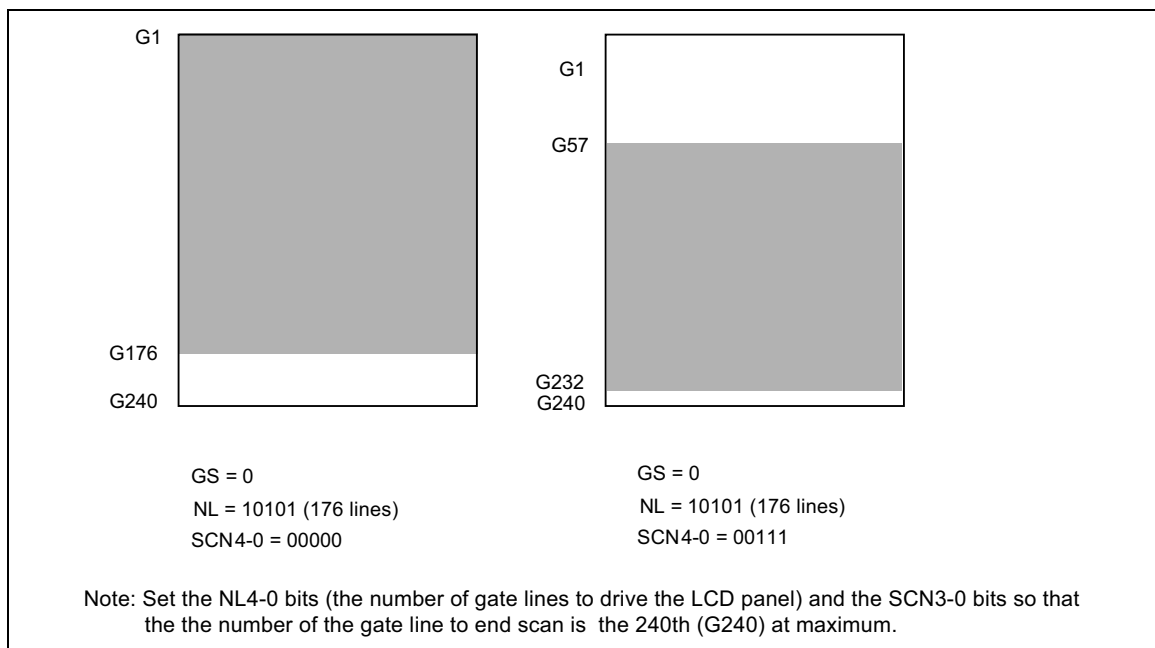
PKP5-0[2:0] : γ fine adjustment register bits for positive polarity
 PRP1-0[2:0] : γ gradient adjustment register bits for positive polarity
 PKN5-0[2:0] : γ fine adjustment register bits for negative polarity
 PRN1-0[2:0] : γ gradient adjustment register bits for negative polarity
 VRP0[3:0]/VRP1[4:0] : amplitude adjustment register bits for positive polarity
 VRN0[3:0]/VRN1[4:0] : amplitude adjustment register bits for negative polarity

For details γ -Correction Function section.

Gate Scan Position (R40h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	0	0	0	SCN4	SCN3	SCN2	SCN1	SCN0

SCN4-0: Specifies the gate line to start scan.

**Figure 21**

R61500

Table 57

SCN4	SCN3	SCN2	SCN1	SCN0	Scan Start Position (Gate line)	
					GS = 0	GS = 1
0	0	0	0	0	G1	G240
0	0	0	0	1	G9	G232
0	0	0	1	0	G17	G224
0	0	0	1	1	G25	G216
0	0	1	0	0	G33	G208
0	0	1	0	1	G41	G200
0	0	1	1	0	G49	G192
0	0	1	1	1	G57	G184
0	1	0	0	0	G65	G176
0	1	0	0	1	G73	G168
0	1	0	1	0	G81	G160
0	1	0	1	1	G89	G152
0	1	1	0	0	G97	G144
0	1	1	0	1	G105	G136
0	1	1	1	0	G113	G128
0	1	1	1	1	G121	G120
1	0	0	0	0	G129	G112
1	0	0	0	1	G137	G104
1	0	0	1	0	G145	G96
1	0	0	1	1	G153	G88
1	0	1	0	0	G161	G80
1	0	1	0	1	G169	G72
1	0	1	1	0	G177	G64
1	0	1	1	1	G185	G56
1	1	0	0	0	G193	G48
1	1	0	0	1	G201	G40
1	1	0	1	0	G209	G32
1	1	0	1	1	G217	G24
1	1	1	0	0	G225	G16

Vertical Scroll Control (R41h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	VL7	VL6	VL5	VL4	VL3	VL2	VL1	VL0

VL7-0: Sets the scrolling amount on the screen in vertical direction. The scrolling amount can be set from 0 line to 240 lines. The start position of display is shifted vertically by the number of lines set with the VL7-0 bits. The part of the image, which is scrolled out from the end line (the 240th line) as a result of scrolling, is displayed from the 1st line of the display. The VL7-0 bits are enabled when either first display vertical scroll enable bit VLE1 or the second display vertical scroll enable bit VLE2 is set to 1. When VLE1-2 = 00, the image on the screen is displayed at the position defined by the SS and SE bits. The vertical scrolling function is disabled in external display interface operation.

Table 58

VL7	VL6	VL5	VL4	VL3	VL2	VL1	VL0	Scrolling lines
0	0	0	0	0	0	0	0	0 line
0	0	0	0	0	0	0	1	1 line
0	0	0	0	0	0	1	0	2 lines
:	:	:	:	:	:	:	:	:
1	1	1	0	1	1	1	0	238 lines
1	1	1	0	1	1	1	1	239 lines

Note: Do not set values representing more than 239 lines.

1st-Screen Drive Position (R42h), 2nd-Screen Drive Position (R43h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	SE17	SE16	SE15	SE14	SE13	SE12	SE11	SE10	SS17	SS16	SS15	SS14	SS13	SS12	SS11	SS10
W	1	SE27	SE26	SE25	SE24	SE23	SE22	SE21	SE20	SS27	SS26	SS25	SS24	SS23	SS22	SS21	SS20

SS17-10: Specifies the gate line from which the first display starts. The R61500 starts LCD drive from the gate line of the number represented by SS17-10 + 1.

SE17-10: Specifies the gate line at which the first display ends. The R61500 ends LCD drive at the gate line of the number represented by SE17-10 + 1. For instance, when SS17-10 = "07"H and SE17-10 = "10"H, the R61500 drives the LCD for the first display from G8 to G17, and gate lines from G1 to G7 and from G18 to the end line are driven for black screen. Make sure that $SS17-10 \leq SE17-10 \leq "EF"H$. For details, see the Partial Display Function section.

SS27-20: Specifies the gate line from which the first display starts. The R61500 starts LCD drive from the gate line of the number represented by SS27-20 + 1. The second display is shown when SPT = 1.

R61500

SE27-20: Specifies the gate line at which the first display ends. The R61500 ends LCD drive at the gate line of the number represented by SE27-20 + 1. For instance, when SPT = 1, and SS27-20 = "20"H, SE27-20 = "4F"H, the R61500 drives the LCD for the second display from G33 to G80.

Make sure that SS17-10 ≤ SE17-10 < SS27-20 ≤ SE27-20 ≤ "EF"H. For details, see the Partial Display Function section.

Horizontal RAM Address Position (R44h), Vertical RAM Address Position (R45h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	HEA7	HEA6	HEA5	HEA4	HEA3	HEA2	HEA1	HEA0	HSA7	HSA6	HSA5	HSA4	HSA3	HSA2	HSA1	HSA0
W	1	VEA7	VEA6	VEA5	VEA4	VEA3	VEA2	VEA1	VEA0	VSA7	VSA6	VSA5	VSA4	VSA3	VSA2	VSA1	VSA0

HSA7-0/HEA7-0: HSA7-0 and HEA7-0 represent the addresses at the start and end of the window address area in horizontal direction respectively. By setting HAS and HEA bits, it is possible to limit the area on the GRAM horizontally to write data. The HSA and HEA bits must be set before starting RAM write operation. In setting these bits, be sure "00"h ≤ HSA7-0 < HEA7-0 ≤ "AF"h.

VSA7-0/VEA7-0: VSA7-0 and VEA7-0 represent the addresses at the start and end of the window address area in vertical direction respectively. By setting VAS and VEA bits, it is possible to limit the area on the GRAM vertically to write data. The VSA and VEA bits must be set before starting RAM write operation. In setting, be sure "00"h ≤ VSA7-0 < VEA7-0 ≤ "EF"h.

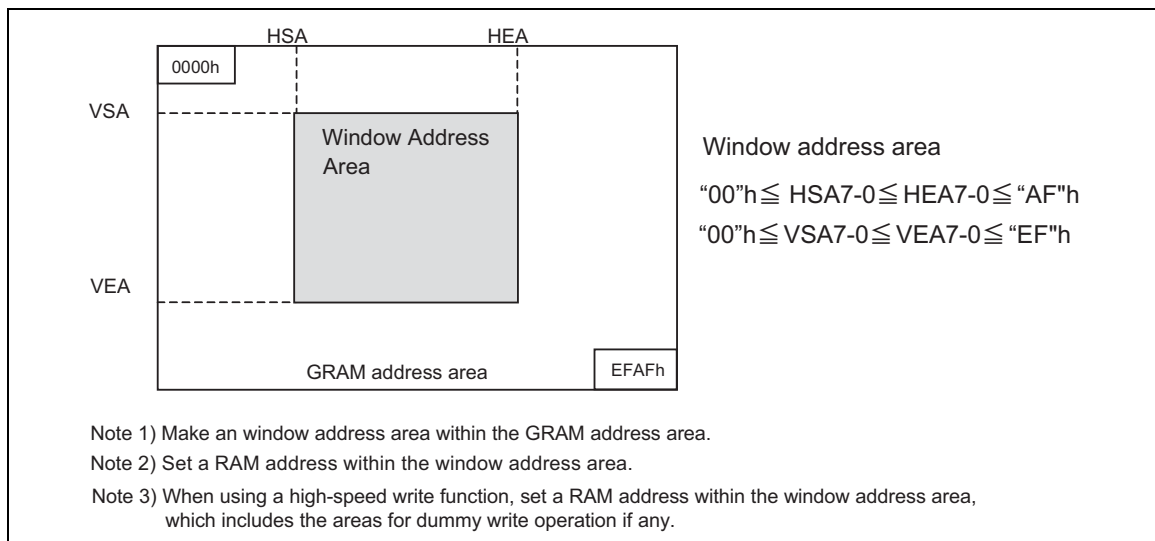


Figure 22 GRAM address and window address area

Main category		Sub category		Upper code								Lower code								Note
Index上位		Index	Command	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0	
-	Index	—	Index	*	*	*	*	*	*	*	*	*	ID6	ID5	ID4	ID3	ID2	ID21	ID0	
SR	Status read	—	Status read	L7	L6	L5	L4	L3	L2	L1	L0	0	0	0	0	0	0	0	0	
0*	Display control	00h	Start oscillation	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	1
			Device code read	0	0	0	1	0	1	0	1	0	0	0	0	0	1	0	0	0
			Driver output	0	VSPL (0)	HSPL (0)	DPL (0)	EPL (0)	SM (0)	GS (0)	SS (0)	0	0	0	NL4 (1)	NL3 (1)	NL2 (1)	NL1 (0)	NL0 (1)	
		02h	LCD driving wave control	LEQ2 (0)	LEQ1 (0)	LEQ0 (0)	0	FLD1 (0)	FLD0 (1)	B/C (0)	EOR (0)	0	0	NW5 (0)	NW4 (0)	NW3 (0)	NW2 (0)	NW1 (0)	NW0 (0)	
		03h	Entry mode	TRI (0)	DFM1 (0)	DFM0 (0)	BGR (0)	0	0	HWM (0)	LHWM (0)	0	0	ID1 (1)	ID0 (1)	AM (0)	LG2 (0)	LG1 (0)	LG0 (0)	
		04h	Compare register (1)	0	0	CP11 (0)	CP10 (0)	CP9 (0)	CP8 (0)	CP7 (0)	CP6 (0)	0	0	CP5 (0)	CP4 (0)	CP3 (0)	CP2 (0)	CP1 (0)	CP0 (0)	
		05h	Compare register (2)	0	0	0	0	0	0	0	0	0	0	CP17 (0)	CP16 (0)	CP15 (0)	CP14 (0)	CP13 (0)	CP12 (0)	
		06h	Setting disabled																	
		07h	Display control (1)	0	0	PT2 (0)	PT1 (0)	PT0 (0)	VLE2 (0)	VLE1 (0)	SPT (0)	0	0	GON 0	DTE (0)	CL (0)	REV (0)	D1 (0)	D0 (0)	
		08h	Display control (2)	0	0	0	0	FP3 (1)	FP2 (0)	FP1 (0)	FP0 (0)	0	0	0	0	BP3 (1)	BP2 (0)	BP1 (0)	BP0 (0)	
		09h	Display control (3)	0	0	0	0	0	0	0	0	0	0	PTG1 (0)	PTG0 (0)	ISC3 (0)	ISC2 (0)	ISC1 (0)	ISC0 (0)	
		0Ah	Setting disabled																	
		0Bh	Frame cycle adjustment	NO1 (0)	NO0 (0)	SDT1 (0)	SDT0 (0)	EQ1 (0)	EQ0 (0)	DIV1 (0)	DIV0 (0)	VEM (0)	0	0	0	0	RTN3 (0)	RTN2 (0)	RTN1 (0)	RTN0 (0)
		0Ch	External display interface control (1)	0	0	ENC1 (0)	ENC0 (0)	0	0	0	RM (0)	0	0	DM1 (0)	DM0 (0)	0	0	RIM1 (0)	RIM0 (0)	
		0Dh	External display interface control (2)	VSEL1 (0)	VSEL0 (0)	0	CPBW 4	CPBW 3	CPBW 2	CPBW 1	CPBW 0	0	0	0	CPAW 4	CPAW 3	CPAW 2	CPAW 1	CPAW 0	
		0Eh	External display interface control (3)	0	0	0	CNBW 4(0)	CNBW 3	CNBW 2	CNBW 1	CNBW 0	0	0	0	CNAW 4	CNAW 3	CNAW 2	CNAW 1	CNAW 0	
		0Fh	External display interface control (4)	0	0	0	0	0	0	0	0	0	0	0	CSTP4 (1)	CSTP3 (0)	CSTP2 (0)	CSTP1 (0)	CSTP0 (0)	
1*	Power control	10h	Power control (1)	0	SAP2 (0)	SAP1 (0)	SAP0 (0)	0	BT2 (0)	BT1 (0)	BT0 (0)	0	AP2 (0)	AP1 (0)	AP0 (0)	0	DK (1)	SLP (0)	STB (0)	
		11h	Power control (2)	0	0	0	0	0	DC12 (0)	DC11 (0)	DC10 (0)	0	DC02 (0)	DC01 (0)	DC00 (0)	0	VC2 (0)	VC1 (0)	VC0 (0)	
		12h	Power control (3)	0	0	0	0	0	0	0	0	0	0	0	PON (0)	VRH3 (0)	VRH2 (0)	VRH1 (0)	VRH0 (0)	
		13h	Power control (4)	0	0	VCOM G	VDV4 (0)	VDV3 (0)	VDV2 (0)	VDV1 (0)	VDV0 (0)	0	0	0	VCM4 (0)	VCM3 (0)	VCM2 (0)	VCM1 (0)	VCM0 (0)	
		14h	Power control (5)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
		15h	Setting disabled																	
		16h	Setting disabled																	
		17h	Setting disabled																	
		18h	Setting disabled																	
		19h	Setting disabled																	
		1Ah	Setting disabled																	
		1Bh	Setting disabled																	
		1Ch	Setting disabled																	
		1Dh	Setting disabled																	
		1Eh	Setting disabled																	
		1Fh	Setting disabled																	
		2*	RAM access	20h	Setting disabled															
21h	RAM address set			AD15 (0)	AD14 (0)	AD13 (0)	AD12 (0)	AD11 (0)	AD10 (0)	AD9 (0)	AD8 (0)	AD7 (0)	AD6 (0)	AD5 (0)	AD4 (0)	AD3 (0)	AD2 (0)	AD1 (0)	AD0 (0)	
22h	Write data to/Read data from GRAM			RAM write/read data (Bit assignment to DB[17:0] differs according to interface)																
23h	RAM write data mask (1)			0	0	WM11 (0)	WM10 (0)	WM9 (0)	WM8 (0)	WM7 (0)	WM6 (0)	0	0	WM5 (0)	WM4 (0)	WM3 (0)	WM2 (0)	WM1 (0)	WM0 (0)	
24h	RAM write data mask (2)			0	0	0	0	0	0	0	0	0	0	WM17 (0)	WM16 (0)	WM15 (0)	WM14 (0)	WM13 (0)	WM12 (0)	
25h	Setting disabled																			
26h	Setting disabled																			
27h	Setting disabled																			
28h	Setting disabled																			
29h	Setting disabled																			
2Ah	Setting disabled																			
2Bh	Setting disabled																			
2Ch	Setting disabled																			
2Dh	Setting disabled																			
2Eh	Setting disabled																			
2Fh	Setting disabled																			
3*	Gamma control			30h	Gamma control (1)	0	0	0	0	0	PKP12 (0)	PKP11 (0)	PKP10 (0)	0	0	0	0	0	PKP02 (0)	PKP01 (0)
		31h	Gamma control (2)	0	0	0	0	0	PKP32 (0)	PKP31 (0)	PKP30 (0)	0	0	0	0	0	PKP22 (0)	PKP21 (0)	PKP20 (0)	
		32h	Gamma control (3)	0	0	0	0	0	PKP52 (0)	PKP51 (0)	PKP50 (0)	0	0	0	0	0	PKP42 (0)	PKP41 (0)	PKP40 (0)	
		33h	Gamma control (4)	0	0	0	0	0	PRP12 (0)	PRP11 (0)	PRP10 (0)	0	0	0	0	0	PRP02 (0)	PRP01 (0)	PRP00 (0)	
		34h	Gamma control (5)	0	0	0	0	0	PKN12 (0)	PKN11 (0)	PKN10 (0)	0	0	0	0	0	PKN02 (0)	PKN01 (0)	PKN00 (0)	
		35h	Gamma control (6)	0	0	0	0	0	PKN32 (0)	PKN31 (0)	PKN30 (0)	0	0	0	0	0	PKN22 (0)	PKN21 (0)	PKN20 (0)	
		36h	Gamma control (7)	0	0	0	0	0	PKN52 (0)	PKN51 (0)	PKN50 (0)	0	0	0	0	0	PKN42 (0)	PKN41 (0)	PKN40 (0)	
		37h	Gamma control (8)	0	0	0	0	0	PRN12 (0)	PRN11 (0)	PRN10 (0)	0	0	0	0	0	PRN02 (0)	PRN01 (0)	PRN00 (0)	
		38h	Gamma control (9)	0	0	0	VRP14 (0)	VRP13 (0)	VRP12 (0)	VRP11 (0)	VRP10 (0)	0	0	0	0	VRP03 (0)	VRP02 (0)	VRP01 (0)	VRP00 (0)	
		39h	Gamma control (10)	0	0	0	VRN14 (0)	VRN13 (0)	VRN12 (0)	VRN11 (0)	VRN10 (0)	0	0	0	0	VRN03 (0)	VRN02 (0)	VRN01 (0)	VRN00 (0)	
		3Ah	Setting disabled																	
		3Bh	Setting disabled																	
		3Ch	Setting disabled																	
		3Dh	Setting disabled																	
		3Eh	Setting disabled																	
		3Fh	Setting disabled																	
		4*	Coordinates control	40h	Gate scan position	0	0	0	0	0	0	0	0	0	0	0	SCN4 (0)	SCN3 (0)	SCN2 (0)	SCN1 (0)
41h	Vertical scroll control			0	0	0	0	0	0	0	0	0	VL7 (0)	VL6 (0)	VL5 (0)	VL4 (0)	VL3 (0)	VL2 (0)	VL1 (0)	VL0 (0)
42h	First screen driving position			SE17 (1)	SE16 (1)	SE15 (1)	SE14 (1)	SE13 (1)	SE12 (1)	SE11 (1)	SE10 (1)	SS17 (0)	SS16 (0)	SS15 (0)	SS14 (0)	SS13 (0)	SS12 (0)	SS11 (0)	SS10 (0)	
43h	Second screen driving position			SE27 (1)	SE26 (1)	SE25 (1)	SE2													

Reset Function

The R61500 is initialized with a RESET input. During a reset period, the R61500 is in a busy state, and neither access to instructions nor to GRAM data from the MPU is accepted. The RESET period must be secured for at least 1ms. In case of power-on reset, wait until the RC oscillation is stabilized (for 10 ms). During this period, neither access to the internal GRAM nor initial instruction setting is allowed.

Initial state of instruction bits

- a. Start oscillator
- b. Driver output control (NL4-0 = 11101, SS = 0, GS = 0, SM = 0, EPL = 0, DPL = 0, HSPL = 0, VSPL = 0)
- c. Liquid crystal alternating drive control (LEQ2-0 = 000, FLD1-0 = 01, B/C = 0, EOR = 0, NW5-0 = 00000)
- d. Entry mode set (HWM = 0, LHWM = 0, I/D1-0 = 11: Increment by 1, AM = 0: Horizontal direction, LG2-0 = 000: Replace mode, BGR = 0, TRI = 0, DFM1-0 = 00)
- e. Compare register (CP17-0: 00_0000_0000_0000_0000)
- f. Display control 1 (PT1-0 = 00, VLE2-1 = 00: No vertical scroll, SPT = 0, DTE = 0, CL = 0: 262,144-color mode, REV = 0, D1-0 = 00: Display OFF, GON = 0)
- g. Display control 2 (BP3-0 = 1000, FP3-0 = 1000)
- h. Display control 3 (PTG1-0 = 00, ISC3-0 = 0000)
- i. Frame cycle control (NO1-0 = 00, SDT1-0 = 00, VEM = 0, EQ1-0 = 00: No equalization, DIV1-0 = 00: 1/1, RTN3-0 = 0000: 16 clocks per 1H period)
- j. External display interface 1 (RIM1-0 = 00: 18-bit RGB interface, DM1-0 = 00: internal clock operation, RM = 0: System interface, ENC1-0 = 00)
- k. Driver output control 1 (VSEL1-0 = 00, CPAW4-0 = 01001, CPBW4-0 = 00100)
- l. Driver output control 2 (CNAW4-0 = 01001, CNBW4-0 = 00100)
- m. External display interface 2 control (CSTP4-0 = 10000)
- n. Power control 1 (SAP2-0 = 000, BT2-0 = 000, AP2-0 = 000: liquid crystal power supply OFF, DK = 1: DCDC1 off, SLP = 0, STB = 0: standby mode off)
- o. Power control 2 (DC12-0 = 000, DC02-00 = 000, VC2-0 = 000)
- p. Power control 3 (PON = 0, VRH3-0 = 0000)
- q. Power control 4 (VCOMG = 0, VDV4-0 = 00000, VCM4-0 = 00000)
- r. RAM address set (AD15-0 = 0000_0000_0000_0000)
- s. RAM write data mask (WM17-0 = 00_0000_0000_0000_0000: No write-mask)
- t. γ control
(PKP02-00 = 000, PKP12-10 = 000, PKP22-20 = 000, PKP32-30 = 000,
PKP42-40 = 000, PKP52-50 = 000, PRP02-00 = 000, PRP12-10 = 000)
(PKN02-00 = 000, PKN12-10 = 000, PKN22-20 = 000, PKN32-30 = 000,
PKN42-40 = 000, PKN52-50 = 000, PRN02-00 = 000, PRN12-10 = 000)
(VRP14-10 = 00000, VRP03-00 = 0000, VRN14-10 = 00000, VRN03-00 = 0000)
- u. Gate scan position (SCN4-0 = 00000)
- v. Vertical scroll (VL7-0 = 00000000)
- w. 1st split screen (SE17-10 = 11111111, SS17-10 = 00000000)
- x. 2nd split screen (SE27-20 = 11111111, SS27-20 = 00000000)
- y. Horizontal RAM address position (HEA7-0 = 10101111, HSA7-0 = 00000000)
- z. Vertical RAM address position (VEA7-0 = 11101111, VSA7-0 = 00000000)

Initial state of GRAM data

The GRAM data is not automatically initialized with a RESET input. Initialize the internal GRAM by software during a display-off period (D1-0 = 00).

Initial state of output pins

- | | |
|--------------------------------|--|
| 1. LCD driver (source outputs) | : All pins output the GND level (Halt) |
| 2. Vcom1/2/11/21 | : Output the GND level (Halt) |
| 3. LCD driver (gate outputs) | : All pins output the Vci (VGH) level (Halt) |
| 4. FLM | : Output the GND level (Halt) |
| 5. Oscillator | : Oscillate |

Interface Specifications

The R61500 supports a system interface to make instruction and other settings and an external display interface for displaying a moving picture. The R61500 allows selecting an optimum interface for the display (moving or still picture) in order to transfer data efficiently.

As the external display interface, the R61500 supports RGB interface and VSYNC interface, which enable data rewrite operation without flickering the moving picture on the screen.

In RGB interface operation, the display operation is performed in synchronization with synchronous signals VSYNC, HSYNC, and DOTCLK. The display data is written to the internal RAM according to the polarity of the data enable signal (ENABLE) via moving picture display data bus DB17-0 in synchronization with VSYNC, HSYNC, and DOTCLK. The R61500 stores all display data in the R61500's GRAM so that the data is transferred when rewriting moving picture frames. By using the window address function, the R61500 rewrite data only within the window address area made on the internal RAM, which enables displaying both moving picture and still picture written in the internal RAM other than the window address area at a time. In order to access RAM in high speed with low power consumption, use a high-speed write function (either HWM = 1 or LHWM = 1).

In VSYNC interface operation, the internal display operation is synchronized with the frame synchronization signal (VSYNC). The VSYNC interface enables the R61500 to display a moving picture via system interface by writing data to the internal GRAM at a calculated minimum or higher speed in synchronization with the falling edge of VSYNC. In this case, there are constraints in speed and method in writing data to the internal RAM.

The R61500 operates in either one of the following 4 modes according to the display operation. The display operation mode setting is made in the external display interface control register. When switching from one mode to another, make sure to follow the sequence (see the sections of RGB interface and VSYNC interface, p.98 and p.110).

R61500

Table 59

Operation Mode	RAM Access Setting (RM)	Display Operation Mode (DM1-0)
Internal operating clock only (Displaying still pictures)	System interface (RM = 0)	Internal operating clock (DM1-0 = 00)
RGB interface (1) (Displaying moving pictures)	RGB interface (RM = 1)	RGB interface (DM1-0 = 01)
RGB interface (2) (Rewriting still pictures while displaying moving pictures)	System interface (RM = 0)	RGB interface (DM1-0 = 01)
VSYNC interface (Displaying moving pictures)	System interface (RM = 0)	VSYNC interface (DM1-0 = 10)

Note 1) Instructions are set only via system interface.

Note 2) RGB interface and VSYNC interface cannot be used simultaneously.

Note 3) Do not make changes to the RGB interface setting (RIM1-0 bits) while it is in operation.

Note 4) Refer to the sections of RGB and VSYNC interfaces for the sequences to follow when switching from one mode to another (p.98 and p.110).

Note 5) Use a high-speed write function (either HWM = 1 or LHWM = 1) when writing data via VSYNC or RGB interface.

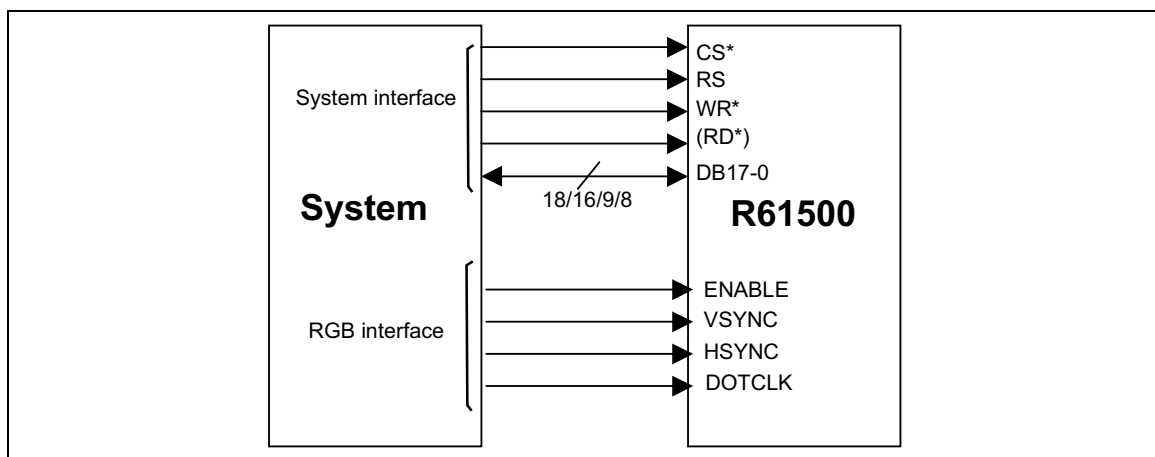


Figure 23 Interfaces and R61500

System Interface

The following are the system interfaces supported by the R61500. The interface is selected by setting the IM3/2/1/0 pins. The system interface is used for setting instructions and RAM access.

Table 60

IM3	IM2	IM1	IM0	MPU-Interface Mode	DB Pin in use
0	0	0	0	Setting disabled	-
0	0	0	1	Setting disabled	-
0	0	1	0	80-system 16-bit interface	DB17 to 10 and 8-to-1
0	0	1	1	80-system 8-bit interface	DB17 to 10
0	1	0	*	Serial interface	SDI, SDO
0	1	1	*	Setting disabled	-
1	0	0	0	Setting disabled	-
1	0	0	1	Setting disabled	-
1	0	1	0	80-system 18-bit interface	DB17 to 0
1	0	1	1	80-system 9-bit interface	DB17 to 9
1	1	*	*	Setting disabled	-

R61500

80-system 16-bit interface

The 80-system 16-bit parallel system interface is selected by setting the IM3/2/1/0 pins to the GND/GND/IOVcc/GND levels respectively.

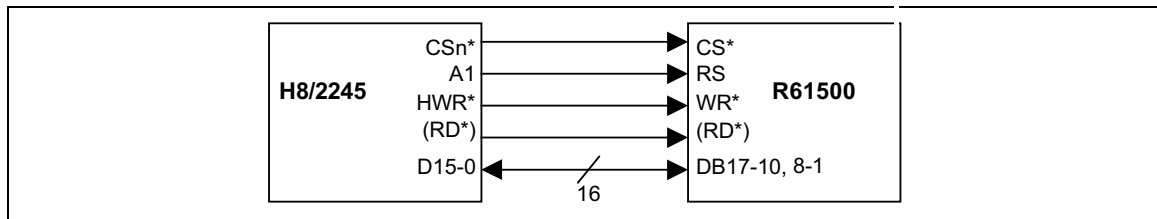


Figure 26 16-bit microcomputer and R61500

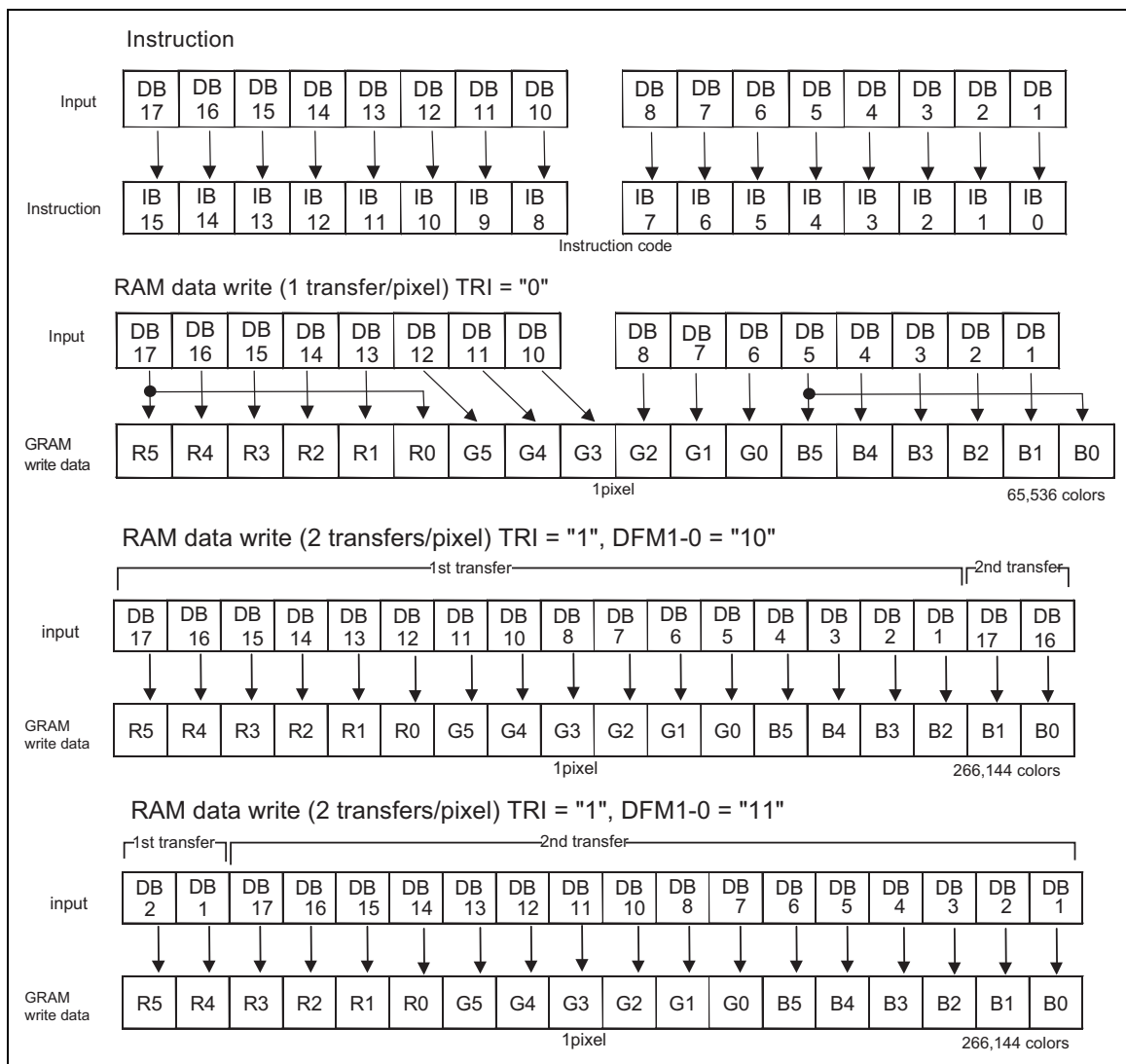


Figure 27 Data format for 16-bit interface

80-system 9-bit interface

The 80-system 9-bit parallel system interface via DB17~DB9 pins is selected by setting the IM3/2/1/0 pins to the IOVcc/GND/IOVcc/IOVcc levels respectively. When transferring a 16-bit instruction, it is divided into upper and lower 8 bits (the LSB is not used), and the upper 8 bits are transferred first. The RAM write data are also divided into the upper and lower 9 bits, and the upper bits are transferred first. The unused DB8-0 pins must be fixed at either IOVcc or GND level. When writing the index register, the upper byte (8 bits) must be written.

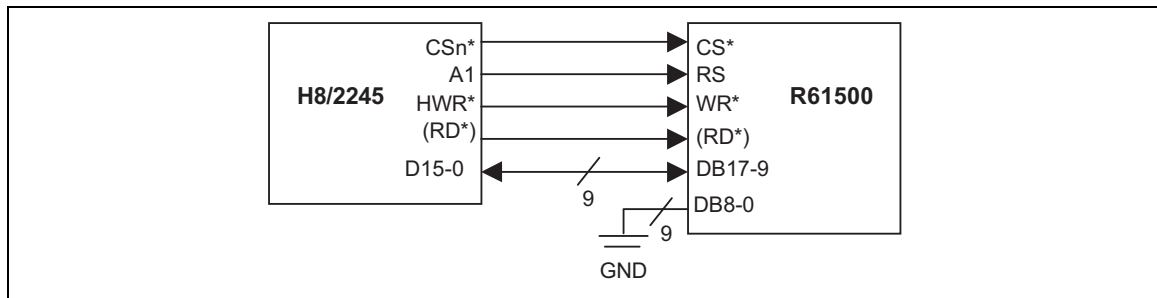


Figure 28 9-bit microcomputer and R61500

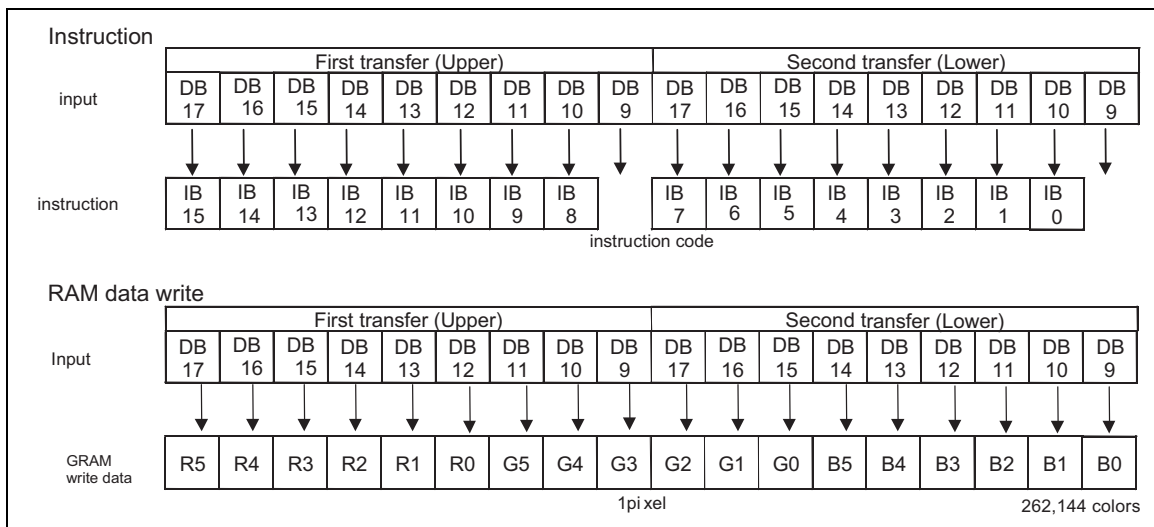


Figure 29 Data format for 9-bit interface

Data transfer synchronizing in 9-bit bus interface mode

The R61500 supports a data transfer synchronization function to reset upper and lower counters counting the number of transfers of upper and lower 9 bits in 9-bit bus interface operation. If a mismatch arises in the numbers of transfers between the upper and lower 9 bit counters due to noise and so on, the "00"H instruction is written 4 times consecutively to reset the upper and lower counters so that data transfer can restart with a upper 9 bit transfer. The synchronization system, when executed periodically, can effectively prevent runaway operation of the display system.

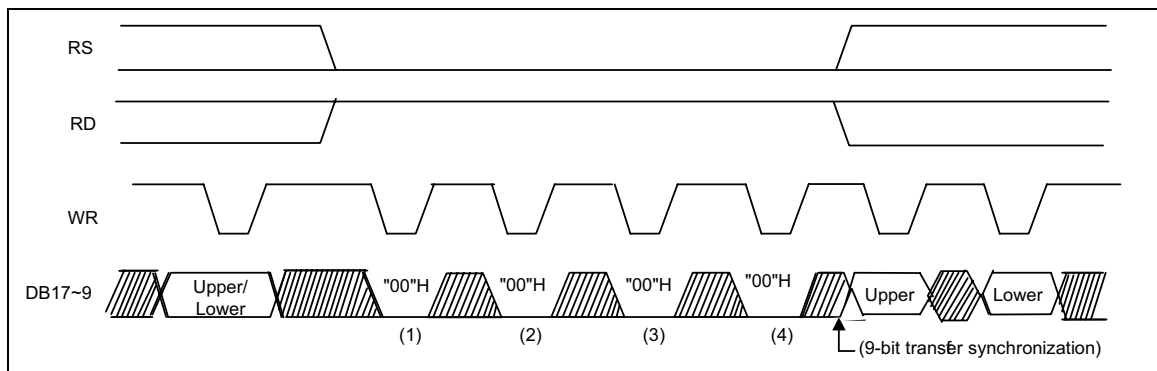


Figure 30 9-bit data transfer synchronization

80-system 8-bit interface

The 80-system 8-bit parallel system interface via DB17~DB10 pins is selected by setting the IM3/2/1/0 pins to the GND/GND/IOVcc/IOVcc levels respectively. When transferring a 16-bit instruction, it is divided into upper and lower 8 bits and the upper 8 bits are transferred first. The RAM data is also divided into the upper and lower 8 bits, and the upper bits are transferred first. The RAM write data are expanded into 18 bits internally (see the figure below). The unused pins DB9-0 must be fixed at either IOVcc or GND level. When writing the index register, the upper byte (8 bits) must be written.

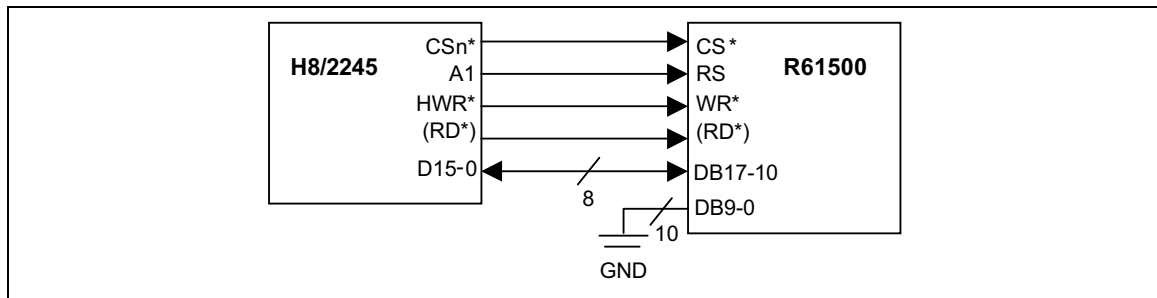


Figure 31 8-bit microcomputer and R61500

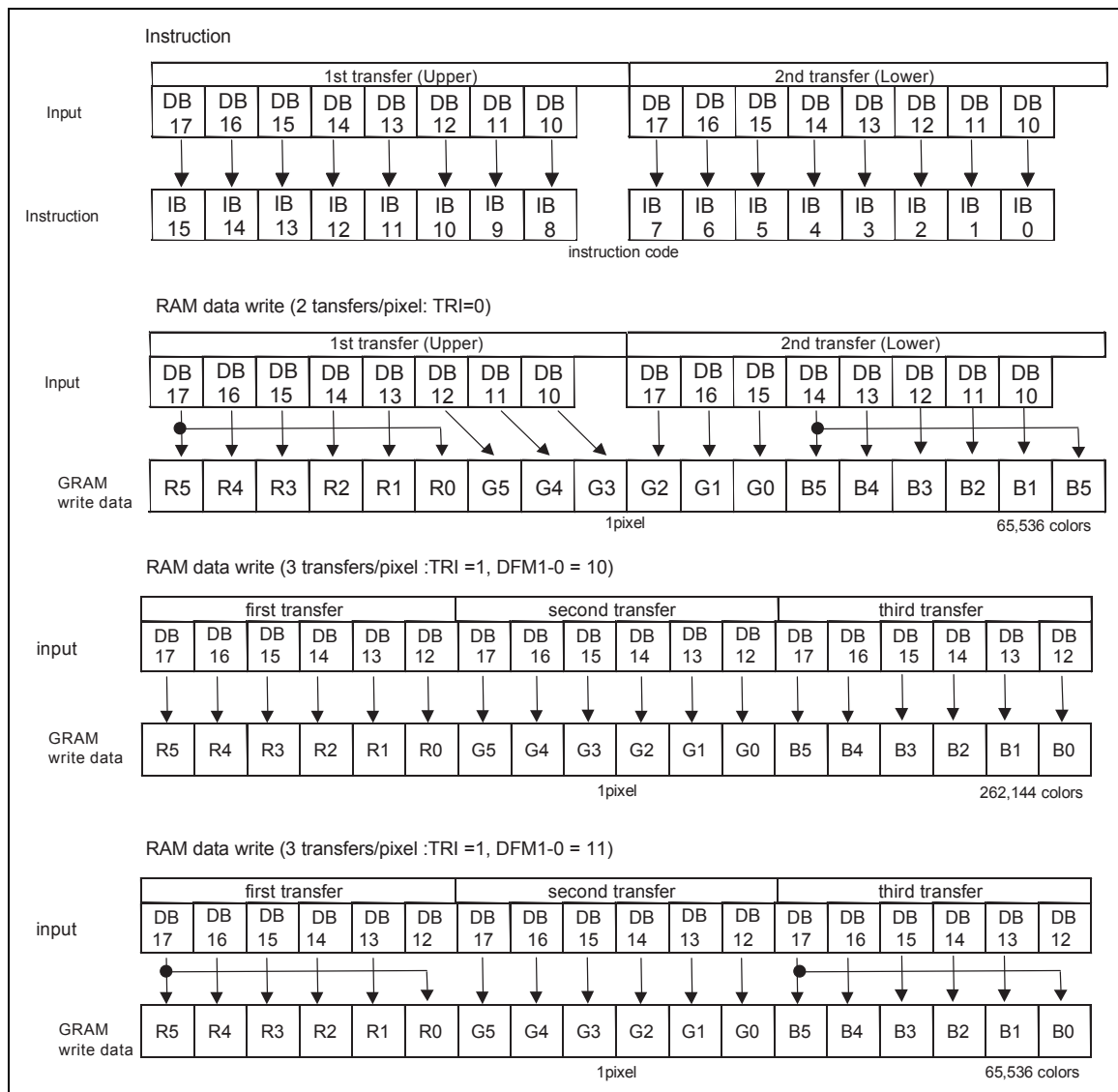


Figure 32 Data format for 8-bit interface

Data transfer synchronization in 8-bit bus interface mode

The R61500 supports a data transfer synchronization function to reset upper and lower counters counting the number of transfers of upper and lower 8 bits in 8-bit bus interface operation. If a mismatch arises in the numbers of transfers between the upper and lower 8 bit counters due to noise and so on, the "00"H instruction is written 4 times consecutively to reset the upper and lower counters so that data transfer can restart with a upper 8 bit transfer. The synchronization system, when executed periodically, can effectively prevent runaway operation of the display system.

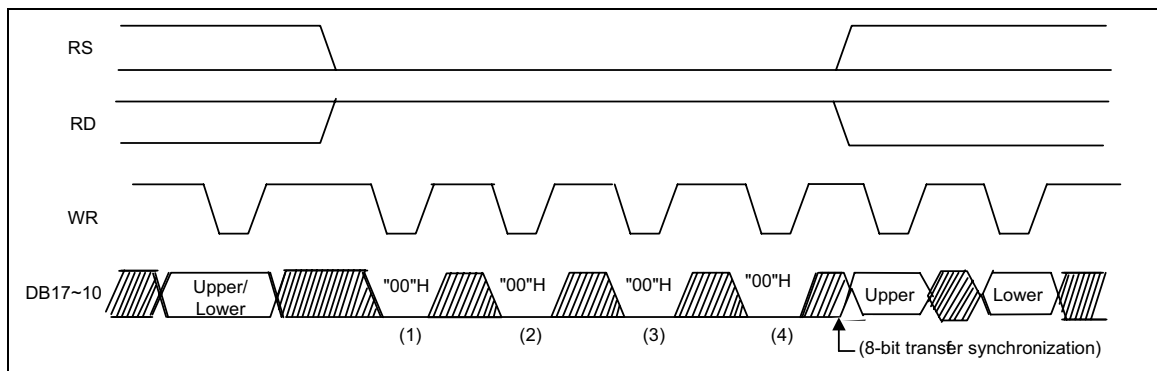


Figure 33 8-bit data transfer synchronization

Serial interface

The serial interface is selected by setting the IM3/2/1 pins to the GND/IOVcc/GND levels respectively. The serial interface is available via chip select line (CS*), serial transfer clock line (SCL), serial data input (SDI), and serial data output (SDO). In serial interface operation, the IM0/ID pin functions as the ID pin and the DB17-2 pins, which are not used, must be fixed at either IOVcc or GND level.

The R61500 recognizes the start of data transfer on the falling edge of CS* input and start transferring the start byte. It recognizes the end of data transfer on the rising edge of CS* input. The R61500 is selected when the 6-bit chip address in the start byte transferred from the transmission unit and the 6-bit device identification code assigned to the R61500 correspond as a result of comparison. When selected, the R61500 starts taking subsequent data. The ID pin sets the least significant bit of the device identification code. Send 01110 to the identification code, which is the five upper bits of the start byte. Two different chip addresses must be assigned to the R61500 because the seventh bit of the start byte is assigned to the register select bit (RS). When RS = 0, either index register write operation or status read operation is executed. When RS = 1, either instruction write operation or RAM read/write operation is executed. The eighth bit of the start byte is to select either read or write operation (R/W bit). The data is received when the R/W bit is 0, and is transferred when the R/W bit is 1.

After receiving the start byte, the R61500 starts transferring or receiving data in units of bytes. The data transfer is executed from the MSB. The R61500's instruction consists of 16 bits and it is executed internally when it is transferred in two bytes (D15-8, D7-0) from the MSB. The GRAM write data is expanded into 18 bits internally. The R61500 recognizes the first and the second bytes transferred after the start byte as the upper and the lower bytes of a 16-bit instruction, respectively.

In serial interface operation, invalid data is sent to the data bus until the R61500 read 5 bytes from the internal GRAM after the start byte. Valid data is read out to the data bus when the R61500 reads the 6th and subsequent bytes from the internal GRAM.

Start Byte Format

Transferred bits	S	1	2	3	4	5	6	7	8
Start byte format	Transfer start	Device ID code						RS	R/W
		0	1	1	1	0	ID		

Note: The ID bit is set by setting the IM0/ID pin.

Table 61

RS	R/W	Function
0	0	Set index register
0	1	Read status
1	0	Write an instruction or RAM data
1	1	Read an instruction or RAM data

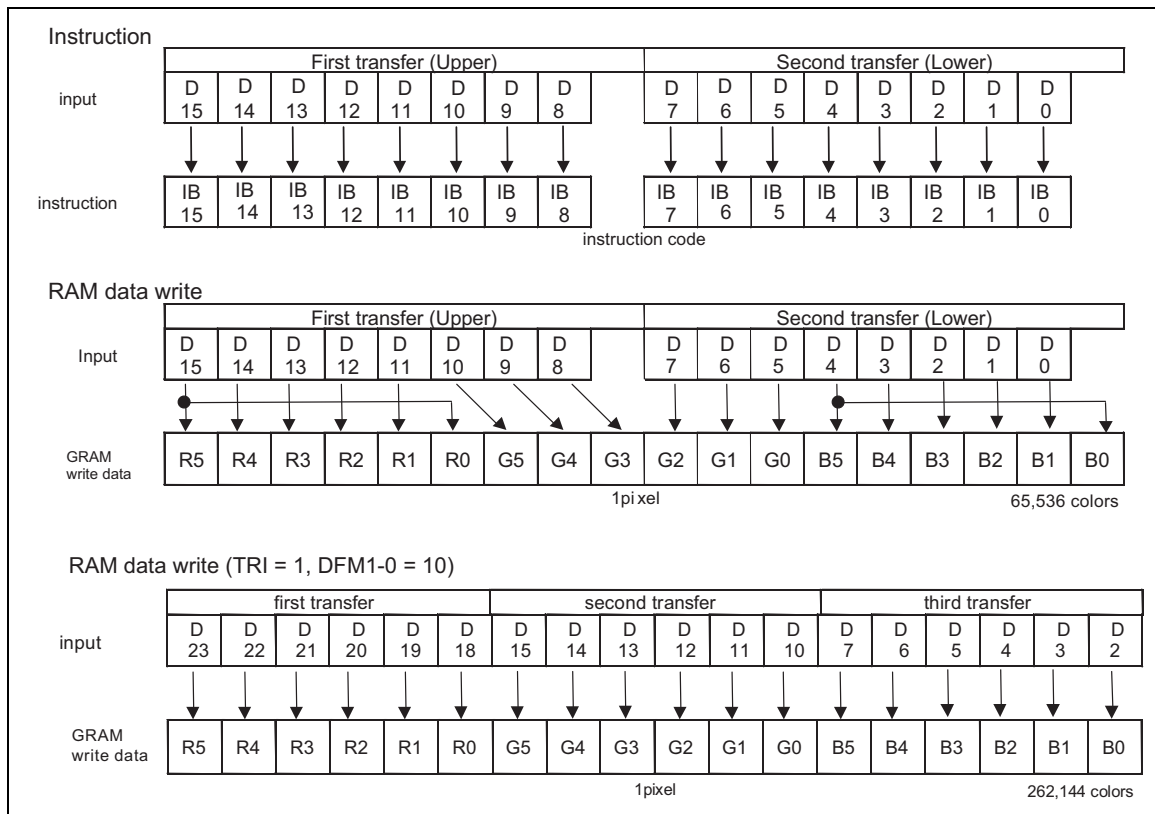
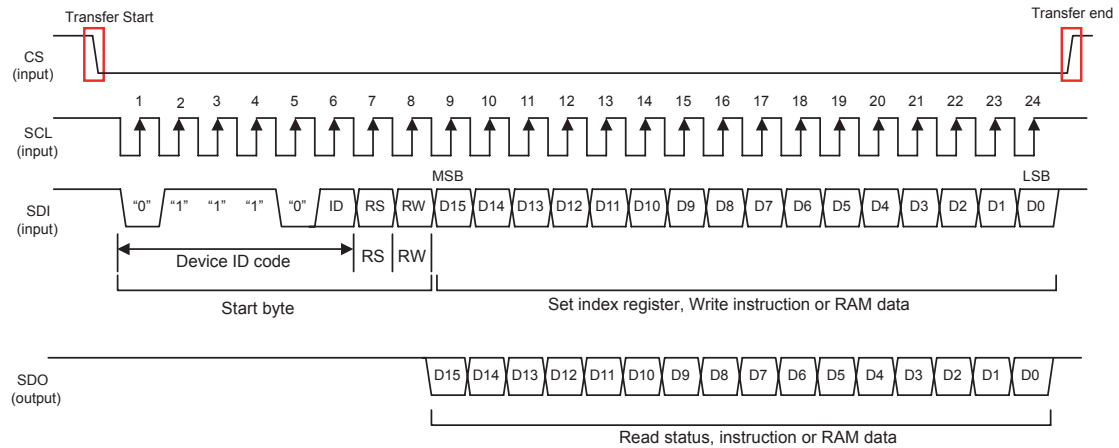
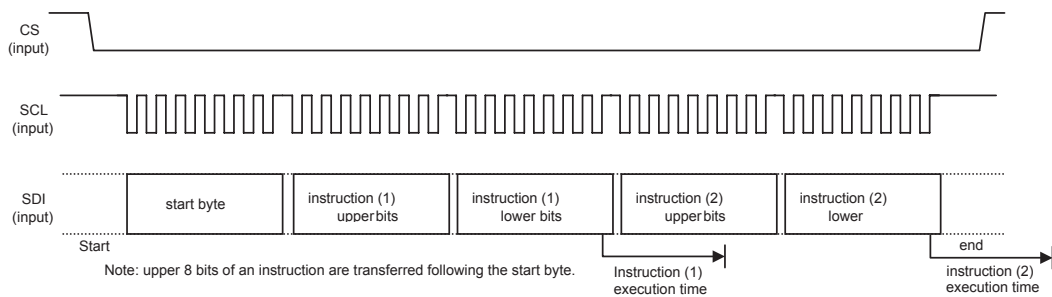


Figure 34 Data format for serial interface

a) Basic transfer via serial interface



b) Consecutive data transfer via serial interface



c) RAM read out transfer

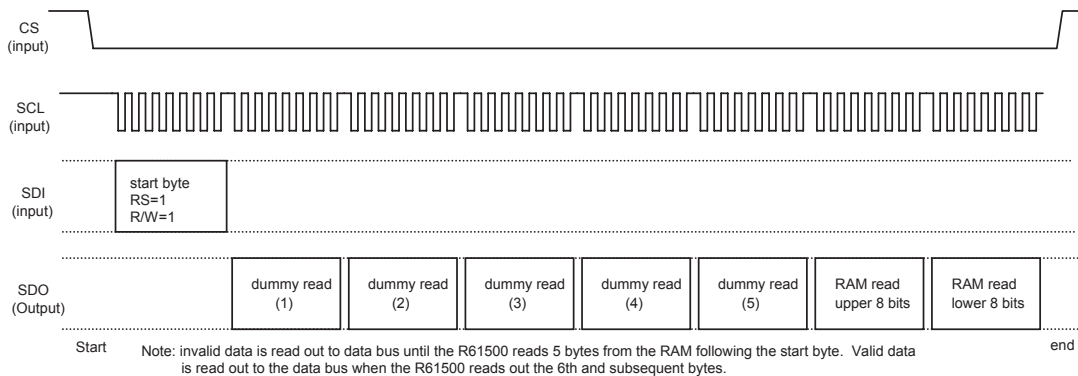
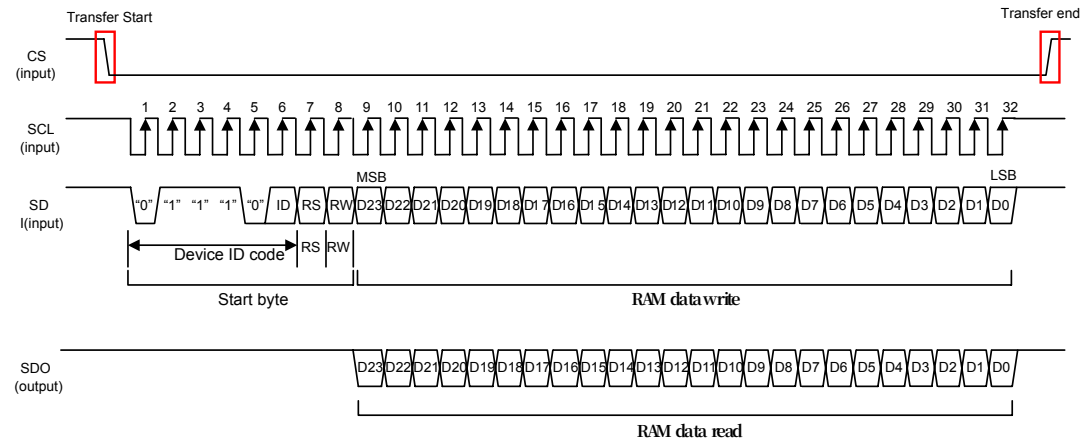


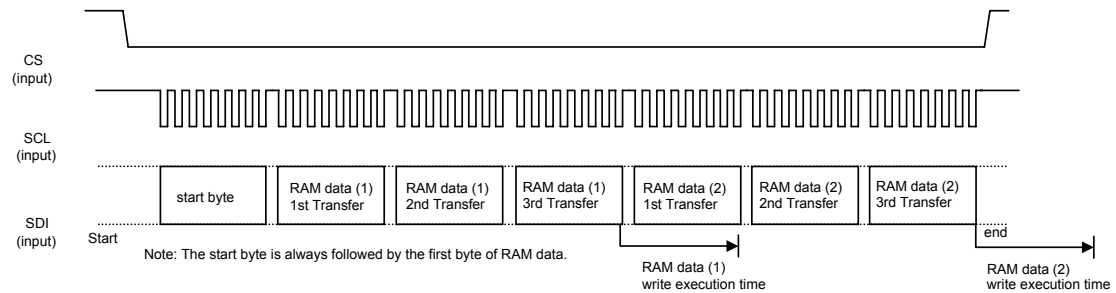
Figure 35 Data transmission via serial interface

RAM data transfer via serial interface when TRI = "1" and DFM1-0 = "10" are as follows.

d) Basic transfer via serial interface



e) Consecutive data transfer via serial interface



f) RAM read out transfer

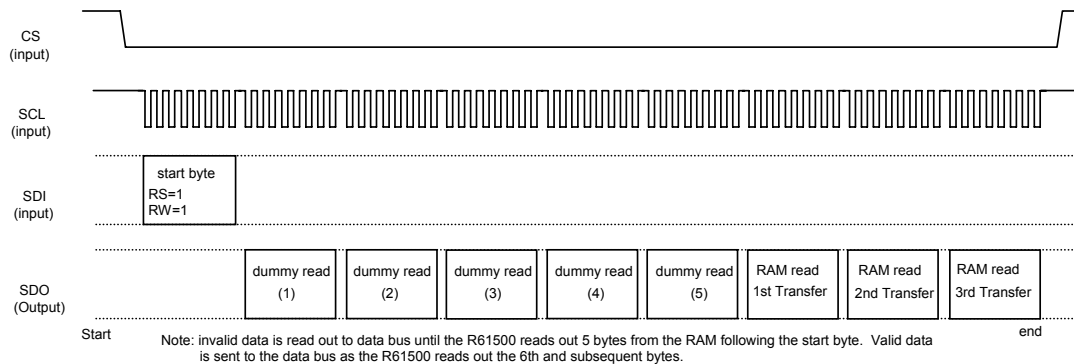
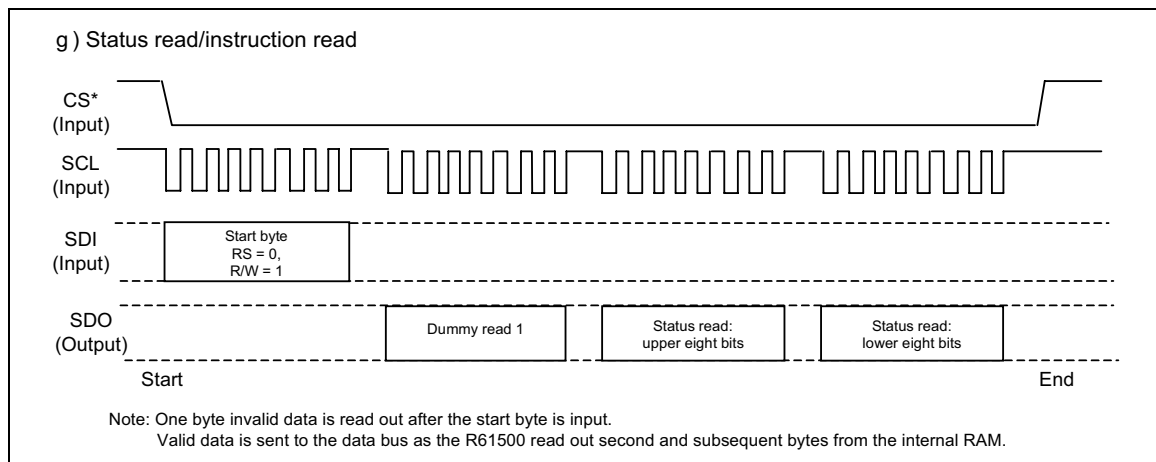


Figure 36 Data transmission via serial interface

**Figure 37 Status/Instruction Read via serial interface**

VSYNC Interface

The R61500 supports VSYNC interface, enabling the R61500 to display a moving picture via system interface by synchronizing with the frame synchronous signal (VSYNC). The VSYNC interface can realize moving picture display with minimum modification to the system interface.

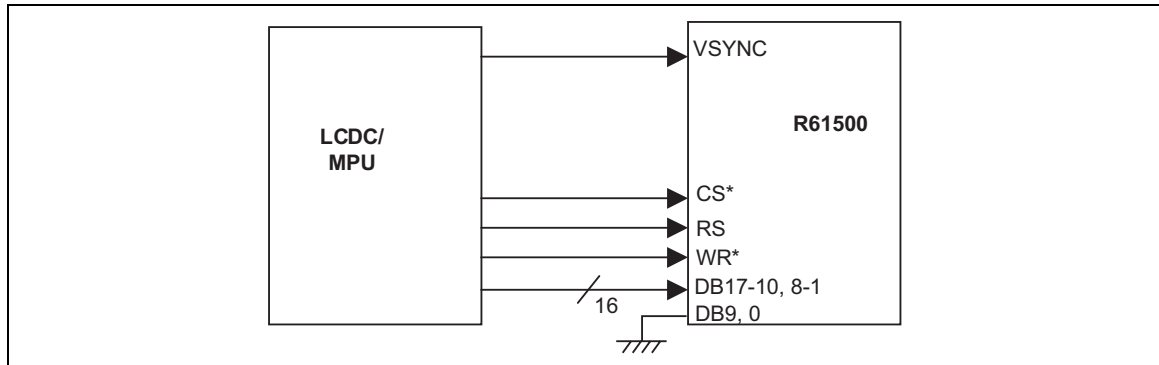


Figure 38 VSYNC interface

The VSYNC interface is selected by setting DM1-0 = 10 and RM = 0. In VSYNC interface operation, the internal display operation is synchronized with the VSYNC signal. By writing data to the internal RAM via system interface at a faster speed to a certain degree than that of internal operation, the VSYNC interface enables the R61500 to display a moving picture via system interface and rewrite moving picture frames without causing flicker on the display.

The VSYNC display operation is synchronized with the internal clock generated from internal oscillators and the VSYNC signal. The R61500 stores all display data in the internal RAM so that the data is transferred only when rewriting moving picture frames in order to minimize the total data transfer for moving picture display. By using a high-speed write function (either HWM = 1 or LHWM = 1), the internal RAM can be accessed in high speed with less power consumption in VSYNC operation.

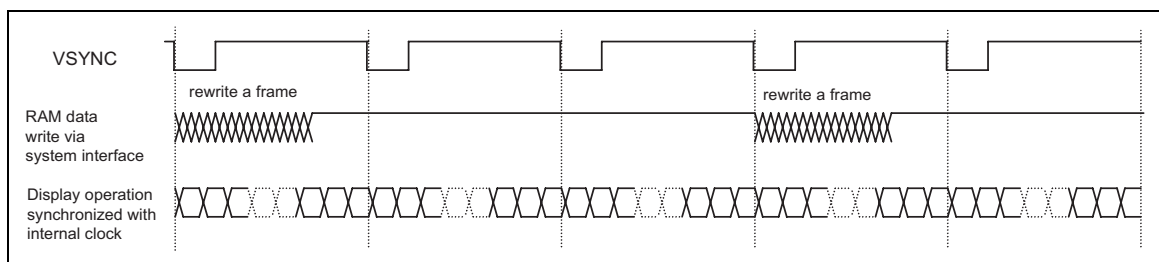


Figure 39 Moving picture data transfer via VSYNC interface

Note: When writing data via VSYNC interface use a high-speed write function (either HWM or LHWM = 1).

R61500

The VSYNC interface operation has the minimum speed and the minimum internal clock frequency in writing data to the internal RAM via system interface, which are calculated from the following formulas.

$$\begin{aligned} & \text{Internal clock frequency (fosc) [Hz]} \\ & = \text{FrameFrequency} \times (\text{DisplayLines(NL)} + \text{FrontPorch(FP)} + \text{BackPorch(BP)}) \times 16(\text{clocks}) \times \text{variance} \\ \\ & \text{RAMWriteSpeed(min.)[Hz]} > \frac{176 \times \text{DisplayLines(NL)}}{(\text{BackPorch(BP)} + \text{DisplayLines(NL)} - \text{margins}) \times 16(\text{clocks}) \times \frac{1}{f_{osc}}} \end{aligned}$$

Note: When RAM write operation is not started on the falling edge of VSYNC, the time from the falling edge of VSYNC until the RAM write operation start must also be taken into account.

The following is an example of calculations for minimum RAM writing speed and internal clock frequency in VSYNC interface operation.

[Example]

Display size	176 RGB × 240 lines
Lines	240 lines (NL = 11101)
Back/front porch	14/2 lines (BP = 1110/FP = 0010)
Frame frequency	60 Hz

$$\begin{aligned} & \text{Internal clock frequency (fosc) [Hz]} \\ & = 60 \text{ Hz} \times (240 + 2 + 14) \text{ lines} \times 16 \text{ Clocks} \times 1.1 / 0.9 = 300 \text{ kHz} \end{aligned}$$

When setting the internal clock frequency, possible causes of variances must also be taken into consideration. In this example, the calculated internal clock frequency with the above register setting allows for a margin of ±10% for variances and ensures completing the display operation within one VSYNC cycle.

In this example, variances attributed to the fabrication process of LSI and room temperature are in consideration. Other possible causes of variances, such as differences in external resistors or voltage changes are not included. It is necessary to allow for an enough margin if these factors must be included.

$$\begin{aligned} & \text{Minimum speed for RAM writing [Hz]} \\ & > 176 \times 240 / \{((14 + 240 - 2) \text{ lines} \times 16 \text{ clock}) / 300 \text{ kHz}\} = 3.14 \text{ MHz} \end{aligned}$$

The above theoretical values are calculated on the premise that the R61500 starts writing data to the internal RAM on the falling edge of VSYNC. There must at least be a margin of 2 lines between the physical display line where the display operation is performed and the RAM line address where the data write operation is performed.

The RAM write speed of 3.14MHz or more on the falling edge of VSYNC will guarantee the completion of RAM write operation before the R61500 starts displaying the RAM data on the screen, enabling rewriting the entire screen without flicker.

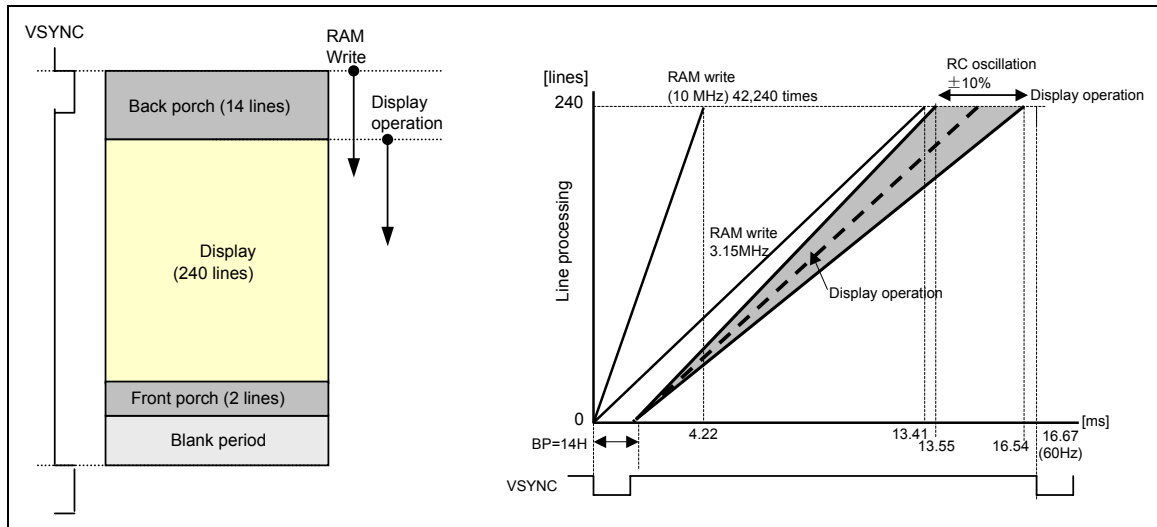


Figure 40 Write/display operation timing via VSYNC interface

Notes to VSYNC interface

1. The above example of calculation gives theoretical values. In an actual setting, other possible causes of variances not counted in the above example such as variances of internal oscillators should also be taken into consideration. It is strongly recommended to allow for an enough margin in setting the RAM writing speed.
2. The above example of calculation gives the values in case of rewriting the entire screen. If a moving picture display area is smaller than that, there will be an extra margin in the range for setting minimum RAM writing speed.

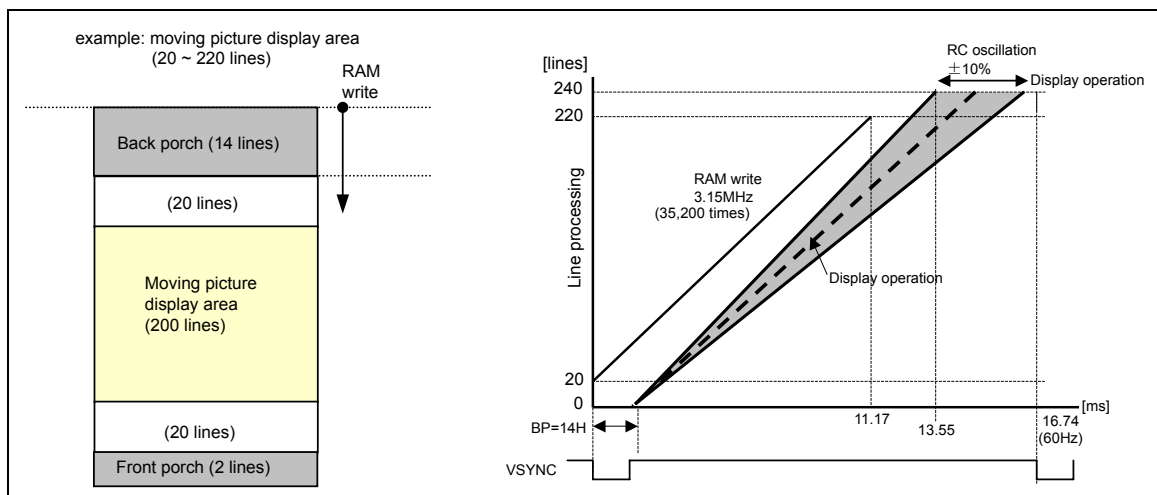


Figure 41 RAM write margin

3. The front porch period continues from the completion of frame to the next VSYNC input.
4. The instruction to switch between the internal clock operation (DM1-0 = 00) and the VSYNC interface operation is enabled not in the frame when the instruction is written but from the next frame.
5. The partial display, vertical scroll, and interlaced scan functions are disabled in VSYNC interface operation.
6. In VSYNC interface operation, set the AM bit to 0 to transfer display data in the method mentioned above.
7. In VSYNC interface mode, use a high-speed write function (either HWM = 1 or LHWM = 1) when writing display data to the internal RAM.

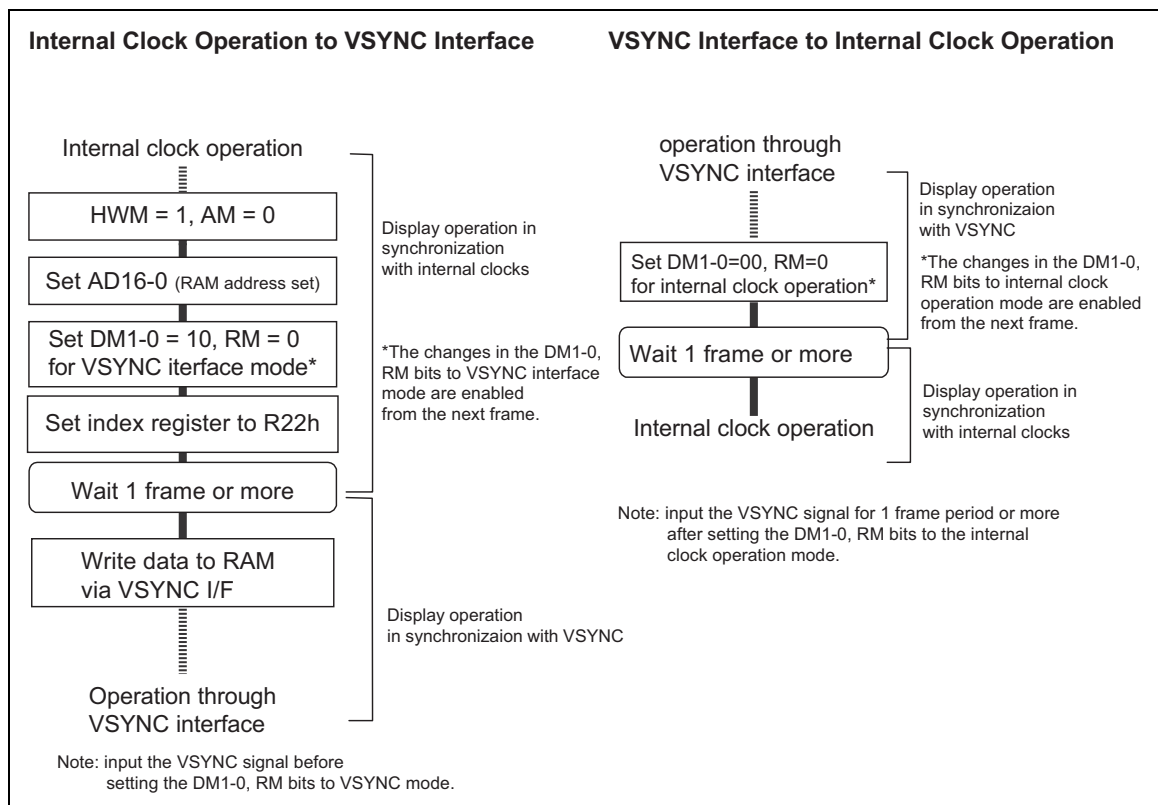


Figure 42 VSYNC/internal clock operation mode switching sequences

External Display Interface

The following are the external display interfaces supported by the R61500. The interface is selected by setting the RIM1-0 bits as follows. The RGB interface is used for RAM access.

Table 62

RIM1	RIM0	RGB Interface	DB Pin
0	0	18-bit RGB interface	DB17-0
0	1	16-bit RGB interface	DB17-10, DB8-1
1	0	6-bit RGB interface	DB17-12
1	1	Setting disabled	

Note: Using more than one RGB interface at the same time is prohibited.

RGB interface

The display operation via RGB interface is synchronized with VSYNC, HSYNC, and DOTCLK. The RGB interface in combination with window address function and high-speed write function enables the R61500 to rewrite only within the specified area made on the internal RAM in high speed with low power consumption. In RGB interface operation, back and front porch periods before and after a display period, must be made respectively.

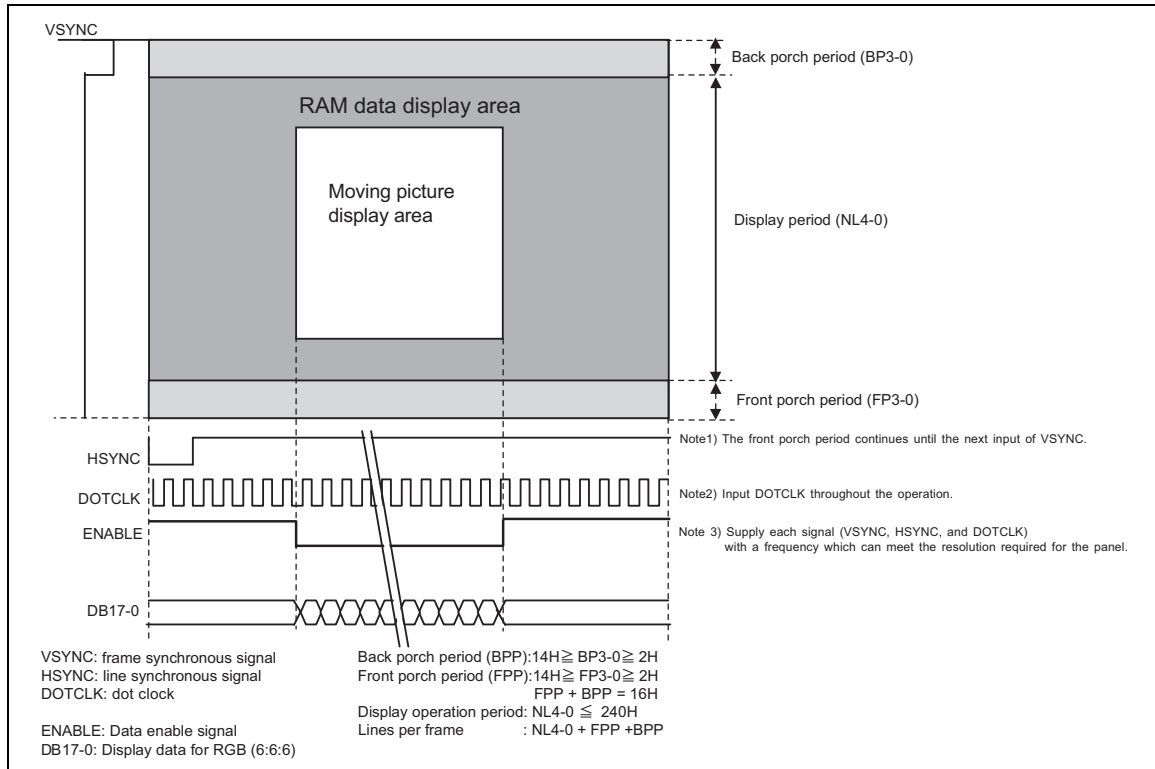


Figure 43 RGB interface

ENABLE signal

The combinations of EPL and ENABLE bits and the functions are as follows. The address is not automatically updated when writing data in the internal RAM only with the ENABLE signal, but the ENABLE signal must be set to Low. The EPL bit can invert the polarity of ENABLE signal.

Table 63

EPL	ENABLE	RAM Write	RAM Address
0	0	Enabled	Updated
0	1	Disenabled	Retained
1	0	Disenabled	Retained
1	1	Enabled	Updated

RGB interface timing

The timing chart of signals in 16/18-bit RGB interface operation is as follows.

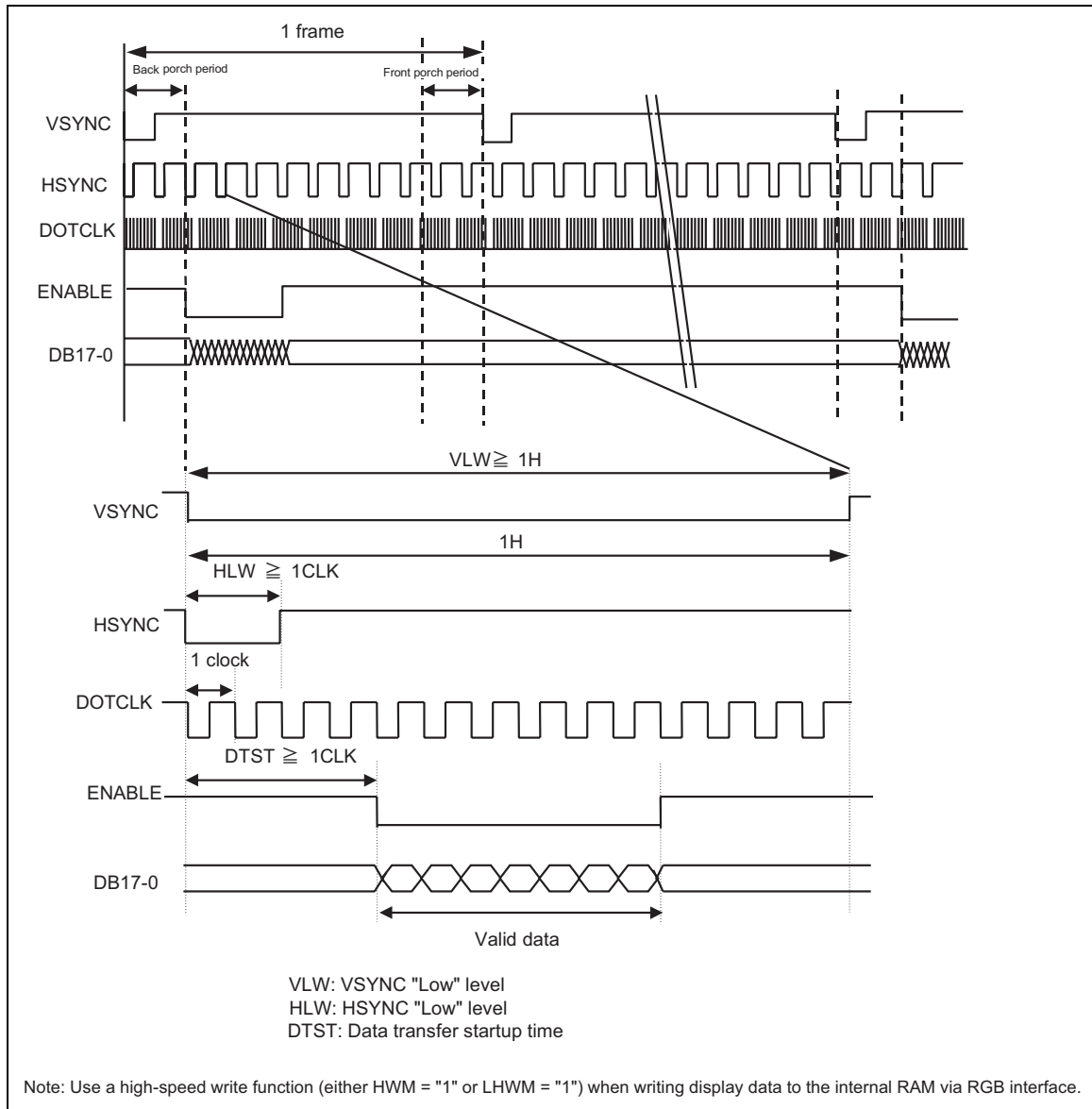


Figure 44 16-/18-bit RGB Interface Timing

The timing chart of signals in 6-bit RGB interface operation is as follows.

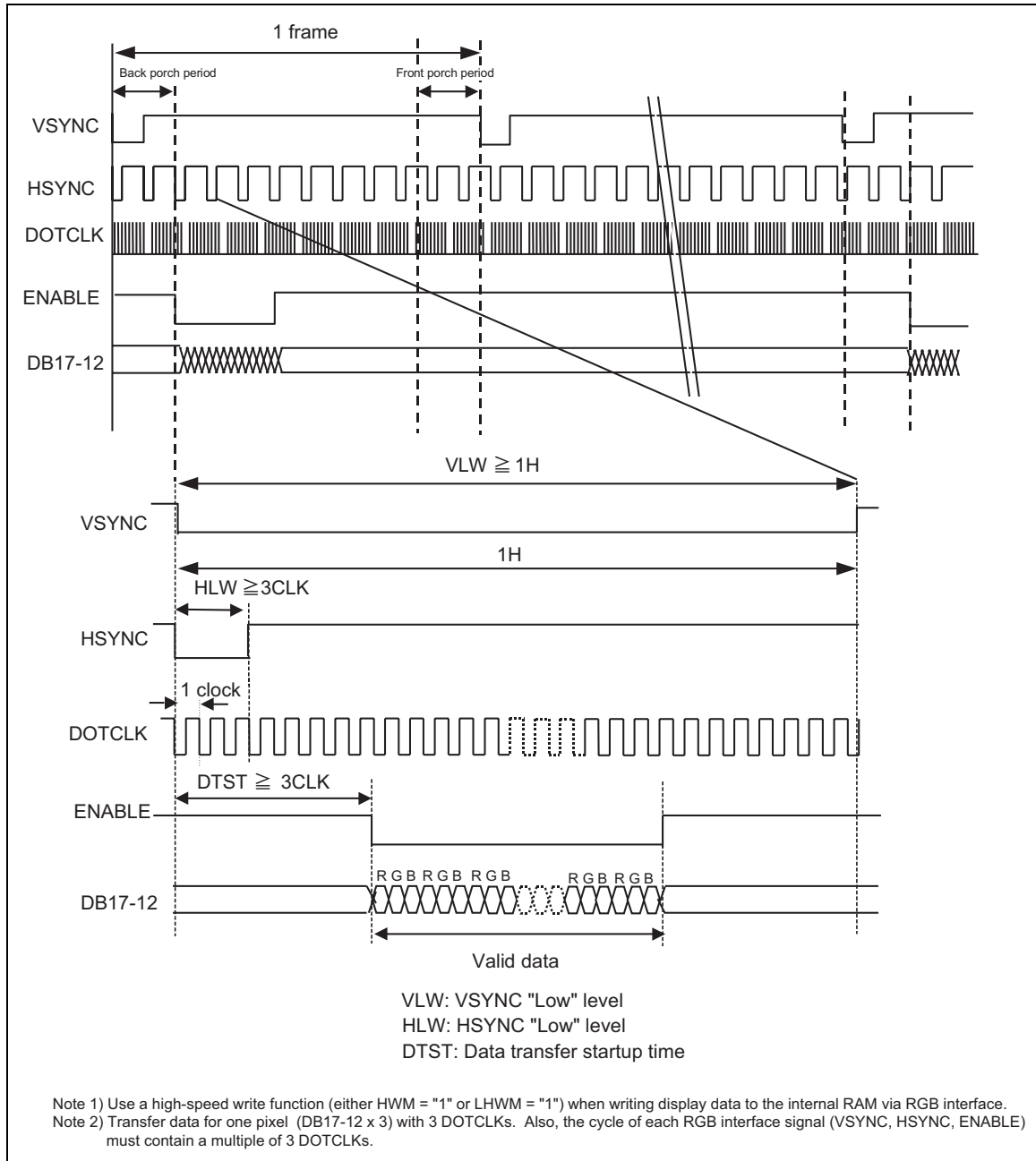


Figure 45 6-bit RGB Interface Timing

Moving picture display

The R61500 supports RGB interface for moving picture display and incorporates RAM for storing moving picture data, which provide the following merits in displaying a moving picture.

- The window address function enables data rewrite operation only within the specified moving picture area
- Enables high-speed, low-power-consumption Ram access with a high-speed write function
- Enables transferring only the necessary data for rewriting moving picture frames
- Contributes to reduction of power consumed by the system as a whole by reducing data transfer
- Allowing the use of system interface to rewrite data, such as icons, in still picture RAM area while displaying a moving picture
- Enables displaying both moving and still pictures at a same time

RAM access via a system interface in RGB-I/F mode

The R61500 allows RAM access via system interface in RGB interface operation. In RGB interface operation, the data is written to the internal RAM in synchronization with DOTCLK while ENABLE is Low. When writing data to the internal RAM via system interface, set ENABLE High to stop writing data via RGB interface. Then set RM = 0 to make RAM accessible via system interface. When restarting RAM access via RGB interface, wait a time for one read/write bus cycle. Then, set RM = 1 and the index register to R22h to start accessing RAM via RGB interface. If RAM accesses via two interfaces conflicts, the data write operation to the internal RAM is not guaranteed.

The following figure illustrates the operation of the R61500 when displaying a moving picture via RGB interface and rewriting data in the still picture RAM area via system interface.

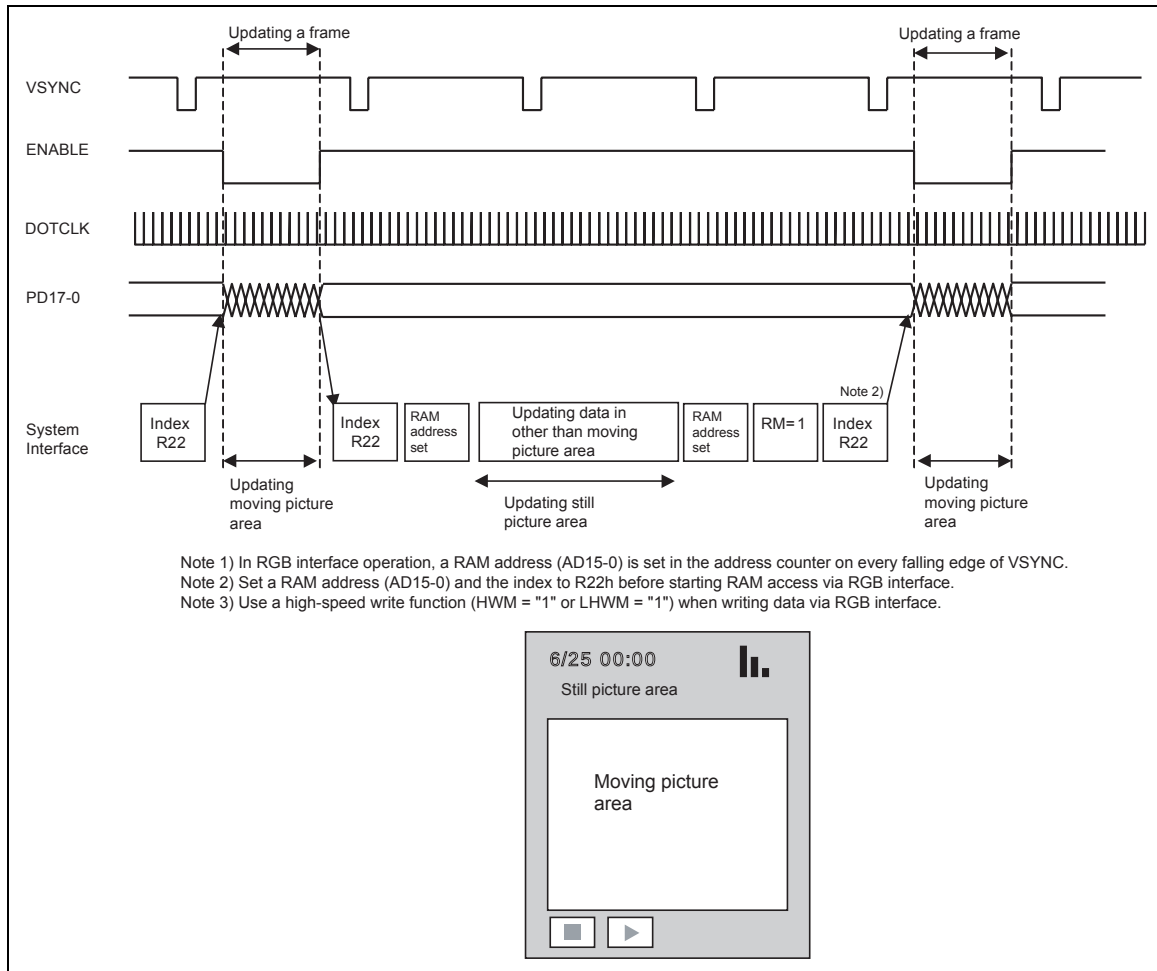


Figure 46

6-bit RGB interface

The 6-bit RGB interface is selected by setting the RIM1-0 bits to 10. The display operation is synchronized with VSYNC, HSYNC, and DOTCLK signals. The display data is transferred to the internal RAM in synchronization with the display operation via 6-bit RGB data bus (DB17-12) according to the polarity of data enable signal (ENABLE). Unused pins (DB11 to 0) must be fixed at either IOVcc or GND level.

Instructions are set only via system interface.

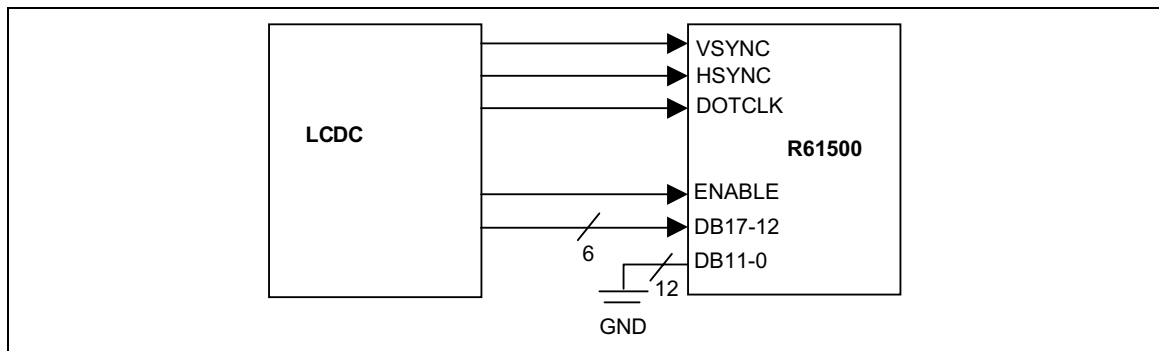


Figure 47 6-bit RGB interface

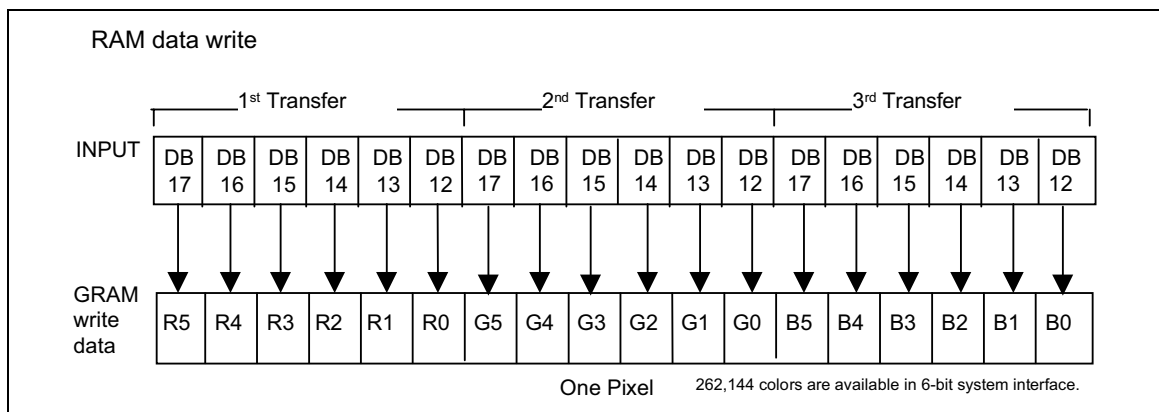


Figure 48 Data format for 6-bit interface

Data transfer synchronization in 6-bit RGB interface mode

The R61500 has data transfer counters for counting the first, second, third data transfers in 6-bit RGB interface operation. The transfer counters are always reset to the state of first data transfer on the falling edge of VSYNC. If a mismatch arises in the number of each data transfer, the counters are reset to the state of first data transfer at the start of the frame (i.e. on the falling edge of VSYNC) to restart data transfer in the correct order from the next frame. This function is useful for moving picture display, which requires consecutive data transfer, in terms of minimizing effects from failed data transfer and enabling the system to return to a normal state.

Note that the internal display operation is performed in units of pixels (RGB: taking 3 inputs of DOTCLK). Accordingly, the number of DOTCLK inputs in one frame period must be a multiple of 3 to complete data transfer correctly. Otherwise, there will be a mismatch and the effect will be carried over to the next frame.

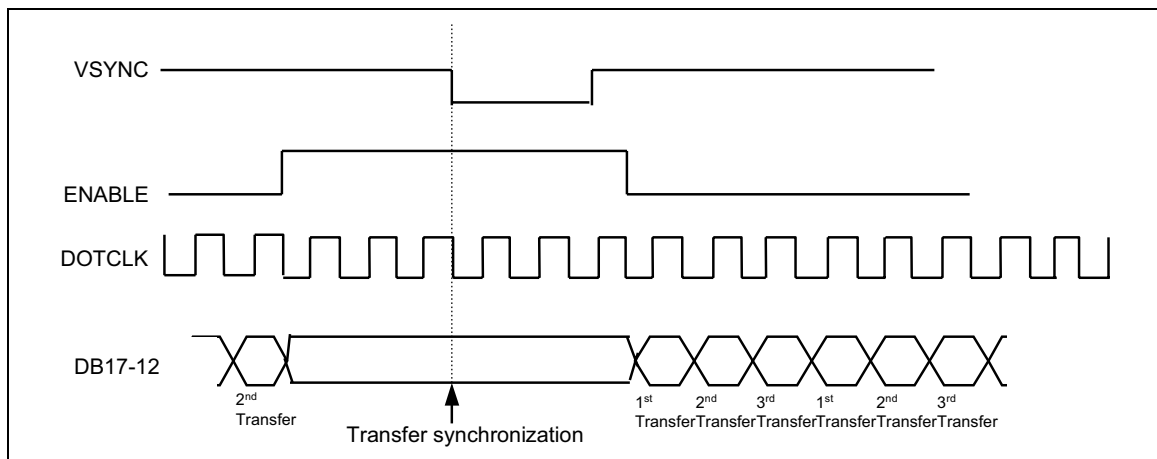


Figure 49 6-bit data transmission synchronization

16-bit RGB interface

The 16-bit RGB interface is selected by setting the RIM1-0 bits to 01. The display operation is synchronized with VSYNC, HSYNC, and DOTCLK signals. The display data is transferred to the internal RAM in synchronization with the display operation via 16-bit RGB data bus (DB17-10, DB8-1) according to the polarity of data enable signal (ENABLE).

Instructions are set only via system interface.

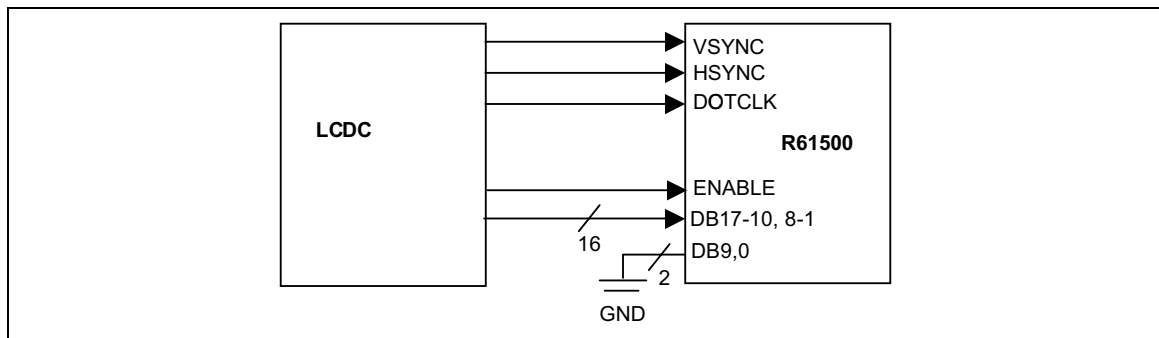


Figure 50 16-bit RGB interface

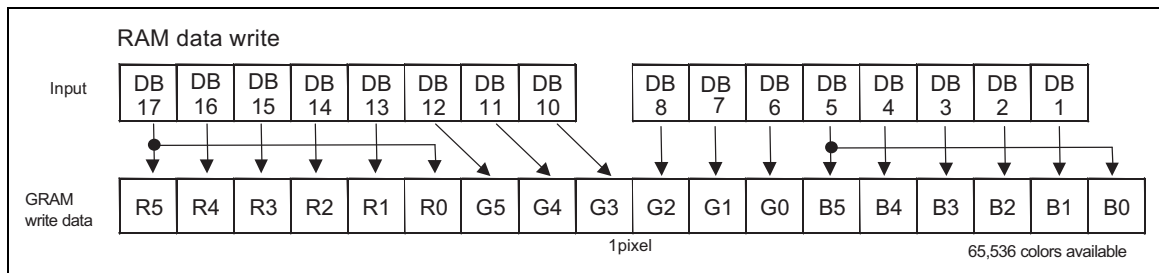


Figure 51 Data format for 16-bit interface

18-bit RGB interface

The 18-bit RGB interface is selected by setting the RIM1-0 bits to 00. The display operation is synchronized with VSYNC, HSYNC, and DOTCLK signals. The display data is transferred to the internal RAM in synchronization with the display operation via 18-bit RGB data bus (DB17-0) according to the polarity of data enable signal (ENABLE).

Instructions are set only via system interface.

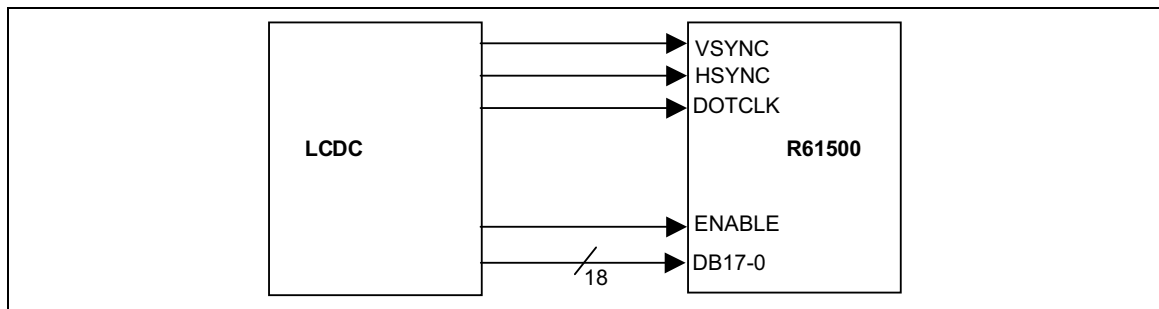


Figure 52 18-bit RGB interface

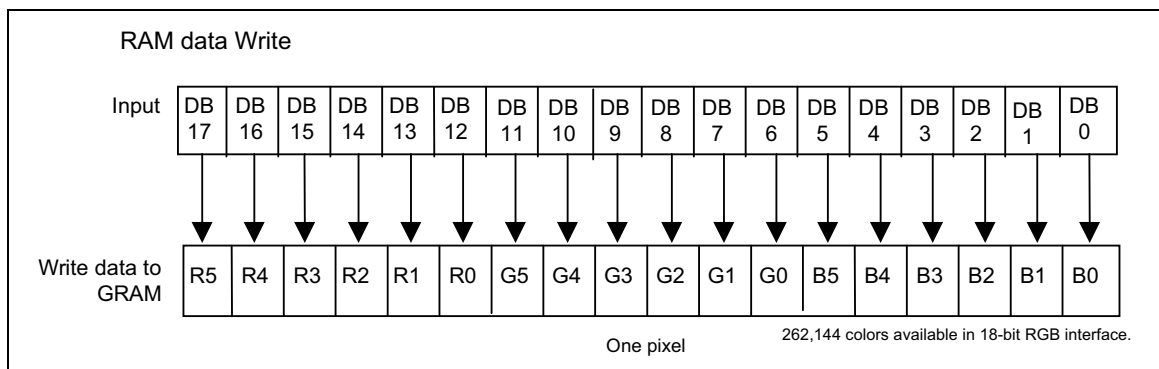


Figure 53 Data format for 18-bit interface

Notes to external display interface

1. The following are the functions disabled in external display interface operation.

Table 64

Function	External Display Interface operation	Internal Display Operation
Partial display	Not available	Available
Scroll function	Not available	Available
Interlaced scan	Not available	Available
Graphics operation function	Not available	Available

2. VSYNC, HSYNC, and DOTCLK signals must be supplied throughout the display operation.
3. The periods set with the NO1-0 bits (gate output non-overlap period), STD1-0 bits (source output delay period) and EQ1-0 bits (equalization period) are not based on the internal clock but based on DOTCLK in RGB interface operation.
4. In 6-bit RGB interface operation, RGB dots are transferred with 3 DOTCLKs. Make sure that data transfer is completed in RGB unit.
5. In 6-bit RGB interface operation, the cycles of RGB interface signals (VSYNC, HSYNC, ENABLE) must be a multiple of 3 DOTCLKs so that data transfer is completed in RGB unit.
6. When switching between the internal operation to the external display interface operation, follow the sequences below.
7. In RGB interface operation, the front porch period continues from the end of a frame to the next VSYNC input.
8. Use a high-speed write function (either HWM = 1 or LHWM = 1) when writing data to the internal RAM via RGB interface.
9. In RGB interface operation, RAM address set (set AD15-0 in the AC) is performed for each frame on the falling edge of VSYNC.

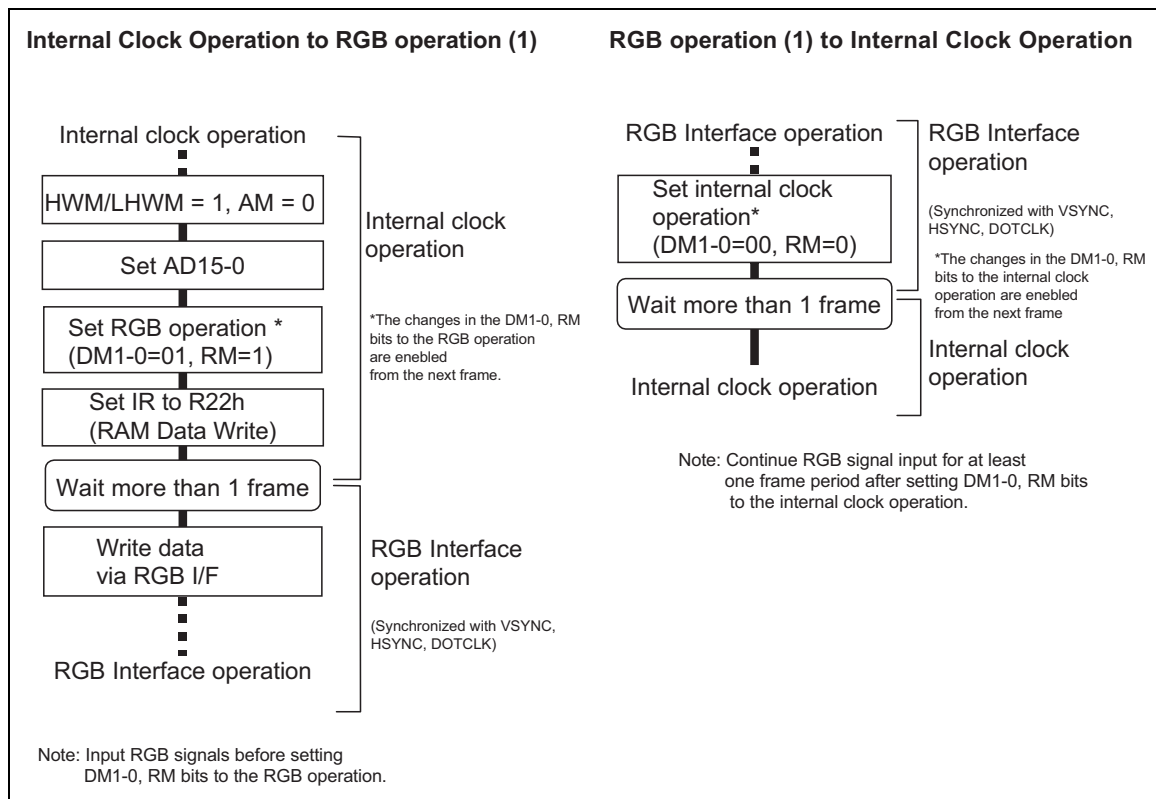


Figure 54 Internal clock operation/RGB interface mode switching sequence

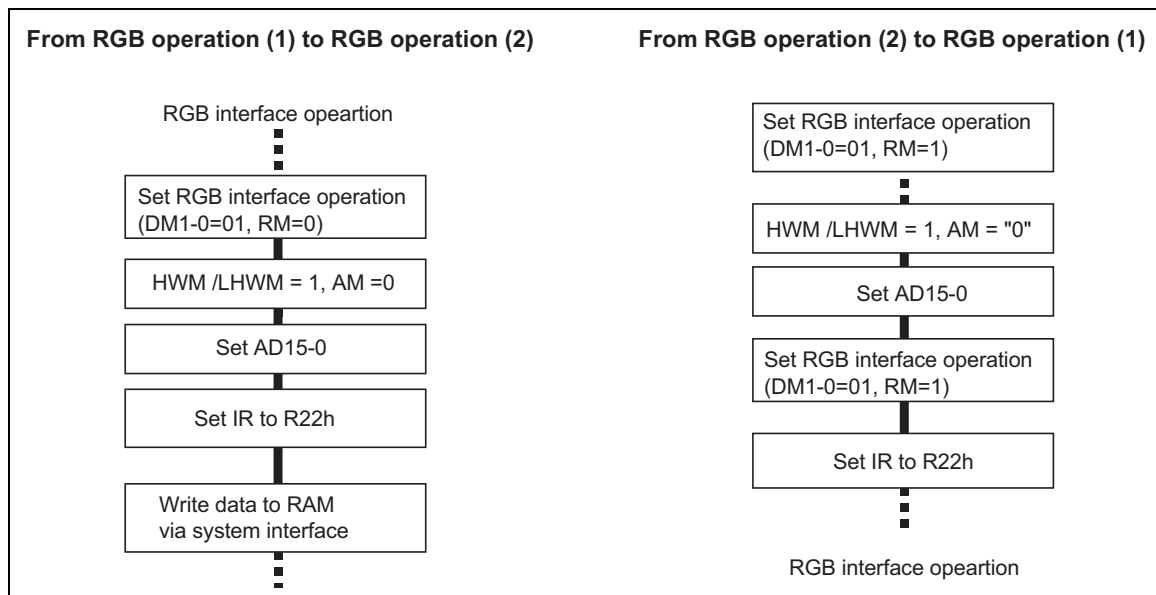


Figure 55 Switching RAM access modes (between system interface and RGB interface modes)

LCD panel Interfacing timing

The following are diagrams of interfacing timing with LCD panel control signals in internal operation and RGB interface operation.

RGB I/F mode

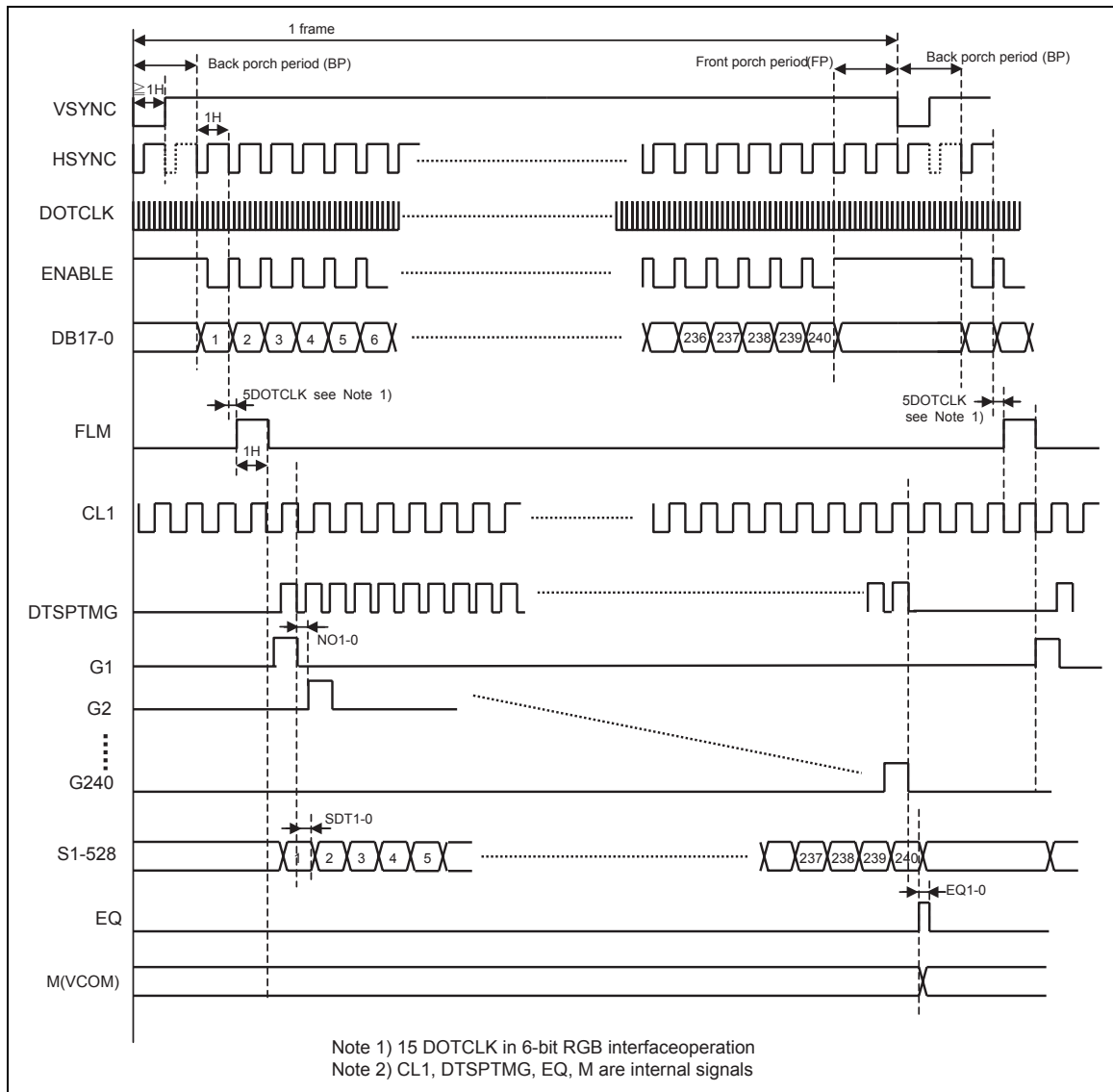


Figure 56

Internal clock operation mode

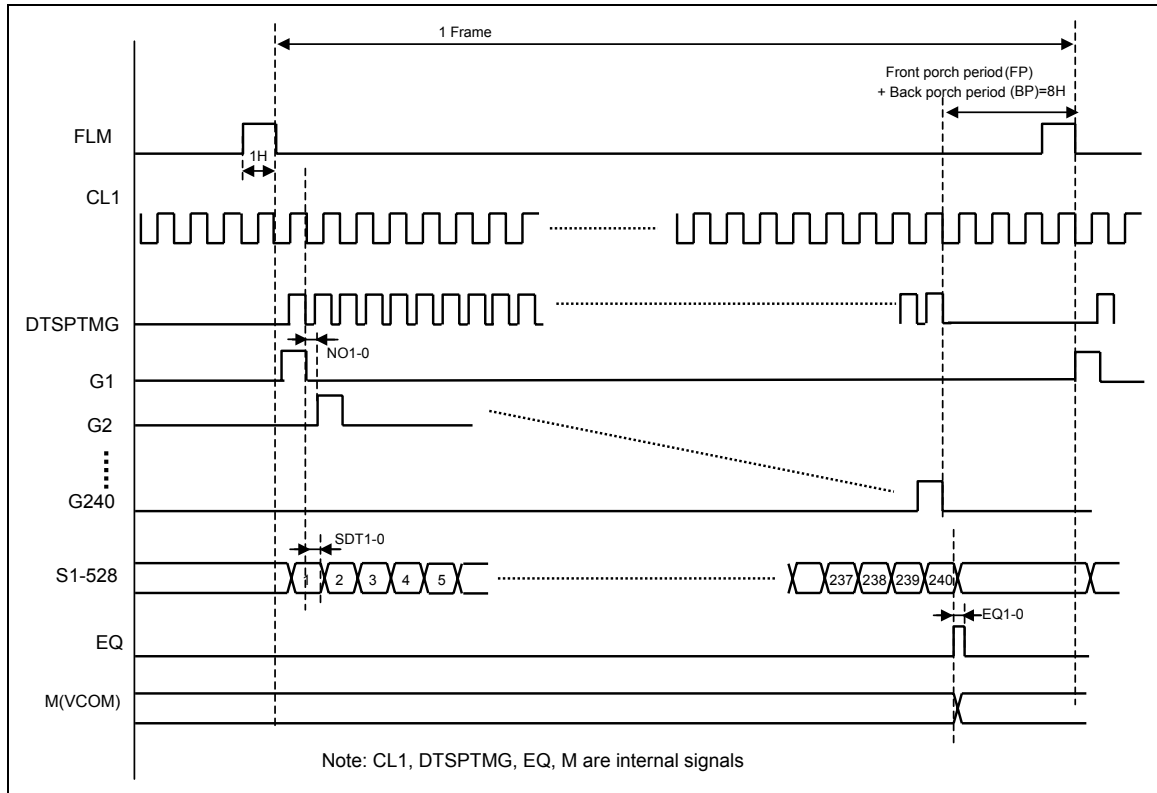


Figure 57

Scan Mode Setting

The R61500 allows for changing the shift direction of gate signal output in the following 4 different ways with combination of SM and GS bits. These combinations allow various connections between the R61500 and the LCD panel.

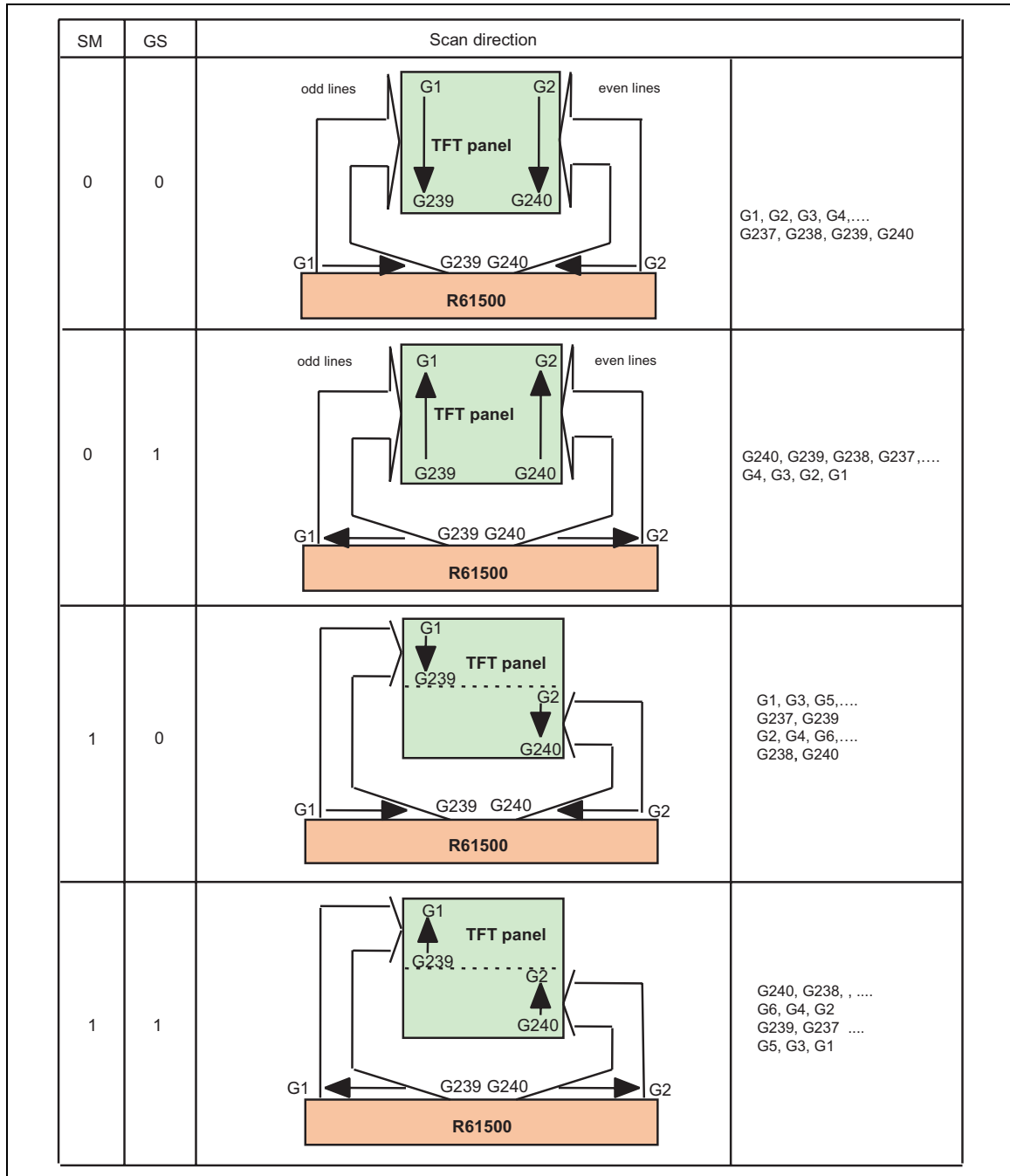


Figure 58

High-Speed RAM Write Function

The R61500 has a function to write data to the internal RAM in high speed, about a quarter of the time of normal RAM write operation. This function makes the R61500 available with applications, which require high-speed, low-power-consumption display data rewrite such as color moving picture display.

When HWM = 1 is selected, the data to be written to the internal RAM is first stored in the internal register of the R61500. The R61500 performs a write operation in units of words. When the write operation to the internal register is performed 4 times, the R61500 writes the 4 words in the register to the internal GRAM at a time. The R61500 allows writing data consecutively to the internal register, while transferring the data from the internal register to the internal RAM. By using this function, the R69400 can be used for moving picture display, which requires high-speed consecutive RAM write

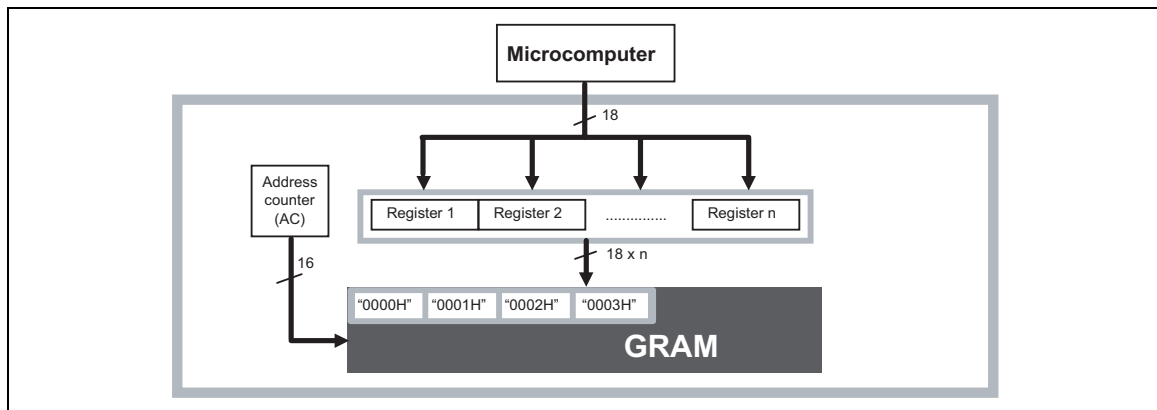


Figure 59 High-Speed RAM Write Operation

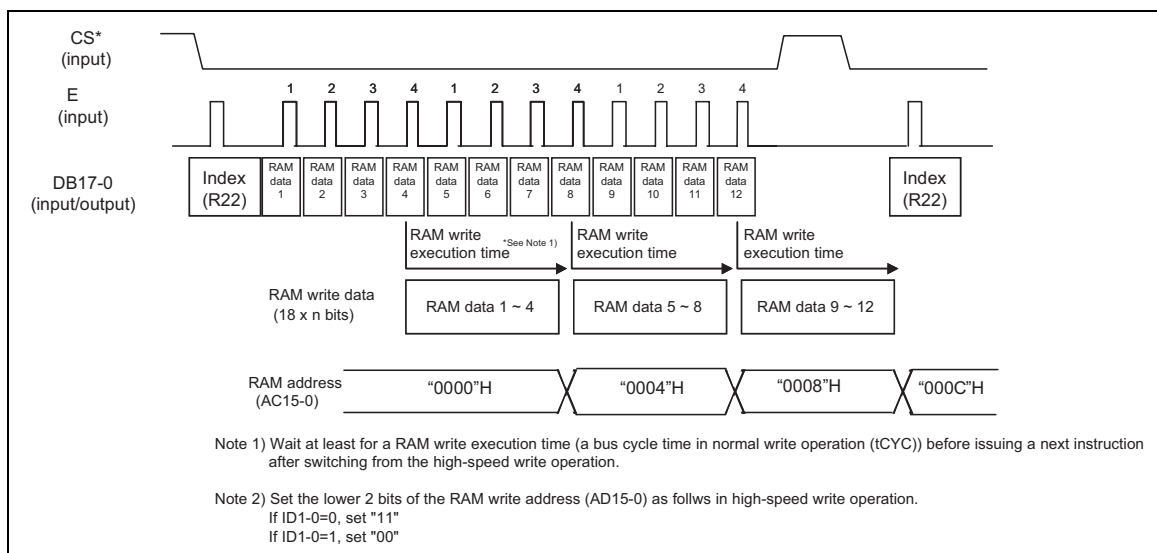


Figure 60 High-Speed Consecutive Write operation to the RAM

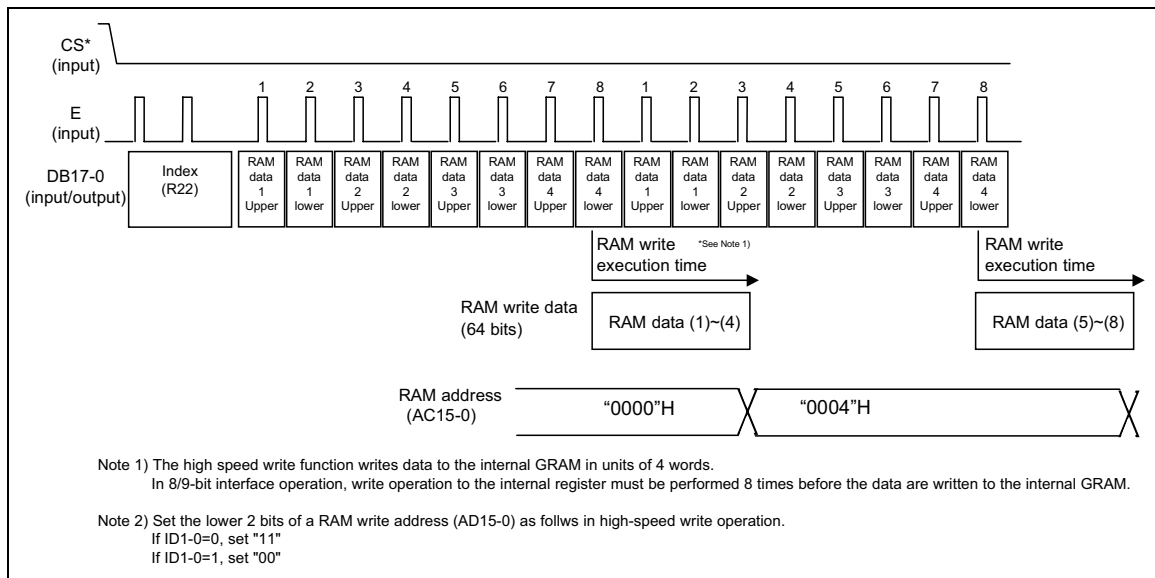


Figure 61 Example of high-speed consecutive write operation (in 8/9-bit interface mode)

Notes to high-speed RAM write function

1. The logical/compare operation is disabled in high-speed write operation.
2. The RAM write operation is performed in units of 4 words. For this reason, set the lower 2 bits of the RAM write address (AD15-0) as follows.

If ID0=0, set 11 to the lower 2 bits of the RAM write address (AD15-0)

If ID0=1, set 00 to the lower 2 bits of the RAM write address (AD15-0)

3. If the data write operation to the internal register is terminated less than 4 times, the data write operation to the internal RAM will fail.
4. When a high-speed write function is enabled, the R61500 always performs the first RAM write operation upon setting the index register to R22h (RAM data read/write). In this case, RAM data read operation is not operable. When reading data from the RAM, make sure to set the HWM and LHWM bits to 0.
5. The high-speed RAM write operation is not operable with the normal RAM write operation. When switching the write mode to the other, make sure to execute RAM address set (AD15-0) after switching the modes.
6. When writing data within a window address area in high-speed write operation, a dummy write operation will be required according to the window address area setting. See High-speed RAM write operation in a window address area for details.

Table 65

	Normal RAM Write (HWM/LHWM = 0)	High-Speed RAM Write (HWM = 1 or LHWM = 1)
Logical operation	Available	Not available
Compare operation	Available	Not available
BGR function	Available	Available
Write mask function	Available	Available
RAM address set	In units of words	ID0 = 0: Set the lower two bits to 11 ID0 = 1: Set the lower two bits to 00
RAM read	In units of words	Not available
RAM write	In units of words	A dummy write operation will be required according to the window address area setting.
Window address	In units of words	Horizontal range (HSA/HSE): more than 4 words Horizontal write operation: 4N (N≥2) times
External display interface	Available	Available
AM bit	AM = 1 or 0	AM = 0

High-speed RAM write operation in a window address area

To rewrite data consecutively in high speed within a rectangular area specified on the internal RAM, the number of dummy write operation must be set in advance, if necessary, to make the total number of write operations a multiple of four times.

The numbers of dummy write operation added at the start and the end of the data write operation are determined when setting the horizontal window address register (HSA1-0, HEA1-0), which specifies the start and the end addresses of the window address area in horizontal direction. Make sure that the number of write operation including dummy write in each horizontal line becomes a multiple of four times.

Table 66 Number of Dummy Write Operation for High-Speed RAM Write (HSA Bits)

HSA1	HSA0	Number of Dummy write operation at start
0	0	0
0	1	1
1	0	2
1	1	3

Table 67 Number of Dummy Write Operation for High-Speed RAM Write (HEA Bits)

HEA1	HEA0	Number of Dummy write operation at end
0	0	3
0	1	2
1	0	1
1	1	0

The number of write operation including dummy operation for each horizontal line must be $4 \times N$ times:
Horizontal write op. = start dummy write op. + data write op. + end dummy write op. = $4 \times N$ (times)

R61500

The following is an example of high-speed RAM write operation within a window address area.

In the following example, HSA1-0 and HEA1-0 are set to 10 and 00, respectively. In this case, two and three dummy write operations are added at the start and the end of the data write operation for each line respectively in order to rewrite the data within the window address area in high speed consecutively.

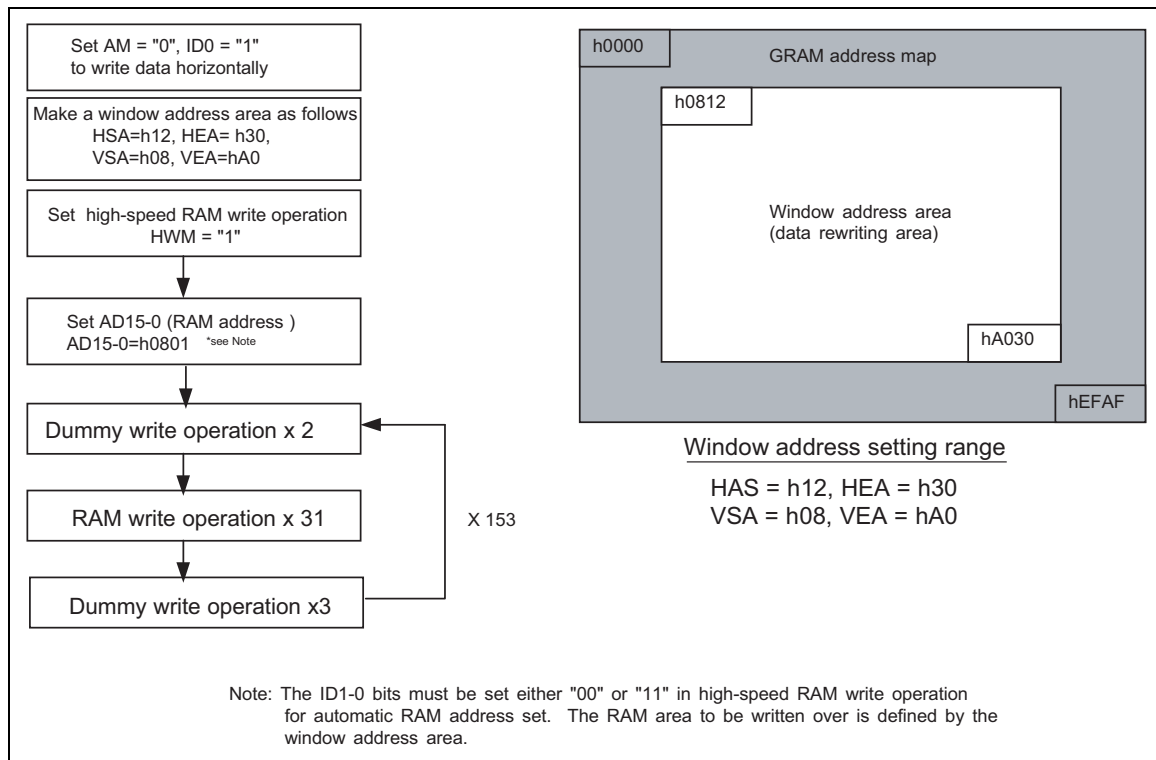


Figure 62 High-Speed RAM Write with Window Address Function

Window Address Function

The window address function enables writing display data consecutively in a rectangular area (a window address area) made on the internal RAM. The window address area is made by setting the horizontal address register (start: HSA7-0, end: HEA 7-0 bits) and the vertical address register (start: VSA7-0, end: VEA7-0 bits). The RAM address transition direction in writing data is determined by setting the AM and I/D1-0 bits (either increment or decrement), which enable the R61500 to write data including image data consecutively without considering the data wrap position.

The window address area must be made within the GRAM address map area. Also, RAM address set must be made within the window address area.

Restrictions

[Window address area]	
(Horizontal direction)	"00"H ≤ HSA7-0 ≤ HSA7-0 ≤ "AF"H
(Vertical direction)	"00"H ≤ VSA7-0 ≤ VEA7-0 ≤ "EF"H
[RAM address set]	
(RAM address)	HSA7-0 ≤ AD7-0 ≤ HEA7-0
	VSA7-0 ≤ AD15-8 ≤ VEA7-0

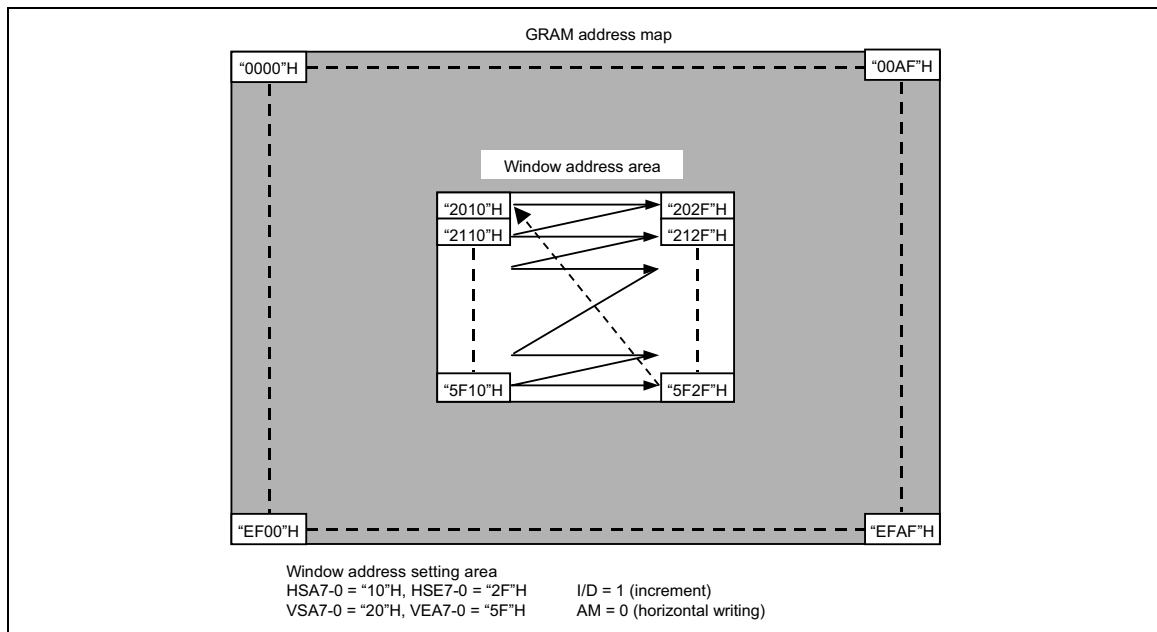


Figure 63

Graphics Operation Function

The R61500 adopts 18-bit bus architecture and supports graphics operation functions in order to reduce the graphics data processing load on the microcomputer. The R61500's graphics operation functions include:

1. Write data mask function for writing only selected bits in the 18-bit write data over the RAM data
2. Conditional rewrite function for writing data sent from the microcomputer only when a certain condition is met as a result of comparing the data with the compare register bits

These graphics operation functions control the write operation of data from the microcomputer, according to the setting in the entry mode register or write data mask register.

Table 68 Graphics operation and related register (bits) setting

Operation Mode	Bit Setting			Function
	I/D	AM	LG2-0	
Write mode 1	0/1	0	000	Horizontal data replacement
Write mode 2	0/1	1	000	Vertical data replacement
Write mode 3	0/1	0	110/111	Conditional horizontal data replacement
Write mode 4	0/1	1	110/111	Conditional vertical data replacement

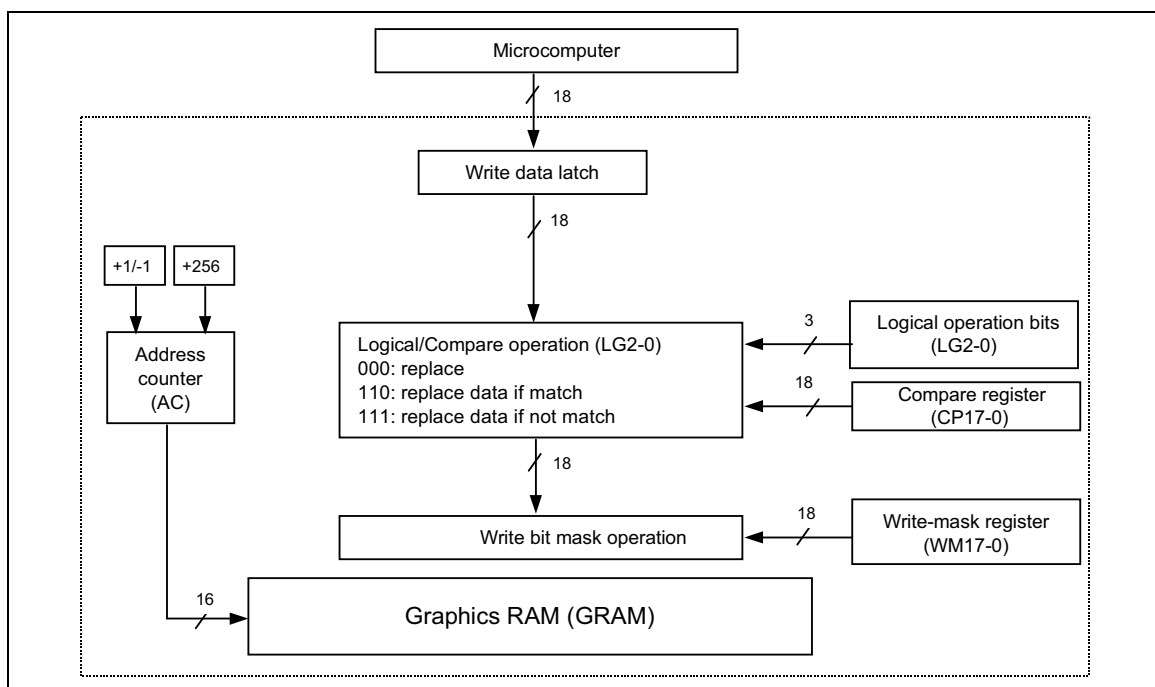


Figure 64 Graphics operation function

Write-data Mask Function

The write data mask function enables the R61500 to control the write operation to the internal GRAM in units of bits. The R61500 expands 16-bit data sent from the microcomputer into 18-bit data internally. In 18-bit interface operation, the data is not expanded.

The write data mask function enables the write operation of bits when the corresponding bits in the write data mask register (WM17-0) are 0 and disables the write operation of bits when the corresponding bits in the write data mask register (WM17-0) are 1. In the latter case, the GRAM data are not overwritten but retained. This function is useful when rewriting only the data of a specific pixel or a particular display color.

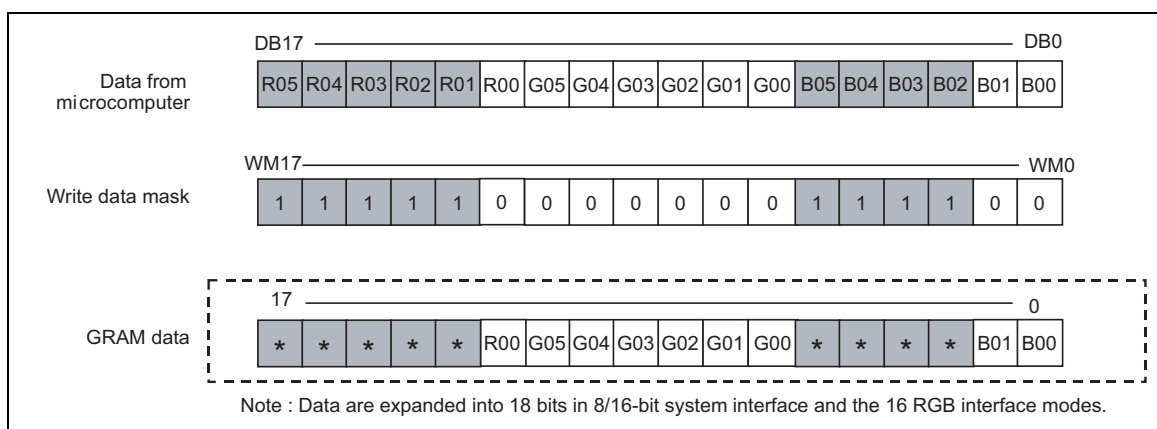


Figure 65 Write data mask function

Graphics Operation Processing Examples

1. Write mode 1: AM = 0, LG2-0 = 000

This mode is used when writing data horizontally in high speed. It can also be used for initializing the graphics RAM (GRAM) or drawing a horizontal line. The write-data mask function (WM17-0) is operable in this operation mode. The address counter (AC) is either automatically incremented (I/D = 1) or decremented by 1 (I/D = 0) as the write operation is performed. When the address comes to either right or left end of the GRAM line address according to the setting, the address in the AC automatically jumps to the one at the opposite end of the line below.

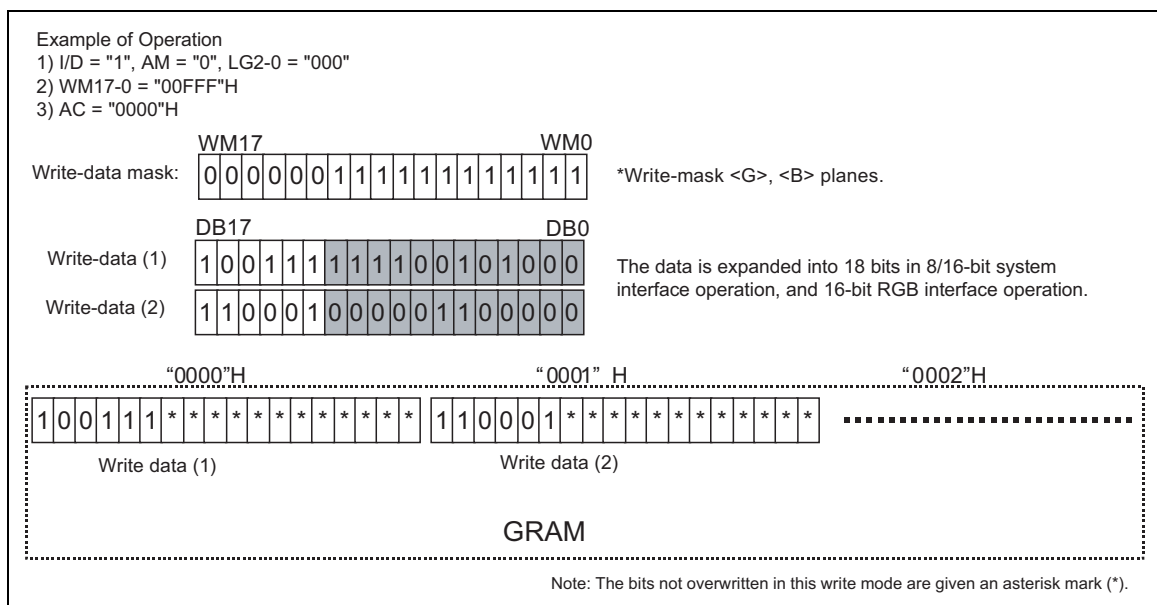


Figure 66 Write Mode 1

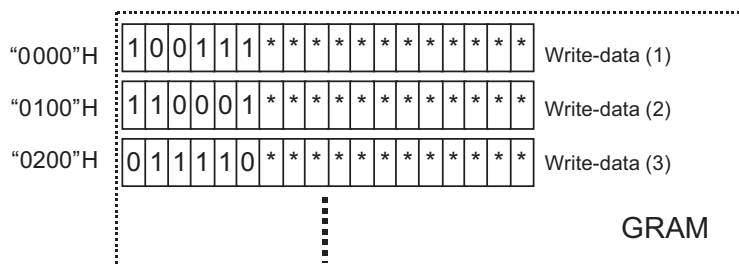
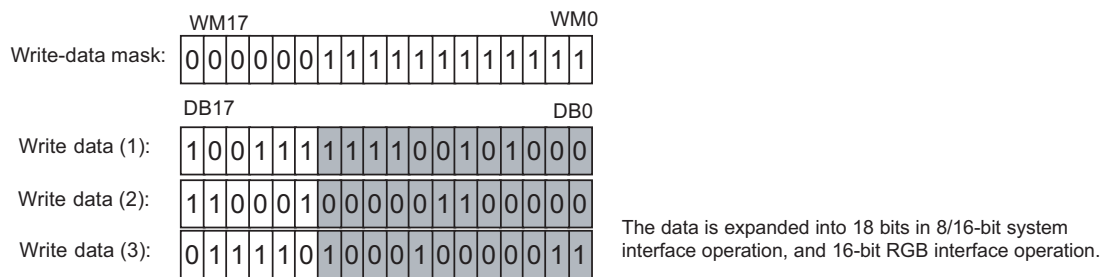
R61500

2. Write mode 2: AM = 1, LG2-0 = 000

This mode is used when writing data vertically in high speed. It can also be used for initializing the graphics RAM (GRAM), developing font patterns or drawing a vertical line. The write-data mask function (WM17-0) is operable in this operation mode. The address counter (AC) is automatically incremented by 256 as the write operation is performed. When the address comes to the bottom of the GRAM address column, the address in the AC automatically jumps to the one at the top of either left or right next column according to the I/D bit (ID = 0: left/ID = 1: right).

Example of Operation

- 1) I/D = "1", AM = "1", LG2-0 = "000"
- 2) WM17-0 = "00FFF"H
- 3) AC = "0000"H



- Note 1) The bits not overwritten in this write mode are given an asterisk mark (*).
- Note 2) When data are written to "EF00"H, the window address jumps to "0001".

Figure 67 Write Mode 2

3. Write mode 3: AM = 0, LG2-0 = 110/111

This mode is used when writing data horizontally as performing a compare operation between the RAM write data and the compare register bits (CP17-0). The data sent from the microcomputer is written to the internal RAM only when a certain condition is met as a result of comparing the data with the compare register bits. The write-data mask function (WM17-0) is operable in this operation mode. The address counter (AC) is either automatically incremented (I/D = 1) or decremented by 1 (I/D = 0) as the write operation is performed. When the address comes to either right or left end of the GRAM line address according to the setting, the address in the AC automatically jumps to the one at the opposite end of the line below.

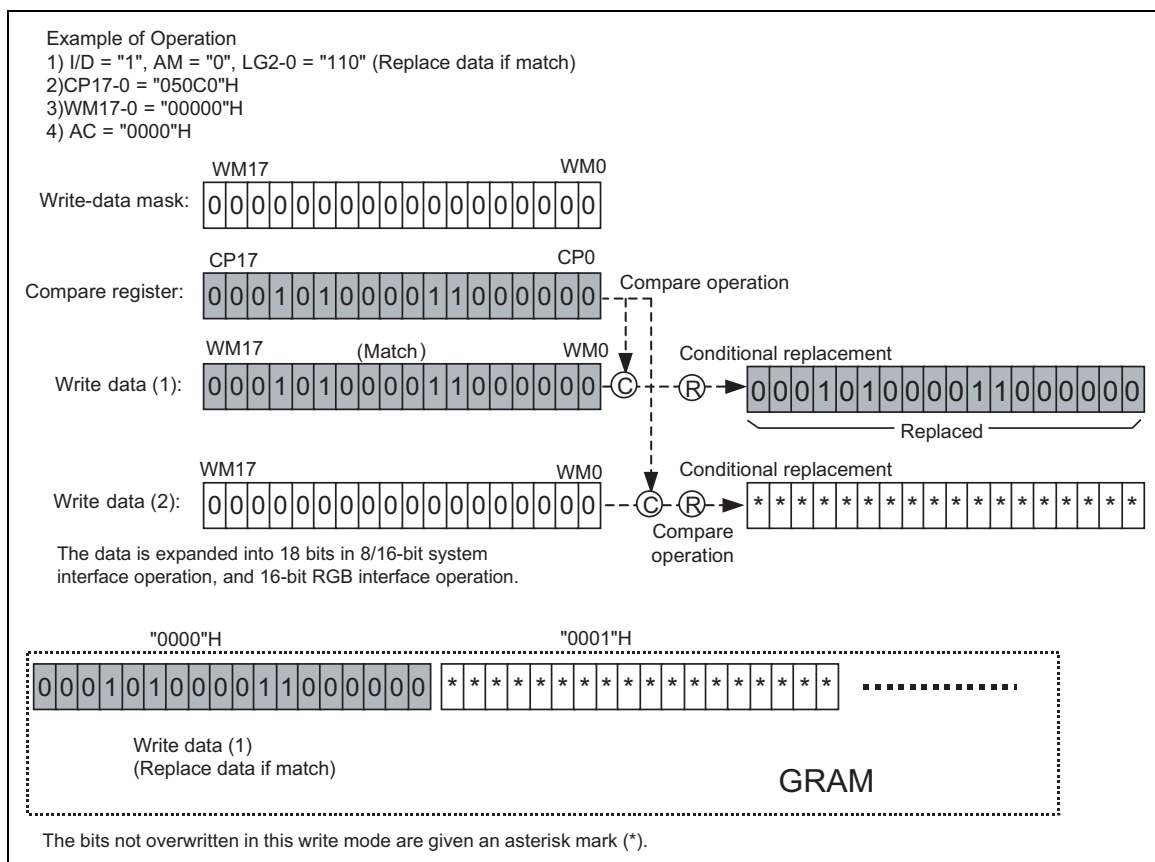


Figure 68 Write Mode 3

4. Write mode 4: AM = 1, LG2-0 = 110/111

This mode is used when writing data vertically as performing a compare operation between the RAM write data and the compare register bits (CP17-0). The data sent from the microcomputer is written to the internal RAM only when a certain condition is met as a result of comparing the data and the compare register bits. The address counter (AC) is automatically incremented by 256 as a write operation is performed. The write-data mask function (WM17-0) is operable in this operation mode. When the address comes to the bottom of the GRAM address column, the address in the AC automatically jumps to the one at the top of either left or right next column according to the I/D bit (ID = 0: left / ID = 1: right).

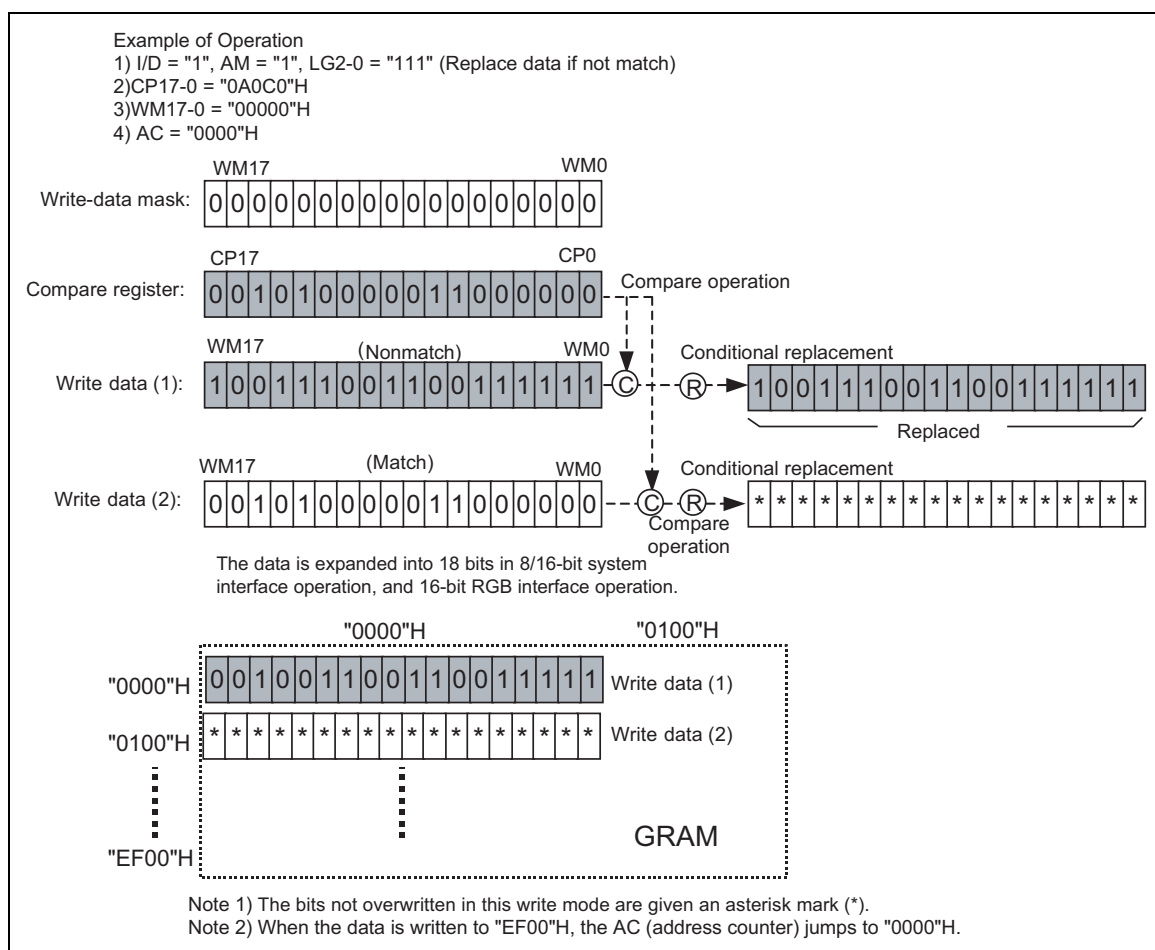


Figure 69 Write Mode 4

γ -Correction Function

The R61500 supports a γ -correction function to display in 262,144 colors simultaneously. The γ -correction is performed with 3 groups of registers determining eight reference grayscale levels, which are gradient adjustment, amplitude adjustment and fine-adjustment registers. The R61500 has these three kinds of registers for both positive and negative polarities, which can be set independently from others to allow setting according to the characteristics of the panel.

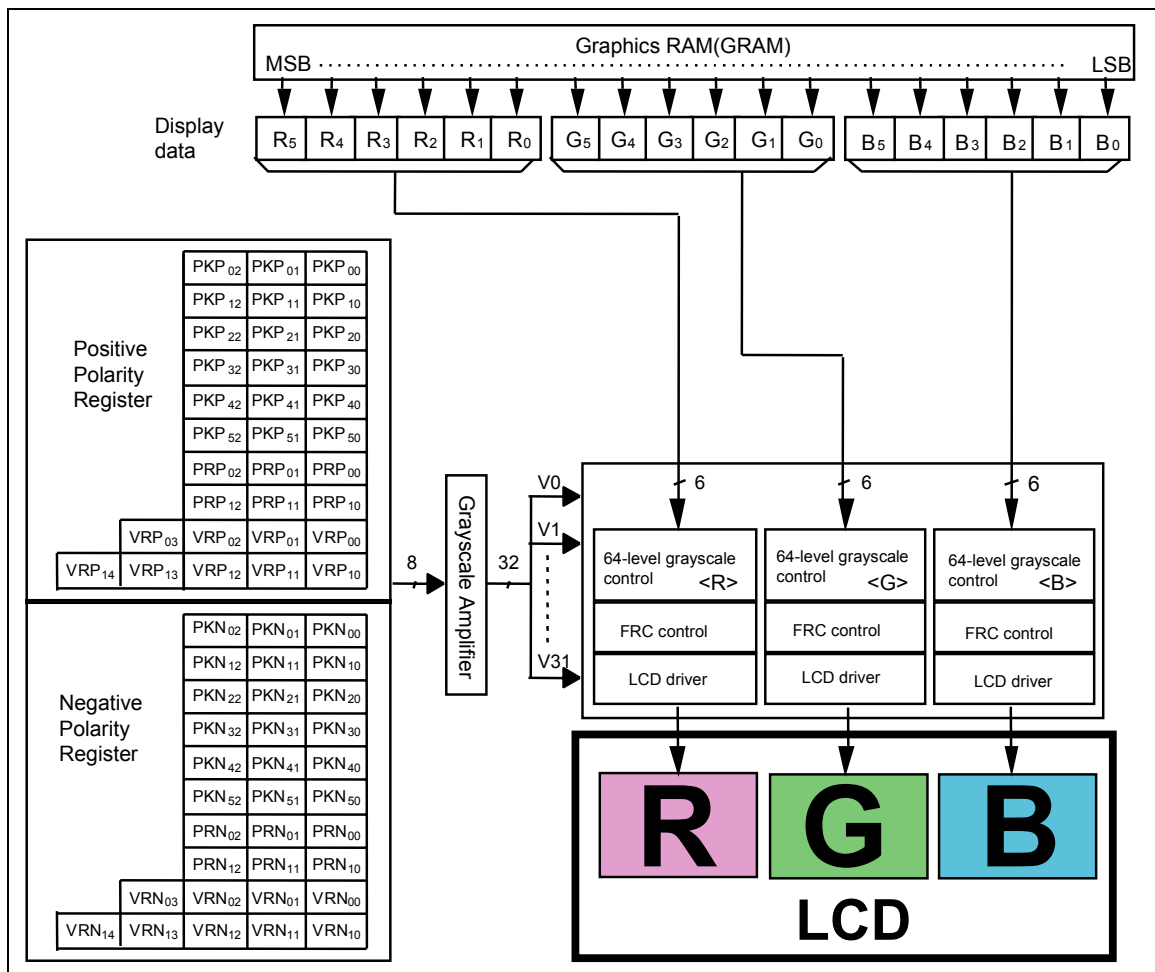


Figure 70

Grayscale Amplifier unit configuration

The following figure illustrates the grayscale amplifier unit of the R61500.

To generate 32 grayscale voltages (V0~V31), the R61500 first generates eight reference grayscale levels (VINP0-7/VINN0-7). The grayscale amplifier unit then divides eight reference grayscale levels with ladder resistors incorporated therein.

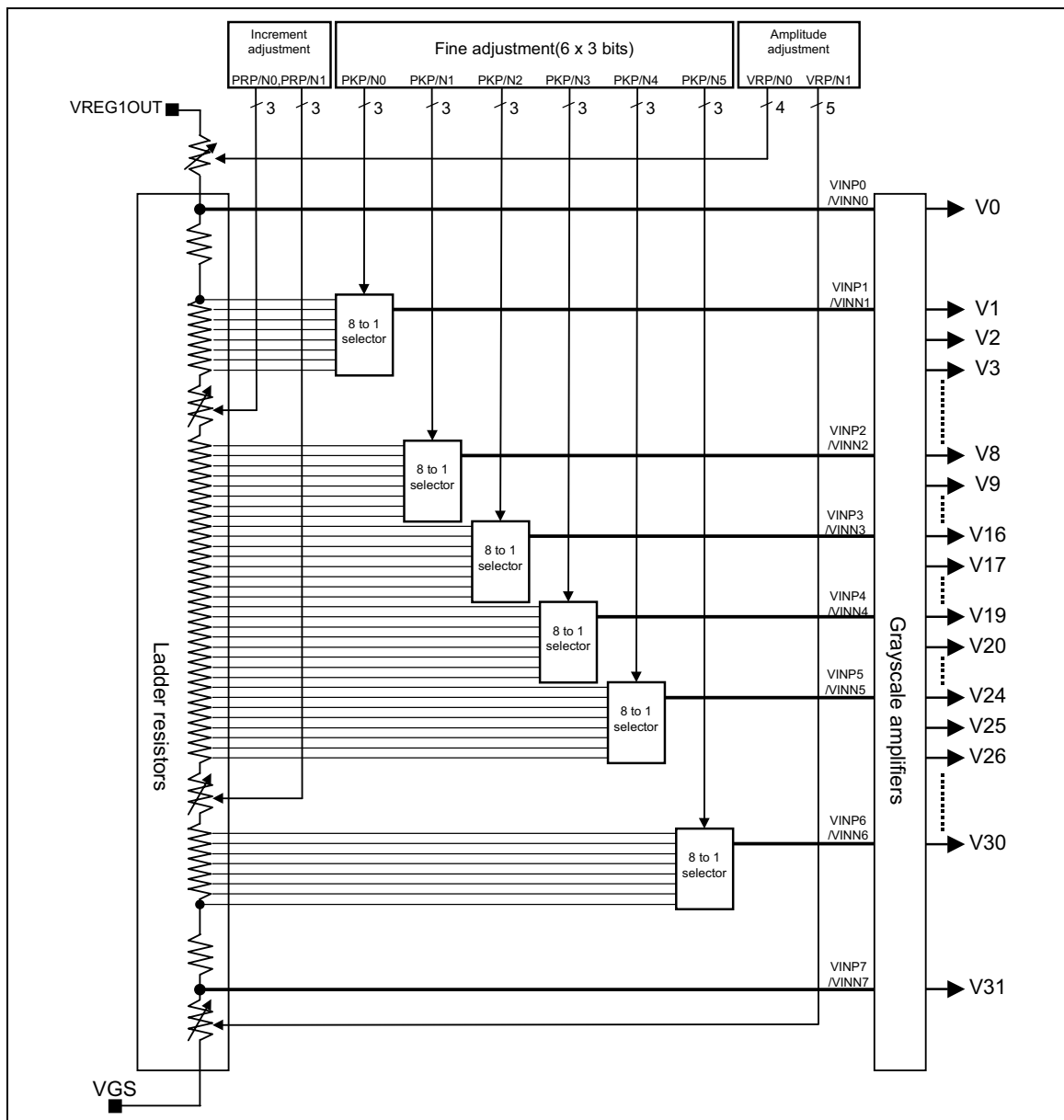


Figure 71 Grayscale amplifier unit

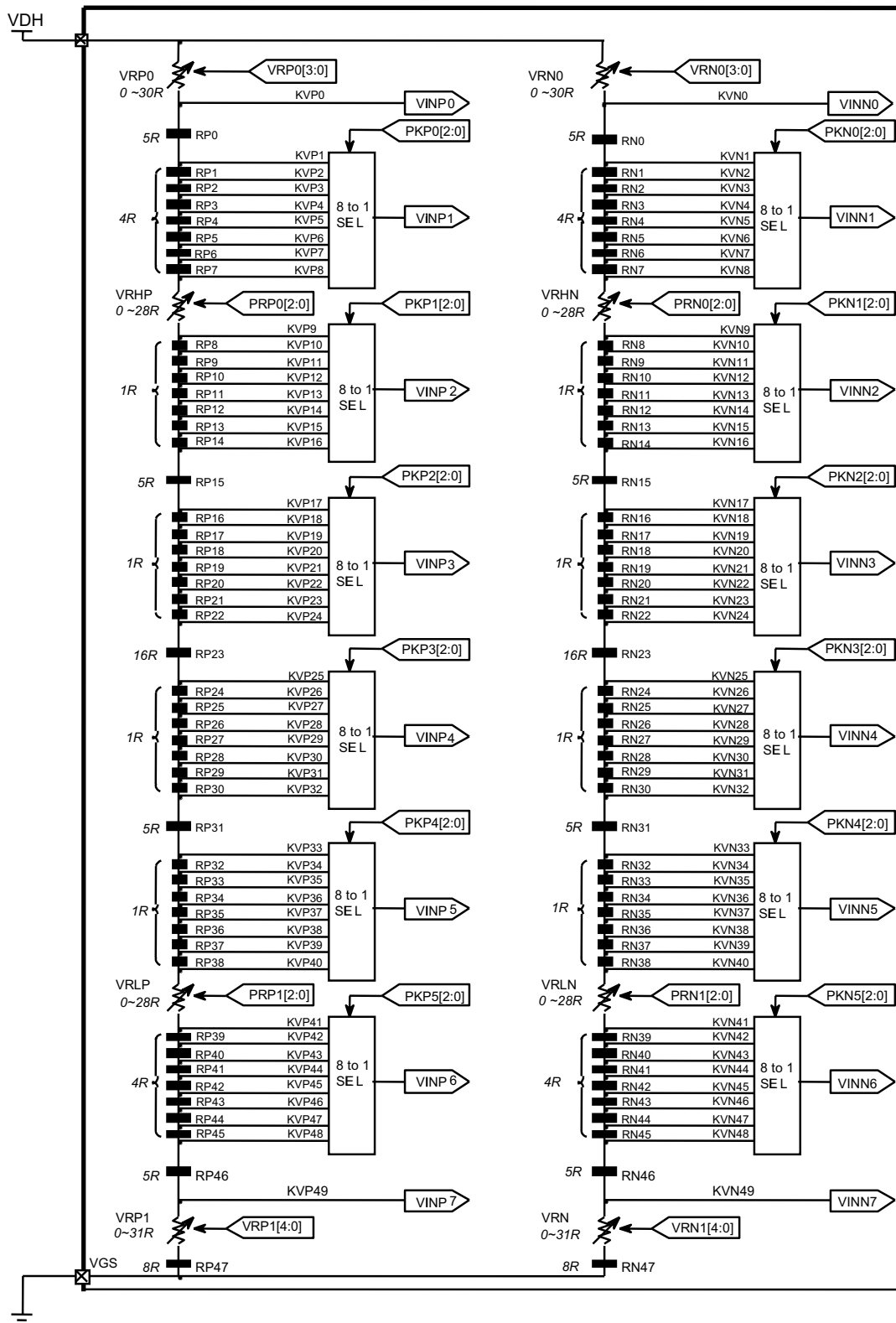


Figure 72 Reference voltage generating block (ladder resistor units and 8-to-1 selectors)

γ -Correction Register

The γ -correction registers of the R69400 consist of gradient adjustment, amplitude adjustment, and fine-adjustment registers. Each register allows for independent settings for positive and negative polarities. The γ -correction register settings and the reference levels are common to all RGB dots.

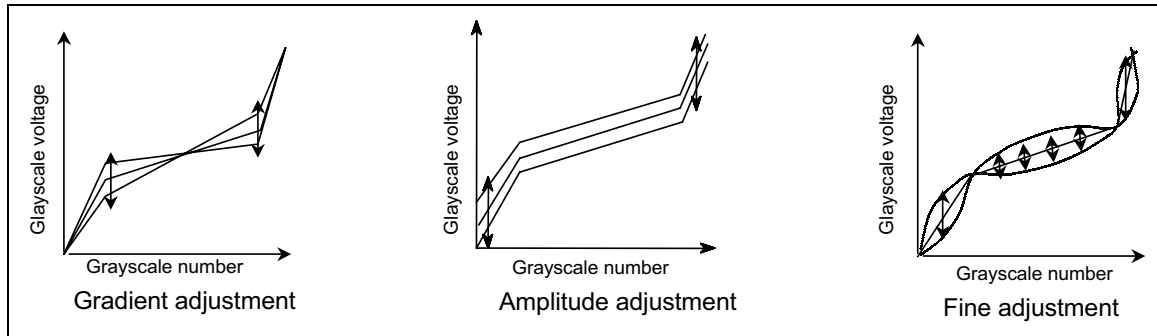


Figure 73

1. Gradient adjustment registers

The gradient adjustment registers are used to adjust the gradient, which represent the grayscale-voltage relationship for the middle grayscale without changing the dynamic range. Adjustments are made by changing the resistance values of the resistors (VRHP(N)/VRLP(N)) in the middle of the ladder resistor unit. The gradient adjustment registers consist of positive and negative polarity registers to allow for asymmetric drive.

2. Amplitude adjustment registers

The amplitude adjustment registers are used to adjust the amplitude of the grayscale voltage by changing the resistance values of the resistors (VRP(N)1/0) at both ends of the ladder resistor unit. Same with the gradient registers, the amplitude adjustment registers consist of positive and negative polarity registers.

3. Fine adjustment registers

The fine adjustment registers are used for minute adjustment of grayscale voltage levels. The fine adjustment register represent one voltage level to be selected in the 8-to-1 selector among 8 levels generated from the ladder resistor unit. Same with other registers, the fine adjustment registers consist of positive and negative polarity registers.

R61500

Table 69

Register Groups	Positive Polarity	Negative Polarity	Description
Gradient adjustment	PRP0 [2:0]	PRN0 [2:0]	Variable resistor VRHP (N)
	PRP1 [2:0]	PRN1 [2:0]	Variable resistor VRLP (N)
Amplitude adjustment	VRP0 [3:0]	VRN0 [3:0]	Variable resistor VRP (N)0
	VRP1 [4:0]	VRN1 [4:0]	Variable resistor VRP (N)1
Fine adjustment	PKP0 [2:0]	PKN0 [2:0]	8-to-1 selector (voltage level of grayscale 1 ~ 3)
	PKP1 [2:0]	PKN1 [2:0]	8-to-1 selector (voltage level of grayscale 4)
	PKP2 [2:0]	PKN2 [2:0]	8-to-1 selector (voltage level of grayscale 10)
	PKP3 [2:0]	PKN3 [2:0]	8-to-1 selector (voltage level of grayscale 21)
	PKP4 [2:0]	PKN4 [2:0]	8-to-1 selector (voltage level of grayscale 27)
	PKP5 [2:0]	PKN5 [2:0]	8-to-1 selector (voltage level of grayscale 28~30)

Ladder resistors and 8-to-1 selector

Block configuration

The ladder resistor and 8-to-1 selector unit as shown in page 128 consists of two ladder resistor unit including variable resistors and 8-to-1 selectors. In this unit, one voltage level is selected in each 8-to-1 selector among 8 levels generated by ladder resistors and it is output as the reference voltage for generating 32 grayscale voltage levels. The γ correction registers represent the resistance values of these resistors in the ladder resistor unit and the reference levels selected in the 8-to-1 selectors (see Table 69). This unit has a pin to connect a volume resistor for adjusting the characteristics of the panel.

Variable resistors

The R61500 uses variable resistors for the following three purposes: gradient adjustment (VRHP(N)/VRLP(N)); amplitude adjustment (1) (VRP(N)0); and amplitude adjustment (2) (VRP(N)1). The resistance values are determined by gradient adjustment and amplitude adjustment registers as below.

Table 70

Gradient adjustment

Contents of Register PRP(N) 0/1[2:0]	Resistance VRHP(N) VRLP(N)
000	0R
001	4R
010	8R
011	12R
100	16R
101	20R
110	24R
111	28R

Table 71

Amplitude adjustment (1)

Contents of Register VRP(N)0[3:0]	Resistance VRP(N)0
0000	0R
0001	2R
0010	4R
:	:
:	:
1101	26R
1111	28R
1111	30R

Table 72

Amplitude adjustment (2)

Contents of Register VRP(N)1[4:0]	Resistance VRP(N)1
00000	0R
00001	1R
00010	2R
:	:
:	:
11101	29R
11110	30R
11111	31R

8 to 1 selector

The 8-to-1 selector selects one voltage level according to the fine adjustment register setting among eight voltage levels generated by ladder resistors, and outputs the selected level as the reference voltage (VINP(N)1~6). The following table shows the correspondence between the selected voltage levels and the settings in the fine adjustment registers for respective reference voltage levels (VINP(N)1~6).

Table 73 **Fine adjustment registers and selected voltage**

The value of Register PKP(N)[2:0]	Selected Voltage					
	VINP(N)1	VINP(N)2	VINP(N)3	VINP(N)4	VINP(N)5	VINP(N)6
000	KVP(N)1	KVP(N)9	KVP(N)17	KVP(N)25	KVP(N)33	KVP(N)41
001	KVP(N)2	KVP(N)10	KVP(N)18	KVP(N)26	KVP(N)34	KVP(N)42
010	KVP(N)3	KVP(N)11	KVP(N)19	KVP(N)27	KVP(N)35	KVP(N)43
011	KVP(N)4	KVP(N)12	KVP(N)20	KVP(N)28	KVP(N)36	KVP(N)44
100	KVP(N)5	KVP(N)13	KVP(N)21	KVP(N)29	KVP(N)37	KVP(N)45
101	KVP(N)6	KVP(N)14	KVP(N)22	KVP(N)30	KVP(N)38	KVP(N)46
110	KVP(N)7	KVP(N)15	KVP(N)23	KVP(N)31	KVP(N)39	KVP(N)47
111	KVP(N)8	KVP(N)16	KVP(N)24	KVP(N)32	KVP(N)40	KVP(N)48

The voltage levels for V0~V31 grayscales are calculated from the following formulas.

Table 74 Formulas for calculating voltage (Positive polarity) (1)

Pin	Formula	Fine adjustment register value	Reference Voltage
KVP0	$VREG1OUT - \Delta V \cdot VRP0 / SUMRP$	-	VINP0
KVP1	$VREG1OUT - \Delta V \cdot (VRP0 + 5R) / SUMRP$	PKP02-00 = "000"	VINP1
KVP2	$VREG1OUT - \Delta V \cdot (VRP0 + 9R) / SUMRP$	PKP02-00 = "001"	
KVP3	$VREG1OUT - \Delta V \cdot (VRP0 + 13R) / SUMRP$	PKP02-00 = "010"	
KVP4	$VREG1OUT - \Delta V \cdot (VRP0 + 17R) / SUMRP$	PKP02-00 = "011"	
KVP5	$VREG1OUT - \Delta V \cdot (VRP0 + 21R) / SUMRP$	PKP02-00 = "100"	
KVP6	$VREG1OUT - \Delta V \cdot (VRP0 + 25R) / SUMRP$	PKP02-00 = "101"	
KVP7	$VREG1OUT - \Delta V \cdot (VRP0 + 29R) / SUMRP$	PKP02-00 = "110"	
KVP8	$VREG1OUT - \Delta V \cdot (VRP0 + 33R) / SUMRP$	PKP02-00 = "111"	
KVP9	$VREG1OUT - \Delta V \cdot (VRP0 + 33R + VRHP) / SUMRP$	PKP12-10 = "000"	VINP2
KVP10	$VREG1OUT - \Delta V \cdot (VRP0 + 34R + VRHP) / SUMRP$	PKP12-10 = "001"	
KVP11	$VREG1OUT - \Delta V \cdot (VRP0 + 35R + VRHP) / SUMRP$	PKP12-10 = "010"	
KVP12	$VREG1OUT - \Delta V \cdot (VRP0 + 36R + VRHP) / SUMRP$	PKP12-10 = "011"	
KVP13	$VREG1OUT - \Delta V \cdot (VRP0 + 37R + VRHP) / SUMRP$	PKP12-10 = "100"	
KVP14	$VREG1OUT - \Delta V \cdot (VRP0 + 38R + VRHP) / SUMRP$	PKP12-10 = "101"	
KVP15	$VREG1OUT - \Delta V \cdot (VRP0 + 39R + VRHP) / SUMRP$	PKP12-10 = "110"	
KVP16	$VREG1OUT - \Delta V \cdot (VRP0 + 40R + VRHP) / SUMRP$	PKP12-10 = "111"	
KVP17	$VREG1OUT - \Delta V \cdot (VRP0 + 45R + VRHP) / SUMRP$	PKP22-20 = "000"	VINP3
KVP18	$VREG1OUT - \Delta V \cdot (VRP0 + 46R + VRHP) / SUMRP$	PKP22-20 = "001"	
KVP19	$VREG1OUT - \Delta V \cdot (VRP0 + 47R + VRHP) / SUMRP$	PKP22-20 = "010"	
KVP20	$VREG1OUT - \Delta V \cdot (VRP0 + 48R + VRHP) / SUMRP$	PKP22-20 = "011"	
KVP21	$VREG1OUT - \Delta V \cdot (VRP0 + 49R + VRHP) / SUMRP$	PKP22-20 = "100"	
KVP22	$VREG1OUT - \Delta V \cdot (VRP0 + 50R + VRHP) / SUMRP$	PKP22-20 = "101"	
KVP23	$VREG1OUT - \Delta V \cdot (VRP0 + 51R + VRHP) / SUMRP$	PKP22-20 = "110"	
KVP24	$VREG1OUT - \Delta V \cdot (VRP0 + 52R + VRHP) / SUMRP$	PKP22-20 = "111"	
KVP25	$VREG1OUT - \Delta V \cdot (VRP0 + 68R + VRHP) / SUMRP$	PKP32-30 = "000"	VINP4
KVP26	$VREG1OUT - \Delta V \cdot (VRP0 + 69R + VRHP) / SUMRP$	PKP32-30 = "001"	
KVP27	$VREG1OUT - \Delta V \cdot (VRP0 + 70R + VRHP) / SUMRP$	PKP32-30 = "010"	
KVP28	$VREG1OUT - \Delta V \cdot (VRP0 + 71R + VRHP) / SUMRP$	PKP32-30 = "011"	
KVP29	$VREG1OUT - \Delta V \cdot (VRP0 + 72R + VRHP) / SUMRP$	PKP32-30 = "100"	
KVP30	$VREG1OUT - \Delta V \cdot (VRP0 + 73R + VRHP) / SUMRP$	PKP32-30 = "101"	
KVP31	$VREG1OUT - \Delta V \cdot (VRP0 + 74R + VRHP) / SUMRP$	PKP32-30 = "110"	
KVP32	$VREG1OUT - \Delta V \cdot (VRP0 + 75R + VRHP) / SUMRP$	PKP32-30 = "111"	
KVP33	$VREG1OUT - \Delta V \cdot (VRP0 + 80R + VRHP) / SUMRP$	PKP42-40 = "000"	VINP5
KVP34	$VREG1OUT - \Delta V \cdot (VRP0 + 81R + VRHP) / SUMRP$	PKP42-40 = "001"	
KVP35	$VREG1OUT - \Delta V \cdot (VRP0 + 82R + VRHP) / SUMRP$	PKP42-40 = "010"	
KVP36	$VREG1OUT - \Delta V \cdot (VRP0 + 83R + VRHP) / SUMRP$	PKP42-40 = "011"	
KVP37	$VREG1OUT - \Delta V \cdot (VRP0 + 84R + VRHP) / SUMRP$	PKP42-40 = "100"	
KVP38	$VREG1OUT - \Delta V \cdot (VRP0 + 85R + VRHP) / SUMRP$	PKP42-40 = "101"	
KVP39	$VREG1OUT - \Delta V \cdot (VRP0 + 86R + VRHP) / SUMRP$	PKP42-40 = "110"	
KVP40	$VREG1OUT - \Delta V \cdot (VRP0 + 87R + VRHP) / SUMRP$	PKP42-40 = "111"	
KVP41	$VREG1OUT - \Delta V \cdot (VRP0 + 87R + VRHP + VRLP) / SUMRP$	PKP52-50 = "000"	VINP6
KVP42	$VREG1OUT - \Delta V \cdot (VRP0 + 91R + VRHP + VRLP) / SUMRP$	PKP52-50 = "001"	
KVP43	$VREG1OUT - \Delta V \cdot (VRP0 + 95R + VRHP + VRLP) / SUMRP$	PKP52-50 = "010"	
KVP44	$VREG1OUT - \Delta V \cdot (VRP0 + 99R + VRHP + VRLP) / SUMRP$	PKP52-50 = "011"	
KVP45	$VREG1OUT - \Delta V \cdot (VRP0 + 103R + VRHP + VRLP) / SUMRP$	PKP52-50 = "100"	
KVP46	$VREG1OUT - \Delta V \cdot (VRP0 + 107R + VRHP + VRLP) / SUMRP$	PKP52-50 = "101"	
KVP47	$VREG1OUT - \Delta V \cdot (VRP0 + 111R + VRHP + VRLP) / SUMRP$	PKP52-50 = "110"	
KVP48	$VREG1OUT - \Delta V \cdot (VRP0 + 115R + VRHP + VRLP) / SUMRP$	PKP52-50 = "111"	
KVP49	$VREG1OUT - \Delta V \cdot (VRP0 + 120R + VRHP + VRLP) / SUMRP$	-	VINP7

SUMRP : Sum of positive ladder resistors = 128R+VRHP+VRLP+VRP0+VRP1
 ΔV : Voltage difference between VREG1OUT and VGS.

Table 75 Formulas for calculating voltage (Positive polarity) (2)

Grayscale Voltage	Formula
V0	VINP0
V1	$V4 + (VINP1 - V4) * (15/24)$
V2	$V4 + (VINP1 - V4) * (8/24)$
V3	$V4 + (VINP1 - V4) * (4/24)$
V4	VINP2
V5	$V10 + (V4 - V10) * (20/24)$
V6	$V10 + (V4 - V10) * (16/24)$
V7	$V10 + (V4 - V10) * (12/24)$
V8	$V10 + (V4 - V10) * (8/24)$
V9	$V10 + (V4 - V10) * (4/24)$
V10	VINP3
V11	$V21 + (V10 - V21) * (21/24)$
V12	$V21 + (V10 - V21) * (19/24)$
V13	$V21 + (V10 - V21) * (17/24)$
V14	$V21 + (V10 - V21) * (15/24)$
V15	$V21 + (V10 - V21) * (13/24)$
V16	$V21 + (V10 - V21) * (11/24)$
V17	$V21 + (V10 - V21) * (9/24)$
V18	$V21 + (V10 - V21) * (7/24)$
V19	$V21 + (V10 - V21) * (5/24)$
V20	$V21 + (V10 - V21) * (3/24)$
V21	VINP4
V22	$V27 + (V21 - V27) * (20/24)$
V23	$V27 + (V21 - V27) * (16/24)$
V24	$V27 + (V21 - V27) * (12/24)$
V25	$V27 + (V21 - V27) * (8/24)$
V26	$V27 + (V21 - V27) * (4/24)$
V27	VINP5
V28	$VINP6 + (V27 - VINP6) * (20/24)$
V29	$VINP6 + (V27 - VINP6) * (16/24)$
V30	$VINP6 + (V27 - VINP6) * (9/24)$
V31	VINP7

Note : Make sure DDVDH - V0 > 0.5V
DDVDH - V4 > 1.1V

Table 76 Formulas for calculating voltage (Negative polarity) (1)

Pin	Formula	Fine adjustment register value	Reference Voltage
KVP0	$VREG1OUT - \Delta V \cdot VRN0 / SUMRN$	-	VINN0
KVN1	$VREG1OUT - \Delta V \cdot (VRN0 + 5R) / SUMRN$	PKN 02-00 = "000"	VINN1
KVN2	$VREG1OUT - \Delta V \cdot (VRN0 + 9R) / SUMRN$	PKN 02-00 = "001"	
KVN3	$VREG1OUT - \Delta V \cdot (VRN0 + 13R) / SUMRN$	PKN 02-00 = "010"	
KVN4	$VREG1OUT - \Delta V \cdot (VRN0 + 17R) / SUMRN$	PKN 02-00 = "011"	
KVN5	$VREG1OUT - \Delta V \cdot (VRN0 + 21R) / SUMRN$	PKN 02-00 = "100"	
KVN6	$VREG1OUT - \Delta V \cdot (VRN0 + 25R) / SUMRN$	PKN 02-00 = "101"	
KVN7	$VREG1OUT - \Delta V \cdot (VRN0 + 29R) / SUMRN$	PKN 02-00 = "110"	
KVN8	$VREG1OUT - \Delta V \cdot (VRN0 + 33R) / SUMRN$	PKN 02-00 = "111"	
KVN9	$VREG1OUT - \Delta V \cdot (VRN0 + 33R + VRHN) / SUMRN$	PKN 12-10 = "000"	VINN2
KVN10	$VREG1OUT - \Delta V \cdot (VRN0 + 34R + VRHN) / SUMRN$	PKN 12-10 = "001"	
KVN11	$VREG1OUT - \Delta V \cdot (VRN0 + 35R + VRHN) / SUMRN$	PKN 12-10 = "010"	
KVN12	$VREG1OUT - \Delta V \cdot (VRN0 + 36R + VRHN) / SUMRN$	PKN 12-10 = "011"	
KVN13	$VREG1OUT - \Delta V \cdot (VRN0 + 37R + VRHN) / SUMRN$	PKN 12-10 = "100"	
KVN14	$VREG1OUT - \Delta V \cdot (VRN0 + 38R + VRHN) / SUMRN$	PKN 12-10 = "101"	
KVN15	$VREG1OUT - \Delta V \cdot (VRN0 + 39R + VRHN) / SUMRN$	PKN 12-10 = "110"	
KVN16	$VREG1OUT - \Delta V \cdot (VRN0 + 40R + VRHN) / SUMRN$	PKN 12-10 = "111"	
KVN17	$VREG1OUT - \Delta V \cdot (VRN0 + 45R + VRHN) / SUMRN$	PKN 22-20 = "000"	VINN3
KVN18	$VREG1OUT - \Delta V \cdot (VRN0 + 46R + VRHN) / SUMRN$	PKN 22-20 = "001"	
KVN19	$VREG1OUT - \Delta V \cdot (VRN0 + 47R + VRHN) / SUMRN$	PKN 22-20 = "010"	
KVN20	$VREG1OUT - \Delta V \cdot (VRN0 + 48R + VRHN) / SUMRN$	PKN 22-20 = "011"	
KVN21	$VREG1OUT - \Delta V \cdot (VRN0 + 49R + VRHN) / SUMRN$	PKN 22-20 = "100"	
KVN22	$VREG1OUT - \Delta V \cdot (VRN0 + 50R + VRHN) / SUMRN$	PKN 22-20 = "101"	
KVN23	$VREG1OUT - \Delta V \cdot (VRN0 + 51R + VRHN) / SUMRN$	PKN 22-20 = "110"	
KVN24	$VREG1OUT - \Delta V \cdot (VRN0 + 52R + VRHN) / SUMRN$	PKN 22-20 = "111"	
KVN25	$VREG1OUT - \Delta V \cdot (VRN0 + 68R + VRHN) / SUMRN$	PKN 32-30 = "000"	VINN4
KVN26	$VREG1OUT - \Delta V \cdot (VRN0 + 69R + VRHN) / SUMRN$	PKN 32-30 = "001"	
KVN27	$VREG1OUT - \Delta V \cdot (VRN0 + 70R + VRHN) / SUMRN$	PKN 32-30 = "010"	
KVN28	$VREG1OUT - \Delta V \cdot (VRN0 + 71R + VRHN) / SUMRN$	PKN 32-30 = "011"	
KVN29	$VREG1OUT - \Delta V \cdot (VRN0 + 72R + VRHN) / SUMRN$	PKN 32-30 = "100"	
KVN30	$VREG1OUT - \Delta V \cdot (VRN0 + 73R + VRHN) / SUMRN$	PKN 32-30 = "101"	
KVN31	$VREG1OUT - \Delta V \cdot (VRN0 + 74R + VRHN) / SUMRN$	PKN 32-30 = "110"	
KVN32	$VREG1OUT - \Delta V \cdot (VRN0 + 75R + VRHN) / SUMRN$	PKN 32-30 = "111"	
KVN33	$VREG1OUT - \Delta V \cdot (VRN0 + 80R + VRHN) / SUMRN$	PKN 42-40 = "000"	VINN5
KVN34	$VREG1OUT - \Delta V \cdot (VRN0 + 81R + VRHN) / SUMRN$	PKN 42-40 = "001"	
KVN35	$VREG1OUT - \Delta V \cdot (VRN0 + 82R + VRHN) / SUMRN$	PKN 42-40 = "010"	
KVN36	$VREG1OUT - \Delta V \cdot (VRN0 + 83R + VRHN) / SUMRN$	PKN 42-40 = "011"	
KVN37	$VREG1OUT - \Delta V \cdot (VRN0 + 84R + VRHN) / SUMRN$	PKN 42-40 = "100"	
KVN38	$VREG1OUT - \Delta V \cdot (VRN0 + 85R + VRHN) / SUMRN$	PKN 42-40 = "101"	
KVN39	$VREG1OUT - \Delta V \cdot (VRN0 + 86R + VRHN) / SUMRN$	PKN 42-40 = "110"	
KVN40	$VREG1OUT - \Delta V \cdot (VRN0 + 87R + VRHN) / SUMRN$	PKN 42-40 = "111"	
KVN41	$VREG1OUT - \Delta V \cdot (VRN0 + 87R + VRHN + VRLN) / SUMRN$	PKN 52-50 = "000"	VINN6
KVN42	$VREG1OUT - \Delta V \cdot (VRN0 + 91R + VRHN + VRLN) / SUMRN$	PKN 52-50 = "001"	
KVN43	$VREG1OUT - \Delta V \cdot (VRN0 + 95R + VRHN + VRLN) / SUMRN$	PKN 52-50 = "010"	
KVN44	$VREG1OUT - \Delta V \cdot (VRN0 + 99R + VRHN + VRLN) / SUMRN$	PKN 52-50 = "011"	
KVN45	$VREG1OUT - \Delta V \cdot (VRN0 + 103R + VRHN + VRLN) / SUMRN$	PKN 52-50 = "100"	
KVN46	$VREG1OUT - \Delta V \cdot (VRN0 + 107R + VRHN + VRLN) / SUMRN$	PKN 52-50 = "101"	
KVN47	$VREG1OUT - \Delta V \cdot (VRN0 + 111R + VRHN + VRLN) / SUMRN$	PKN 52-50 = "110"	
KVN48	$VREG1OUT - \Delta V \cdot (VRN0 + 115R + VRHN + VRLN) / SUMRN$	PKN 52-50 = "111"	
KVN49	$VREG1OUT - \Delta V \cdot (VRN0 + 120R + VRHN + VRLN) / SUMRN$	-	VINN7

SUMRN : Sum of negative ladder resistors = 128R+VRHN+VRLN+VRN0+VRN1
 ΔV : Voltage difference between VREG1OUT and VGS

Table 77 Formulas for calculating voltage (Negative polarity) (2)

Grayscale Voltage	Formula
V0	VINN0
V1	$V4 + (VINN1 - V4) \times (15/24)$
V2	$V4 + (VINN1 - V4) \times (8/24)$
V3	$V4 + (VINN1 - V4) \times (4/24)$
V4	VINN2
V5	$V10 + (V4 - V10) \times (20/24)$
V6	$V10 + (V4 - V10) \times (16/24)$
V7	$V10 + (V4 - V10) \times (12/24)$
V8	$V10 + (V4 - V10) \times (8/24)$
V9	$V10 + (V4 - V10) \times (4/24)$
V10	VINN3
V11	$V21 + (V10 - V21) \times (21/24)$
V12	$V21 + (V10 - V21) \times (19/24)$
V13	$V21 + (V10 - V21) \times (17/24)$
V14	$V21 + (V10 - V21) \times (15/24)$
V15	$V21 + (V10 - V21) \times (13/24)$
V16	$V21 + (V10 - V21) \times (11/24)$
V17	$V21 + (V10 - V21) \times (9/24)$
V18	$V21 + (V10 - V21) \times (7/24)$
V19	$V21 + (V10 - V21) \times (5/24)$
V20	$V21 + (V10 - V21) \times (3/24)$
V21	VINN4
V22	$V27 + (V21 - V27) \times (20/24)$
V23	$V27 + (V21 - V27) \times (16/24)$
V24	$V27 + (V21 - V27) \times (12/24)$
V25	$V27 + (V21 - V27) \times (8/24)$
V26	$V27 + (V21 - V27) \times (4/24)$
V27	VINN5
V28	$VINN6 + (V27 - VINN6) \times (20/24)$
V29	$VINN6 + (V27 - VINN6) \times (16/24)$
V30	$VINN6 + (V27 - VINN6) \times (9/24)$
V31	VINN7

Note : Make sure DDVDH - V0 > 0.5V
DDVDH - V4 > 1.1V

Relationship between RAM data and voltage output levels

The relationship between RAM data and source output voltage levels is as follows. See also Table 56
GRAM data and LCD output level.

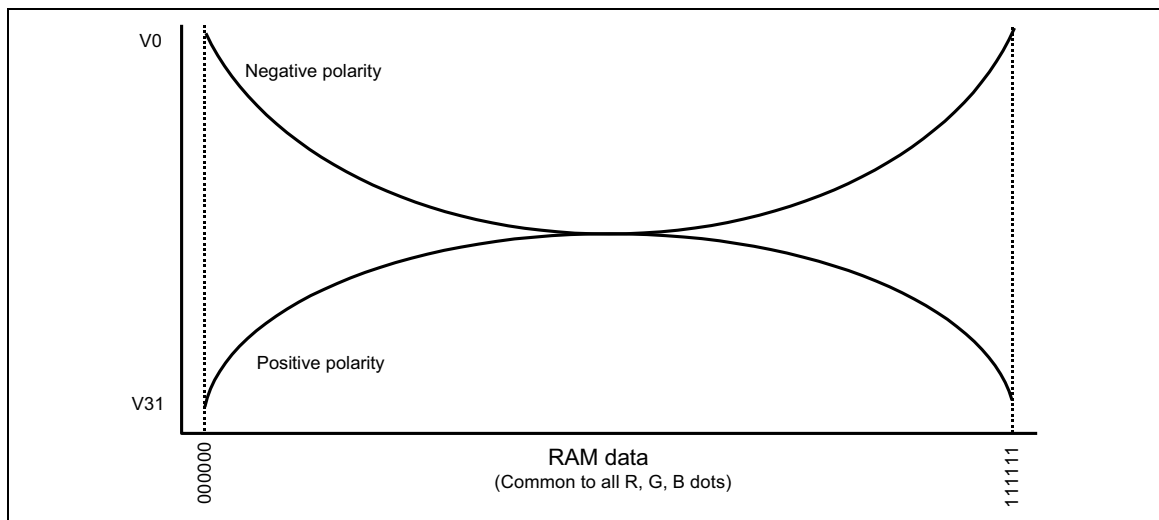


Figure 74 RAM data and the output voltage (REV = 0)

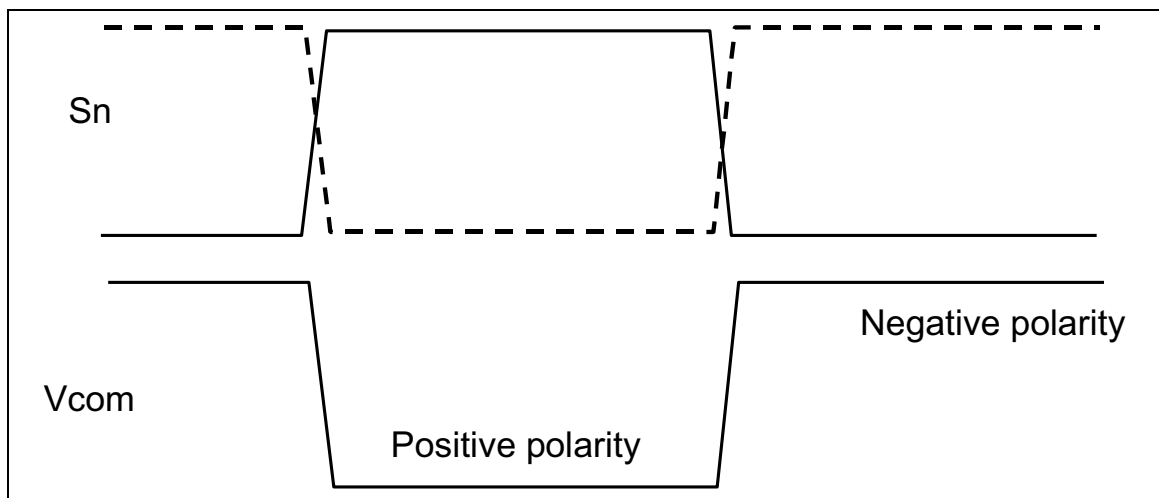


Figure 75 Source output and Vcom

8-color Display Mode

The R61500 supports a function to display in 8colors. In 8-color mode, available grayscale levels are V0 and V31, and the power supplies to other grayscales (V1-V30) are halted to reduce power consumption.

The γ - correction registers, PKP0-PKP5 and PKN0-PKN5, are disabled in 8-color display mode. The display data is written only with the MSB.

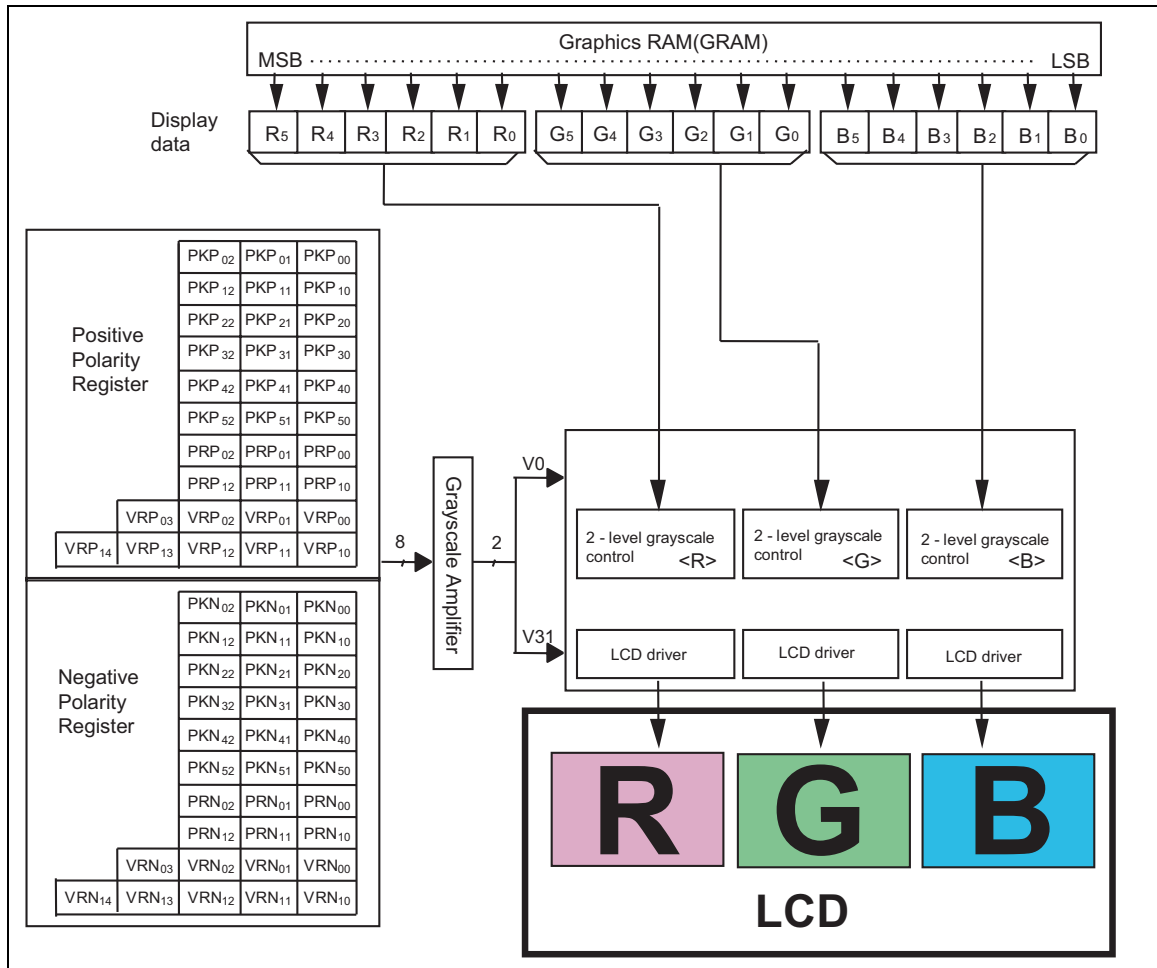


Figure 76

To switch between the 262, 144-color mode and the 8-color mode, follow the sequence below.

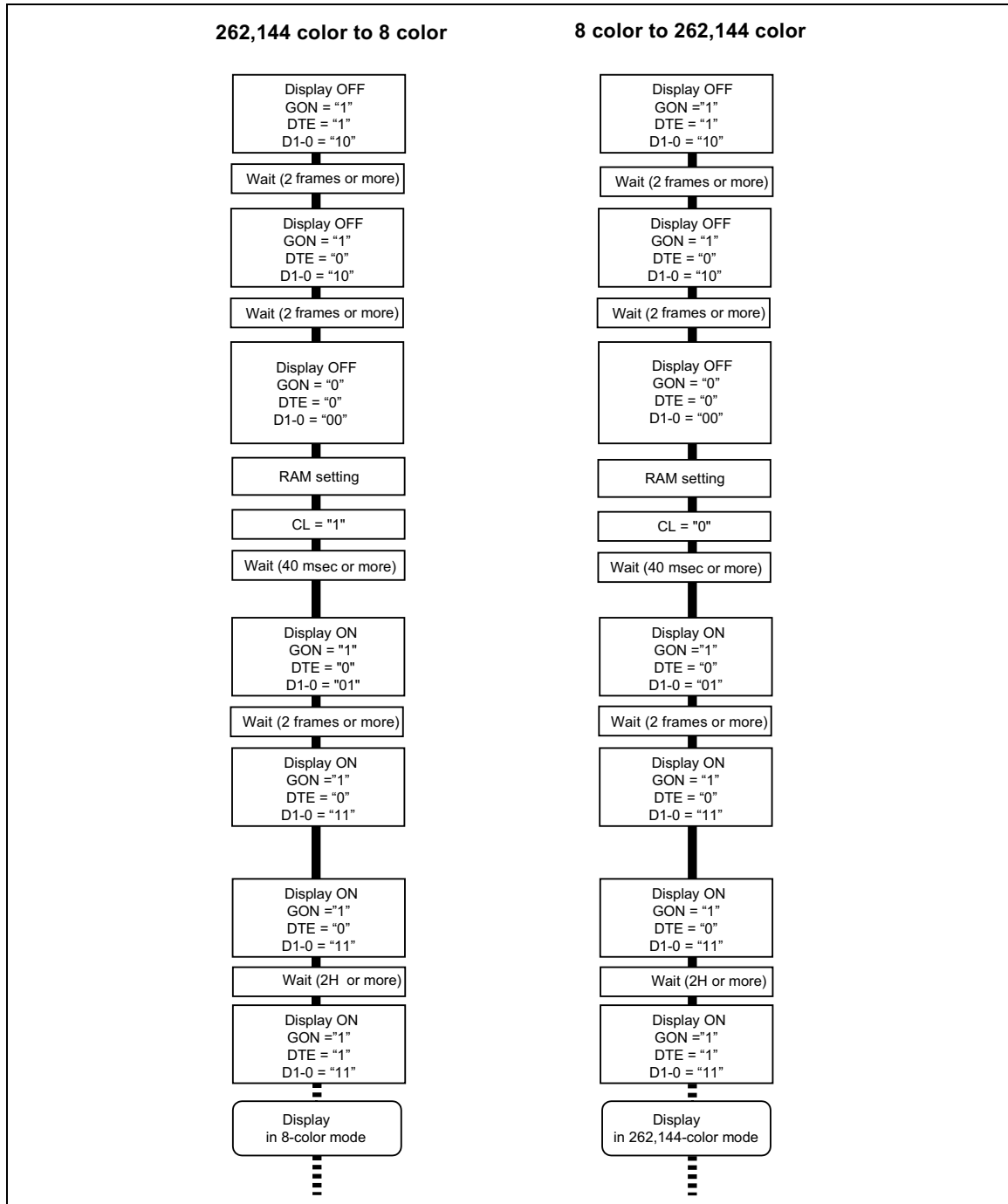


Figure 77

Power supply circuit Configuration

The following are the configurations of power supply circuit to generate liquid crystal panel driving levels.

Power supply connection example 1 ($V_{ci1} = V_{ciOUT}$): The V_{ciOUT} level is adjusted using the V_{ciOUT} output circuit.

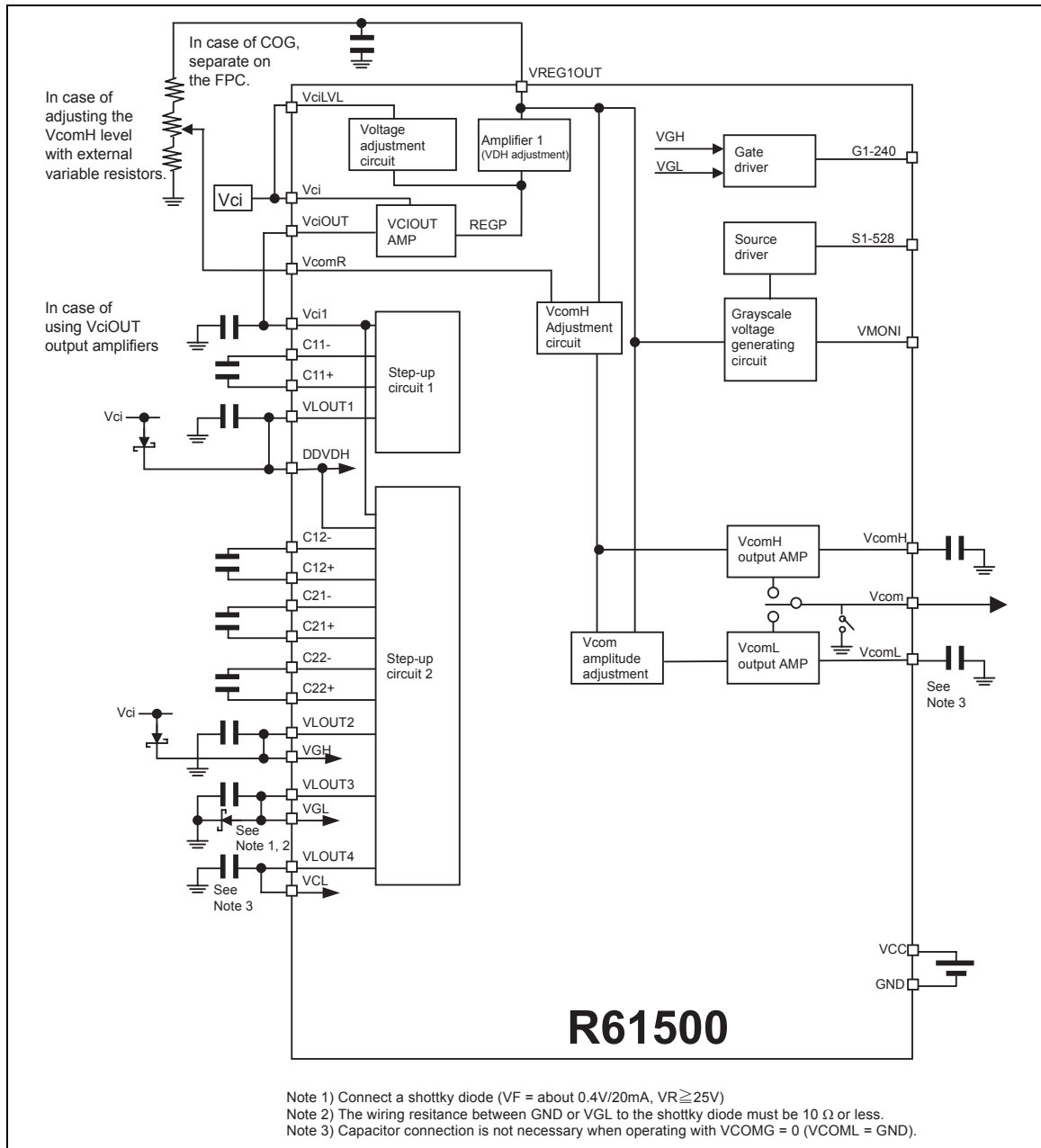


Figure 78

R61500

Power supply connection example 2 (Vci1 = Vci direct input): The Vci level can be directly input to the Vci1. In this case, the VciOUT level adjustment is disabled but step-up circuit efficiency is enhanced.

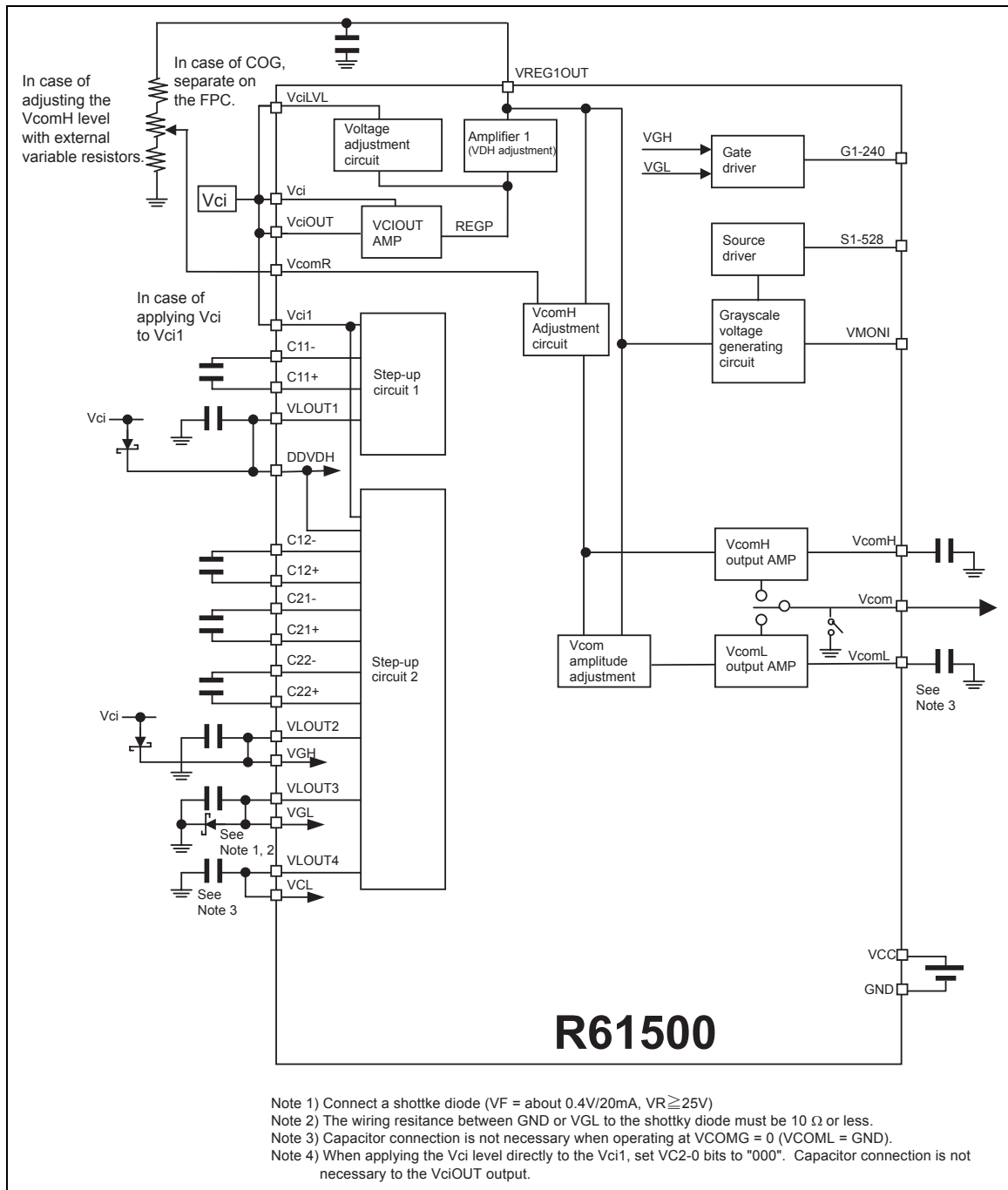


Figure 79

R61500

Specification of External Elements Connected to the R61500 Power Supply

The following table shows specifications of external elements connected to the R61500's power supply circuit.

Table 78 Capacitor

Capacity	Recommended voltage	Pin connection
1 μ F (B characteristics)	6V	VREG1OUT, VciOUT, VOUT4 ^{*see note} , VcomH, VcomL ^{*see note} , C11+/-, C12+/-
	10V	VLOUT1, C21+/-, C22+/-
	25V	VLOUT2, VLOUT3

Note: There are cases capacitor connection is not necessary according to the mode setting.

Table 79 Schottky diode

Feature	Pin connection
$V_F < 0.4V$ / 20mA at 25°C, $V_R \geq 25V$ (Recommended diode: HSC226)	GND – VGL (Vci – VGH) (Vci – DDVDH)

Table 80 Variable resistor

Feature	Pin connection
$> 200 \text{ k}\Omega$	VcomR

Instruction Setting

When setting the following instructions, follow respective sequences below.

Display ON/OFF

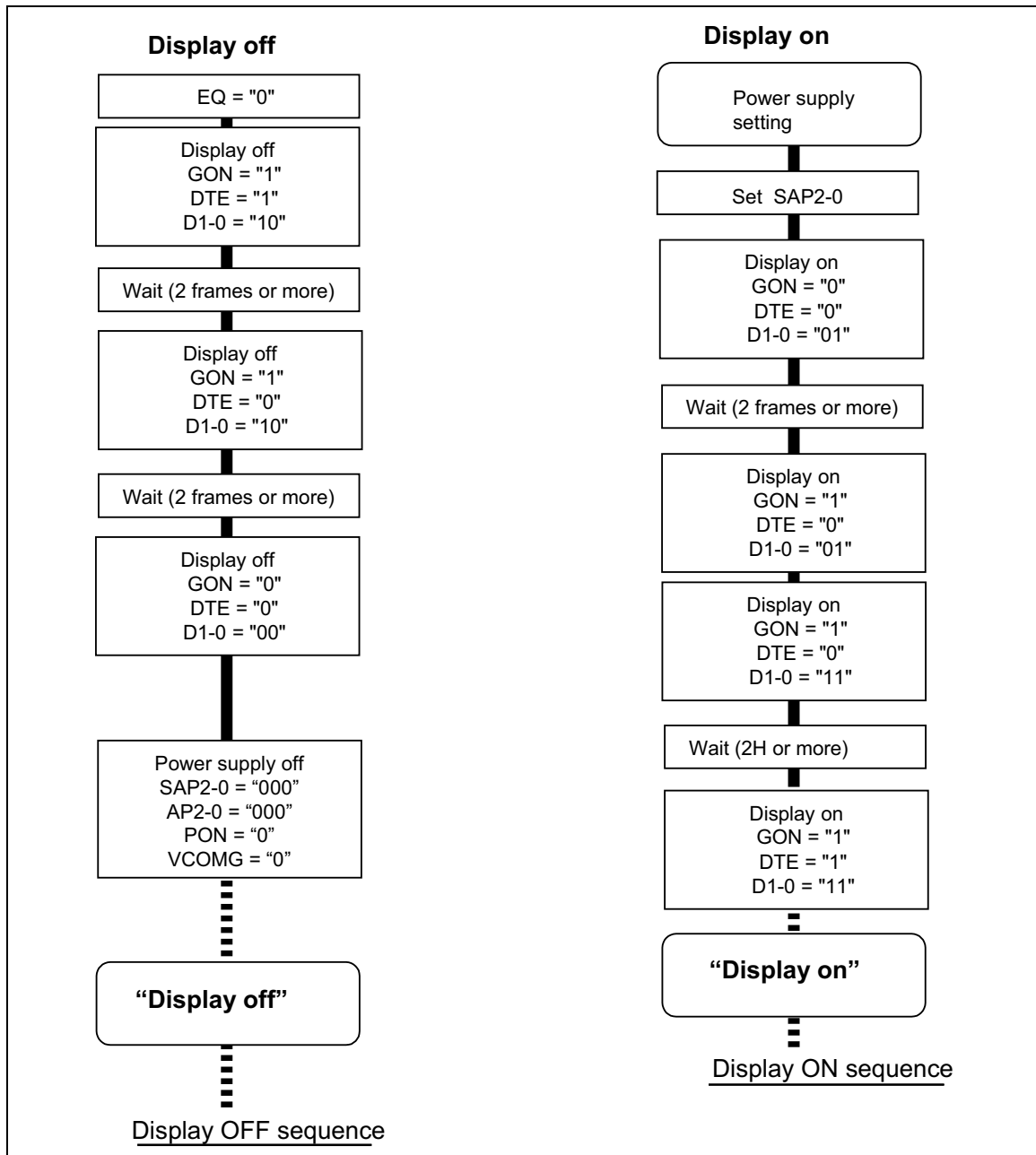


Figure 80

Standby and Sleep mode

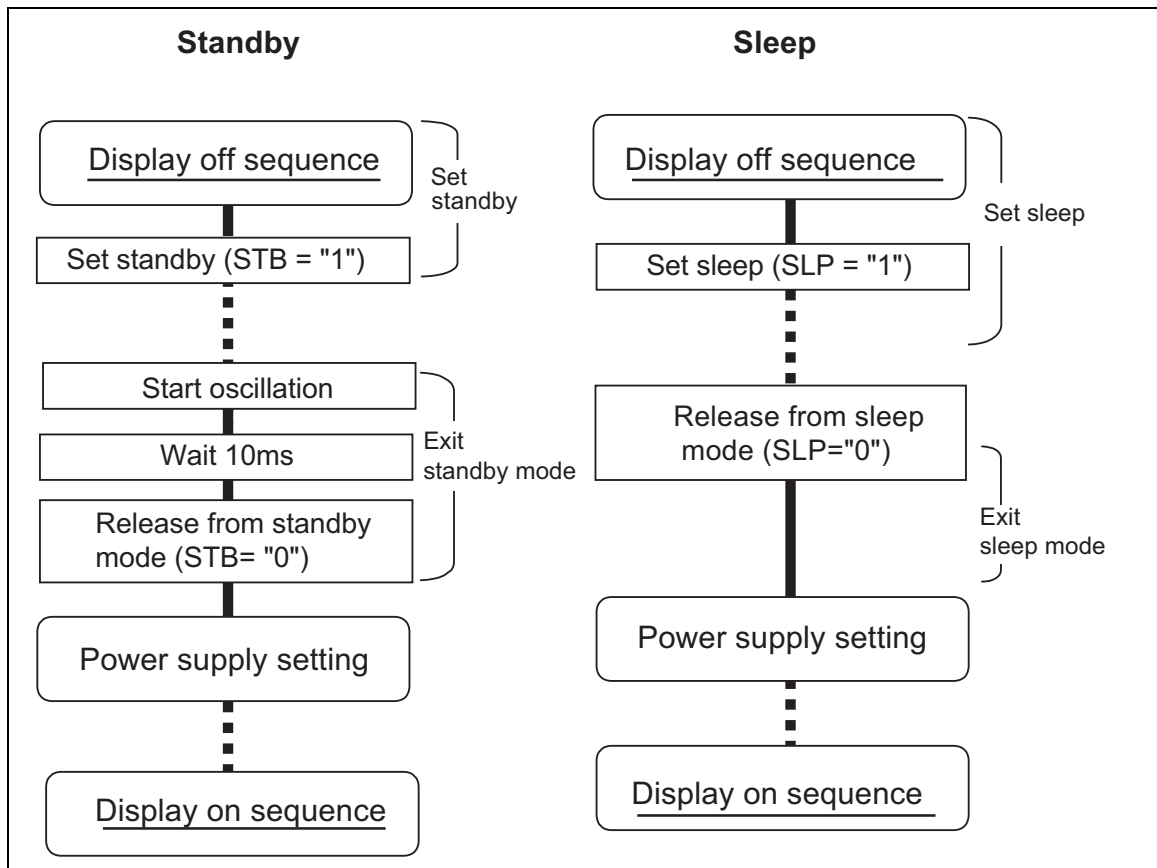


Figure 81

Power Supply Setting

The following are Power ON/OFF sequences.

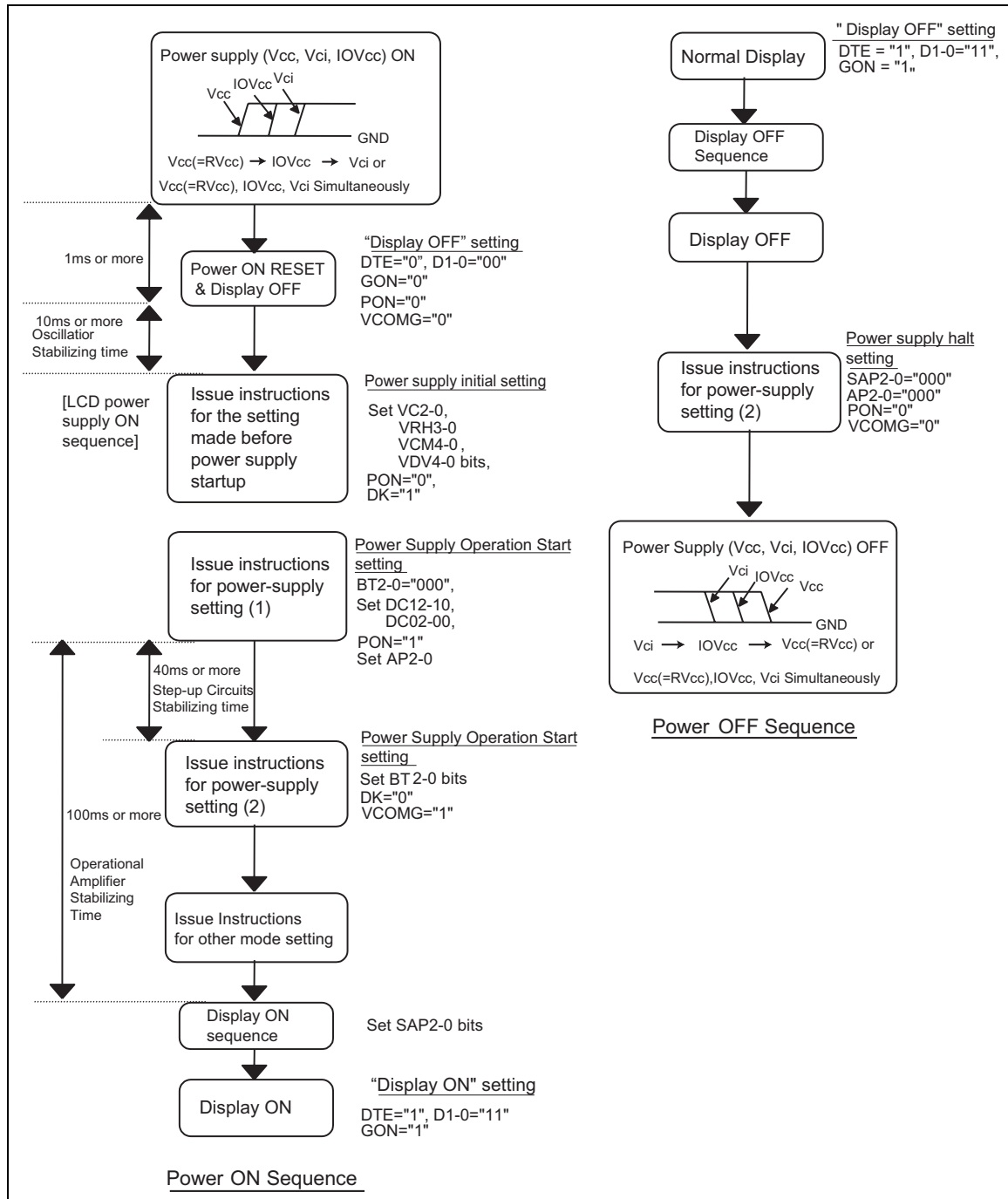


Figure 82

Pattern Diagram for Voltage Setting

The pattern diagram for setting the voltages and the waveforms of the voltages of the R61500 are as follows.

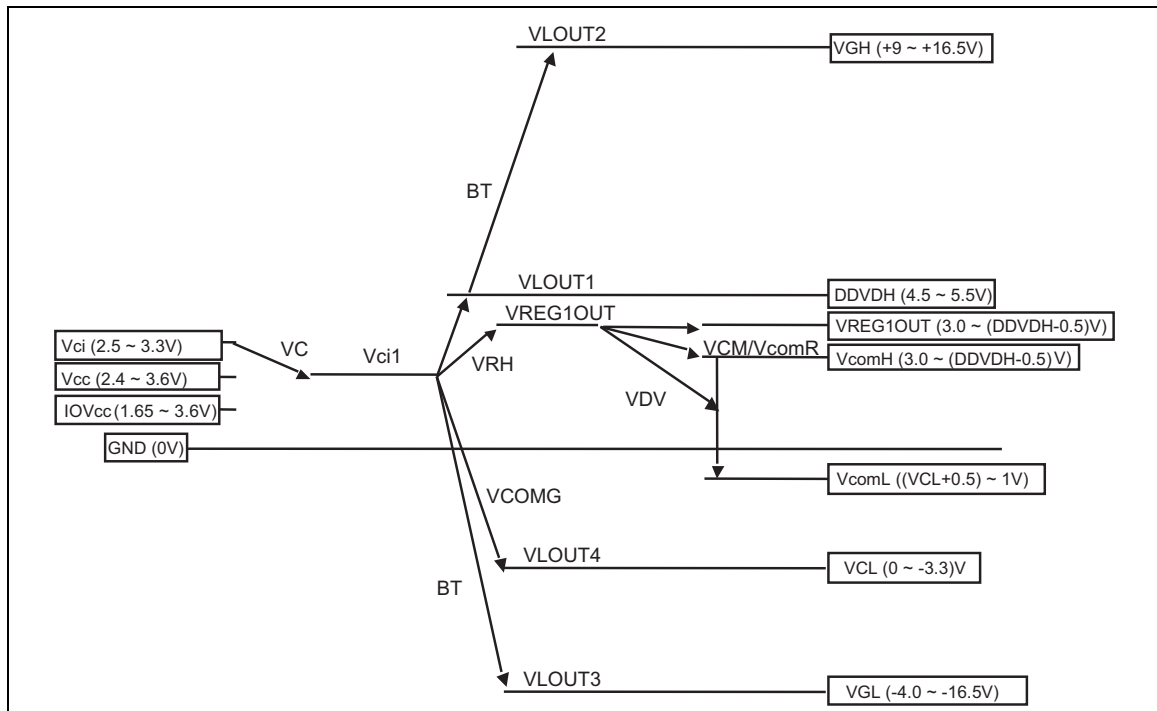


Figure 83 Pattern diagram for voltage setting

Note: The DDVDH, VGH, VGL, and VCL output voltage levels are lower than their theoretical levels (ideal voltage levels) due to current consumption at respective outputs. The voltage levels in the following relationships $(DDVDH - VREG1OUT) > 0.5V$, $(VcomL1 - VCL) > 0.5V$, $(VcomL2 - VCL) > 0.5V$ are the actual voltage levels. When the alternating cycles of Vcom1 and Vcom2 are high (e.g. the polarity inverts every line cycle), current consumption is large. In this case, check the voltage before use.

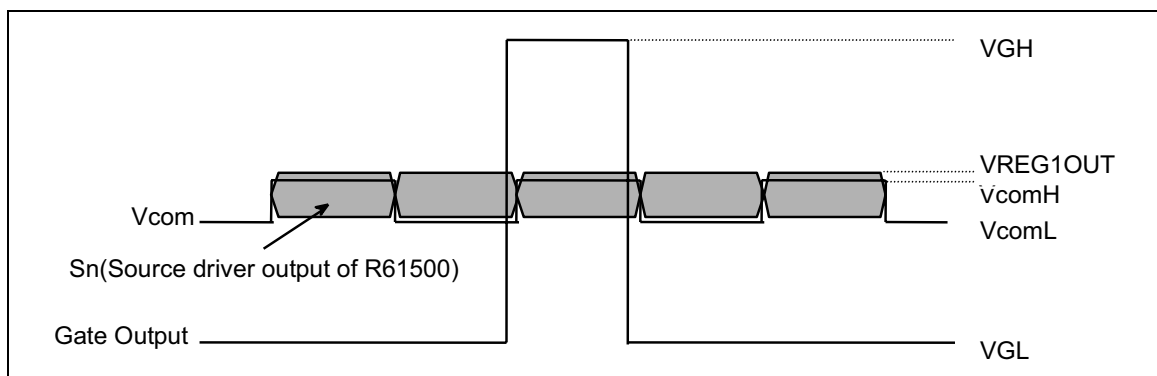


Figure 84 Applied voltage to the TFT display

Oscillator

The R61500 generates oscillation with the internal RC oscillator connected to an external oscillation resistor between the OSC1 and OSC2 pins. The oscillation frequency depends on the resistance value of external resistor, wiring distance, and the operating supply voltage. For example, when connecting an R_f resistor of a larger resistance value, or lowering the supply voltage level, the oscillation frequency will be reduced. See the Notes to Electrical Characteristics section for the relationship between the resistance value of R_f resistor and the oscillation frequency.

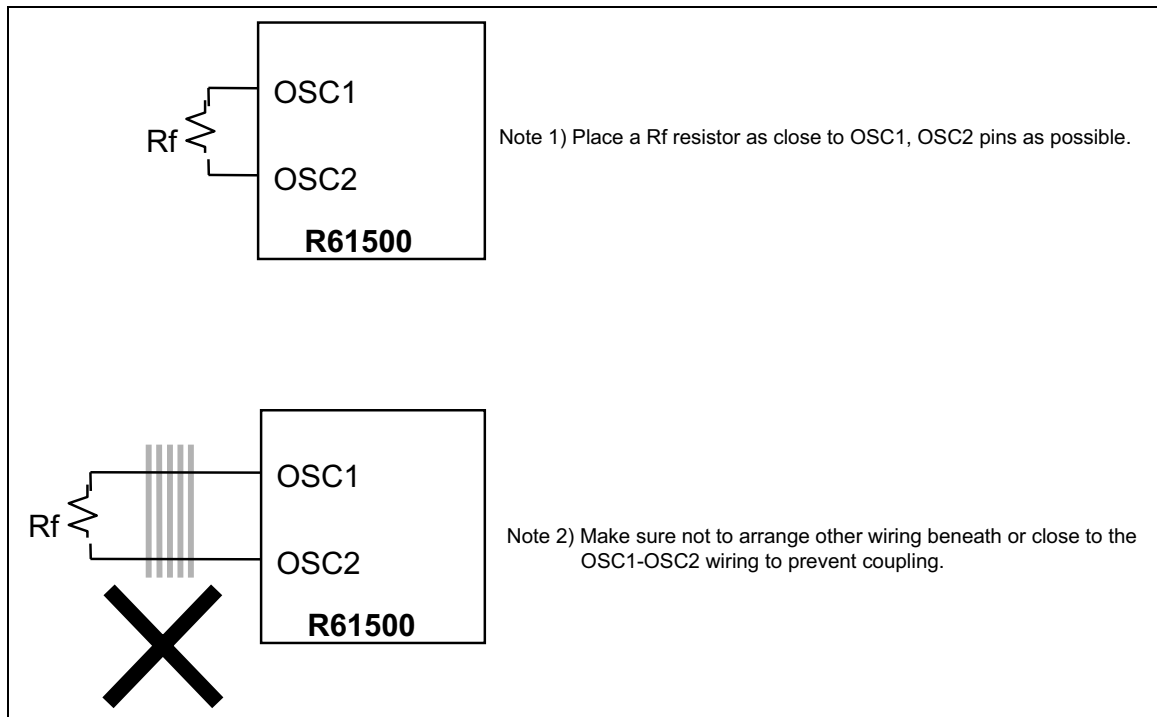


Figure 85

n-line inversion AC drive

The R61500, in addition to the frame-inversion liquid crystal AC drive, supports the n-line inversion AC drive, in which the polarity of liquid crystal is inverted in units of n lines, where n takes a number from 1 to 64. The quality of display can be improved by using n-line inversion.

In determining the number of lines for alternating cycle (n: the value represented by NW bits +1), check the quality of display on the liquid crystal panel in use. Note that setting a smaller number of lines will raise the frequency of liquid crystal polarity inversion and increase charging/discharging current on liquid crystal cells.

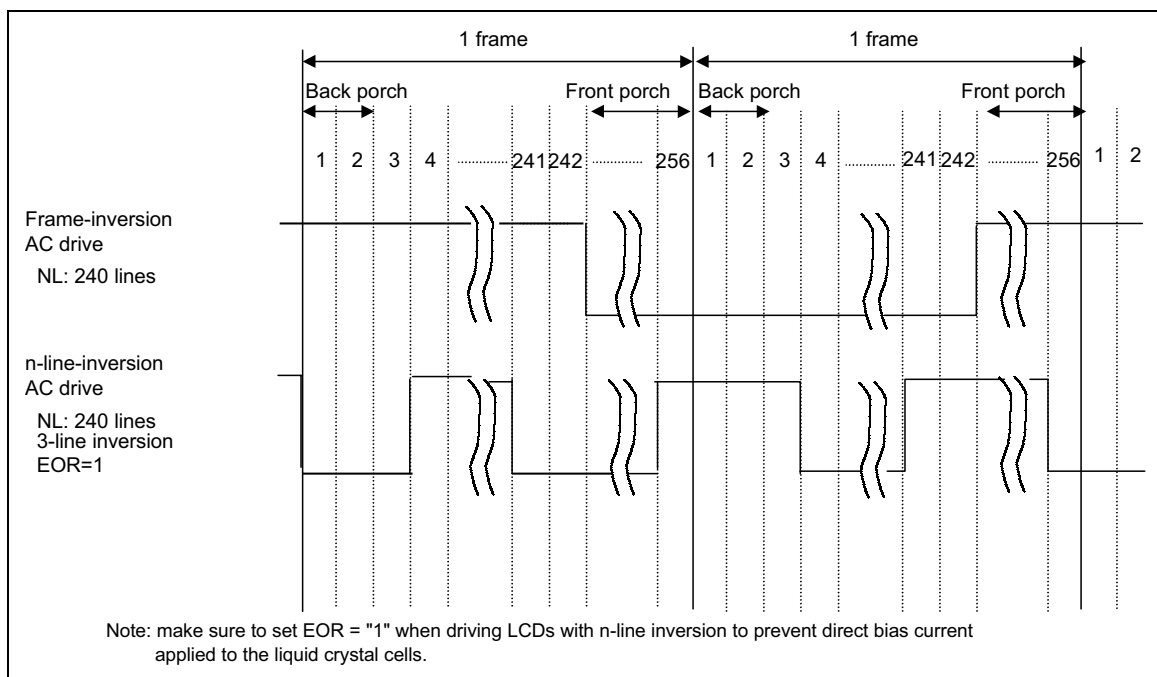


Figure 86

Interlaced scan

The R61500 supports interlaced scan to drive the LCD by splitting a frame into n fields in order to prevent flicker.

To determine the number of fields (n : value set with the FLD bits), check the quality of display on the liquid crystal panel in use. The following table shows the scanned (gate) lines in each field. When FLD1-0 = 01, the number of fields in one frame is one. When FLD1-0 = 11, the number of fields in one frame is three. The figure illustrates the output waveforms of 3-field interlaced scan.

Table 81

GS = 0					
FLD1-0	01	11			
Field	-	1	2	3	
Gate					
G1	*	*			
G2	*		*		
G3	*			*	
G4	*	*			
G5	*		*		
G6	*			*	
G7	*	*			
G8	*		*		
G9	*			*	
⋮	⋮	⋮	⋮	⋮	
G237	*			*	
G238	*	*			
G239	*		*		
G240	*			*	

GS = 1					
FLD1-0	01	11			
Field	-	1	2	3	
Gate					
G240	*	*			
G239	*		*		
G238	*			*	
G237	*	*			
G236	*		*		
G235	*			*	
G234	*	*			
G233	*		*		
G232	*			*	
⋮	⋮	⋮	⋮	⋮	
G4	*			*	
G3	*	*			
G2	*		*		
G1	*			*	

*: Scanned gate lines

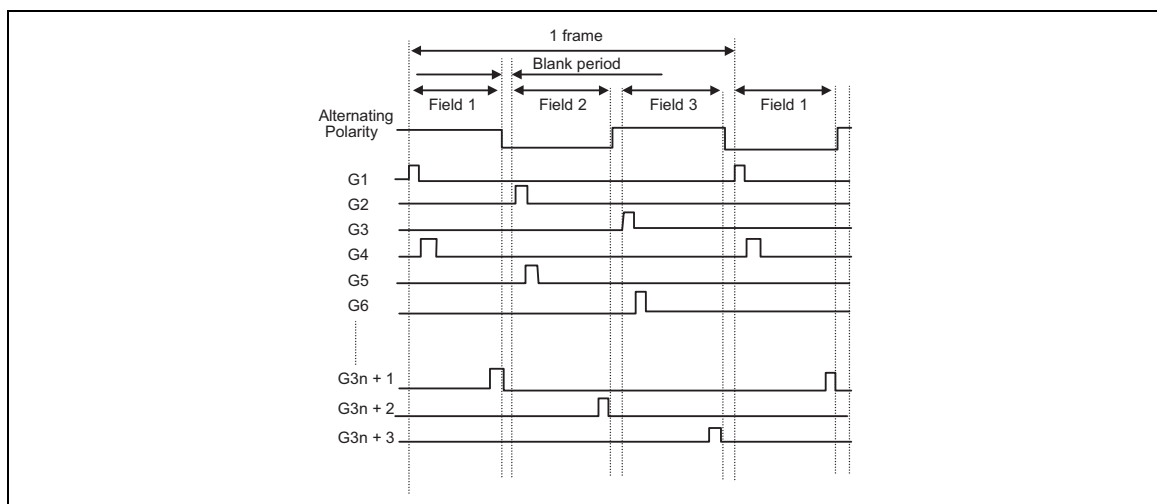


Figure 87 Gate output timing of 3-field interlaced scan

Alternating Timing

The following figure shows the timing of liquid crystal polarity inversion in different driving methods. In case of frame-inversion AC drive, the polarity is inverted after drawing one frame, followed by a blank period (all output VGL level) lasting for 16H periods. In case of n-field interlaced scan, the polarity is inverted after drawing one field, followed by a blank period, which amounts to 16H periods in one frame. In case of n-line inversion AC drive, the polarity is inverted as drawing n lines, and a blank period lasting for 16H periods is inserted after drawing one frame.

In the interlaced scan, set the periods of back and front porches to 3 line periods (BP = 3) and 5 line periods (FP = 5), respectively.

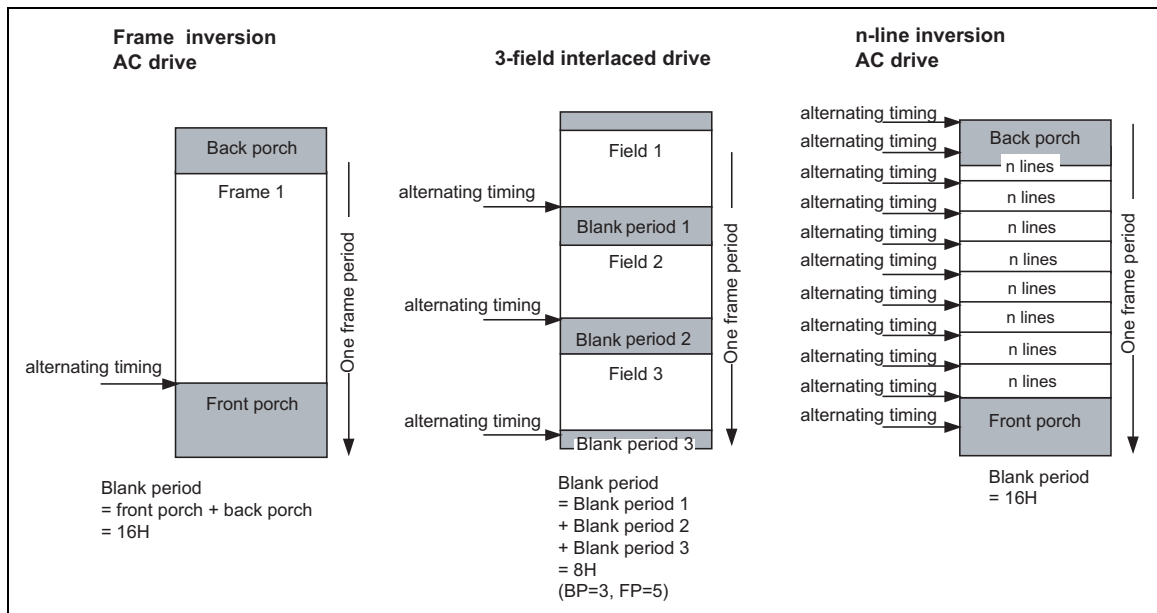


Figure 88

Frame-Frequency Adjustment Function

The R61500 supports a frame frequency adjustment function. The frame frequency for driving the LCD can be adjusted by setting the DIV, RTN bits without changing the oscillation frequency.

To switch frame frequencies between when displaying a moving picture and when displaying a still picture, set a high oscillation frequency in advance. By doing so, it becomes possible to set a low frame frequency for saving power consumption when displaying a still picture and set a high frame frequency when displaying a moving picture.

Relationship between Liquid Crystal Drive Duty and Frame Frequency

The relationship between the liquid crystal drive duty and the frame frequency is calculated from the following formula. The frame frequency is adjusted by instruction using the 1H period adjustment bits (RTN bits) and the operation clock division bits (DIV bits).

(Formula to calculate frame frequency)

$$\text{Frame frequency} = \frac{f_{\text{osc}}}{\text{Clock cycles per line} \times \text{division ratio} \times (\text{Line} + \text{BP} + \text{FP})} \quad [\text{Hz}]$$

f_{osc} : RC oscillation frequency

Line: number of lines to drive the LCD (NL bits)

Clock cycle per line: RTN bits

Division ratio: DIV bits

Number of lines for front porch: FP

Number of lines for back porch: BP

Example of Calculation: when maximum frame frequency = 60 Hz

Number of lines to drive the LCD: 240 lines

1H period: 16 clock cycle (RTN3-0 = 0000)

Operational clock division ratio: 1/1

$$f_{\text{osc}} = 60 \text{ Hz} \times (0 + 16) \text{ clock} \times 1/1 \times (240 + 16) \text{ lines} = 246 \text{ (kHz)}$$

In this case, the RC oscillation frequency is 246 kHz. Adjust the external resistor of the RC oscillator to 246 kHz.

Partial Display Function

The R61500 allows selectively driving the LCD for two displays on the screen at arbitrary positions specified by the screen-drive position registers (R42 and R43). Only the lines necessary to display two images are selectively driven in order to reduce current consumption.

The first display drive position register (R42) includes the start line setting bits (SS17-10) and the end line setting bits (SE17-10) for displaying the first image. The second display drive position register (R43) includes the start line setting bits (SS27-20) and the end line setting bits (SE27-20) for displaying the second image. The second display control is effective when the SPT bit is set to 1. The total number of lines driven for displaying the first and second displays must be less than the number of lines set with the NL4-0 bits.

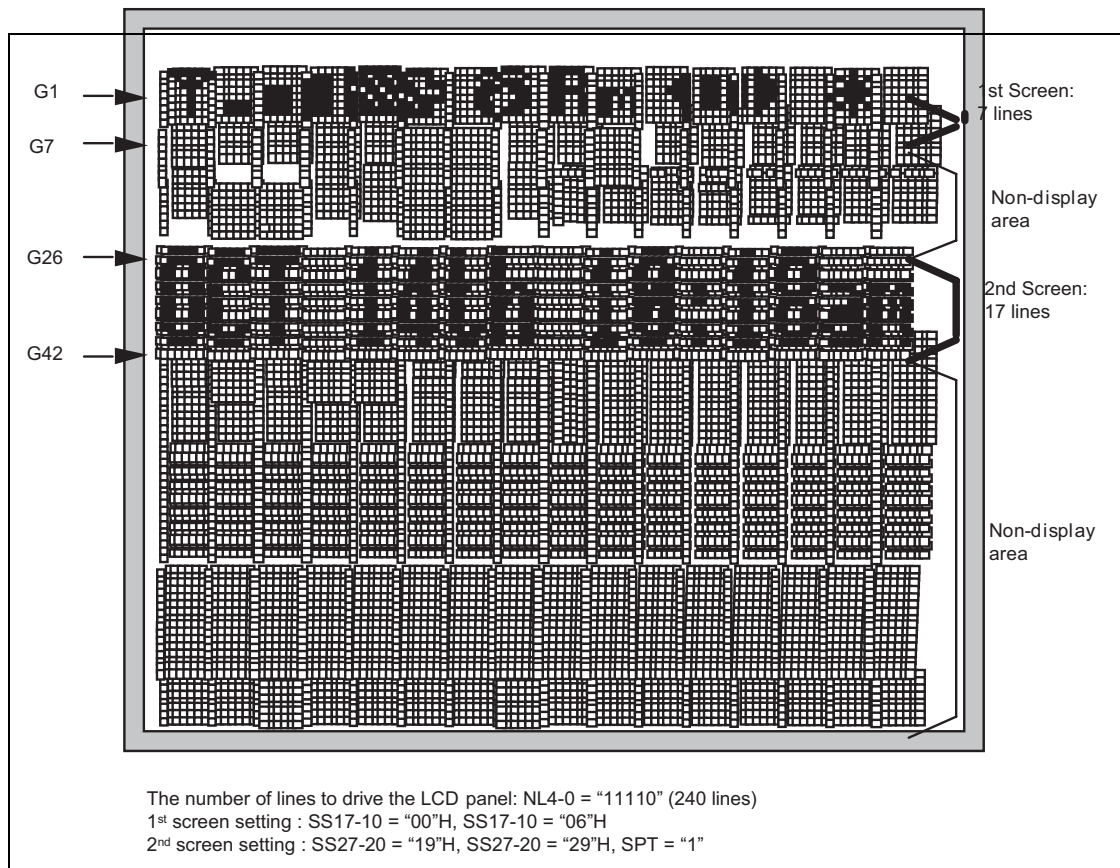


Figure 89

Constraints in Setting the 1st/2nd Screen Drive Position Register

When setting the start line setting bits (SS17-10) and the end line setting bits (SE17-10) of the first display drive position register (R42), and the start line setting bits (SS27-20) and the end line setting bits (SE27-20) of the second display drive position register (R43), it is necessary to satisfy the following conditions to display correctly.

Table 82 One Screen Drive (SPT = 0)

Register Settings	Display Operation
$(SE17-10) - (SS17-10) = NL$	Full screen display The area of $(SE17-10) - (SS17-10)$ is normally displayed.
$(SE17-10) - (SS17-10) < NL$	Partial screen display The area $(SE17-10) - (SS17-10)$ is normally displayed. The rest of the area becomes blank in white irrespective of data in RAM.
$(SE17-10) - (SS17-10) > NL$	Setting disabled

Table 83 Two Screen Drive (SPT = 1)

Register Settings	Display Operation
$((SE17-10) - (SS17-10)) + ((SE27-20) - (SS27-20)) = NL$	Full screen display The area of $(SE27-20) - (SS17-10)$ is normally displayed.
$((SE17-10) - (SS17-10)) + ((SE27-20) - (SS27-20)) < NL$	Partial screen display The area of $(SE27-10) - (SS17-10)$ is normally displayed. The rest of the area becomes blank in white irrespective of data in RAM.
$((SE17-10) - (SS17-10)) + ((SE27-20) - (SS27-20)) > NL$	Setting disabled

Note 1) Make sure that $SS17-10 \leq SE17-10 < SS27-20 \leq SE27-20 \leq "EF"H$.

Note 2) Make sure that $((SE27-20) - (SS17-10)) \leq NL$.

R61500

The output levels from the source and gate drivers in non-display area in partial display operation can be changed as follows. Select the appropriate source output level according to the characteristics of the display panel.

Table 84 **Source output level in non-display area**

PT2	PT1	PT0	Source Output Level		Operating amplifier in non-display area
			Positive Polarity	Negative Polarity	
0	0	0	V31	V0	V0 ~ V31
0	0	1	Setting disabled		-
0	1	0	GND	GND	V0 ~ V31
0	1	1	High-impedance	High-impedance	V0 ~ V31
1	0	0	V31	V0	V0, V31
1	0	1	Setting disabled		-
1	1	0	GND	GND	V0, V31
1	1	1	High-impedance	High-impedance	V0, V31

Table 85 **Gate output level in non-display area**

PTG 1	PTG 0	Gate output level	Source output level	Vcom Output level
0	0	Normal scan	Set with the PT2-0 bits	VcomH-VcomL amplitude
0	1	VGL (fixed)	Set with the PT2-0 bits	VcomH-VcomL amplitude
1	0	Interval scan	Set with the PT2-0 bits	VcomH-VcomL amplitude
1	1	Setting disabled	-	-

Follow the sequences below when making a partial display setting.

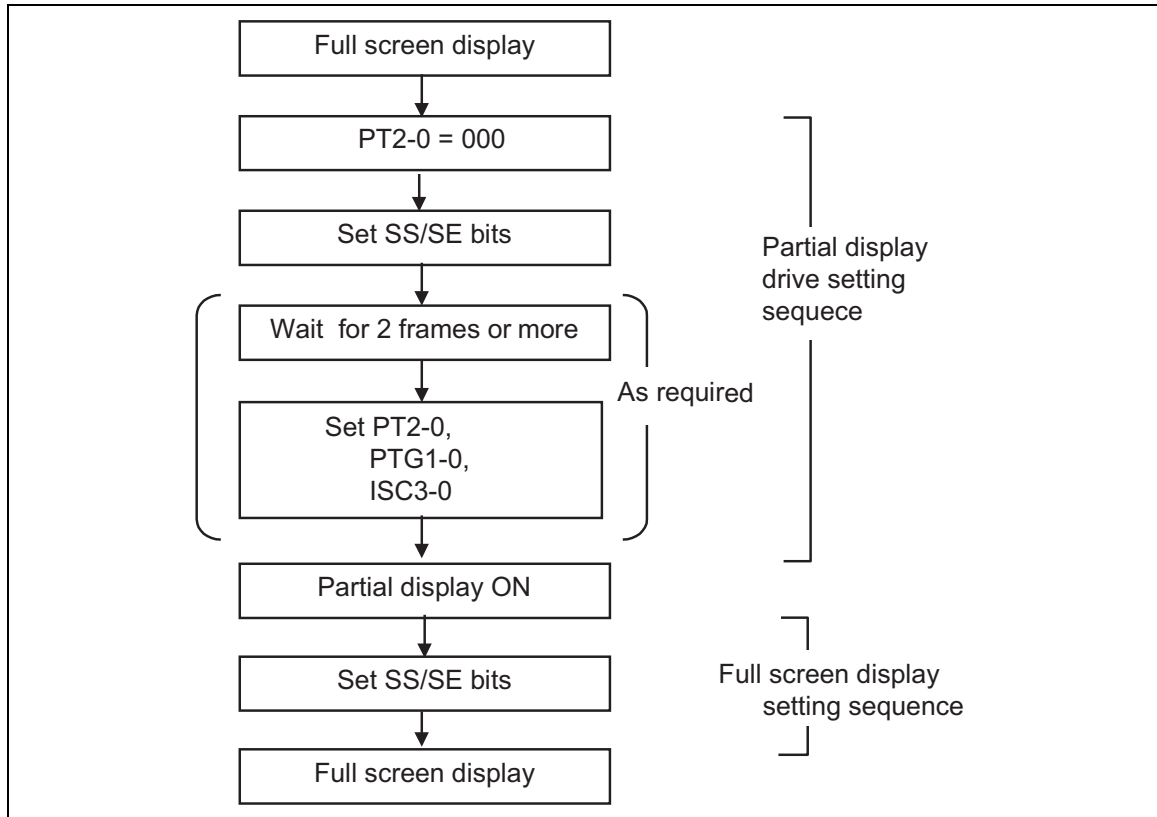
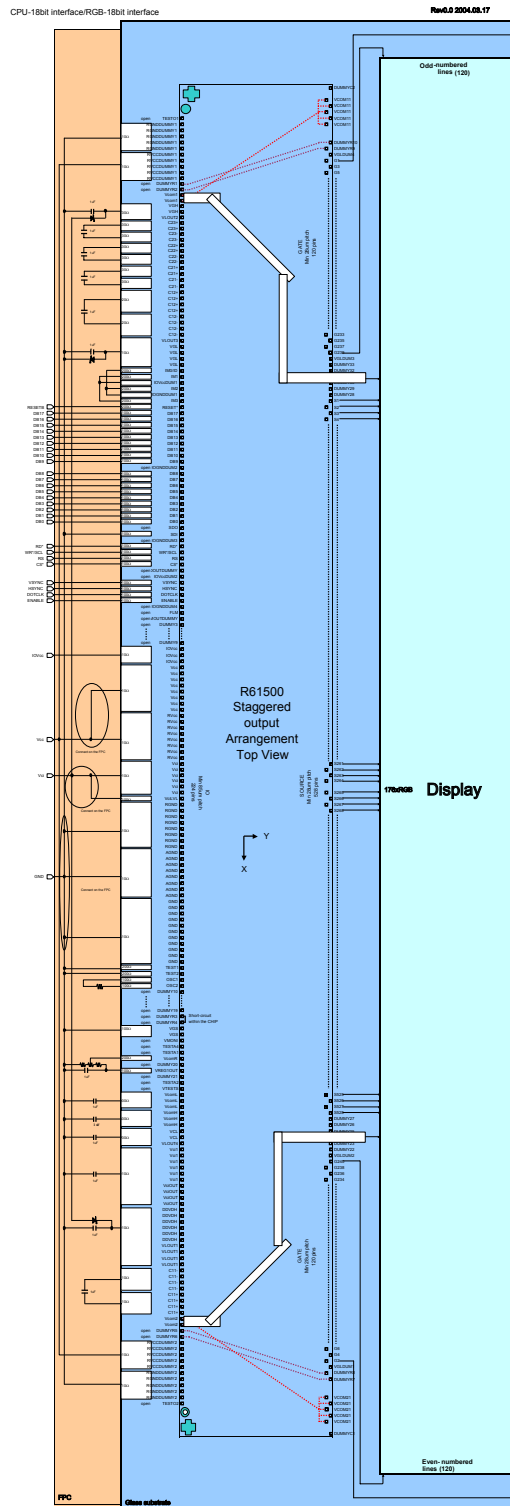


Figure 90

Panel wiring example



Absolute Maximum Ratings

Table 86

Item	Symbol	Unit	Value	Notes
Power supply voltage (1)	Vcc, RVcc, IOVcc	V	-0.3 ~ + 4.6	1, 2
Power supply voltage (2)	Vci - AGND	V	-0.3 ~ + 4.6	1, 3
Power supply voltage (3)	DDVDH - AGND	V	-0.3 ~ + 6.0	1, 4
Power supply voltage (4)	AGND - VCL	V	-0.3 ~ + 4.6	1,
Power supply voltage (5)	DDVDH - VCL	V	-0.3 ~ + 9.0	1, 5
Power supply voltage (6)	VGH - AGND	V	-0.3 ~ + 18.5	1, 6
Power supply voltage (7)	AGND - VGL	V	-0.3 ~ + 18.5	1, 7
Input voltage	Vt	V	-0.3 ~ Vcc + 0.3	1
Operating temperature	Topr	°C	-40 ~ + 85	1, 8
Storage temperature	Tstg	°C	-55 ~ + 110	1

Note 1) If used beyond the absolute maximum ratings, the LSI may permanently be damaged. It is strongly recommended to use the LSI at a condition within the electrical characteristics for normal operation. Exposure to a condition not within the electrical characteristics may affect the reliability of the device.

Note 2) Make sure (High) (RVcc = Vcc) $\geq$ GND (Low), (High) IOVcc $\geq$ GND (Low).

Note 3) Make sure (High) Vci $\geq$ GND (Low).

Note 4) Make sure (High) DDVDH $\geq$ AGND (Low).

Note 5) Make sure (High) DDVDH $\geq$ VCL (Low).

Note 6) Make sure (High) VGH $\geq$ AGND (Low).

Note 7) Make sure (High) AGND $\geq$ VGL (Low).

Note 8) The DC/AC characteristics of die and wafer products are guaranteed at 85 °C.

Electrical Characteristics

DC Characteristics

Table 87

Item	Symbol	Unit	Test Condition	Min.	Typ.	Max.	Notes
Input high-level voltage	V_{IH}	V	$IOV_{CC} = 1.65 \sim 3.6$ V	$0.8 IOV_{CC}$	-	IOV_{CC}	2, 3
Input low-level voltage	V_{IL}	V	$IOV_{CC} = 1.65 \sim 3.6$ V	-0.3	-	$0.2 IOV_{CC}$	2, 3
Output high-level voltage (1) (DB17-0, SDO, FLM)	V_{OH}	V	$IOV_{CC} = 1.65 \sim 3.6$ V $I_{OH} = -0.1$ mA	$0.8 IOV_{CC}$	-	-	2
Output low-level voltage (1) (DB17-0, SDO, FLM)	V_{OL}	V	$IOV_{CC} = 1.65 \sim 3.6$ V $I_{OL} = 0.1$ mA	-	-	$0.2 IOV_{CC}$	2
I/O leakage current	I_{Li}	μ A	$V_{in} = 0 \sim IOV_{CC}$	-1	-	1	4
Current consumption: normal operation mode, (IOV_{CC} -GND)+(V_{CC} -GND)+(RV_{CC} -GND)	I_{OP1}	μ A	RC oscillation, $f_{osc} = 315$ kHz (240 lines), $IOV_{CC}=V_{CC}=RV_{CC}= 3.0$ V, $T_a=25^{\circ}\text{C}$, RAM data: 0000h	-	155	260	5, 6
Current consumption: 8-color mode, 24-line partial (IOV_{CC} -GND)+(V_{CC} -GND)+(RV_{CC} -GND)	I_{OP2}	μ A	RC oscillation, $f_{osc} = 315$ kHz (24-line partial), $IOV_{CC}=V_{CC}=RV_{CC}= 3.0$ V, $T_a=25^{\circ}\text{C}$, RAM data: 0000h	-	70	-	6
Current consumption: RAM access mode (1) Normal write mode (HWM = 0) (IOV_{CC} -GND)+(V_{CC} -GND)+(RV_{CC} -GND)	I_{RAM1}	μ A	$t_{CYCW}=250$ ns, $IOV_{CC}=V_{CC}=RV_{CC}=2.40$ V, $T_a=25^{\circ}\text{C}$, 80-8bit I/F, $TRI=0$, Consecutive RAM access during display	-	5.0	-	6
Current consumption: RAM access mode (2) High-speed write mode (HWM = 1) (IOV_{CC} -GND)+(V_{CC} -GND)+(RV_{CC} -GND)	I_{RAM2}	μ A	$t_{CYCW}=250$ ns, $IOV_{CC}=V_{CC}=RV_{CC}=2.40$ V, $T_a=25^{\circ}\text{C}$, 80-8bit I/F, $TRI=0$, Consecutive RAM access during display	-	1.5	-	6
Current consumption: Standby mode (IOV_{CC} -GND)+(V_{CC} -GND)+(RV_{CC} -GND)	I_{ST}	μ A	$IOV_{CC}=V_{CC}=RV_{CC}=V_{ci}=3.0$ V, $T_a \leq 50^{\circ}\text{C}$	-	0.4	10	5
			$IOV_{CC}=V_{CC}=RV_{CC}=V_{ci}=3.0$ V, $T_a > 50^{\circ}\text{C}$	-	-	30	
LCD power supply current (V_{ci} -GND) 260k color display	I_{ci1}	mA	$IOV_{CC}=V_{CC}=RV_{CC}=3.0$ V, $V_{ci}=3.0$ V, $f_{osc}=315$ kHz (240 lines) $f_{FLM}=70$ Hz, $T_a=25^{\circ}\text{C}$, RAM data: "0000" h, REV=0, SAP=100, AP=100, DC0=000, DC1=010, VRN14-00=0, VRP14-00=0, PKP52-00=0, PRP12-00=0, PKN52-00=0, PRN12-00=0, B/C=0, BT=001, VC=001, VRH=0011, VCM=10011, VDV=10000, VCOMG=1, CL=0, No load on panel	-	0.7	1.2	5, 6

R61500

DC Characteristics

Item	Symbol	Unit	Test Condition	Min.	Typ.	Max.	Notes
LCD power supply current (Vci-GND) 8-color mode, 24-line partial	I _{ci2}	mA	IOVcc=Vcc=RVcc=Vci=3V, fosc=315kHz, fFLM=40Hz, Ta=25°C, RAM data: "0000"h, REV=0, SAP=010, AP=010, DC0=001, DC1=011, VRN14-00=0, VRP14-00=0, PKP52-00=0, PRP12-00=0, PKN52-00=0, PRN12-00=0, B/C=0, BT=001, VC=001, VRH=0011, VCM=10011, VDV=01110, VCOMG=1, CL=1, PTG=10, ISC=0111, No load on panel	-	0.2	-	5, 6
Output voltage difference	ΔV_o	mV	-	-	5	-	7
Average output voltage variance	ΔV	mV	-	-	-	35	8

Step-up circuit Characteristics

Table 88

Item		Unit	Test Condition	Min	Typ	Max	Notes
Step-up output voltage	VLOUT1	V	IOVcc=Vcc=RVcc=3.0V, Vci=3.0V, fOSC=400kHz, Ta=25°C, VC=001, AP=100, SAP=100, BT=000, DC0=001, C11=C12=C21=C22=C23=1uF/B characteristics VLOUT1=VLOUT2=VLOUT3=VLOUT4 =1uF/B characteristics No load on panel, Iload=-1mA	5.0	5.2	-	
	VLOUT2	V	IOVcc=Vcc=RVcc=3.0V, Vci=3.0V, fosc=400kHz, Ta=25°C, VC=001, AP=100, SAP=100, BT=000, DC0=000, DC1=010, C11=C12=C21=C22=C23=1uF/B characteristics VLOUT1=VLOUT2=VLOUT3=VLOUT4 =1uF/B characteristics No load on panel, Iload=-100μA	15.7	16.2	-	
	VLOUT3	V	IOVcc=Vcc=RVcc=3.0V, Vci=3.0V, fosc=260kHz, Ta=25°C, VC=001, AP=100, SAP=100, BT=000, DC0=000, DC1=010, C11=C12=C21=C22=C23=1uF/B characteristics VLOUT1=VLOUT2=VLOUT3=VLOUT4 =1uF/B characteristics No load on panel, Iload=+100μA	-12.9	-13.5	-	
	VLOUT4	V	IOVcc=Vcc=RVcc=3.0V, Vci=3.0V, fosc=260kHz, Ta=25°C, VC=001, AP=100, SAP=100, BT=000, DC0=000, DC1=010, C11=C12=C21=C22=C23=1uF/B characteristics VLOUT1=VLOUT2=VLOUT3=VLOUT4 =1uF/B characteristics No load on panel, Iload=+200μA	-2.1	-2.6	-	
Input voltage	Vci	V		2.5	-	3.3	

AC Characteristics

(V_{CC} = RV_{CC} = 2.40 ~ 3.60V, IOV_{CC} = 1.65 ~ 3.60V, Ta = -40 ~ +85°C*) * see Note 1

Table 89 Clock Characteristics

Item	Symbol	Unit	Test Condition	Min	Typ	Max	Notes
RC oscillation clock	f _{osc}	kHz	Rf = 150kΩ, Vcc = 3.0V	244	305	366	9

80-system Bus Interface Timing Characteristics (18/16-bit interface mode)

Table 90 Normal Write mode (HWM/LHWM= 0),
IOVcc = 1.65 ~ 3.60V, Vcc = RVcc = 2.40 ~ 3.60V

Item		Symbol	Unit	Timing diagram	Min	Typ	Max
Bus cycle time	Write	t_{CYCW}	ns	Figure 91	120	-	-
	Read	t_{CYCR}			4000	-	-
Write Low level pulse width	Write	PW_{LW}	ns	Figure 91	40	-	-
Read Low level pulse width	Read	PW_{LR}			200	-	-
Write High level pulse width	Write	PW_{HW}	ns	Figure 91	50	-	-
Read High level pulse width	Read	PW_{HR}			200	-	-
Write/Read rise/fall time		t_{WRr}, t_{WRf}	ns	Figure 91	-	-	25
Setup time	Write (RS ~ CS*, WR*)	t_{AS}	ns	Figure 91	0	-	-
	Read (RS ~ CS*, RD*)				10	-	-
Address hold time		t_{AH}	ns	Figure 91	2	-	-
Write data setup time		t_{DSW}	ns	Figure 91	25	-	-
Write data hold time		t_H	ns	Figure 91	5	-	-
Read data delay time		t_{DDR}	ns	Figure 91	-	-	100
Read data hold time		t_{DHR}	ns	Figure 91	5	-	-

Table 91 High-speed Write mode (HWM/LHWM= 1),
IOVcc = 1.65 ~ 3.60V, Vcc = RVcc = 2.40 ~ 3.60V

Item		Symbol	Unit	Timing diagram	Min	Typ	Max
Bus cycle time	Write	t_{CYCW}	ns	Figure 91	80	-	-
	Read	t_{CYCR}			400	-	-
Write Low level pulse width	Write	PW_{LW}	ns	Figure 91	40	-	-
Read Low level pulse width	Read	PW_{LR}			200	-	-
Write High level pulse width	Write	PW_{HW}	ns	Figure 91	30	-	-
Read High level pulse width	Read	PW_{HR}			200	-	-
Write/Read rise/fall time		t_{WRr}, t_{WRf}	ns	Figure 91	-	-	25
Setup time	Write (RS ~ CS*, WR*)	t_{AS}	ns	Figure 91	0	-	-
	Read (RS ~ CS*, RD*)				10	-	-
Address hold time		t_{AH}	ns	Figure 91	2	-	-
Write data setup time		t_{DSW}	ns	Figure 91	25	-	-
Write data hold time		t_H	ns	Figure 91	5	-	-
Read data delay time		t_{DDR}	ns	Figure 91	-	-	100
Read data hold time		t_{DHR}	ns	Figure 91	5	-	-

R61500

80-system Bus Interface Timing Characteristics (8/9-bit interface mode)

Table 92 Normal/High-speed write mode (HWM/LHWM= 0/1),
IOV_{CC} = 1.65 ~ 3.60V, V_{CC} = RV_{CC} = 2.40 ~ 3.60V

Item		Symbol	Unit	Timing diagram	Min	Typ	Max
Bus cycle time	Write	t _{CYCW}	ns	Figure 91	70	-	-
	Read	t _{CYCR}			400	-	-
Write Low level pulse width	Write	PW _{LW}	ns	Figure 91	40	-	-
Read Low level pulse width	Read	PW _{LR}			200	-	-
Write High level pulse width	Write	PW _{HW}	ns	Figure 91	30	-	-
Read High level pulse width	Read	PW _{HR}			200	-	-
Write/Read rise/fall time		t _{WRr} , t _{WRf}	ns	Figure 91	-	-	25
Setup time	Write (RS ~ CS*, WR*)	t _{AS}	ns	Figure 91	0	-	-
	Read (RS ~ CS*, RD*)				10	-	-
Address hold time		t _{AH}	ns	Figure 91	2	-	-
Write data setup time		t _{DSW}	ns	Figure 91	25	-	-
Write data hold time		t _H	ns	Figure 91	5	-	-
Read data delay time		t _{DDR}	ns	Figure 91	-	-	100
Read data hold time		t _{DHR}	ns	Figure 91	5	-	-

Serial interface timing characteristics

Table 93 Normal/High-speed write mode (HWM/LHWM= 0/1),
IOV_{CC} = 1.65 ~ 3.60V, V_{CC} = RV_{CC} = 2.40 ~ 3.60V

Item		Symbol	Unit	Timing diagram	Min	Typ	Max
Serial clock cycle time	Write (received)	t _{SCYC}	ns	Figure 92	100	-	20,000
	Read (transmitted)				350	-	20,000
Serial clock High level pulse width	Write (received)	t _{SCH}	ns	Figure 92	40	-	-
	Read (transmitted)				150	-	-
Serial clock Low level pulse width	Write (received)	t _{SCL}	ns	Figure 92	40	-	-
	Read (transmitted)				150	-	-
Serial clock rise/fall time		t _{scr} , t _{scf}	ns	Figure 92	-	-	20
Chip select setup time		t _{CSU}	ns	Figure 92	20	-	-
Chip select hold time		t _{CH}	ns	Figure 92	60	-	-
Serial input data setup time		t _{SISU}	ns	Figure 92	30	-	-
Serial input data hold time		t _{SIH}	ns	Figure 92	30	-	-
Serial output data delay time		t _{SOD}	ns	Figure 92	-	-	130
Serial output data hold time		t _{SOH}	ns	Figure 92	5	-	-

Reset Timing Characteristics**Table 94** (IOV_{CC} = 1.65 ~ 3.60V, V_{CC} = RV_{CC} = 1.80 ~ 3.60V)

Item	Symbol	Unit	Timing diagram	Min	Typ	Max
Reset Low level width	t _{RES}	ms	Figure 93	1	-	-
Reset rise time	t _{rRES}	μs	Figure 93	-	-	10

R61500

RGB Interface timing characteristics

Table 95 18/16-bit I/F, HWM/LHWM= 1, IOV_{CC} = 1.65 ~ 3.60V, V_{CC} = RV_{CC} = 2.40 ~ 3.60V

Item	Symbol	Unit	Timing diagram	Min	Typ	Max
VSYNC/HSYNC setup time	tSYNCS	clocks	Figure 94	0	-	1
ENABLE setup time	tENS	ns	Figure 94	10	-	-
ENABLE hold time	tENH	ns	Figure 94	20	-	-
DOTCLK Low level pulse width	PW _{DL}	ns	Figure 94	40	-	-
DOTCLK High level pulse width	PW _{DH}	ns	Figure 94	40	-	-
DOTCLK cycle time	tCYCD	ns	Figure 94	100	-	-
Data setup time	tPDS	ns	Figure 94	10	-	-
Data hold time	tPDH	ns	Figure 94	40	-	-
DOTCLK, VYSNC, HSYNC rise/fall time	Trgbr, trgbf	ns	Figure 94	-	-	25

Table 96 6-bit I/F, HWM = 1, IOV_{CC} = 1.65 ~ 3.30V, V_{CC} = RV_{CC} = 2.40 ~ 3.30V

Item	Symbol	Unit	Timing diagram	Min	Typ	Max
VSYNC/HSYNC setup time	tSYNCS	clocks	Figure 94	0	-	1
ENABLE setup time	tENS	ns	Figure 94	10	-	-
ENABLE hold time	tENH	ns	Figure 94	20	-	-
DOTCLK Low level pulse width	PW _{DL}	ns	Figure 94	30	-	-
DOTCLK High level pulse width	PW _{DH}	ns	Figure 94	30	-	-
DOTCLK cycle time	tCYCD	ns	Figure 94	80	-	-
Data setup time	tPDS	ns	Figure 94	10	-	-
Data hold time	tPDH	ns	Figure 94	30	-	-
DOTCLK, VYSNC, HSYNC rise/fall time	Trgbr, trgbf	ns	Figure 94	-	-	25

LCD driver output Characteristics

Table 97

Item	Symbol	Unit	Test condition	Min	Typ	Max	Note
Driver output delay time	t_{DD1}	μs	V _{CC} =3.0V, DDVDH=5.5V, VREG1OUT=5.0V, RC oscillation: fosc =315kHz (driving 240 lines), Ta=25°C REV=0, SAP=010, AP=010, VRN14-00=0, VRP14-00=0, PKP52-00=0, PKN52-00=0, PRP12-00=0, PRN12-00=0, Load resistance R=10k Ω , Load capacitance C=20pF Time to reach the target voltage level $\pm 35\text{mV}$ from a same grayscale level on all source pins	-	25	-	11
	t_{DD2}	μs		-	15	-	

Notes to Electrical Characteristics

1. The DC/AC electrical characteristics of bare die and wafer products are guaranteed at 85°C.
2. The following are the configurations of I pin, I/O pin, and O pin.

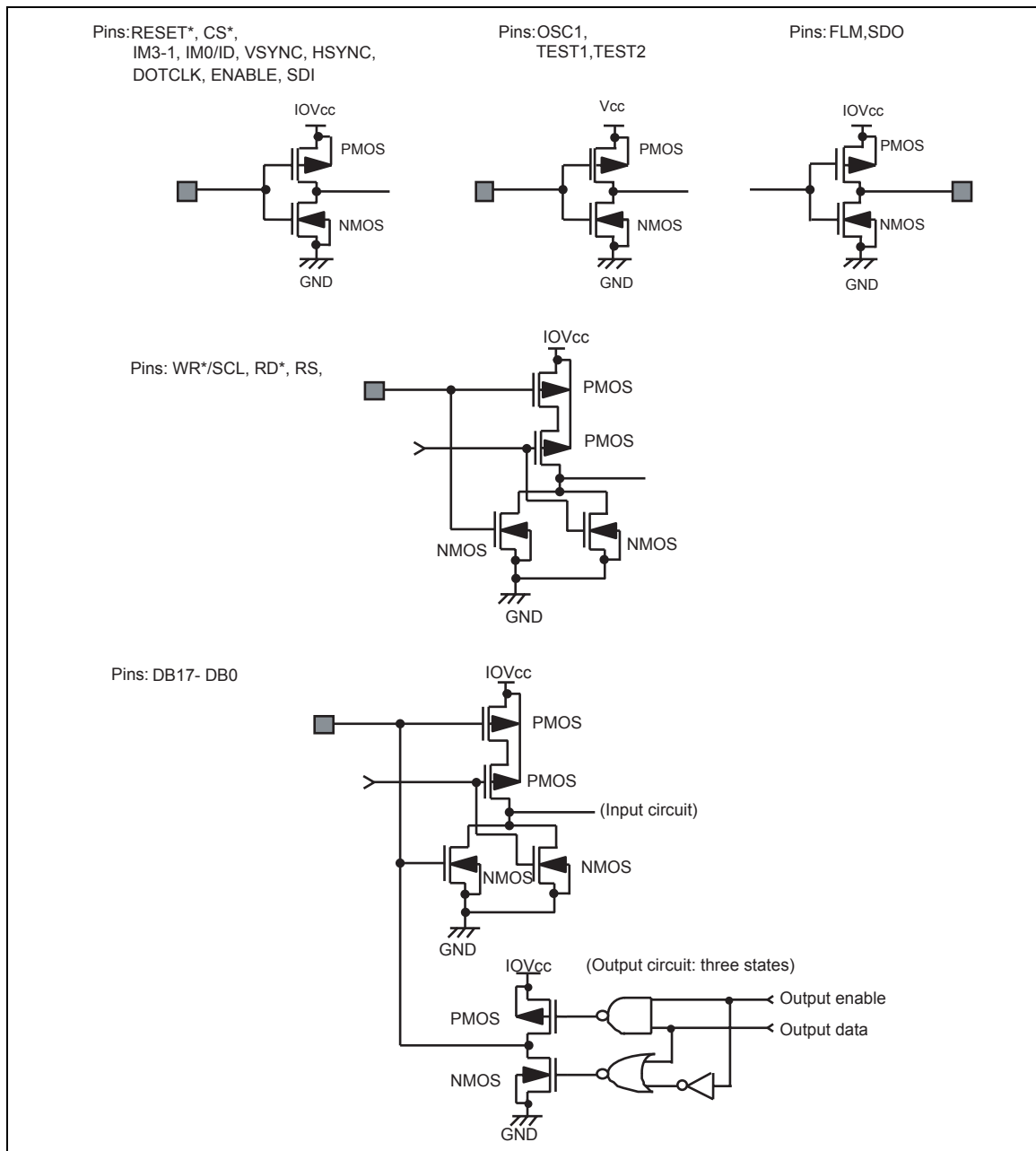


Figure 91

R61500

3. The TEST1 and TEST2 pins must be grounded (GND). The IM3/2/1, IM0/ID pins must be fixed at either GND or IOVcc level.
4. This excludes currents through the output drive MOS.
5. This excludes currents conducted through input/output units. Make sure that input levels are fixed to prevent increase in through currents in input units when a CMOS input level takes a medium range level. While not accessing via interface pins, current consumption will not change whether the CS* pin is set to "High" or "Low".
6. The relationships between the voltages and current consumption are as follows.

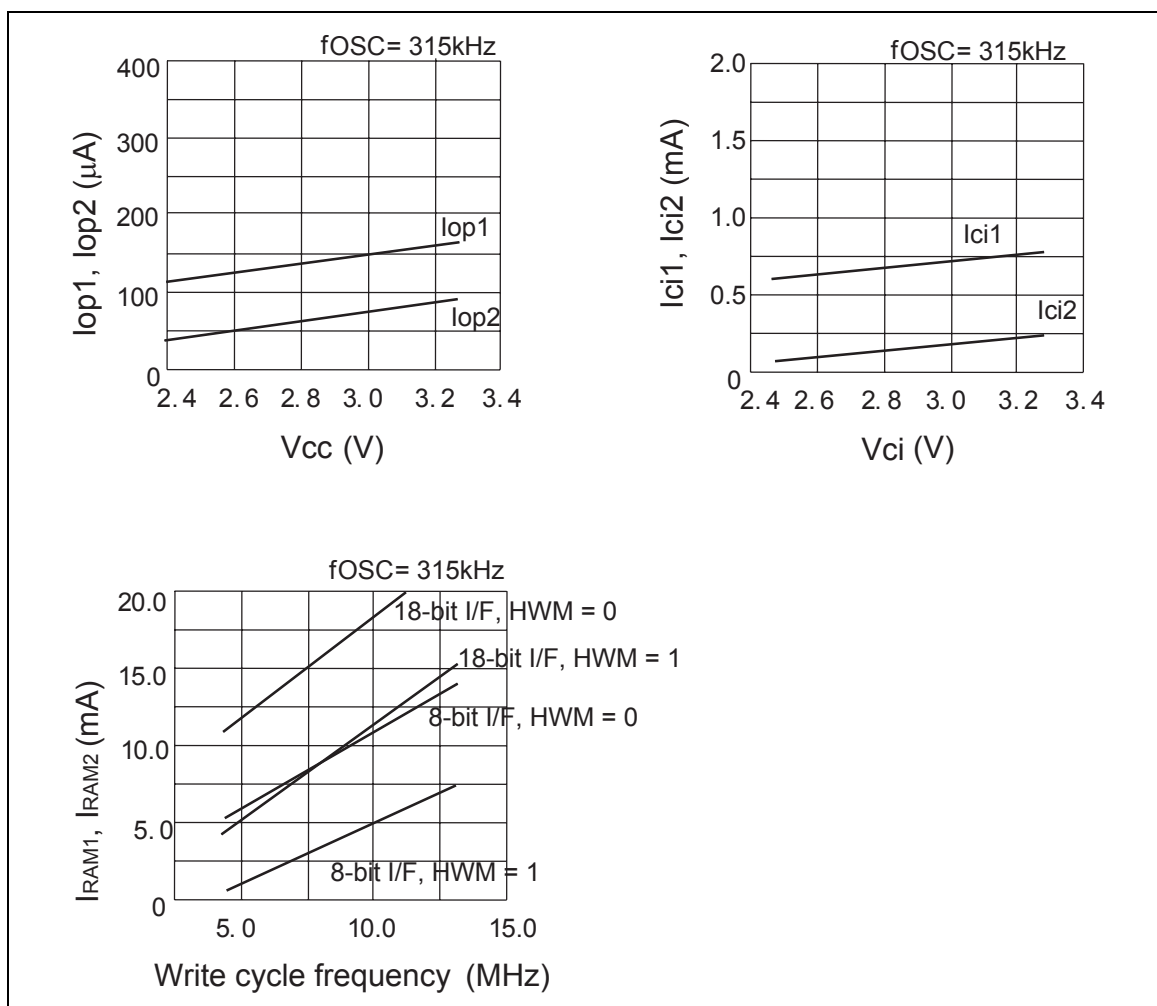


Figure 92

7. The output voltage difference is the difference in voltage levels for a same grayscale, which are output from adjacent source pins. This value is for reference.

R61500

8. The average output voltage variance is the difference in the average source output voltages of the same product. The average source output voltage is measured when all output pins output the voltage for a same grayscale.
9. This is the case when an external oscillation resistor R_f is used.

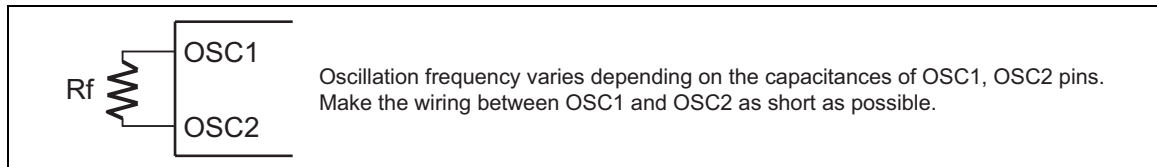


Figure 93

Table 98 Reference data, $T_a=25^\circ\text{C}$

Oscillation Resistance ($\text{k}\Omega$)	RC oscillation frequency: f_{osc} (kHz)		
	$V_{\text{CC}} = 2.4\text{V}$	$V_{\text{CC}} = 3.0\text{V}$	$V_{\text{CC}} = 3.3\text{V}$
110 $\text{k}\Omega$	372	401	411
150 $\text{k}\Omega$	284	305	311
180 $\text{k}\Omega$	243	258	263
200 $\text{k}\Omega$	222	235	240
240 $\text{k}\Omega$	188	198	202
270 $\text{k}\Omega$	169	177	181
300 $\text{k}\Omega$	153	161	163
390 $\text{k}\Omega$	121	126	128
430 $\text{k}\Omega$	110	115	116

10. The COG wiring resistance is not considered. No load is applied except pins for measurement.
11. The LCD driver output delay time varies due to the difference in LCD panel load. Check the quality of display on the actual panel when setting a frame frequency and a cycle per line.

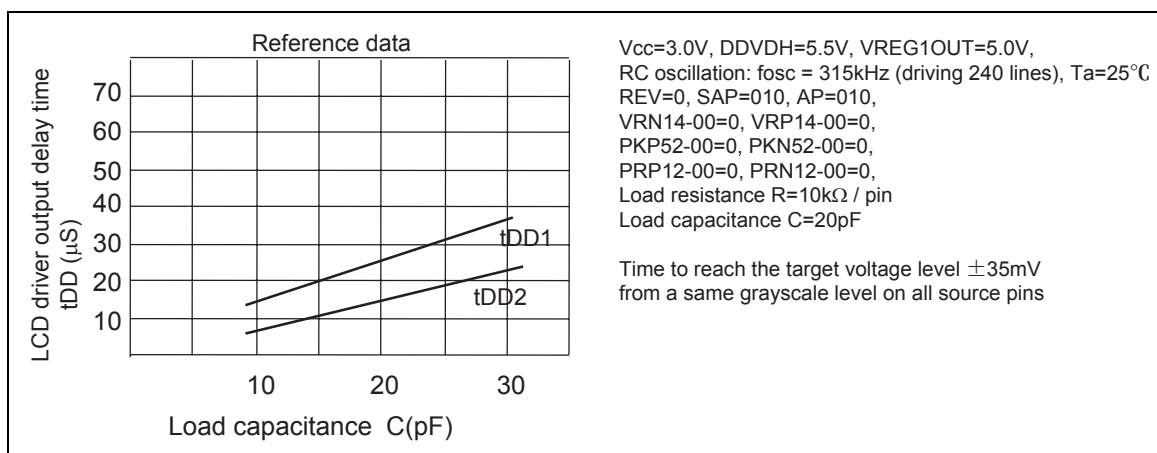


Figure 94

Example of test circuits

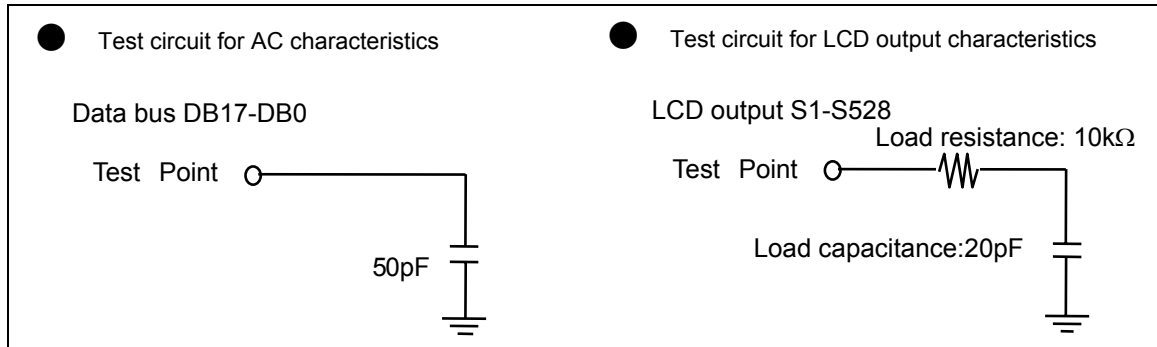


Figure 95

Timing characteristics diagram

80-system bus interface operation

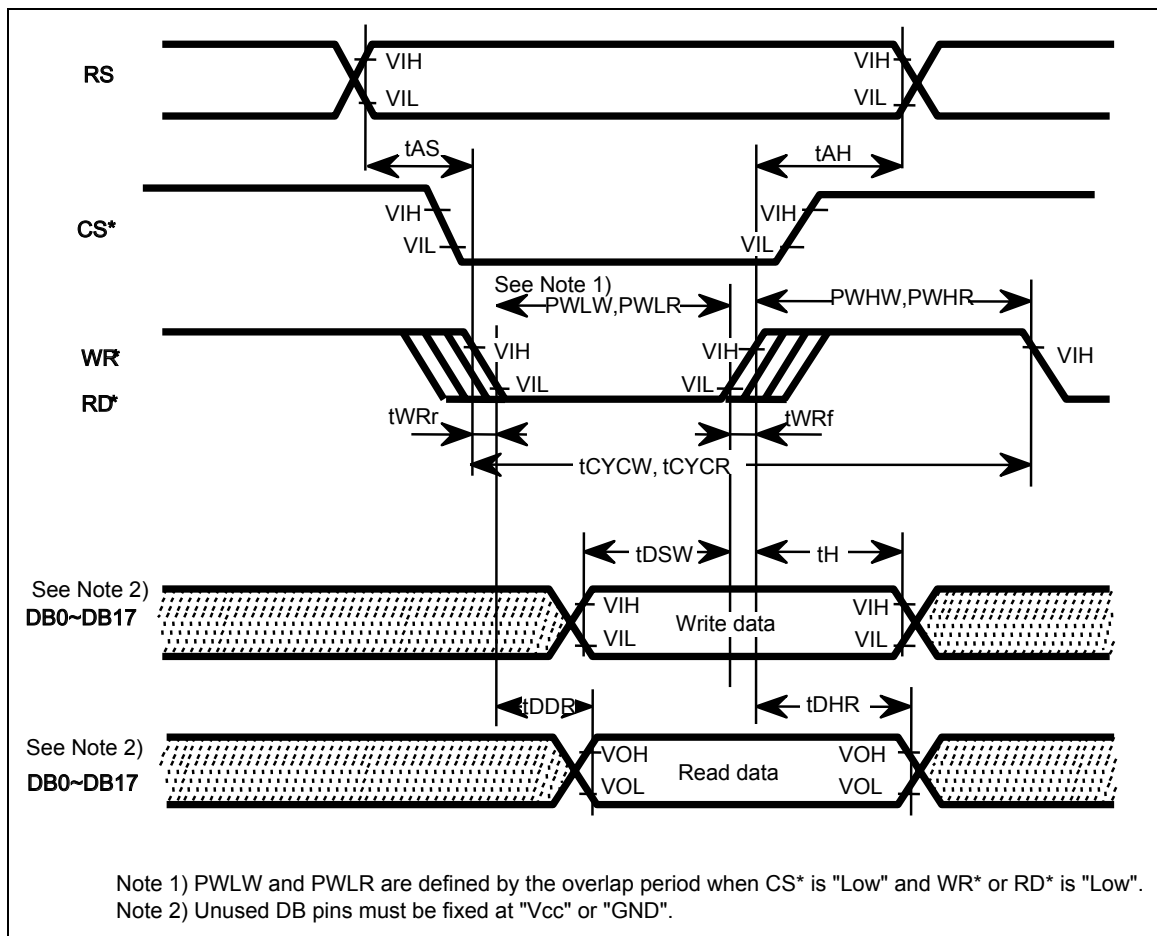


Figure 96

serial interface Operation

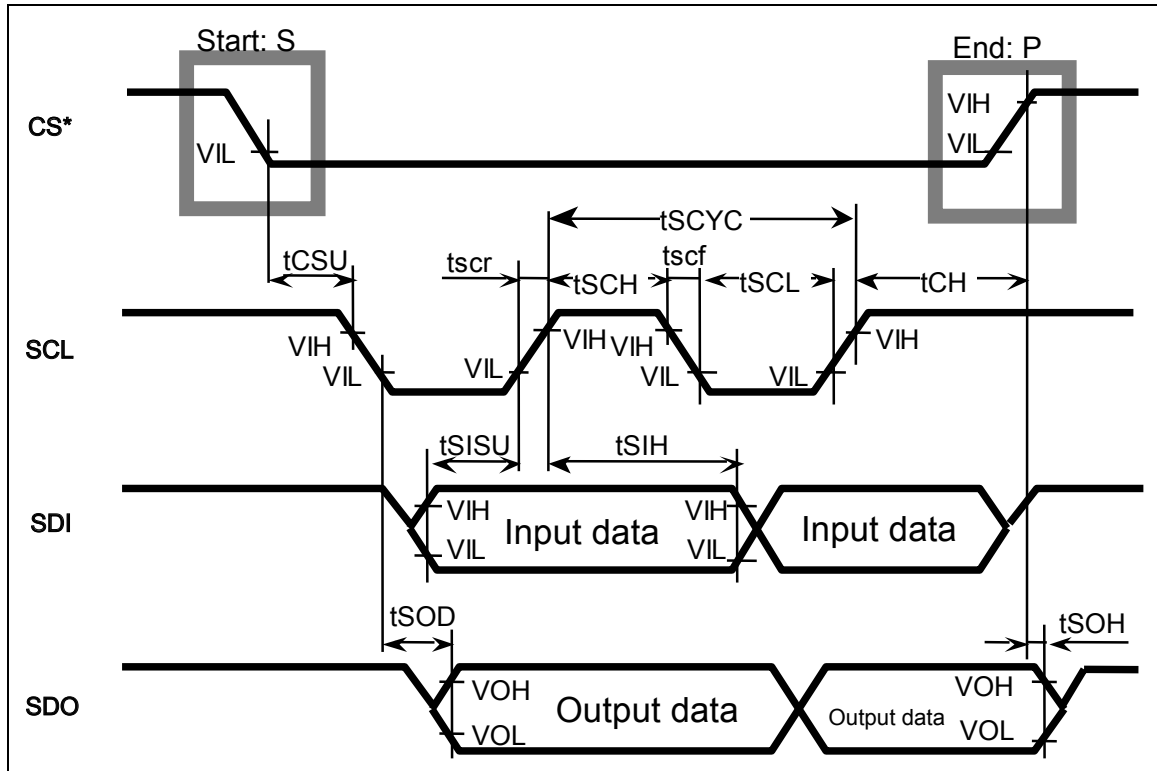


Figure 97

Reset operation

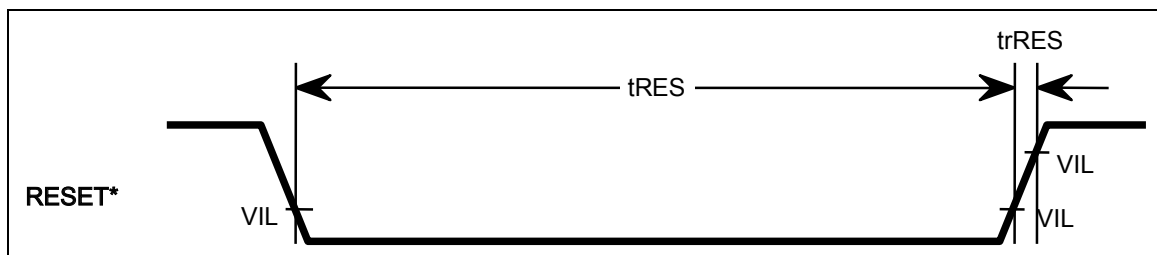


Figure 98

RGB interface

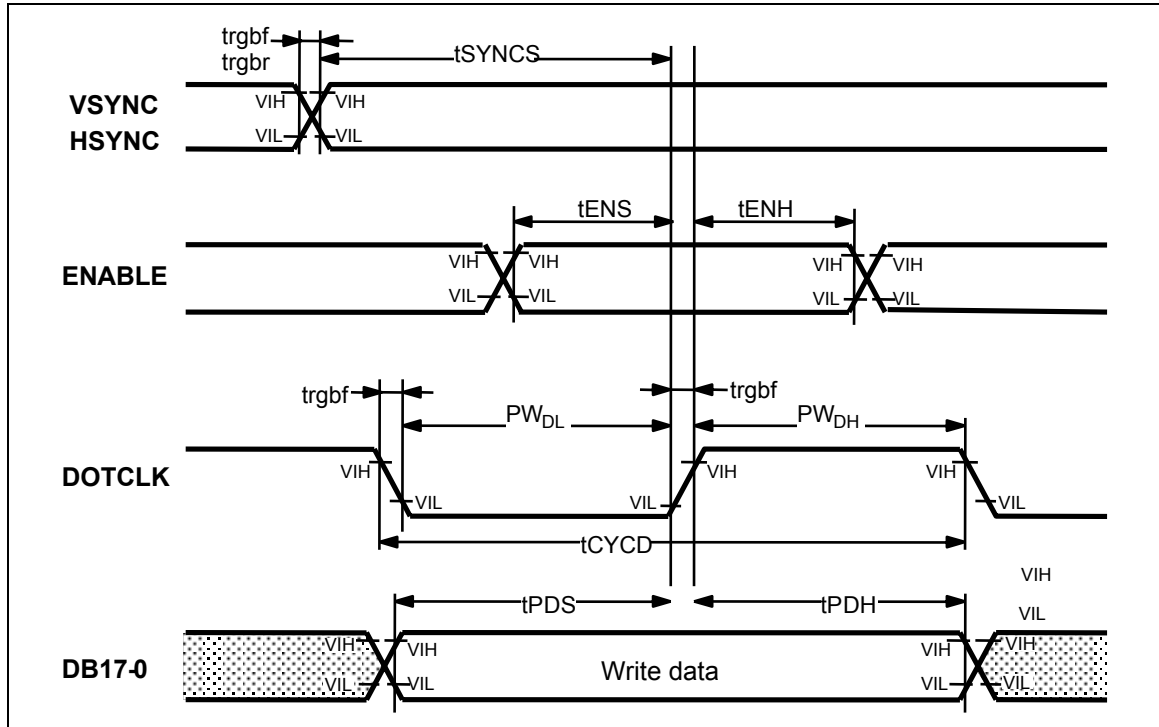


Figure 99

LCD driver outputs

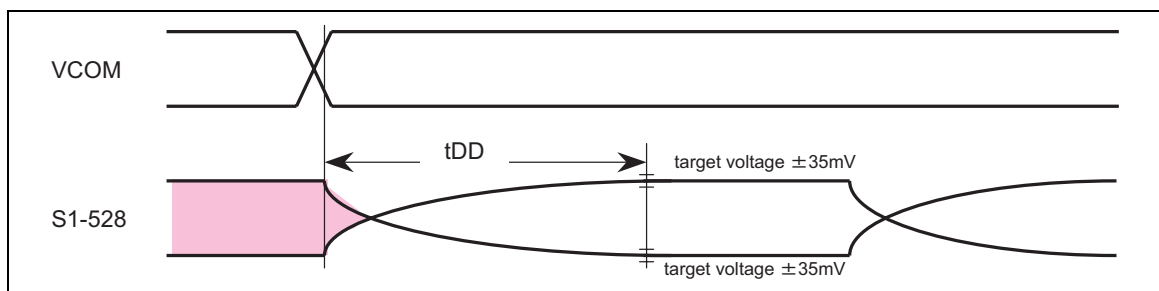


Figure 100

Renesas Technology Corp. Sales Strategic Planning Div. Nippon Bldg., 2-6-2, Ohte-machi, Chiyoda-ku, Tokyo 100-0004, Japan

Keep safety first in your circuit designs!

1. Renesas Technology Corporation puts the maximum effort into making semiconductor products better and more reliable, but there is always the possibility that trouble may occur with them. Trouble with semiconductors may lead to personal injury, fire or property damage.
Remember to give due consideration to safety when making your circuit designs, with appropriate measures such as (i) placement of substitutive, auxiliary circuits, (ii) use of nonflammable material or (iii) prevention against any malfunction or mishap.

Notes regarding these materials

1. These materials are intended as a reference to assist our customers in the selection of the Renesas Technology Corporation product best suited to the customer's application; they do not convey any license under any intellectual property rights, or any other rights, belonging to Renesas Technology Corporation or a third party.
2. Renesas Technology Corporation assumes no responsibility for any damage, or infringement of any third-party's rights, originating in the use of any product data, diagrams, charts, programs, algorithms, or circuit application examples contained in these materials.
3. All information contained in these materials, including product data, diagrams, charts, programs and algorithms represents information on products at the time of publication of these materials, and are subject to change by Renesas Technology Corporation without notice due to product improvements or other reasons. It is therefore recommended that customers contact Renesas Technology Corporation or an authorized Renesas Technology Corporation product distributor for the latest product information before purchasing a product listed herein.

The information described here may contain technical inaccuracies or typographical errors.

Renesas Technology Corporation assumes no responsibility for any damage, liability, or other loss rising from these inaccuracies or errors.

Please also pay attention to information published by Renesas Technology Corporation by various means, including the Renesas Technology Corporation Semiconductor home page (<http://www.renesas.com>).

4. When using any or all of the information contained in these materials, including product data, diagrams, charts, programs, and algorithms, please be sure to evaluate all information as a total system before making a final decision on the applicability of the information and products. Renesas Technology Corporation assumes no responsibility for any damage, liability or other loss resulting from the information contained herein.
5. Renesas Technology Corporation semiconductors are not designed or manufactured for use in a device or system that is used under circumstances in which human life is potentially at stake. Please contact Renesas Technology Corporation or an authorized Renesas Technology Corporation product distributor when considering the use of a product contained herein for any specific purposes, such as apparatus or systems for transportation, vehicular, medical, aerospace, nuclear, or undersea repeater use.
6. The prior written approval of Renesas Technology Corporation is necessary to reprint or reproduce in whole or in part these materials.
7. If these products or technologies are subject to the Japanese export control restrictions, they must be exported under a license from the Japanese government and cannot be imported into a country other than the approved destination.

Any diversion or reexport contrary to the export control laws and regulations of Japan and/or the country of destination is prohibited.

8. Please contact Renesas Technology Corporation for further details on these materials or the products contained therein.
-



<http://www.renesas.com>

Copyright © 2003. Renesas Technology Corporation, All rights reserved. Printed in Japan.
Colophon 0.0

R61500

Revision Record

Rev.	Date	Contents of Modification	Drawn by	Approved by
1.0a	2004.11.18	First issue		
